The Alaoglu–Erdős Integer-Exponent Conjecture

Machine-Checked Partial Results and Exact Conditional Structure, Determinant, Transcendence, and Approximation Methods, and the Remaining Barrier

A unified report on the corpus in

https://github.com/VladimirReshetnikov/ProveIt

Bottom line. The conjecture — that $2^x, 3^x \in \mathbb{Z}$ forces $x \in \mathbb{Z}$ — remains open, and the reduction to the four exponentials conjecture explains why no routine combination of present methods can close it. What the corpus does establish is unusually complete. Kernel-verified in Lean: a zero-free region $2^x < 4096$; transcendence of every hypothetical counterexample; multiplicative rank at most two with a canonical primitive odd core; and — conditional on a single counterexample — the *exact* classification

$$\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\}, \qquad \mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\},$$

with unique coordinates, exact dyadic block counts, a sharp quadratic cumulative asymptotic, and vanishing block Fourier modes. These reduce the problem to one narrow statement: no primitive odd $w > 1$ has a positive power of w^ϑ in $\mathbb{Z}[1/3]$.

This report adds, at paper level, two determinant methods — an arbitrary-node Vandermonde repulsion inequality and a generalized-Vandermonde hierarchy over the mixed exponent semigroup $\mathbb{N}_0 + \mathbb{N}_0\vartheta$, the latter giving an independent explicit $O((\log X)^2)$ dyadic bound — together with a diagnosis of exactly why they stop short: archimedean interpolation needs $\asymp (\log X)^2$ nodes at scale X, while a counterexample supplies only $\asymp \log X$. It also records a new sufficient condition of a different kind: if the products $\log p \log q$ of two prime logarithms are $\mathbb{Q}$-linearly independent, the conjecture follows. The present continuation further identifies the conjecture with a fixed-base case of the Matrix Coefficient Conjecture, proves an exact $K^2 - 1$ Newton-polytope interpolation threshold on the conditional $K \times K$ grid, and extends the directed finite exclusion through 2^{29}. The fourth continuation proves an all-degree obstruction to the generic normalized rational-polynomial strategy, an exact contact-equals-multiplicity theorem, and an arithmetic Müntz certificate, leaving global adelic cancellation as the viable approximation target.

August 21, 2026

Abstract

Let $\vartheta = \log_2 3$. The Alaoglu–Erdős conjecture asserts that a real x with $2^x, 3^x \in \mathbb{Z}$ must be an integer; equivalently, the only positive integers m with $m^\vartheta \in \mathbb{Z}$ are the powers of two. It is the first unresolved 2×2 instance of the four exponentials problem, while its three-base analogue follows from six exponentials and is formalized here by an interpolation determinant.

This unified report has four purposes. First, it catalogues the machine-checked corpus with exact Lean identifiers: zero-free regions, the Gelfond–Schneider dichotomy, multiplicative rank two, the canonical primitive odd core, the rational-power index and radical degree, localized-radical equivalences, arbitrary-order arithmetic-progression obstructions, shrinking-target Diophantine bounds, fractional-part gap equivalences, and the exact conditional structure theory — the free monoid $\mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0\beta$ with a unique least generator, exact block counts, and the sharp quadratic asymptotic. Second, it develops two new determinant methods. Arbitrary candidate tuples on the lattice curve $y = x^\vartheta$ satisfy

$$\prod_{0 \le i < j \le r} (m_j - m_i) \;\ge\; \frac{r!}{|\,(\vartheta)_{\underline{r}}\,|}\, m_0^{r-\vartheta},$$

with no arithmetic-progression hypothesis; and the mixed exponent semigroup supplies positive integer generalized Vandermonde determinants whose repulsion exponent tends to one, yielding the explicit unconditional bound $\#(\mathcal{C} \cap [X, 2X]) \le 10000 \max\{1, \log X\}^2$ by a route independent of the prime-valuation rank theorem. Third, it quantifies the remaining gap: the determinant hierarchy misses the conditional population by exactly one factor of $\log X$. Fourth, it records new equivalent and sufficient conditions — block-growth dichotomies (including the observation that the conjecture is equivalent to the dyadic block count being 1 for *infinitely many* exponents), a compact synchronized S-integral orbit formulation, and the prime-log-product independence criterion — and closes with a prioritized research program and a formalization blueprint.

A third continuation connects the problem to the Matrix Coefficient Conjecture of Dasgupta–Kakde. On the conditional $K \times K$ multiplicative grid it proves that the minimum Bernstein–Kushnirenko degree of a nonzero rational vanishing polynomial is exactly $K^2 - 1$; in the support-based Matrix Coefficient strategy this exposes an exact factor-four gap. The directed-rounding computation is independently extended through 2^{29}, so every hypothetical nonintegral solution satisfies $x > 29$ at the stated software-trust level.

A fourth continuation investigates uniform, rational, integer-valued, Müntz, Padé, and Hermite–Padé approximation with every coefficient denominator included in the arithmetic budget. It proves that the generic normalized rational-polynomial forcing inequality is impossible in every degree for dyadic blocks $T \ge 4$; that rational contact with the irrational-power curve at a positive rational point equals ordinary algebraic multiplicity; and that a norm-below-one integral Müntz polynomial gives an exact candidate-count certificate. Direct-scale approximation is shown to require degree linear in T, far above the conditional block population. The recent 2024–2026 literature on constrained Remez algorithms, effective Hermite–Padé methods, transcendence measures, and arithmetic holonomy is audited. No complete proof is claimed: the remaining approximation target is a genuinely adelic auxiliary residual whose smallness comes from global arithmetic cancellation.

Contents

1 Executive summary

1.1 What is proved, and at what level of trust

The full conjecture remains open. Several routes were pushed to their exact failure points: the three-base determinant method does not shrink to two bases without proving a case of the four exponentials conjecture; equidistribution of multiples does not defeat exponentially shrinking integrality gaps; local curvature obstructions cannot reach the sparse global structure; rational approximation to a hypothetical least generator stalls at the same transcendence input; and every successful reduction terminates at the same transcendence-theoretic wall, isolated in Section 6. Two lines of attack opened after that survey – interpolation determinants on arbitrary candidate nodes, and generalized Vandermonde determinants over the full semigroup of integral mixed monomials – do produce new unconditional partial results, and a third, the prime-log-product criterion of Section 12, isolates a clean sufficient condition of a different flavor. None of them closes the final gap.

To keep unlike kinds of evidence separate, every important statement in this report carries one of five labels.

[kernel-verified Lean]
A proof source is present in the repository, lies on the audited `ExponentialIdentities.lean` import surface, and has been accepted by the Lean kernel in direct or full builds of the cited modules (toolchain 4.31/4.32 series, Mathlib pinned by the repository manifest). No `sorry`, no extra axioms, no `native_decide`.

[paper proof in this report]
A conventional mathematical proof is supplied in this document. It is intended to be suitable for formalization, but no claim of kernel checking is made.

[Lean draft, not yet accepted]
A Lean source for the statement exists in the repository tree, but it has *not* yet been accepted by the kernel at the pinned state. A draft is weaker evidence than **[paper proof in this report]** prose backed by a written proof, not stronger: it records intent to formalize, nothing more.

[interval computation]
A finite statement was checked by outward-rounded interval arithmetic. The script, parameters, worst margins, and manifest hashes are recorded in Section 25 and Appendix F. The trusted base includes CPython and `mpmath`; this is not interchangeable with a Lean certificate.

[classical/open background]
A classical theorem quoted from the literature.

An earlier audit, performed against `423ac24`, correctly observed that three modules cited by the survey — `FiniteCheck4096`, `OddCore` and `FractionalPartDensity` — were absent then. That discrepancy is resolved: the three modules were subsequently committed in `98646b455` and are kernel-verified; the survey's temporary pending-verification remark was removed in `716617d72`. Statements from these modules are therefore labeled [kernel-verified Lean] below. The kernel-checked feature wave also includes

`PrimitiveOddCore`, `LeastSolution`, `ThreeDenominatorNormalization`, `RationalPowerIndex`, `PrimitiveGenerator`, `ExactCounting`, `QuadraticAsymptotic`, `BlockWeyl`, `BlockEquidistribution`, `OutputNormalization`, `RationalThirdBase`, `RadicalReformulation`, `PrimitiveRadical`, `RadicalDegree`, `SharperProgressions`, `ThirdDifference`, `HigherDifference`.

The determinant and reformulation modules that earlier editions of this report listed as drafts have since been accepted, and a further round added

`CandidateLattice`, `KernelDichotomy`, `SecondIterateKernel`, `TowerRigidity`, `SymmetricTowerConstants`, `PrimeLogSpaceOperator`, `ArbitraryNodeRepulsion`, `SemigroupGapBound`.

Public headline theorems throughout were audited with `##print axioms`; only the standard Mathlib axioms `propext`, `Classical.choice`, and `Quot.sound` occur, and the audit for the most recent round is itself a module, `AuditNewModules`. The two analytic inputs still missing from the determinant chain are carried as named hypotheses of the theorems that use them, not as axioms. The per-statement ledger is Appendix A and the module inventory is Appendix B.

1.2 Headline results

1. **Verified zero-free region.** If $2^x, 3^x \in \mathbb{Z}$ and $2^x < 4096$, then $x \in \mathbb{Z}$; equivalently every nonintegral solution has $x \geq 12$. [kernel-verified Lean] A directed-rounding scan, including the third continuation, extends the finite barrier through $2^x < 2^{29}$, i.e. $x > 29$. [interval computation]
2. **Transcendence.** Every nonintegral solution is transcendental; for x with $2^x \in \mathbb{Z}$, algebraicity of x is *equivalent* to integrality. $\vartheta = \log_2 3$ is transcendental. [kernel-verified Lean]
3. **Exact conditional structure.** Conditional on a counterexample there is a unique gcd-normalized odd core $w > 1$, every candidate has unique coordinates $2^i w^j$, and the solution set has the exact free-monoid classification $\mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0\beta$ with a unique least nonintegral generator β, which is affinely primitive. Candidate counts below N are at most $c\log_2 N \cdot c\log_3 N$, each dyadic block $[2^T, 2^{T+1})$ holds at most $c\log_3(2^{T+1})$ candidates, the exact block count is $\lfloor (N+1)/\beta \rfloor$, the cumulative count satisfies $C(T)/T^2 \to 1/(2\beta)$, and the conjecture is equivalent to subquadratic growth along 2^T. Exact block averages of every finite trigonometric polynomial converge to its constant Fourier coefficient. [kernel-verified Lean] The extension to all continuous tests and interval indicators remains paper-level. [paper proof in this report]
4. **Unconditional gap equivalences.** The conjecture is equivalent to the existence of any nonempty open subinterval of $(0, 1)$ free of fractional parts $\{x\}$ of nonintegral solutions, and to those fractional parts being bounded away from 0, or from 1; the convergent denominators of ϑ obey $q_{n+1} < 2m^{q_n}$. [kernel-verified Lean]

5. **Sharper local combinatorics.** For every $r \geq 2$, the sharp criterion $|(\vartheta)_{\underline{r}}|\, h^r < n^{r-\vartheta}$ forbids $r+1$ candidates in arithmetic progression, uniformly in the progression length. The underlying arbitrary-order forward-difference mean-value identity is formalized as well, and for the three-term case the rational effective criterion $h^{400} \leq n^{83}$ (step exponent $83/400 = 0.2075$) is verified independently. **[kernel-verified Lean]**

6. **New: arbitrary-node repulsion and the semigroup hierarchy.** Any $r+1$ distinct natural candidates – with no arithmetic-progression hypothesis – obey a quantitative product-of-gaps lower bound, giving explicit local packing bounds and a best ordinary-power exponent $0.241503\ldots$ at determinant order 4 (Section 19). Enlarging the exponent set to the mixed semigroup $\mathbb{N}_0 + \mathbb{N}_0\vartheta$ yields arbitrarily large positive integer generalized Vandermonde determinants, a near-linear repulsion exponent $1 - O(r^{-1/2})$, and the unconditional dyadic bound $\#(\mathcal{C} \cap [X, 2X]) \leq 10000 \max\{1, \log X\}^2$ (Section 20). This route is independent of the repository's prime-valuation rank theorem. **[paper proof in this report]** Lean sources `ArbitraryNodeDeterminant`, `MixedExponentSemigroup`, `SemigroupDeterminant`, `FallingFactorialBound`, and `SemigroupWeightBound` exist but are not yet accepted. **[Lean draft, not yet accepted]**

7. **New: a sufficient condition from products of prime logarithms.** Suppose $2^x = A \in \mathbb{Z}$ and $3^x = M \in \mathbb{Z}$. Eliminating x gives the exact identity

$$\log A \, \log 3 = \log M \, \log 2.$$

Expanding A and M into prime factorizations turns this into an integer-coefficient linear form in the pairwise products $\log p \, \log q$ of prime logarithms, and the form is nontrivial precisely when $x \notin \mathbb{Z}$. Hence: *if the products* $\log p \, \log q$*, over unordered pairs of primes, are linearly independent over* $\mathbb{Q}$*, the conjecture holds* (Section 12). That independence is a consequence of Schanuel's conjecture. It is different in kind from the four exponentials conjecture – it constrains a bilinear lattice of logarithm products rather than a 2×2 logarithm determinant – so it supplies a genuinely separate route to the same conclusion, not a reformulation of the same wall. **[paper proof in this report]** A Lean module `LogProductIndependence` is in preparation and has not yet been accepted by the kernel. **[Lean draft, not yet accepted]**

8. **The irreducible remainder.** The conjecture is equivalent to: no odd integer $u > 1$ satisfies $u^\vartheta \in \mathbb{Z}[1/3]$, with a symmetric base-swapped formulation and an equivalent reduction to canonical power-primitive odd bases. The least radical index gives an irreducible binomial and the exact algebraic degree. **[kernel-verified Lean]** The exclusion itself remains open even for $u = 3$.

1.3 The strongest new conclusions

Determinant hierarchy (Sections 19 and 20). For every $r \geq 2$, any $r+1$ distinct natural candidates obey a quantitative product-of-gaps lower bound, with no arithmetic-progression hypothesis. Enlarging the exponent set from ordinary powers to the mixed semigroup $\mathbb{N}_0 + \mathbb{N}_0\vartheta$ supplies arbitrarily large positive integer generalized Vandermonde determinants; their archimedean upper bounds give a near-linear repulsion exponent $1 - O(r^{-1/2})$ and, unconditionally,

$$\#(\mathcal{C} \cap [X, 2X]) \ll (\log X)^2,$$

with the explicit constant 10000 proved below. Paper proof in this report; a Lean formalization is drafted but not yet accepted. **[paper proof in this report] [Lean draft, not yet accepted]**

The one-logarithm deficit (Section 22). To make the semigroup determinant nearly force a gap comparable with X, one needs $r \asymp (\log X)^2$ candidate nodes. Conditional on a counterexample, the exact ProveIt classification supplies only $\asymp \log X$ candidates in $[X, 2X]$. Purely archimedean interpolation, at its present strength, therefore misses the conditional population by exactly one power of $\log X$. A successful refinement must either reduce the determinant's node requirement to $O(\log X)$ or extract additional nonarchimedean size from the same nodes. **[paper proof in this report]**

1.4 Why this is not a proof

The conjecture is already a special integral instance of the real four exponentials conjecture. In the natural 2×2 matrix

$$\begin{pmatrix} 1 \cdot \log 2 & 1 \cdot \log 3 \\ x \log 2 & x \log 3 \end{pmatrix},$$

the four exponentials are $2, 3, 2^x, 3^x$. For a nonintegral solution, both the row pair and the column pair are rationally linearly independent, yet all four exponentials are algebraic. The conclusion needed here is therefore exactly what four exponentials predicts and what the known six exponentials theorem does not supply in a 2×2 configuration.

The additional integrality hypotheses are stronger than mere algebraicity, and they do yield the new determinant inequalities of Sections 19 and 20 and the bilinear criterion of Section 12. They do not, by themselves, produce enough nodes, enough lower-bound divisibility, or the required independence of logarithm products to cross the four-exponentials threshold. Conjecture 2.1 therefore remains marked open throughout this report.

2 The problem and its equivalent forms

2.1 Original statement

Conjecture 2.1 (Alaoglu–Erdős, two-base integer form)**.** *For every $x \in \mathbb{R}$,*

$$2^x \in \mathbb{Z} \quad \textit{and} \quad 3^x \in \mathbb{Z} \qquad \Longrightarrow \qquad x \in \mathbb{Z}.$$

Status: recorded in Lean as the proposition `AlaogluErdosConjecture`, without being asserted.

Since 2^x and 3^x are positive, the hypotheses actually put them in $\mathbb{N}$; and $2^x \in \mathbb{N}$ forces $x \geq 0$, since a negative exponent gives $0 < 2^x < 1$. All solutions therefore live in $[0, \infty)$ (`nonneg_of_two_rpow_integer`), and "integer" may be replaced throughout by "positive natural number."

Historical placement. Questions of this type go back to the 1944 study of colossally abundant numbers by Alaoglu and Erdős [1], where the arithmetic nature of simultaneous rational powers p^x, q^x of distinct primes controls the quotients of consecutive colossally abundant numbers — a theme reaching back to Ramanujan's superior highly composite numbers [2]. Alaoglu and Erdős isolated the two-prime assertion explicitly, regarded it as very likely, could not prove it, and reported a communicated three-prime result of Siegel. In modern language Siegel's case is supplied by the six exponentials theorem, whereas the two-base statement sits at the unresolved 2×2 threshold of the four exponentials conjecture (Schneider's first problem [9]; see Waldschmidt [10, 13]). The elementary-looking integrality question is thus one of the canonical concrete tests of the gap between six and four exponentials, not an isolated recreational puzzle. As of the dated literature check for this report, the four exponentials conjecture remains open.

2.2 Candidate-base normalization

Put

$$\vartheta := \log_2 3 = \log_2 3 = 1.584962500721156\ldots$$

and define the *candidate set*

$$\mathcal{C} := \{m \in \mathbb{N} : m > 0,\ m^\vartheta \in \mathbb{N}\}.$$

Lemma 2.2 (Exponent–candidate bijection)**.** *The maps $x \mapsto 2^x$ and $m \mapsto \log_2 m$ are mutually inverse bijections between $\mathcal{S} := \{x \in \mathbb{R} : 2^x, 3^x \in \mathbb{Z}\}$ and $\mathcal{C}$. Under the bijection, $3^x = m^\vartheta$.*

Proof. If $x \in \mathcal{S}$, set $m = 2^x \in \mathbb{N}$. Then $m^\vartheta = \exp(\log m \cdot \log_2 3) = \exp(x \log 3) = 3^x \in \mathbb{N}$. Conversely, if $m \in \mathcal{C}$ and $x = \log_2 m$, then $2^x = m$ and the same identity gives $3^x = m^\vartheta \in \mathbb{N}$. $\square$

Status: [kernel-verified Lean] — `exists_twoBaseNaturalCandidate_of_two_three_rpow_integer`, `TwoBaseNaturalCandidate.integer_powers`.

Integral solutions correspond exactly to powers of two, by the identifier

`twoBaseCandidateExponent_isInteger_iff_isPowerOfTwo`,

so

$$\boxed{\text{Conjecture 2.1} \iff \mathcal{C} = \{2^n : n \in \mathbb{N}_0\}.} \tag{1}$$

This equivalence is itself formalized, as `alaogluErdosConjecture_iff_candidates_are_powers_of_two`.

The candidate formulation is well suited to unique factorization, finite certification, finite differences of t^{ϑ}, multiplicative-rank arguments, and the determinant constructions of this report, and it is the formulation in which most of the corpus is stated.

Two closure properties of $\mathcal{S}$ are elementary but essential: $\mathcal{S}$ is closed under addition and under multiplication by positive naturals, so it is an additive submonoid of $[0,\infty)$ containing $\mathbb{N}$ (`rpow_integer_add`, `rpow_integer_nat_mul`).

2.3 The logarithmic determinant

Writing

$$M = 2^x, \qquad A = 3^x,$$

a solution is equivalently a positive integer pair satisfying

$$\log M\ \log 3 - \log A\ \log 2 = 0, \tag{2}$$

and the conjecture says that all positive integer solutions of (2) are

$$(M, A) = (2^n, 3^n), \qquad n \in \mathbb{N}_0.$$

Thus the problem is the vanishing of a 2×2 determinant of logarithms of algebraic numbers,

$$\det \begin{pmatrix} \log M & \log 2 \\ \log A & \log 3 \end{pmatrix} = 0,$$

with the additional information that two of the four entries are logarithms of *unknown integers*. This is not merely an analogy with the four exponentials conjecture; it is its exact local geometry.

2.4 One-parameter subgroup form

Let

$$\Gamma = \{(2^t, 3^t) : t \in \mathbb{R}\} \subset \mathbb{R}^2_{>0}.$$

Then Conjecture 2.1 is the statement

$$\Gamma \cap \mathbb{N}^2 = \{(2^n, 3^n) : n \in \mathbb{N}_0\}.$$

The curve Γ is a real one-parameter subgroup of the multiplicative torus $\mathbb{G}_m^2(\mathbb{R})$, and the assertion is that it meets the integer lattice points of the torus only at the obvious places. The difficulty is that its direction vector $(\log 2, \log 3)$ is transcendental rather than algebraic, so standard algebraic-subgroup intersection theorems do not apply directly.

2.5 The four-exponentials reduction

Suppose x is nonintegral and $2^x, 3^x$ are algebraic. The rational-exponent argument (Lemma 2.3) shows that x cannot be rational, so $1, x$ are $\mathbb{Q}$-linearly independent; and $\log 2, \log 3$ are $\mathbb{Q}$-linearly independent since $2^a 3^b = 1$ forces $a = b = 0$. The four exponentials built from these two pairs are

$$e^{1\cdot\log 2} = 2, \quad e^{1\cdot\log 3} = 3, \quad e^{x\log 2} = 2^x, \quad e^{x\log 3} = 3^x,$$

all four algebraic — contrary to the four exponentials conjecture. This reduction is formalized: `alaogluErdosConjecture_of_realFourExponentialsConjecture`. **[kernel-verified Lean]**

The reduction is a diagnostic, not just context: any proposed elementary argument must exploit integrality, positivity, or arithmetic structure beyond bare algebraicity of those four values, or else it would prove a substantial case of the four exponentials conjecture by another name.

2.6 Rational and algebraic exponents

Lemma 2.3 (Rational exponent)**.** *If $x \in \mathbb{Q}$ and $2^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$. Consequently every nonintegral solution is irrational.*

Proof. Write $x = a/b$ in lowest terms with $b > 0$. If $2^{a/b} = m \in \mathbb{N}$, then $2^a = m^b$; comparing the exponent of 2 gives $a = b\, v_2(m)$, so $b \mid a$ and $b = 1$. ☐

Status: **[kernel-verified Lean]** —
`IntegerExponent.integer_of_rational_of_two_rpow_integer`,
`irrational_of_not_integer_of_two_rpow_integer`.

Gelfond–Schneider upgrades "irrational" to "transcendental" and removes the algebraic irrational case entirely.

Lemma 2.4 (Transcendence dichotomy)**.** *If $2^x \in \mathbb{Z}$, then either $x \in \mathbb{Z}$ or x is transcendental. In particular every counterexample to Conjecture 2.1 is a transcendental number, and $\vartheta = \log_2 3$ is itself transcendental.*

Status: **[kernel-verified Lean]** —
`transcendental_of_not_integer_of_two_three_rpow_integer`,
`transcendental_logThreeDivLogTwo`; the supporting Gelfond–Schneider material is discussed in Section 4.

Together with the finite verification these lemmas already remove several whole regimes. In summary form, the pinned corpus proves the following unconditional restrictions on a hypothetical counterexample x.

- x is irrational, and in fact transcendental. **[kernel-verified Lean]**
- ϑ is transcendental. **[kernel-verified Lean]**
- $x \geq 12$, by the kernel-checked zero-free region $2^x < 4096$. **[kernel-verified Lean]**
- $2^x > 2^{29}$, hence $x > 29$, by the directed-rounding scans through the third continuation; this raises the barrier only under the stated software trust boundary and is not a kernel certificate. **[interval computation]**
- The real four exponentials conjecture implies Conjecture 2.1. **[kernel-verified Lean]**

These facts eliminate the rational, algebraic-irrational, small, and routine three-exponentials cases. They also explain why an elementary manipulation of logarithms is unlikely to finish the proof: the unknown lies exactly where present transcendence theory changes from theorem to conjecture.

2.7 The three-base theorem and its determinant architecture

By contrast with the two-base problem, three multiplicatively independent bases fall to the six exponentials mechanism, and the special case needed here is fully formalized.

Theorem 2.5 (Three-base theorem). *If $2^x, 3^x, 5^x \in \mathbb{Z}$ for a real x, then $x \in \mathbb{Z}$.*

Status: [kernel-verified Lean] — modules `IntegerExponent`, `IntegerExponent.DeterminantBound`, `IntegerExponent.ExponentialKernel`, `IntegerExponent.Grid`, following Waldschmidt's square-determinant proof [12].

In outline, for irrational x one forms a large square matrix $\mathcal{A}_{ij} = \exp(a_i b_j)$, where the row arguments a_i encode monomials in $2, 3, 5$ and the column arguments b_j encode affine forms in $1, x$. Irrationality makes the column parameters pairwise distinct and multiplicative independence makes the row parameters pairwise distinct; a total-positivity argument gives $\det \mathcal{A} \neq 0$; integrality of all entries forces $|\det \mathcal{A}| \geq 1$; and an interpolation estimate with explicit size choices gives $|\det \mathcal{A}| < 1$, a contradiction.

The dimension count is exactly where the argument stops short of Conjecture 2.1. With three independent row directions and two column directions the analytic upper bound decays faster than the arithmetic lower bound 1 permits, and the contradiction closes. Shrinking to two bases removes one row direction, and the same balance then lands precisely on the classical boundary where the decay below one fails. Nothing in the proof is wasteful enough to be recovered by sharper constants: a successful two-base determinant of this shape would be a new four-exponentials argument, which is why Section 19 and Section 20 enlarge the *node set* rather than tune the estimate.

A denominator-controlled rational third output. The rational-output extension now formalized is deliberately conditional. Let $a > 1$ and assume the monomial map

$$(i, j, k) \longmapsto 2^i 3^j a^k$$

is injective. If $2^x = m_2$ and $3^x = m_3$ are natural numbers, $a^x = q \in \mathbb{Q}$, and for some $u, v \in \mathbb{N}_0$ the reduced denominator satisfies

$$\mathrm{den}(q) \mid m_2^u m_3^v,$$

then $x \in \mathbb{Q}$, and in fact $x \in \mathbb{Z}$ because 2^x is integral. A concrete corollary obtains the monomial injectivity when a has a prime divisor different from 2 and 3.

On the determinant side, a nonzero $m \times m$ real determinant whose entries become integers after multiplication by a common positive denominator Q satisfies $|\det A| \geq Q^{-m}$. The explicit analytic estimate also retains a denominator margin: for $m = 2r$, $r > 2$, $192C \leq r$, and $bs + 2r < r^2$, the verified upper-bound expression remains strictly below 1 even after multiplication by b^s.

Status: [kernel-verified Lean] — `one_div_pow_le_abs_det_of_common_denominator`, `exponential_det_numeric_bound_with_denominator`, `rational_of_two_three_a_rpow_rational_of_den_dvd_outputs`, and `integer_of_two_three_a_rpow_rational_of_prime_factor_of_den_dvd_outputs` in `RationalThirdBase`. These are denominator-divisibility and explicit-margin results; the unrestricted determinant grid for an arbitrary rational third-base output has *not* been completed.

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3 Repository timeline and verification status

The formal development advanced in three waves of Lean work, interleaved with four documentation milestones. All commits are in the repository [16]; the mathematical role of each is summarized below.

Commit	Role
`f492c18f1`	Three-base theorem via interpolation determinants (`IntegerExponent` and submodules).
`5ed4e6185`	Merge of the three-base branch.
`a37d97f9d`	Basic two-base module: the conjecture as a proposition, closure under natural multiples, irrationality of nonintegral solutions, endpoint gaps, numerical bounds for ϑ, the $2^x < 10$ exclusion, second-difference obstruction, aligned-block 2/3 counting.
`4a9b03d12`	Major corpus: finite checks through $2^x < 256$; arithmetic descent; multiplicative rank; candidate structure; zero density; shrinking targets; the 3-smooth theorem; the four-exponentials reduction; transcendence consequences; and a complete Gelfond–Schneider port.
`18a85e6b8`	Merge of the two-base branch.
`423ac24f0`	First survey document (`two-base-partial-results.tex`) with rendered PDF.
`98646b455`	Three further modules, kernel-verified: `FractionalPartDensity` (density dichotomy and gap equivalences), `OddCore` (common odd core, improved counting, dyadic blocks), `FiniteCheck4096` (zero-free region to $2^x < 4096$).
`716617d72`	Survey finalized after kernel acceptance of the 4096 region.
`11ac3093f`	First independent research report (audit at `423ac24f0`, exact conditional structure, interval scan to 2^{20}); one of the sources merged into the present document.
`83ef82ab1`	Merge of the two earlier documents into a single expanded report.
`46f5b67e3`	Localization, sharp arithmetic progressions and an exact generator: `Localization`, `SharpProgression`, and the first form of `PrimitiveGenerator`.
`5d3982ceb`	Merge of the parallel conditional-structure and asymptotic formalizations: canonical primitive generator and exact solution monoid, least solution, exact dyadic counting, the quadratic cumulative asymptotic, block Weyl sums and block equidistribution, simultaneous output normalization and power-of-three denominator normalization, the rational-power index, the radical modules (radical reformulation, primitive radical, exact radical degree), the higher-difference obstruction, and denominator-controlled rational third-base outputs.
`8c9c59224`	Denominator-growth constraint $q_{n+1} < 2m^{q_n}$ on the continued fraction of a hypothetical solution (`DenominatorGrowth`), together with the report analysis naming the obstruction as a vanishing integer-coefficient linear form in products of two prime logarithms.

Commit	Role
`2ece309cc`	Second research report: arbitrary-node Vandermonde repulsion, the semigroup generalized-Vandermonde hierarchy and its dyadic $O((\log X)^2)$ bound, the one-logarithm deficit analysis, and the compact-orbit and block-growth reformulations; the other source merged into the present document.

The audit performed in `11ac3093f` treated results of the then-uncommitted modules as article-level claims; with `98646b455` in the tree and kernel-verified, those results are now cited as committed Lean. The interval scan to 2^{20} remains at the **[interval computation]** level by design. The feature modules merged in `5d3982ceb` prove the cumulative quadratic asymptotic and the finite-trigonometric-polynomial block averages; they do *not* yet formalize the stronger continuous-test-function or interval-counting equidistribution theorem, and that distinction is maintained throughout this report.

3.1 Evidence convention

The evidence convention of this report is source-tree plus build-level, not rebuild-on-demand. Every claim marked **[kernel-verified Lean]** was checked against the cited Lean source and theorem statement at the repository state above, and the corresponding module is on the aggregate import surface `ExponentialIdentities.lean`, where it is recorded as accepted by the Lean kernel. The label does not assert that a separate fresh build was run while preparing this document; it asserts that the theorem exists, in the stated form, on an import surface the kernel has accepted.

Three weaker labels are used deliberately. **[interval computation]** marks a finite interval computation with an explicit software trust boundary and no kernel certificate; **[paper proof in this report]** marks a conventional proof given in this report, written with formalization in mind but not yet translated; and **[Lean draft, not yet accepted]** marks a Lean module that exists in the tree as a draft but has *not* been accepted by the kernel, and therefore carries no more weight than **[paper proof in this report]**. The complete label-by-label inventory is in Appendix A, and the per-module inventory in Appendix B.

4 The kernel-verified corpus

Throughout this section x denotes a hypothetical nonintegral solution, $m = 2^x \in \mathbb{N}$, $n = 3^x \in \mathbb{N}$. Every statement here is **[kernel-verified Lean]** unless labeled otherwise.

4.1 Fractional-part gaps and shrinking targets

An integral power caught strictly between consecutive integral powers of its base must clear both by at least one full unit; dividing through normalizes this to the fractional part.

Theorem 4.1 (Fractional-part gaps)**.** *Let $x \in (N, N+1)$ with $N \in \mathbb{N}$, $t = x - N$, and suppose $2^x, 3^x \in \mathbb{Z}$. Then*

$$1 + 2^{-N} \le 2^t \le 2 - 2^{-N}, \qquad 1 + 3^{-N} \le 3^t \le 3 - 3^{-N}.$$

Status: [kernel-verified Lean] — `two_three_fractional_part_gaps`, `two_three_rpow_gaps_near_nat`, from `rpow_integer_gap_between_consecutive_powers`.

Near the endpoints the excluded zones have width $\asymp 2^{-N}$: they *shrink* as the integer part grows. Applied to all multiples kx (which are again solutions), this yields explicit Diophantine lower bounds.

Theorem 4.2 (Effective irrationality bound)**.** *Let $x \notin \mathbb{Z}$ satisfy $2^x \in \mathbb{Z}$ and write $m = 2^x$. Then for all integers p and all $k \geq 1$,*

$$\left|x - \frac{p}{k}\right| \;\geq\; \frac{1}{2\,k\,m^k}.$$

Exact (slightly stronger) radii, and two-base versions taking the better of the base-2 and base-3 radii, are formalized as well.

Status: [kernel-verified Lean] — `rational_approximation_exponential_lower_bound`, `two_three_multiple_distance_to_integer_lower_bound`, `exists_natural_base_rational_approximation_lower_bound` in `ShrinkingTarget`.

In shorthand, $\|kx\|_{\mathbb{Z}} \gtrsim 2^{-kx}$. This bound is compatible with ordinary Diophantine approximation: Dirichlet guarantees infinitely many k with $\|kx\|_{\mathbb{Z}} < 1/k$, and the shrinking radius 2^{-kx} is incomparably smaller. The interplay between this exponential gap and equidistribution is discussed in Section 26.

The bound dualizes into a growth constraint on the continued fraction of a solution: if $|x - p/q| < 1/(qQ)$ with $q, Q \geq 1$, then $Q < 2m^q$. Applied to consecutive convergent denominators, which satisfy $|x - p_n/q_n| < 1/(q_n q_{n+1})$, this gives

$$q_{n+1} \;<\; 2\,m^{\,q_n} :$$

the partial quotients of a hypothetical nonintegral solution are capped by a single exponential in the preceding denominator, with base its own output $m = 2^x$.

Status: [kernel-verified Lean] — `approximation_quality_lt_of_noninteger_solution` in `DenominatorGrowth`.

4.2 Verified zero-free regions and certificate architecture

Theorem 4.3 (Zero-free region)**.** *If $2^x, 3^x \in \mathbb{Z}$ and $2^x < 4096$, then $x \in \mathbb{Z}$. Equivalently, every nonintegral solution satisfies $x \geq 12$, and every candidate that is not a power of two is at least 4096.*

Status: [kernel-verified Lean] — `integer_of_two_three_rpow_integer_of_two_rpow_lt_4096`, `twelve_le_of_not_integer_of_two_three_rpow_integer`, `le_of_twoBaseNonintegerCandidate_4096` in `FiniteCheck4096`, extending `FiniteCheck` ($2^x < 100$) and `FiniteCheck256` ($2^x < 256$, hence $x \geq 8$).

The certificate architecture is worth recording, both because it is the repository's model of a trustworthy large computation and because scaling it is a named research direction. For each non-power-of-two m the certificate is a pair of exact integer power comparisons against rational enclosures $L_p/L_q < \vartheta < U_p/U_q$ drawn from continued-fraction convergents of ϑ: from

$$a^{L_q} < m^{L_p} \quad \text{and} \quad m^{U_p} < (a+1)^{U_q}$$

one traps $a < m^\vartheta < a+1$, so $m^\vartheta \notin \mathbb{Z}$. The $2^x < 256$ stage verifies per-value certificates by `norm_num` with three enclosure tiers (exponent sizes ~500, ~1500, ~25000). The extension to 4096 verifies 3836 certificates through a single kernel-replayed `decide` over a table generated by PARI/GP and revalidated independently with exact big-integer arithmetic in Python. Five tiers suffice, with tier populations 78, 488, 3241, 28, 1:

tier	lower convergent L_p/L_q	upper convergent U_p/U_q
0	569/359	485/306
1	1054/665	1539/971
2	25781/16266	24727/15601
3	50508/31867	125743/79335
4	176251/111202	301994/190537

Only the value $m = 2762$, whose ϑ-power is within $3 \cdot 10^{-9}$ of an integer in logarithmic scale, needs tier 4; its certificate compares integers of roughly 90,000 digits, which the kernel's exact arithmetic replays without difficulty. No axioms beyond the Lean kernel are involved; in particular no `native_decide`.

4.3 Arithmetic descent

Theorem 4.4 (Affine descent)**.** *If the two integral outputs factor simultaneously as $2^x = 2^r u^k$ and $3^x = 3^r v^k$ with $r \geq 0$, $k \geq 1$, then $(x-r)/k$ is again a solution. Combined with the $x \geq 8$ region this gives the unconditional constraint*

$$r + 8k \leq x$$

for nonintegral x; in particular common perfect-power shapes of the two outputs are strongly restricted.

Status: [kernel-verified Lean] — `affine_descent_integral_powers`, `base_depth_add_eight_mul_degree_le`, `min_output_base_factorization_add_eight_le` in `ArithmeticRigidity`. With the 4096 region the constant 8 improves to 12 by bookkeeping; with the interval scan of Section 25 it would improve to 20 at the [interval computation] level.

For the *least* counterexample the same descent later becomes an exact primitivity statement in Section 5: no nontrivial matched decomposition exists at all.

4.4 Local progression obstructions and zero density

The function $t \mapsto t^\vartheta$ is strictly convex with second differences at moderate steps lying strictly between 0 and 1 — hence never integers.

Theorem 4.5 (Committed AP obstruction). *Let $n, h \geq 1$ with $h^5 \leq n$. Then $n, n+h, n+2h$ are not all candidates. In particular no three consecutive naturals are candidates, and among the positive integers up to $3N$ at most $2N$ are candidates.*

Status: [kernel-verified Lean] — `not_three_arithmetic_progression_twoBaseNaturalCandidates`, `not_three_consecutive_logThreeDivLogTwo_rpow_integers`, `twoBaseNaturalCandidate_Icc_card_le`. The sharper criterion $\vartheta(\vartheta-1)h^2 < n^{2-\vartheta}$ is kernel-verified in `SharperProgressions`, and the analogous four-term criterion is kernel-verified in `ThirdDifference`. The arbitrary-order statement for every $r \geq 2$ is `not_arithmetic_progression_twoBaseNaturalCandidates_of_curvature` in `HigherDifference`, which formalizes a generic iterated forward-difference mean-value identity and applies uniformly in the order r.

For finite verification the same curvature mechanism is also available in a purely rational effective form: if $h^{400} \leq n^{83}$ — step exponent $83/400 = 0.2075$, against the $1/5$ of Theorem 4.5 — then $n, n+h, n+2h$ are not all candidates.

Status: [kernel-verified Lean] — `not_three_arithmetic_progression_twoBaseNaturalCandidates_sharp` in `SharpProgression`, with the kernel-replayed enclosure $\vartheta < 317/200$ obtained from the exact integer comparison $3^{200} < 2^{317}$.

The hypothesis bounding h is essential: for steps comparable to n the second difference is large and the obstruction vanishes. This locality is a genuine barrier — a *global* progression obstruction would, for example, forbid candidates with odd part $2^s \pm 1$ via progressions such as $(2^i, 2^{i+t-1}, 2^i(2^t-1))$, but no such global statement is available; under a counterexample the exact monoid of Section 5 shows sparse candidates whose additive relations the curvature argument cannot see.

Feeding the local obstruction into the theory of Roth numbers gives quantitative sparsity.

Theorem 4.6 (Zero density). *For every $H \geq 1$ the number of candidates below N is at most $H^5 + \lceil (N-H^5)/H \rceil\, r_3(H)$, where $r_3(H)$ is the maximal size of a 3-AP-free subset of $\{0, \ldots, H-1\}$. Consequently the candidate counting function is $o(N)$: the candidates, and a fortiori the values 2^x of nonintegral solutions, have asymptotic density zero.*

Status: [kernel-verified Lean] — `twoBaseNaturalCandidatesBelow_card_le`, `twoBaseNaturalCandidatesBelow_isLittleO_id`, `tendsto_twoBaseNonintegerCandidatesBelow_density_zero` in `ZeroDensity` and `NonintegerDensity`.

4.5 Multiplicative rank two and the common odd core

Theorem 4.7 (Rank two). *The prime-exponent vectors $\mathbf{v}(a) = (v_p(a))_p$ of all candidates span a rational space of dimension at most two. Consequently at most $(\mathrm{clog}_2 N)^2$ candidates lie below N.*

Status: [kernel-verified Lean] — `finrank_span_primeExponentVectors_le_two`, `twoBaseNaturalCandidatesBelow_card_le_clog_sq` in `MultiplicativeRank`. The mechanism abstracts the determinant method: three candidates with independent exponent vectors would give three multiplicatively independent bases, contradicting the three-base theorem of Section 2 through the irrationality of ϑ (`irrational_logThreeDivLogTwo`).

Since 2 is itself a candidate, all odd parts lie on at most one further rational direction. Pairwise this yields the cross-valuation identity, and globally a common odd core.

Lemma 4.8 (Cross-valuation identity). *Let $a, b \in \mathcal{C}$ and let p be an odd prime with $v_p(a) > 0$. Then for every odd prime q,*

$$v_p(a)\ v_q(b) = v_p(b)\ v_q(a)\,.$$

Status: [kernel-verified Lean] — `odd_factorization_cross_mul` in `CandidateStructure`; the fixed-power relations `fixed_power_relation_of_odd_prime_factor` follow.

Theorem 4.9 (Common odd core). *There is a single odd $w > 1$ such that every candidate equals $2^i w^j$ for some $i, j \in \mathbb{N}_0$; candidates that are not powers of two have $j \geq 1$. In exponent coordinates: every solution has the form $x = i + j \log_2 w$ with $i, j \in \mathbb{N}_0$.*

Status: [kernel-verified Lean] — `exists_common_odd_core`, `exists_common_odd_core_split`, `exists_common_odd_core_exponent` in `OddCore`. The canonical (gcd-normalized) version with uniqueness of w and of each coordinate pair is `existsUnique_primitive_common_odd_core` in `PrimitiveOddCore`.

Corollary 4.10 (Improved counting). *The number of candidates below N is at most $\mathrm{clog}_2 N \cdot \mathrm{clog}_3 N$, and each dyadic block $[2^T, 2^{T+1})$ contains at most $\mathrm{clog}_3(2^{T+1}) \approx 0.631\,(T+1)$ candidates. In exponent coordinates, the number of solutions with $x \in [T, T+1)$ grows at most linearly in T, against the quadratic global bound.*

Status: [kernel-verified Lean] — `twoBaseNaturalCandidatesBelow_card_le_clog_mul_clog`, `twoBaseNonintegerCandidatesBelow_card_le_clog_mul_clog`, `twoBaseNaturalCandidatesInDyadicBlock_card_le` in `OddCore`.

The logarithmic-square order resists improvement by counting alone: if a candidate with an odd prime factor exists, multiplicative closure generates injective boxes $2^i a^j$, so the count below $2^k a^l$ is at least kl (`twoBaseNaturalCandidatesBelow_card_ge_box`, `twoBaseNonintegerCandidatesBelow_card_ge_box`) — a conditional lower bound of the same quadratic-in-logarithm shape, made exact in Section 5.

4.6 Transcendence

The corpus contains a complete Lean proof of the Gelfond–Schneider theorem (Hilbert's seventh problem), following Hua's exposition [4], ported from Michail Karatarakis's Mathlib formalization (Mathlib PR #42911, Apache 2.0).

Theorem 4.11 (Gelfond–Schneider)**.** *If α, β are algebraic, $\alpha \neq 0, 1$, and β is irrational, then α^β is transcendental.*

Status: [kernel-verified Lean] —
`GelfondSchneider.transcendental_cpow_of_isAlgebraic_of_irrational`.

Corollary 4.12 (Algebraic–integral dichotomy)**.** *If $2^x \in \mathbb{Z}$ then x is either an integer or transcendental; algebraicity of such x is equivalent to integrality. Every nonintegral two-base solution is transcendental, and $\vartheta = \log_2 3$ is transcendental.*

Status: [kernel-verified Lean] —
`transcendental_of_not_integer_of_two_rpow_integer`,
`isAlgebraic_iff_integer_of_two_rpow_integer`,
`transcendental_of_not_integer_of_two_three_rpow_integer`,
`transcendental_logThreeDivLogTwo`.

The dichotomy does not resolve the conjecture — the unknown exponent is allowed to be transcendental — but it constrains every construction below: the primitive generator β of Section 5 is transcendental, as is $\log_2 w$ for the odd core.

4.7 Classification of auxiliary bases: the 3-smooth theorem

Theorem 4.13 (3-smooth classification)**.** *Suppose $x \notin \mathbb{Z}$ and $2^x, 3^x \in \mathbb{Z}$. Then for a positive natural a, the power a^x is an integer if and only if $a = 2^u 3^v$ is 3-smooth. In particular a third base with any prime factor other than 2 and 3 forces $x \in \mathbb{Z}$.*

Status: [kernel-verified Lean] —
`rpow_integer_iff_eq_two_pow_mul_three_pow_of_not_integer`,
`integer_of_two_three_a_rpow_integer_of_prime_factor` in `ThreeSmooth`.

This is an exact classification of bases for a *fixed* hypothetical counterexample. It should not be confused with the classification of candidate values 2^x as x ranges over all solutions, which is the subject of the next section. Note also the scope limitation explained in Section 26: the theorem controls the quantity b^x, not divisibility of the integer 2^x by b.

4.8 Unconditional gap equivalences

Because natural multiples of a solution are solutions, one nonintegral solution generates infinitely many, with irrational generator; Dirichlet approximation and a rotation-stepping argument spread their fractional parts densely.

Theorem 4.14 (Density dichotomy)**.** *If some $x \notin \mathbb{Z}$ satisfies $2^x, 3^x \in \mathbb{Z}$, then for every $t \in [0,1]$ and $\varepsilon > 0$ there is a nonintegral solution y with $|\{y\} - t| < \varepsilon$; and the set of nonintegral solutions is infinite.*

Theorem 4.15 (Gap equivalences)**.** *Each of the following is equivalent to Conjecture 2.1:*

(i) *some nonempty open interval $(u, v) \subseteq (0, 1)$ contains no fractional part of a nonintegral solution;*

(ii) *the fractional parts of nonintegral solutions are bounded away from 1;*

(iii) *the fractional parts of nonintegral solutions are bounded away from 0.*

Status: **[kernel-verified Lean]** — `dense_fract_of_noninteger_solution`, `infinite_noninteger_solutions_of_exists`, `alaogluErdosConjecture_iff_fract_avoids_interval`, `alaogluErdosConjecture_iff_fract_bounded_away_one`, `alaogluErdosConjecture_iff_fract_bounded_away_zero` in `FractionalPartDensity`, built on the reusable `exists_pos_nat_fract_close_of_irrational`.

Remark 4.16 (Consistency pincer)*.* Theorem 4.1 bounds $\{x\}$ away from 0 and 1 by margins $\asymp 2^{-\lfloor x \rfloor}$ that decay with the height of the solution, while Theorem 4.14 produces fractional parts arbitrarily close to 0 and 1 along multiples of unbounded height. The two results meet edge-to-edge without contradiction, and Theorem 4.15 shows the decay is essential: *any* height-independent gap, however small and wherever located, is exactly as strong as the full conjecture. The elementary gap bounds are therefore of optimal shape, and the fractional-part axis measures the entire remaining distance to Conjecture 2.1.

These equivalences are *unconditional* biconditionals. Conditional on failure, the exact structure of the next section gives kernel-verified convergence for every finite trigonometric polynomial and the paper-level strengthening to full blockwise Weyl equidistribution (Section 5).

5 Exact conditional structure of all solutions

The most consequential fact in the current corpus is not another exclusion range. It is the complete description of *all* solutions under the hypothesis that even one nonintegral solution exists. Conditional on failure of Conjecture 2.1, the solution set is not merely sparse or contained in some rank-two set: it is a free rank-two additive monoid with an explicit canonical generator, exactly countable in every dyadic block, and equidistributed blockwise on the circle.

Throughout this section we write

$$\mathcal{S} := \{x \in \mathbb{R} : 2^x \in \mathbb{N} \text{ and } 3^x \in \mathbb{N}\}, \qquad \mathcal{C} := \{2^x : x \in \mathcal{S}\}, \qquad \mathcal{P} := \{(2^x, 3^x) : x \in \mathcal{S}\},$$

and we say a counterexample exists when $\mathcal{S} \not\subseteq \mathbb{N}_0$. The canonical primitive odd core, the rational-power index, the power-of-three denominator normalization, the unique least solution, the simultaneous primitive output normalization, the exact free-monoid classification, the exact unit-block and dyadic counts, the cumulative quadratic asymptotic, and the finite-Fourier block averages are all **[kernel-verified Lean]**. Only the extension of the block averages from finite trigonometric polynomials to arbitrary continuous tests or interval indicators (Theorem 5.10) remains **[paper proof in this report]**.

5.1 The primitive common odd core

Theorem 5.1 (Primitive common odd core). *Assume $\mathcal{C}$ contains a non-power of two. Then there is a unique odd integer $w > 1$ that is power-free in the sense*

$$\gcd\{v_p(w) : p \mid w\} = 1,$$

and every candidate $b \in \mathcal{C}$ has a unique representation

$$b = 2^i w^j, \qquad i, j \in \mathbb{N}_0.$$

Proof. Choose a non-power-of-two candidate a. Its odd valuation vector $A = (A_q)_{q \neq 2}$, $A_q = v_q(a)$, is nonzero and finitely supported. Let

$$g = \gcd\{A_q : A_q > 0\}, \qquad c_q = A_q/g, \qquad w = \prod_{q \neq 2} q^{c_q}.$$

Then w is odd, greater than one, and the positive c_q have gcd one.

Let $b \in \mathcal{C}$ and set $B_q = v_q(b)$. Fix an odd prime p with $A_p > 0$. By the cross-valuation identity of Section 4, $A_p B_q = B_p A_q$ for all odd q. If $A_q = 0$, this forces $B_q = 0$. On the support of A it says

$$B_q = \frac{B_p}{A_p} A_q = \Big(\frac{g B_p}{A_p}\Big) c_q.$$

Put $t = gB_p/A_p \in \mathbb{Q}_{\geq 0}$. Every tc_q is an integer. Since the c_q have gcd one, there are integers r_q, finitely many nonzero, with $\sum r_q c_q = 1$; therefore $t = \sum_q r_q(tc_q) \in \mathbb{Z}$. Writing $j = t \in \mathbb{N}_0$, the odd part of b is exactly w^j, so $b = 2^i w^j$ with $i = v_2(b)$.

Uniqueness of (i, j) for fixed w follows from the odd valuations and the 2-adic valuation. If w' is another primitive common core, the odd valuation vectors of w and w' generate the same rational ray and are both primitive integer vectors, hence equal. □

Status: [kernel-verified Lean]

`existsUnique_primitive_common_odd_core` in `PrimitiveOddCore`. The module defines `oddPrimeValuationGCD`, proves its equivalence with power-theoretic primitivity for $w > 1$, and kernel-checks both uniqueness of the normalized core and uniqueness of every pair (i, j). The earlier non-normalized existence statement `exists_common_odd_core` recorded in Section 4 is subsumed by it.

The gcd normalization is exactly what makes w canonical: it prevents a hidden perfect power from being absorbed into the exponent j, so that both the core and the coordinates are determined by $\mathcal{C}$ alone.

Set

$$\alpha := \log_2 w, \qquad E := 3^\alpha = w^\vartheta.$$

Every candidate exponent then has the form $x = i + j\alpha$. The number α is irrational (otherwise the rational-exponent lemma applied to $2^\alpha = w$ would make it integral, forcing the odd $w > 1$ to be a power of two), and in fact transcendental by the algebraic–integral dichotomy of Section 4. The candidate condition becomes

$$3^{i+j\alpha} = 3^i E^j \in \mathbb{N},$$

and at this stage the admissible pairs (i, j) still form an unknown subset of $\mathbb{N}_0^2$. The next subsection shows that this subset is completely rigid.

5.2 The rational-power index and the denominator normalization

Because a non-power-of-two candidate exists, some $j > 0$ has $E^j \in \mathbb{Q}$. Let

$$d := \min\{j \in \mathbb{N}_{>0} : E^j \in \mathbb{Q}\}.$$

Lemma 5.2 (All rational powers are multiples of the least one)**.** *For every $j \in \mathbb{Z}$: $E^j \in \mathbb{Q} \iff d \mid j$.*

Proof. The set $J := \{j \in \mathbb{Z} : E^j \in \mathbb{Q}\}$ is an additive subgroup of $\mathbb{Z}$ (it contains 0, products add exponents, and reciprocals negate them), and every nonzero subgroup of $\mathbb{Z}$ equals $d\mathbb{Z}$ for its least positive element d. $\square$

Status: [kernel-verified Lean]

`rationalPowerIndices`, `AddSubgroup.Int.exists_least_positive_generator`, and `exists_rationalPowerIndex` in `RationalPowerIndex`.

Write the positive rational E^d in lowest terms. Its denominator has no prime other than 3: taking a candidate with odd-core exponent $j = dk > 0$ and 2-exponent i, integrality of $3^i(E^d)^k$ shows that any other prime in the reduced denominator would survive in the k-th power. Hence there are unique integers $a \geq 0$ and $A \geq 1$ with

$$E^d = \frac{A}{3^a}, \qquad \text{and} \qquad 3 \nmid A \quad \text{when } a > 0, \tag{3}$$

so that $a = 0$ or $3 \nmid A$ in every case. Define

$$\beta := a + d\alpha = a + d\log_2 w, \qquad M := 2^\beta = 2^a w^d, \qquad A = 3^\beta = 3^a E^d. \tag{4}$$

Thus β is itself a solution, and it is irrational because α is. (Where the denominator normalization is written with a named integer numerator c, the identity $E^d = c/3^a$ with $a = 0$ or $3 \nmid c$ is exactly (3) with $c = A = 3^\beta$.)

Status: [kernel-verified Lean]

The denominator-support step is `exists_three_pow_denominator_of_rational_power` in `ThreeDenominatorNormalization`; the normalized integrality criterion is `three_pow_mul_normalized_pow_integer_iff` in `RationalPowerIndex`.

Equation (3) also has a field-theoretic reading: E is a root of the binomial $X^d - A/3^a$, which is the minimal polynomial of E over $\mathbb{Q}$, so $[\mathbb{Q}(E) : \mathbb{Q}] = d$. That computation, together with the equivalent one-number and localized-radical formulations of the conjecture that follow from it, is developed in Section 6; we do not repeat it here.

5.3 The exact primitive-generator theorem

Theorem 5.3 (Exact primitive generator and solution monoid)**.** *Assume Conjecture 2.1 is false. With w, d, a, β, M, A as above:*

(i) *every solution has a unique representation* $x = n + k\beta$ *with* $n, k \in \mathbb{N}_0$*, that is,*

$$\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\};$$

(ii) *every candidate has a unique representation* $m = 2^n M^k$ *with* $n, k \in \mathbb{N}_0$*, that is,*

$$\mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\};$$

(iii) *every output pair has a unique representation* $(2^x, 3^x) = (2^n M^k, 3^n A^k)$*;*

(iv) β *is the unique least nonintegral solution.*

Proof. Let $x \in \mathcal{S}$. By Theorem 5.1, $2^x = 2^i w^j$ for unique $i, j \in \mathbb{N}_0$, so $x = i + j\alpha$. From $3^x = 3^i E^j \in \mathbb{N}$ we get $E^j \in \mathbb{Q}$, so Lemma 5.2 gives $j = dk$ for some $k \in \mathbb{N}_0$. Using (3),

$$3^x = 3^i (E^d)^k = 3^{\,i-ak} A^k.$$

Because $A/3^a$ is reduced, this is an integer exactly when $i \geq ak$. Put $n = i - ak \in \mathbb{N}_0$; then

$$x = i + dk\alpha = n + k(a + d\alpha) = n + k\beta.$$

Conversely, for any $n, k \in \mathbb{N}_0$,

$$2^{\,n+k\beta} = 2^n M^k \in \mathbb{N}, \qquad 3^{\,n+k\beta} = 3^n A^k \in \mathbb{N},$$

so $n + k\beta \in \mathcal{S}$. If $n + k\beta = n' + k'\beta$ with $k \neq k'$, then $\beta = (n' - n)/(k - k') \in \mathbb{Q}$, a contradiction; hence the representation is unique. This proves (i), and (ii)–(iii) follow by exponentiation. A nonintegral solution has $k \geq 1$; the least such value is attained at $(n, k) = (0, 1)$ and equals β, uniquely so by uniqueness of the representation. $\square$

Status: [kernel-verified Lean]
`exists_canonical_primitiveGenerator_of_not_alaogluErdosConjecture` in `PrimitiveGenerator`, built from
`exists_exact_solution_monoid_of_fixed_common_oddCore` in `RationalPowerIndex`.
The Lean statement supplies the canonical odd core w, the rational-power index d, the depth a, the numerator A, and the unique-coordinate clauses in one package. Conditional on failure, $\mathcal{S}$ is not merely contained in a rank-two additive set: every solution has a unique pair of coordinates in the free additive monoid on 1 and the irrational least generator β. The set of occurring odd-core exponents is exactly $d\,\mathbb{N}_0$, with the power of two governed by one linear threshold.

Packaged algebraically, Theorem 5.3 says that

$$(\mathbb{N}_0^2, +) \xrightarrow{\sim} (\mathcal{S}, +), \quad (n, k) \mapsto n + k\beta, \qquad (\mathbb{N}_0^2, +) \xrightarrow{\sim} (\mathcal{C}, \cdot), \quad (n, k) \mapsto 2^n M^k,$$

are monoid isomorphisms, and that the output-pair monoid $\mathcal{P}$ is freely generated by $(2, 3)$ and (M, A). Every multiplicative relation among candidates is the obvious coordinate relation; there is no room for an unexpected coincidence. This exact conditional model is the benchmark against which any future density, progression, or descent argument must be tested.

The phrase “least nonintegral” is materially stronger than choosing an arbitrary counterexample. It forces affine primitivity: no nontrivial common root can be extracted from the two output integers while preserving the same base-adic offset. It is the arithmetic analogue of choosing a primitive lattice direction, and the next subsection turns it into exact exclusions.

5.4 Primitivity of the least output pair

Corollary 5.4 (No matched base depth)**.** *For the primitive pair* $(M, A) = (2^\beta, 3^\beta)$ *we have* $\min\{v_2(M)\,, v_3(A)\} = 0$*; that is,* M *is odd or* $3 \nmid A$ *(possibly both).*

Proof. If both valuations were positive, then $2^{\beta-1} = M/2$ and $3^{\beta-1} = A/3$ would be integers, so $\beta - 1$ would be a smaller nonintegral solution. ☐

Corollary 5.5 (No common perfect-power degree)**.** *There is no* $k \geq 2$ *for which both* M *and* A *are perfect* k*-th powers.*

Proof. If $M = U^k$ and $A = V^k$, then $2^{\beta/k} = U$ and $3^{\beta/k} = V$, so β/k is a smaller nonintegral solution. ☐

Corollary 5.6 (Exact affine primitivity)**.** *If* $M = 2^r U^k$ *and* $A = 3^r V^k$ *with* $r \in \mathbb{N}_0$*,* $k \in \mathbb{N}_{>0}$*, and* $U, V \in \mathbb{N}$*, then* $r = 0$ *and* $k = 1$*.*

Proof. The affine descent theorem of Section 4 produces the solution $(\beta - r)/k$, which is nonintegral since β is irrational; if $r > 0$ or $k > 1$ it is strictly smaller than β, contradicting leastness. ☐

Status: [kernel-verified Lean]

`exists_least_twoBaseNonintegerSolution`,
`IsLeastTwoBaseNonintegerSolution.unique`,
`IsLeastTwoBaseNonintegerSolution.not_both_base_dvd_outputs`,
`IsLeastTwoBaseNonintegerSolution.no_common_perfect_power`, and
`IsLeastTwoBaseNonintegerSolution.affine_primitive` in `LeastSolution`.

The odd-core normalization of M has a symmetric counterpart on the other output, and the two can be imposed simultaneously.

Theorem 5.7 (Simultaneous primitive output normalization)**.** *If the conjecture fails, its least nonintegral solution* β *admits simultaneous factorizations*

$$M = 2^\beta = 2^a w^d, \qquad A = 3^\beta = 3^b v^e,$$

where $w > 1$ *is odd and power-primitive,* $v > 1$ *is power-primitive with* $3 \nmid v$*, and* $d, e > 0$*. Moreover* $a = 0$ *or* $b = 0$*; the exponents* a *and* b *are the exact base-adic depths;* d *and* e *are the gcds of the prime valuations of the corresponding base-free parts; and*

$$\gcd(\gcd(d, e),\, |b - a|) = 1.$$

Since every counterexample satisfies $\beta \geq 12$*, these normalizations also obey the size dichotomy*

$$w^d \geq 4096 \qquad \textit{or} \qquad v^e \geq 3^{12}.$$

Status: [kernel-verified Lean]

`exists_simultaneous_primitive_output_normalization`, `IsLeastTwoBaseNonintegerSolution.simultaneousCoordinateGCD_eq_one`, and `exists_large_primitiveCore_of_not_alaogluErdosConjecture` in `OutputNormalization`. The gcd-one statement simultaneously controls the two outer power degrees and the difference of the two base-adic depths; it is strictly stronger than merely excluding a common perfect-power degree (Corollary 5.5).

These restrictions are strong, but they still admit infinitely many formal integer parameter patterns. The unresolved analytic content is untouched: the pair (M, A) must satisfy

$$\frac{\log M}{\log 2} = \frac{\log A}{\log 3},$$

and no known local valuation argument turns the depth dichotomy $a = 0$ or $b = 0$, or the gcd-one condition, into a contradiction. Every constraint proved so far is compatible with the existence of such a pair; what fails to exist must fail for a transcendence reason, not a valuation-theoretic one.

5.5 Exact counting

The free-monoid description makes the population of a hypothetical counterexample completely explicit, block by block.

Theorem 5.8 (Exact unit-block and dyadic counts)**.** *Assume a counterexample exists and let β be the primitive generator. For every $N \in \mathbb{N}_0$, the nonintegral solutions in $[N, N+1)$ are precisely*

$$x_{N,k} = \lceil N - k\beta \rceil + k\beta, \qquad 1 \le k \le \left\lfloor \frac{N+1}{\beta} \right\rfloor,$$

so their number is exactly $\lfloor (N+1)/\beta \rfloor$. Consequently

$$\#(\mathcal{S} \cap [N, N+1)) = \#(\mathcal{C} \cap [2^N, 2^{N+1})) = 1 + \left\lfloor \frac{N+1}{\beta} \right\rfloor.$$

Proof. For fixed $k \ge 1$ there is at most one integer n with $N \le n + k\beta < N+1$; such an $n \ge 0$ exists if and only if $k\beta < N+1$, and then $n = \lceil N - k\beta \rceil$. Irrationality of β makes $(N+1)/\beta$ nonintegral, which gives the stated range. The unique integral solution in the block is N itself, whence the additional 1; passing through $m = 2^x$ turns the exponent block $[N, N+1)$ into the magnitude block $[2^N, 2^{N+1})$ bijectively. □

Status: [kernel-verified Lean]

`ExactCounting` proves the explicit ceiling-image parametrization, `twoBaseNonintegerSolutionsInUnitBlock_ncard`, `twoBaseNaturalCandidatesInDyadicBlock_card`, and the exact finite-sum identity `cumulativeTwoBaseCandidateCount_eq`. `QuadraticAsymptotic` identifies that cumulative count with the candidate count below 2^T and proves the sharp bounds and the limit in (5).

In particular a failed conjecture would produce only linearly many candidates in the logarithmic block index,

$$\#(\mathcal{C} \cap [X, 2X]) \asymp \log X,$$

and the kernel-verified zero-free region gives $\beta \geq 12$, hence $\beta > 12$ because β is irrational. The block count is therefore at most $\lfloor N/12 \rfloor$ (and at most $\lfloor N/20 \rfloor$ under the interval scan, which is **[interval computation]**). Summing over blocks, the cumulative candidate count below 2^T is

$$C(T) = T + \sum_{r=1}^{T} \left\lfloor \frac{r}{\beta} \right\rfloor = \frac{T^2}{2\beta} + O(T). \tag{5}$$

More precisely, Lean proves the two-sided bound and the limit

$$\frac{T(T+1)}{2\beta} \leq C(T) \leq \frac{T(T+1)}{2\beta} + T, \qquad \frac{C(T)}{T^2} \longrightarrow \frac{1}{2\beta}.$$

This is conditionally sharper than the unconditional verified bounds ($(\mathrm{clog}_2 N)^2$ and $\mathrm{clog}_2 N \cdot \mathrm{clog}_3 N$ of Section 4) and matches them in leading order: with $\beta > 12$ the leading coefficient $1/(2\beta)$ is below $1/24$.

There is also an unconditional growth reformulation, obtained by comparing the two cases. Under the conjecture the count below 2^T is exactly T; under failure the preceding limit is the strictly positive constant $1/(2\beta)$. Hence

$$\text{Conjecture 2.1 holds} \iff \frac{\#(\mathcal{C} \cap [1, 2^T))}{T^2} \longrightarrow 0, \tag{6}$$

which is `alaogluErdosConjecture_iff_candidate_count_below_two_pow_subquadratic`. **[kernel-verified Lean]**

This exact population is the correct benchmark for every proposed density or determinant argument. A method that proves only $O((\log X)^2)$ candidates in $[X, 2X]$, for instance, is a valid unconditional theorem but cannot contradict Theorem 5.8: it is weaker than the conditional truth it would need to refute.

5.6 Blockwise irrational rotation and equidistribution

By Theorem 5.8 the fractional part of $x_{N,k}$ is $\{k\beta\}$, so a high unit block contains exactly an initial segment

$$\{\beta\}, \ \{2\beta\}, \ \ldots, \ \{K_N\beta\}, \qquad K_N = \left\lfloor \frac{N+1}{\beta} \right\rfloor,$$

of one irrational rotation orbit, and $K_N \to \infty$.

Theorem 5.9 (Finite Fourier tests on exact blocks)**.** *Assume a counterexample exists and let β be its primitive generator. For every finitely supported coefficient function $c : \mathbb{Z} \to \mathbb{C}$, put*

$$P_c(t) = \sum_{h \in \mathbb{Z}} c(h) e^{2\pi i h t}.$$

Then, on the exact ceiling-parametrized points $x_{N,k} = \lceil N - k\beta \rceil + k\beta$,

$$\frac{1}{K_N} \sum_{k=1}^{K_N} P_c(x_{N,k}) \longrightarrow c(0) \qquad (N \to \infty).$$

Status: [kernel-verified Lean]

`BlockWeyl` proves vanishing of every nonzero normalized Fourier mode on the exact block points, and `BlockEquidistribution` proves `tendsto_primitiveUnitBlockTrigonometricAverage` for every finitely supported coefficient function, with limit `c 0`. This is convergence against finite trigonometric polynomials; it is not yet a theorem for arbitrary continuous tests.

Theorem 5.10 (Blockwise Weyl equidistribution)**.** *Assume a counterexample exists. For every interval $I \subseteq [0,1]$ whose boundary has measure zero,*

$$\frac{\#\{x \in (\mathcal{S} \setminus \mathbb{N}_0) \cap [N, N+1) : \{x\} \in I\}}{\#\big((\mathcal{S} \setminus \mathbb{N}_0) \cap [N, N+1)\big)} \longrightarrow |I| \qquad (N \to \infty).$$

Proof. By Theorem 5.8 the relevant fractional parts are $\{\beta\}, \{2\beta\}, \dots, \{K_N\beta\}$ with $K_N = \lfloor (N+1)/\beta \rfloor \to \infty$. For every nonzero integer h the Weyl sum is a finite geometric series,

$$\frac{1}{K}\sum_{k=1}^{K} e^{2\pi i h k\beta} = \frac{e^{2\pi i h\beta}\big(1 - e^{2\pi i hK\beta}\big)}{K\big(1 - e^{2\pi i h\beta}\big)} \longrightarrow 0,$$

since β is irrational and hence $e^{2\pi i h\beta} \neq 1$. Weyl's criterion concludes. □

Status: [paper proof in this report]

The finite-trigonometric-polynomial input is kernel-verified by Theorem 5.9, but the Stone–Weierstrass and Weyl-criterion extension to all continuous tests, and then to interval indicators, has not been formalized. No effective discrepancy rate follows without further Diophantine input on β: the geometric-sum proof controls each fixed Fourier mode but not uniformly over many modes.

This regularity is attractive, and it is worth stating plainly why it does not by itself decide anything. The integrality condition being tested is not a fixed positive-measure target in the circle; it is an exponentially shrinking target, since the relevant approximation quality must improve like the reciprocal of the integer sizes involved. Equidistribution at a fixed scale is simply the wrong instrument for a target whose measure tends to zero geometrically, and that mismatch recurs in the analytic approaches examined later in this report.

6 Equivalent one-number and radical formulations

The exact conditional structure of Section 5 compresses a statement quantified over all real x into a statement about a single real number at a time. This section records that compression in its two kernel-verified forms — a localization at the prime 3 and a radical-algebraicity normalization — and then names the exact arithmetic obstruction that remains.

6.1 Localization at the prime three

Theorem 6.1 (Localization reformulation)**.** *Conjecture 2.1 is false if and only if there exists an odd integer $u > 1$ with*

$$u^{\vartheta} \in \mathbb{Z}[1/3],$$

equivalently, an odd $u > 1$ and $a \in \mathbb{N}_0$ with $3^a u^\vartheta \in \mathbb{N}$.

Proof. If x is a counterexample, write $2^x = 2^a u$ with u odd; since 2^x is not a power of two, $u > 1$. Using $(2^a)^\vartheta = 3^a$,

$$u^\vartheta = \frac{(2^x)^\vartheta}{(2^a)^\vartheta} = \frac{3^x}{3^a} \in \mathbb{Z}[1/3].$$

Conversely, if $u^\vartheta = B/3^a$ with $B \in \mathbb{N}$, set $x = a + \log_2 u$; then $2^x = 2^a u \in \mathbb{N}$ and $3^x = 3^a u^\vartheta = B \in \mathbb{N}$, and $x \notin \mathbb{Z}$ since otherwise the odd $u > 1$ would be a power of two. □

Status: [kernel-verified Lean] — `alaogluErdosConjecture_iff_odd_rpow_not_localized` in `Localization` and `alaogluErdosConjecture_iff_oddLocalizedRadicalExclusion` in `RadicalReformulation`; the candidate-level variant `alaogluErdosConjecture_iff_no_odd_multiple_candidate`, also in `Localization`, is proved independently and phrases the same content in terms of admissible candidate bases. The equivalence is kernel-verified; the localized-radical exclusion on its right-hand side remains the open conjecture. The allowance of 3-power denominators is exactly the freedom created by shifting the exponent by an integer.

The denominator restriction to powers of 3 is not cosmetic. Only the prime 3 can occur in the denominator, because the exponent shift $x \mapsto x + 1$ multiplies 2^x by 2 and 3^x by 3; the odd part u absorbs the 2-adic freedom completely and $\mathbb{Z}[1/3]$ absorbs the remaining 3-adic freedom. Everything else has been eliminated.

6.2 Reduction to power-primitive odd bases

An odd $u > 1$ that is a proper power, $u = w^d$ with w odd and power-primitive, satisfies $u^\vartheta = (w^\vartheta)^d$. Unique primitive-power decomposition therefore lets the quantifier in Theorem 6.1 range over power-primitive bases only, at the cost of an extra exponent $d \geq 1$.

Status: [kernel-verified Lean] — `PrimitiveRadical` proves unique primitive-power decomposition of every integer $u > 1$ together with the exact equivalence `alaogluErdosConjecture_iff_primitiveOddLocalizedRadicalExclusion`. Proper-power choices of u introduce no genuinely new cases.

6.3 Canonical radical algebraicity

In the notation of the least generator of Section 5, a counterexample produces natural numbers w, d, a, A with $w > 1$ odd and power-primitive, $d > 0$, and

$$\beta = a + d \log_2 w, \qquad 2^\beta = 2^a w^d, \qquad 3^\beta = A,$$

normalized so that $a = 0$ or $3 \nmid A$. Setting

$$E := w^\vartheta = 3^{\log_2 w},$$

one has $E^d = A/3^a \in \mathbb{Q}$, and d is the least positive exponent for which E^d is rational. Minimality of d has an exact algebraic consequence.

Proposition 6.2 (Exact radical degree)**.** *The binomial $X^d - A/3^a$ is irreducible over $\mathbb{Q}$, it is the minimal polynomial of E, and E is algebraic of exact degree d.*

Proof. For a positive rational q, the binomial $X^d - q$ can be reducible only if q is a p-th power in $\mathbb{Q}$ for some prime $p \mid d$ (the additional $4 \mid d$ exceptional case concerns -4 times a fourth power and is irrelevant for $q > 0$). If $q = r^p$ with $p \mid d$, the positive real $E^{d/p}$ is the positive p-th root of q, hence rational, contradicting minimality of d. □

Status: [kernel-verified Lean] — `irreducible_X_pow_sub_C_of_least_rational_power` in `RadicalDegree` proves the general norm argument, and `oddCoreRpow_exact_radical_degree` specializes it to the least normalized radical index. It identifies the minimal polynomial with $X^d - A/3^a$ and proves that the degree is exactly d.

A counterexample therefore produces the very special algebraic number $E = 3^{\log_2 w}$, reached from a transcendental exponent. Taking logarithms of $E^d = A/3^a$:

$$d \log w \log 3 = (\log A - a \log 3) \log 2. \tag{7}$$

This is a *bilinear* relation among logarithms of integers: each side is a product of two logarithms, not a linear form in logarithms with algebraic coefficients. Baker's theorem [3] bounds linear forms $\sum \alpha_i \log \gamma_i$ with algebraic α_i away from zero, and no substitution turns (7) into such a form, because the coefficient of $\log w$ on the left is $d \log 3$, itself a logarithm rather than an algebraic number. The four exponentials conjecture does forbid the relation. Equation (7) is thus a precise description of the transcendence-theoretic wall, not a disguised elementary factorization problem.

6.4 Where the wall stands in the smallest case

The smallest structural question already meets the open core. A 3-smooth candidate $m = 2^a 3^b$ with $b \geq 1$ would require $3^a E_3^b \in \mathbb{N}$ for

$$E_3 = 3^{\log_2 3} = e^{(\log 3)^2/\log 2} = 5.7045224946911176\ldots,$$

and even the irrationality of E_3 is open: it is a "three-logarithms" quantity outside the reach of Gelfond–Schneider, Baker's method, and the six exponentials theorem. No currently known technique excludes the odd core $w = 3$; this is why the corpus emphasizes counting, structure, and verified finite regions rather than infinite families of exclusions, and why Conjecture 6.3 below is the honest name of the obstruction.

Conjecture 6.3 (Radical four-exponentials target)**.** *For every odd integer $u > 1$, $u^{\log_2 3} \notin \mathbb{Z}[1/3]$. Equivalently, for every power-primitive odd $w > 1$ and every $d \geq 1$, $(3^{\log_2 w})^d \notin \mathbb{Z}[1/3]$.*

By Theorem 6.1 and the primitive reduction above, both formulations are exactly equivalent to Conjecture 2.1; neither is proved. A proof of the special case $u = 3$ — even of the weaker assertion that $3^{\log_2 3}$ is irrational — by any genuinely new mechanism would already be a meaningful breakthrough.

7 The exact four-exponentials barrier

7.1 The two-by-two exponential matrix

Suppose x is a nonintegral solution. Consider the row vector of parameters $(1, x)$ and the column vector of parameters $(\log 2, \log 3)$. The four exponentials formed from the products are

$$\exp(1 \cdot \log 2) = 2, \quad \exp(1 \cdot \log 3) = 3, \quad \exp(x \log 2) = 2^x, \quad \exp(x \log 3) = 3^x,$$

all four algebraic — indeed positive integers. The two row parameters $1, x$ are $\mathbb{Q}$-linearly independent because x is irrational: a rational nonintegral exponent is excluded by the rational-exponent argument of Section 2. The two column parameters $\log 2, \log 3$ are $\mathbb{Q}$-linearly independent because $2^a 3^b = 1$ forces $a = b = 0$, that is, no nonzero powers of 2 and 3 agree. The real four exponentials conjecture predicts that at least one of the four displayed values must be transcendental, a contradiction [10].

Status: [kernel-verified Lean] — `alaogluErdosConjecture_of_realFourExponentialsConjecture` in `FourExponentialsReduction` formalizes exactly this implication. It does not assume four exponentials as an axiom: the conjectural statement is defined as a proposition and the implication is proved conditionally on it. Four exponentials itself is open, so this is context and diagnosis, not a proof of Conjecture 2.1.

The reduction is a diagnostic and not merely background. Any proposed elementary argument must exploit integrality, positivity, or arithmetic structure beyond bare algebraicity of those four values, or else it would prove a substantial case of the four exponentials conjecture by another name.

7.2 Why six exponentials does not specialize

The six exponentials theorem applies to two independent row parameters together with three independent column parameters, or to three rows and two columns. Here only two canonical logarithmic directions are available. Adding a third base 5 supplies the missing direction and yields the kernel-verified three-base theorem

$$2^x, 3^x, 5^x \in \mathbb{Z} \implies x \in \mathbb{Z},$$

proved by an interpolation determinant. In the two-base problem, any additional base built from powers of 2 and 3 has logarithm in the rational span of $\log 2$ and $\log 3$ and therefore does not create an independent column.

The dimension count is what decides the outcome. With three independent row directions and two column directions the analytic size estimate beats the arithmetic lower bound $|\det| \geq 1$; in the 2×2 configuration the same balance lands exactly on the classical boundary, where the required decay below one fails. A successful two-base determinant argument would, by construction, be a new four-exponentials argument.

The classification of auxiliary bases makes the obstruction precise: `ThreeSmooth` and the denominator-controlled results in `RationalThirdBase` show that an auxiliary base with a prime factor outside $\{2, 3\}$ would supply a genuinely new column, but its corresponding exponential value is not forced to be an integer — or even algebraic — by the original hypotheses. Bases supported only on 2 and 3 do not increase the transcendence rank.

7.3 Integrality is stronger than algebraicity, but not by enough

Four exponentials asks only for algebraicity. The values here are integers, so there is extra discrete structure available, and the determinant machinery of Section 19 and Section 20 extracts some of it: the relevant determinants have integer entries, so a nonzero determinant is a nonzero integer and therefore satisfies $|\det| \geq 1$.

That inequality is the whole of the extra archimedean information. A nonzero integer has absolute value at least one, and nothing in the integrality hypotheses improves the constant 1 by itself. To cross the four-exponentials threshold one would need a strictly stronger arithmetic input, for instance:

- a proof that the determinants are divisible by a large integer, replacing $|\det| \geq 1$ by $|\det| \geq D$ with D growing in the size parameters; or
- a proof that a suitable common denominator is anomalously small, improving the arithmetic side rather than the analytic side.

Both routes ask for divisibility rather than size, which is precisely why the p-adic direction is given priority in the research program of Section 27.

Status: [paper proof in this report] for the determinant hierarchy cited here: the arbitrary-node repulsion theorem and the semigroup generalized-Vandermonde bounds are paper proofs in this report. **[Lean draft, not yet accepted]** — `ArbitraryNodeDeterminant`, `SemigroupDeterminant`, and `MixedExponentSemigroup` contain Lean drafts that the kernel has not yet accepted. The reduction of Conjecture 2.1 to four exponentials is **[kernel-verified Lean]**, but four exponentials is open and the conjecture remains open with it.

8 Rational-function rigidity and closure properties

The exact primitive-generator theorem of Section 5 determines the additive structure of $\mathcal{S}$ completely under failure of Conjecture 2.1. It has a consequence that deserves to be isolated: a hypothetical solution monoid is not merely additively free of rank two, it is *rigid* against every nonlinear rational operation over the algebraic numbers. Addition and multiplication by naturals are the only operations that stay inside $\mathcal{S}$, and every apparent exception collapses to an identity in one indeterminate. The results below are exact equivalences, several of them unconditional, and they translate directly into closure reformulations of the conjecture itself.

8.1 Evaluation injectivity at the generator

Throughout this section, β denotes the canonical least nonintegral solution supplied by the generator theorem of Section 5, so that

$$\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\}$$

with unique coordinates n, k. The transcendence dichotomy of Section 4 makes β transcendental over $\mathbb{Q}$. The rational-function statements need the slightly stronger form over the algebraic closure.

Lemma 8.1 (Evaluation is injective over $\overline{\mathbb{Q}}$)**.** *Let $\gamma \in \mathbb{C}$ be transcendental over $\mathbb{Q}$. Then no nonzero $P \in \overline{\mathbb{Q}}[T]$ satisfies $P(\gamma) = 0$; equivalently, the evaluation homomorphism $\overline{\mathbb{Q}}[T] \to \mathbb{C}$, $P \mapsto P(\gamma)$, is injective.*

Proof. If $P(\gamma) = 0$ for some nonzero $P \in \overline{\mathbb{Q}}[T]$, then γ is algebraic over $\overline{\mathbb{Q}}$. Since $\overline{\mathbb{Q}}/\mathbb{Q}$ is an algebraic extension, the tower property makes γ algebraic over $\mathbb{Q}$, contradicting the hypothesis. Injectivity follows by applying this to differences. ☐

Status: **[paper proof in this report]** — the transcendence of β itself is **[kernel-verified Lean]** (Section 4, `Transcendence`); the only new content here is the standard tower argument that upgrades transcendence over $\mathbb{Q}$ to transcendence over $\overline{\mathbb{Q}}$.

8.2 The rational-function intersection theorem

Theorem 8.2 (Rational-function rigidity at the primitive generator)**.** *Assume Conjecture 2.1 is false and let β be the canonical generator. Let $R(T) \in \overline{\mathbb{Q}}(T)$ have no pole at β. Then*

$$\boxed{R(\beta) \in \mathcal{S} \quad \Longleftrightarrow \quad R(T) = n + kT \ \text{ in } \overline{\mathbb{Q}}(T) \text{ for some } n, k \in \mathbb{N}_0.}$$

Proof. The reverse implication is additive closure of $\mathcal{S}$ together with closure under multiplication by naturals: $R(\beta) = n + k\beta \in \mathcal{S}$.

For the forward implication write $R = P/Q$ with $P, Q \in \overline{\mathbb{Q}}[T]$ and $Q(\beta) \neq 0$. If $R(\beta) \in \mathcal{S}$, the exact coordinates give $R(\beta) = n + k\beta$ for unique $n, k \in \mathbb{N}_0$, hence

$$P(\beta) - (n + k\beta)\, Q(\beta) = 0.$$

The left-hand side is the value at β of the polynomial $P(T) - (n + kT)Q(T) \in \overline{\mathbb{Q}}[T]$. By Lemma 8.1 that polynomial is identically zero, so $P(T) = (n + kT)Q(T)$ and $R(T) = n + kT$ in $\overline{\mathbb{Q}}(T)$. ☐

Status: **[paper proof in this report]**

The theorem allows poles away from β, permits arbitrary cancellation, and takes the full algebraic closure as coefficient field. It can be read as the exact intersection

$$\mathcal{S} \cap \overline{\mathbb{Q}}(\beta) = \mathbb{N}_0 + \mathbb{N}_0\beta, \tag{8}$$

with the extra information that every rational expression realizing an element of that intersection is *identically* the corresponding affine expression — there is no accidental representation.

Corollary 8.3 (Rational self-maps of the solution set)**.** *Under failure, the rational functions over $\overline{\mathbb{Q}}$ that are regular on $\mathcal{S}$ and map $\mathcal{S}$ into itself are exactly the maps*

$$T \longmapsto n + kT, \qquad n, k \in \mathbb{N}_0.$$

Proof. Such a function in particular sends β into $\mathcal{S}$, so Theorem 8.2 identifies it as $n + kT$ with $n, k \in \mathbb{N}_0$. Conversely, translation by a natural and dilation by a natural preserve $\mathbb{N}_0 + \mathbb{N}_0\beta$. ☐

Status: [paper proof in this report]

Proposition 8.4 (Multivariate reduction to one indeterminate)**.** *Assume failure, let $x_i = n_i + k_i\beta \in \mathcal{S}$ for $i = 1, \ldots, r$, and let $F \in \overline{\mathbb{Q}}(X_1, \ldots, X_r)$ be regular at $(x_1, \ldots, x_r)$. Then*

$$F(x_1, \ldots, x_r) \in \mathcal{S}$$

if and only if the one-variable specialization

$$G(T) := F(n_1 + k_1T, \ldots, n_r + k_rT)$$

is identically $p + qT$ in $\overline{\mathbb{Q}}(T)$ for some $p, q \in \mathbb{N}_0$.

Proof. Write $F = P/Q$ with $P, Q \in \overline{\mathbb{Q}}[X_1, \ldots, X_r]$ and $Q(x_1, \ldots, x_r) \neq 0$. The specialized denominator $Q(n_1 + k_1T, \ldots, n_r + k_rT)$ is a polynomial in T that does not vanish at $T = \beta$, hence is not the zero polynomial. Therefore G is a well-defined element of $\overline{\mathbb{Q}}(T)$ with no pole at β, and $G(\beta) = F(x_1, \ldots, x_r)$. Now apply Theorem 8.2. □

Status: [paper proof in this report]

Every rational operation on finitely many hypothetical solutions is thus reduced to an elementary identity between polynomials in a single indeterminate, with coordinates in $\mathbb{N}_0$. No analytic input is required beyond the transcendence already verified by the kernel.

8.3 Exact multiplication criterion

The next statement needs no assumption at all: the branch in which the conjecture holds is trivial, and the branch in which it fails is governed by the generator.

Theorem 8.5 (Products of two solutions)**.** *For all $x, y \in \mathcal{S}$, unconditionally,*

$$\boxed{xy \in \mathcal{S} \iff x \in \mathbb{N}_0 \text{ or } y \in \mathbb{N}_0.}$$

Proof. If one factor, say x, lies in $\mathbb{N}_0$, then xy is a natural multiple of a solution and therefore a solution; this direction is unconditional.

For the converse, split on Conjecture 2.1. If the conjecture holds then $\mathcal{S} = \mathbb{N}_0$ and the right-hand side is automatically true, so there is nothing to prove. Under failure, write the unique coordinates

$$x = n + k\beta, \qquad y = m + \ell\beta, \qquad n, k, m, \ell \in \mathbb{N}_0.$$

If neither factor is integral then $k, \ell > 0$ and

$$xy = nm + (n\ell + mk)\beta + k\ell\,\beta^2,$$

with $k\ell > 0$. Membership $xy \in \mathcal{S}$ would force $xy = r + s\beta$ for some $r, s \in \mathbb{N}_0$, producing a nonzero polynomial of degree exactly two over $\mathbb{Q}$ vanishing at the transcendental number β. Equivalently, apply Theorem 8.2 to $R(T) = (n + kT)(m + \ell T)$, which is not affine. Both routes are impossible. □

Status: [paper proof in this report] — note that only transcendence over $\mathbb{Q}$ is used, which is [kernel-verified Lean]; the appeal to $\overline{\mathbb{Q}}$ in Theorem 8.2 is a convenience, not a necessity.

Corollary 8.6 (Finite products). *Let $x_1, \dots, x_r \in \mathcal{S}$ be positive. Then*

$$\prod_{i=1}^{r} x_i \in \mathcal{S} \quad \iff \quad \text{at most one } x_i \text{ is nonintegral.}$$

If a zero factor is allowed, the product is $0 \in \mathcal{S}$ and the criterion is vacuous.

Proof. If the conjecture holds, every x_i is integral and both sides are true. Under failure, the positive integral factors multiply to a positive natural, so after removing them the product is a natural multiple of a product of nonintegral solutions. Writing each remaining factor as $n_i + k_i\beta$ with $k_i > 0$ and expanding, the result is a polynomial in β whose degree equals the number of nonintegral factors and whose leading coefficient is positive. Degree at least two is excluded by Theorem 8.2, while degree zero or one is preserved by natural dilation and translation. □

Status: [paper proof in this report]

Corollary 8.7 (Internal power test). *For every $x \in \mathcal{S}$ and every integer $r \geq 2$,*

$$x^r \in \mathcal{S} \quad \iff \quad x \in \mathbb{N}_0.$$

In particular $x \in \mathbb{N}_0$ holds exactly when $x^2 \in \mathcal{S}$: a single squaring test detects integrality.

Proof. Apply Theorem 8.5 with $y = x$ and induct, or apply Corollary 8.6 to r equal factors, noting that $x = 0$ satisfies both sides. □

Status: [paper proof in this report]

8.4 Exact quotient criterion

The same evaluation argument classifies division. Division is more delicate than multiplication because even the true solution set $\mathbb{N}_0$ is not closed under arbitrary quotients, so the criterion cannot be a clean dichotomy; it is instead an exact divisibility condition on coordinates.

Theorem 8.8 (Quotients of solutions). *Assume Conjecture 2.1 is false and write the unique coordinates*

$$x = n + k\beta, \qquad y = m + \ell\beta > 0.$$

Exactly one of the two cases $\ell = 0$ and $\ell > 0$ applies, and in that case $x/y \in \mathcal{S}$ if and only if the corresponding condition holds:

(a) *$\ell = 0$, so that $y = m$ is a positive natural, and $m \mid n$ and $m \mid k$;*

(b) *$\ell > 0$, and $x = qy$ for some $q \in \mathbb{N}_0$, equivalently $n = qm$ and $k = q\ell$.*

In particular, if x and y are both nonintegral, then

$$\frac{x}{y} \in \mathcal{S} \quad \Longleftrightarrow \quad \frac{x}{y} \in \mathbb{N}_0.$$

Proof. Suppose $x/y = r + s\beta$ with $r, s \in \mathbb{N}_0$. Clearing the denominator gives

$$n + k\beta = mr + (ms + \ell r)\beta + \ell s\,\beta^2.$$

The difference of the two sides is a polynomial in β with rational coefficients, so transcendence of β forces every coefficient to vanish:

$$\ell s = 0, \qquad k = ms + \ell r, \qquad n = mr.$$

If $\ell = 0$ then $y = m$ and $y > 0$ gives $m \geq 1$; the equations read $n = mr$ and $k = ms$, which is precisely $m \mid n$ and $m \mid k$. If $\ell > 0$ then $s = 0$, so $n = mr$ and $k = \ell r$, that is $x = ry$ with $r \in \mathbb{N}_0$. The converses follow by direct substitution.

For the final claim, both x and y nonintegral means $k, \ell > 0$, so case (b) applies and $x/y = r \in \mathbb{N}_0$. □

Status: **[paper proof in this report]**

Corollary 8.9 (Reciprocals)**.** *Unconditionally, for $x \in \mathcal{S}$ with $x > 0$,*

$$\frac{1}{x} \in \mathcal{S} \quad \Longleftrightarrow \quad x = 1.$$

Proof. If the conjecture holds then $\mathcal{S} = \mathbb{N}_0$ and the claim is the elementary fact that a positive natural with natural reciprocal equals 1. Under failure apply Theorem 8.8 with numerator $1 = 1+0\cdot\beta$ and denominator $y = x$. If $x = m$ is integral, case (a) gives $m \mid 1$, so $m = 1$. If x is nonintegral, case (b) gives $1 = qx$ for some $q \in \mathbb{N}_0$; here $q = 0$ is impossible, and $q \geq 1$ would make $x = 1/q$ rational, contradicting the irrationality of every nonintegral solution. □

Status: **[paper proof in this report]**

A hypothetical solution monoid is therefore additively rich — a free additive monoid of rank two — and almost maximally hostile to every multiplicative operation: products, powers, reciprocals and quotients all escape $\mathcal{S}$ except in the degenerate cases forced by the additive structure itself.

8.5 Closure formulations of the conjecture

Because the multiplication criterion is unconditional, it converts directly into equivalent statements of the conjecture in terms of closure.

Theorem 8.10 (Multiplicative closure equivalences)**.** *The following are equivalent.*

(i) *Conjecture 2.1 is true.*

(ii) *$\mathcal{S}$ is closed under multiplication.*

(iii) *$\mathcal{S}$ is closed under squaring.*

(iv) *For every real x,*

$$2^x, 3^x \in \mathbb{Z} \quad \Longrightarrow \quad 2^{x^2}, 3^{x^2} \in \mathbb{Z}.$$

Proof. If the conjecture holds then $\mathcal{S} = \mathbb{N}_0$, which is closed under multiplication, so (i) implies (ii); and (ii) trivially implies (iii). Statements (iii) and (iv) are the same assertion, written once for $\mathcal{S}$ and once for the defining integrality conditions. Finally, assume (iii). For any $x \in \mathcal{S}$ we get $x^2 \in \mathcal{S}$, so Corollary 8.7 forces $x \in \mathbb{N}_0$. Hence $\mathcal{S} = \mathbb{N}_0$, which is Conjecture 2.1. □

Status: [paper proof in this report]

The candidate formulation of Section 2 carries the same content in multiplicative language. For positive reals define the commutative operation

$$m \star n := 2^{(\log_2 m)(\log_2 n)} = m^{\log_2 n} = n^{\log_2 m}. \tag{9}$$

If $m = 2^x$ and $n = 2^y$, then $m \star n = 2^{xy}$, so $\star$ is exactly the transport of multiplication of exponents through the exponent–candidate bijection of Section 2.

Corollary 8.11 (Candidate star-product)**.** *For $m, n \in \mathcal{C}$,*

$$m \star n \in \mathcal{C} \quad \Longleftrightarrow \quad m \text{ is a power of } 2 \text{ or } n \text{ is a power of } 2.$$

Consequently Conjecture 2.1 is equivalent both to closure of $\mathcal{C}$ under $\star$ and to the apparently weaker diagonal condition

$$m \in \mathcal{C} \quad \Longrightarrow \quad m \star m \in \mathcal{C}.$$

Proof. Put $x = \log_2 m$ and $y = \log_2 n$, so that $x, y \in \mathcal{S}$ and $m \star n = 2^{xy}$. Membership $m \star n \in \mathcal{C}$ is equivalent to $xy \in \mathcal{S}$, and m is a power of two exactly when $x \in \mathbb{N}_0$; now apply Theorem 8.5. The two closure equivalences are the image of Theorem 8.10(ii) and (iii) under the same bijection. □

Status: [paper proof in this report]

These are genuine equivalences and not a proof. There is at present no independent mechanism forcing nonlinear closure: nothing in the archimedean, valuation-theoretic, or determinant toolkits of Sections 4, 19 and 20 supplies a reason for $\mathcal{C}$ to be closed under $\star$. Their value is diagnostic — they identify the missing property as a single crisp failure of semiring-like structure, rather than as a quantitative deficit.

8.6 Formalization cost

The multiplication, power and quotient criteria are inexpensive Lean targets, and a route exists that avoids building a general rational-function library. For a new module — `RationalRigidity` would be a natural name — the plan is:

1. case split on `AlaogluErdosConjecture`, discharging the true branch by $\mathcal{S} = \mathbb{N}_0$;

2. in the failure branch obtain the canonical generator and the unique coordinates from the packaged statement

```
exists_canonical_primitiveGenerator_of_not_alaogluErdosConjecture
```

in the module `PrimitiveGenerator`;

3. expand the relevant equality as a polynomial of degree at most two in β with rational coefficients;

4. invoke the kernel-verified transcendence in `Transcendence` to force every coefficient to vanish;

5. discharge the remaining natural-number divisibility and coordinate conditions with `omega`.

Only the full rational-function theorem, Theorem 8.2, needs evaluation injectivity over $\overline{\mathbb{Q}}$ and hence Lemma 8.1; the multiplication theorem, the power test, the quotient criterion and the closure equivalences need nothing beyond transcendence over $\mathbb{Q}$, which is already accepted by the kernel.

Status: Planned; no draft yet. The mathematics is **[paper proof in this report]** throughout this section, and the prerequisites are **[kernel-verified Lean]**.

8.7 What the rigidity statements buy

The practical payoff is that they sharply restrict how a counterexample could be manufactured from a smaller one. A natural strategy for producing new solutions from old ones is to apply some algebraic operation — square a solution, multiply two of them, take a ratio, invert, or substitute solutions into a rational function — and hope to land back in $\mathcal{S}$, either to run a descent or to build an infinite family with better arithmetic properties than the canonical generator. Equation (8) closes that door completely: the only rational functions over $\overline{\mathbb{Q}}$ that map solutions to solutions are the affine maps $T \mapsto n + kT$ with natural coefficients, which reproduce nothing that the additive monoid $\mathbb{N}_0 + \mathbb{N}_0\beta$ did not already contain. Any construction of a new counterexample from an old one must therefore be transcendental in an essential way, or must leave the solution set and return by a route that is not rational over the algebraic numbers. In the other direction, the closure equivalences of Theorem 8.10 and Corollary 8.11 convert the conjecture into a purely structural question — is $\mathcal{C}$ closed under $\star$? — which is attractive precisely because it has no quantitative parameter to optimize, and which is a warning that no counting or equidistribution bound alone will produce the answer. Finally, all of these statements are short, elementary consequences of results already accepted by the kernel, which makes them the cheapest genuine additions to the verified corpus that this report proposes.

9 Three-smooth outputs and transcendental product exponents

Call a positive integer *three-smooth* when it has no prime factor outside $\{2, 3\}$, that is, when it equals 2^u3^v for some $u, v \in \mathbb{N}_0$. The kernel-verified classification of auxiliary bases recorded in Section 4 settles which bases B satisfy $B^x \in \mathbb{Z}$ for a *fixed* nonintegral solution x: exactly the three-smooth ones. This section turns that classification onto the outputs $2^x, 3^x$ themselves, and then onto second-level exponents xy formed from two solutions.

Two distinct things come out, and they have different provenance. The first is an unconditional integrality test: a solution is integral precisely when its own two outputs are three-smooth. It uses only the kernel-verified transcendence of ϑ and unique factorization, so it is cheap to add to the Lean development. The second is an upgrade of "not an integer" to "transcendental", obtained from the six exponentials theorem. That upgrade converts several apparently weaker *algebraicity* statements into exact reformulations of Conjecture 2.1, at the cost of a classical input that the repository has not formalized in general form.

9.1 The two outputs cannot both be three-smooth

Theorem 9.1 (Three-smooth output test)**.** *For every $x \in \mathcal{S}$ the following are equivalent:*

(i) $x \in \mathbb{N}_0$*;*

(ii) *both* 2^x *and* 3^x *are three-smooth;*

(iii) 6^x *is three-smooth.*

Proof. Condition (i) gives (ii) immediately, since 2^x and 3^x are then powers of 2 and 3 respectively. Conditions (ii) and (iii) are equivalent because 2^x and 3^x are positive integers with product 6^x, and a product of positive integers is three-smooth exactly when both factors are.

For (ii)⇒(i), write

$$2^x = 2^p 3^q, \qquad 3^x = 2^r 3^s$$

with $p, q, r, s \in \mathbb{N}_0$. Taking logarithms and dividing by $\log 2$ and $\log 3$ respectively gives

$$x = p + q\vartheta, \qquad x = s + \frac{r}{\vartheta},$$

where $\vartheta = \log_2 3$. Eliminating x and clearing the denominator yields the quadratic relation

$$q\,\vartheta^2 + (p - s)\,\vartheta - r = 0 \tag{10}$$

with rational — indeed integer — coefficients. The repository proves ϑ transcendental, so every coefficient in (10) vanishes: $q = 0$, $p = s$, and $r = 0$. Hence $2^x = 2^p$ and $x = p \in \mathbb{N}_0$. □

Status: [paper proof in this report]

The statement is unconditional and does not use the conditional generator, the exact solution monoid, or any six-exponentials input. Its only nonelementary ingredient is the kernel-verified transcendence of ϑ,

`transcendental_logThreeDivLogTwo`

in `Transcendence`, together with unique factorization. A Lean proof is expected to be short; see Section 28 and the skeleton at the end of this section.

The test is sharp in a useful direction: it rules out one of the four a priori smoothness patterns of the least output pair.

Corollary 9.2 (Primitive-core trichotomy)**.** *Assume Conjecture 2.1 fails, and use the simultaneous least-output normalization of Section 5,*

$$M = 2^\beta = 2^a w^d, \qquad A = 3^\beta = 3^b v^e,$$

with $w > 1$ odd and power-primitive, $v > 1$ power-primitive and coprime to 3, and $d, e > 0$. Then at most one of $w = 3$ and $v = 2$ holds, and exactly one of the following three patterns occurs:

M	A	*primitive cores*
three-smooth	*not three-smooth*	$w = 3,\ v \neq 2$
not three-smooth	*three-smooth*	$w \neq 3,\ v = 2$
not three-smooth	*not three-smooth*	$w \neq 3,\ v \neq 2$

The missing fourth pattern, both outputs three-smooth, is impossible.

Proof. An odd, three-smooth, power-primitive integer greater than one is a power of 3, and primitivity forces the exponent to be 1; hence M is three-smooth exactly when $w = 3$. Symmetrically a three-smooth power-primitive integer greater than one that is coprime to 3 is exactly 2, so A is three-smooth exactly when $v = 2$. The least nonintegral solution β lies in $\mathcal{S} \setminus \mathbb{N}_0$, so Theorem 9.1 forbids both outputs from being three-smooth, which is the exclusion $\neg(w = 3 \wedge v = 2)$. □

Status: [paper proof in this report]

The normalization itself, including power-primitivity of w and v, the depth dichotomy $a = 0$ or $b = 0$, and the gcd-one condition, is **[kernel-verified Lean]** in `OutputNormalization` (Section 5). Only the exclusion of the fourth pattern is new here.

The two exceptional rows are precisely the cases the corpus already identifies as hardest: $w = 3$ makes 3^ϑ the sharp radical target of Section 6, and $v = 2$ is its base-swapped mirror with $2^{1/\vartheta} = 2^{\log_3 2}$. The third row is generic and carries two independent outside-prime supports.

9.2 Third-base transcendence from six exponentials

The remaining statements of this section rest on one published transcendence theorem, which we state in the form we use.

Theorem 9.3 (Six exponentials theorem; Siegel, Lang, Ramachandra)**.** *Let u_1, u_2 be complex numbers linearly independent over $\mathbb{Q}$, and let v_1, v_2, v_3 be complex numbers linearly independent over $\mathbb{Q}$. Then at least one of the six numbers $e^{u_i v_j}$, $1 \leq i \leq 2$, $1 \leq j \leq 3$, is transcendental.*

Status: [classical/open background] — classical, not formalized in this repository in general form.

See Waldschmidt [10, 11]. Every consequence below is tagged **[classical/open background]** for the same reason: the repository formalizes only the integer-entry specialization of the underlying interpolation determinant, which yields the 3-smooth classification of Section 4 but not the transcendence conclusion.

Lemma 9.4 (Third-base transcendence)**.** *Let $y \in \mathcal{S} \setminus \mathbb{N}_0$ and let $B \in \mathbb{N}_{>0}$ be not three-smooth. Then B^y is transcendental.*

Proof. The numbers $\log 2, \log 3, \log B$ are linearly independent over $\mathbb{Q}$: a nontrivial rational relation among them would give a nontrivial multiplicative relation among 2, 3, and B, which unique factorization forbids because B has a prime factor outside $\{2, 3\}$. Also 1 and y are linearly independent over $\mathbb{Q}$, since y is irrational — a rational solution is integral by the kernel-verified rational-exponent lemma of Section 4, and $y \notin \mathbb{N}_0$.

Apply Theorem 9.3 with $u_1 = 1$, $u_2 = y$ and $v_1 = \log 2$, $v_2 = \log 3$, $v_3 = \log B$. The six values are

$$2, \quad 3, \quad B, \quad 2^y, \quad 3^y, \quad B^y,$$

and the first five are integers, hence algebraic: $2, 3, B$ by hypothesis and $2^y, 3^y$ because $y \in \mathcal{S}$. Therefore the sixth, B^y, is transcendental. □

Status: [classical/open background] — consequence of the six exponentials theorem (Theorem 9.3).

Combining Lemma 9.4 with the kernel-verified 3-smooth classification of Section 4 gives an exact algebraicity trichotomy for integer bases: for every $y \in \mathcal{S} \setminus \mathbb{N}_0$ and every $B \in \mathbb{N}_{>0}$,

$$B^y \in \overline{\mathbb{Q}} \iff B^y \in \mathbb{Z} \iff B = 2^u 3^v \text{ for some } u, v \in \mathbb{N}_0. \tag{11}$$

There is no algebraic-irrational middle case. Any base that is not three-smooth produces a transcendental value outright.

9.3 Products of noninteger solutions

The set $\mathcal{S}$ is closed under addition and under multiplication by natural numbers, but it is not known to be closed under multiplication of two of its own nonintegral elements. The next theorem says exactly what happens at the product exponent xy.

Theorem 9.5 (Product-exponent dichotomy)**.** *Let $x, y \in \mathcal{S}$.*

(a) *If $x \in \mathbb{N}_0$ or $y \in \mathbb{N}_0$, then $2^{xy} \in \mathbb{N}$ and $3^{xy} \in \mathbb{N}$.*

(b) *If $x \notin \mathbb{N}_0$ and $y \notin \mathbb{N}_0$, then each of 2^{xy} and 3^{xy} is either an integer or transcendental, at least one of the two is transcendental, and*

$$6^{xy} \text{ is transcendental.}$$

Proof. Part (a) is natural-number dilation: if $y = n \in \mathbb{N}_0$ then $xy = nx$ is a natural multiple of a solution and therefore a solution, and symmetrically in x.

For part (b) put $M_x = 2^x$ and $A_x = 3^x$, both positive integers, and note

$$2^{xy} = M_x^y, \qquad 3^{xy} = A_x^y.$$

If M_x is three-smooth, the 3-smooth classification of Section 4 makes M_x^y an integer; if M_x is not three-smooth, Lemma 9.4 makes it transcendental. The same alternative holds for A_x^y. By

Theorem 9.1 applied to the nonintegral solution x, the two outputs M_x and A_x are not both three-smooth, so at least one of $2^{xy}, 3^{xy}$ is transcendental.

Finally $6^x = M_x A_x$ is a positive integer, and it is not three-smooth, since a prime outside $\{2, 3\}$ dividing one factor divides the product. Applying Lemma 9.4 with base 6^x and exponent y gives $(6^x)^y = 6^{xy}$ transcendental. □

Status: [**classical/open background**] for part (b) — via Theorem 9.3.

Part (a) is elementary and its underlying closure property, that natural multiples of a solution are solutions, is [**kernel-verified Lean**].

Corollary 9.6 (Diagonal transcendence). *Let $x \in \mathcal{S} \setminus \mathbb{N}_0$. Then 6^{x^2} is transcendental and at least one of 2^{x^2}, 3^{x^2} is transcendental. More precisely,*

$$2^{x^2} \in \mathbb{N} \iff 2^x \text{ is three-smooth},$$
$$3^{x^2} \in \mathbb{N} \iff 3^x \text{ is three-smooth},$$

and failure of either equivalence's right-hand side makes the corresponding value transcendental.

Proof. Take $y = x$ in Theorem 9.5(b). The two displayed equivalences are the alternative used in that proof, applied to the bases $M_x = 2^x$ and $A_x = 3^x$: a three-smooth base gives an integer by the 3-smooth classification of Section 4, and a base that is not three-smooth gives a transcendental value by Lemma 9.4. □

Corollary 9.7 (Mixed three-smooth bases). *Let $x, y \in \mathcal{S} \setminus \mathbb{N}_0$ and let $u, v \geq 1$ be integers. Then $(2^u 3^v)^{xy}$ is transcendental.*

Proof. The positive integer

$$(2^u 3^v)^x = (2^x)^u (3^x)^v$$

is divisible by every prime occurring in either output, because $u, v \geq 1$. By Theorem 9.1 at least one output has a prime factor outside $\{2, 3\}$, so this base is not three-smooth. Apply Lemma 9.4 with exponent y. □

Status: [**classical/open background**] for both corollaries — via Theorem 9.3.

9.4 A global second-level trichotomy

The exact generator theorem of Section 5 makes the preceding dichotomy uniform: which of the three patterns occurs is a property of the counterexample as a whole, not of the particular pair (x, y).

Theorem 9.8 (Uniform second-level pattern). *Assume Conjecture 2.1 fails and let $M = 2^\beta$, $A = 3^\beta$ for the primitive generator β. For every $x, y \in \mathcal{S} \setminus \mathbb{N}_0$, exactly one of the following three patterns holds, and the pattern does not depend on x or y:*

condition on (M, A)	2^{xy}	3^{xy}
M *three-smooth,* A *not*	*integer*	*transcendental*
A *three-smooth,* M *not*	*transcendental*	*integer*
neither three-smooth	*transcendental*	*transcendental*

Proof. By the exact primitive-generator theorem of Section 5, write $x = n + k\beta$ with $n, k \in \mathbb{N}_0$ and $k \geq 1$, so that

$$2^x = 2^n M^k, \qquad 3^x = 3^n A^k.$$

Multiplying by a power of the distinguished smooth prime cannot remove a prime factor outside $\{2, 3\}$, and $k \geq 1$ keeps every such prime of M present in M^k; hence 2^x is three-smooth exactly when M is, and likewise 3^x is three-smooth exactly when A is. Now apply Theorem 9.5(b) together with the alternative used in its proof. The fourth pattern, both M and A three-smooth, is excluded by Theorem 9.1 applied to β. ☐

Status: [classical/open background] — via Theorem 9.3.

The structural input, namely $\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\}$ with unique coordinates, is **[kernel-verified Lean]** in `PrimitiveGenerator` (Section 5).

This suggests a clean three-case research split, matching Corollary 9.2. The two one-smooth-output cases reduce to the exceptional primitive cores $w = 3$ and $v = 2$, where the target radicals are 3^{ϑ} and $2^{1/\vartheta}$ respectively and where Section 6 already locates the wall. In the generic case both second iterates are transcendental and two independent outside-prime supports are available, which is the configuration a mixed archimedean and non-archimedean determinant would exploit; see Section 27.

9.5 Algebraicity formulations equivalent to the conjecture

Theorem 9.9 (Algebraicity tests)**.** *Each of the following is equivalent to Conjecture 2.1.*

(i) *For every* $x \in \mathcal{S}$, $6^{x^2} \in \overline{\mathbb{Q}}$.

(ii) *For every* $x \in \mathcal{S}$, *both* $2^{x^2} \in \overline{\mathbb{Q}}$ *and* $3^{x^2} \in \overline{\mathbb{Q}}$.

(iii) *For every* $x, y \in \mathcal{S}$, $6^{xy} \in \overline{\mathbb{Q}}$.

(iv) *For one fixed pair of integers* $u, v \geq 1$, *every* $x \in \mathcal{S}$ *satisfies* $(2^u 3^v)^{x^2} \in \overline{\mathbb{Q}}$.

(v) *For every* $x \in \mathcal{S}$, *the integer* 6^x *is three-smooth.*

Proof. If Conjecture 2.1 holds, then $\mathcal{S} = \mathbb{N}_0$, every exponent appearing in (i)–(iv) is a nonnegative integer, and all the listed values are positive integers, hence algebraic; condition (v) is immediate because $6^x = 2^x 3^x$ is then a power of 6.

Conversely, suppose the conjecture fails and fix $x \in \mathcal{S} \setminus \mathbb{N}_0$. Then (i) and (ii) fail by Corollary 9.6, (iii) fails by taking $y = x$ in Theorem 9.5(b), (iv) fails by Corollary 9.7 with $y = x$, and (v) fails by Theorem 9.1. ☐

Status: [classical/open background] for (i)–(iv) — via Theorem 9.3. **[paper proof in this report]** for (v).

Equivalence (v) needs only Theorem 9.1, hence only the kernel-verified transcendence of ϑ; it is therefore the one item in this list that is within immediate reach of the Lean development.

Items (i)–(iv) replace integrality by the apparently much weaker target of algebraicity, and the six exponentials theorem makes the replacement lossless. That sharpens the statement of what is missing: one would need an independent reason why some nonlinear iterate such as 6^{x^2} is algebraic, derived from the two original integer values 2^x and 3^x alone. No mechanism in the present corpus produces algebraicity at the second level, which is consistent with the four-exponentials diagnosis of Section 7: the two-base configuration supplies only two independent logarithmic directions, and squaring the exponent does not create a third.

9.6 Formalization boundary

The division of labour in this section is unusually clean, and it marks exactly where the repository can advance without new transcendence theory.

Theorem 9.1, Corollary 9.2, part (a) of Theorem 9.5, and equivalence (v) of Theorem 9.9 use only kernel-verified ingredients: transcendence of ϑ, unique factorization, the closure of $\mathcal{S}$ under natural multiples, the simultaneous output normalization, and the 3-smooth classification in `ThreeSmooth`. A proposed module `OutputSmoothness` would host them. With the predicate

$$\mathrm{ThreeSmooth}(m) :\Longleftrightarrow \exists\, u, v \in \mathbb{N}_0,\ m = 2^u 3^v,$$

the headline statement is a biconditional between integrality of the exponent and joint smoothness of the two outputs:

```
theorem integer_iff_outputs_threeSmooth
    {x : R} (hx : TwoBaseIntegralSolution x) :
    x in Set.range ((^) : Z -> R) <->
      ThreeSmooth (natOutputTwo hx) /\
      ThreeSmooth (natOutputThree hx)
```

Listing 1: Proposed output-smoothness signature.

The reverse direction is the interesting one, and it follows the proof of Theorem 9.1: take logarithms of the two smooth factorizations, assemble (10) as a polynomial of degree at most two in ϑ with integer coefficients, apply

`transcendental_logThreeDivLogTwo`

to force every coefficient to vanish, and discharge the remaining natural-number arithmetic with `omega`. Mathlib's polynomial evaluation interface is the natural way to build $qX^2 + (p - s)X - r$ and invoke the definition of transcendence.

Everything else in this section — Lemma 9.4, part (b) of Theorem 9.5, Corollaries 9.6 and 9.7, Theorem 9.8, and items (i)–(iv) of Theorem 9.9 — needs the six exponentials theorem in general form. The repository's interpolation determinant is currently specialized to integer entries: a candidate matrix has integer entries, a nonzero integer determinant satisfies $|D| \geq 1$, and that inequality is what drives the 3-smooth classification. The natural formal route to Lemma 9.4 is to replace the bound $|D| \geq 1$ by a number-field norm bound for a nonzero *algebraic* determinant, retaining the same analytic upper bound from the interpolation estimate. Packaging that as an "algebraic entries of bounded height" interface would recover the transcendence conclusion while reusing the existing determinant infrastructure, and would place the general six exponentials theorem within the same framework. This is a substantial addition, not a refactor, and it is listed among the transcendence-theoretic priorities of Section 27.

10 Perfect-power content, affine descent, and primitive-output statistics

Section 5 closes the conditional structure question: if Conjecture 2.1 fails, then $\mathcal{S}$ is the free rank-two additive monoid generated by 1 and the canonical least nonintegral solution β, and the least output pair $(M, A) = (2^\beta, 3^\beta)$ is affinely primitive. Both facts are **[kernel-verified Lean]**. What that corpus does not yet record is the *exact* perfect-power content of the output pair at an arbitrary point of the monoid: the kernel-verified corollaries exclude matched decompositions at β, but say nothing about which matched decompositions actually occur elsewhere.

This section supplies that classification. It determines exactly which simultaneous factorizations

$$2^x = 2^r U^d, \qquad 3^x = 3^r V^d$$

can occur for $x \in \mathcal{S}$, identifies the maximal common perfect-power degree of the output pair with a gcd of coordinates, and then converts the classification into exact limiting statistics for primitive output pairs. Everything here is conditional on failure and rests on the kernel-verified monoid classification; the classification and the statistics themselves are **[paper proof in this report]**.

Throughout, assume failure and fix the canonical generator β of Section 5, with

$$M = 2^\beta, \qquad A = 3^\beta.$$

For $x = n + k\beta$ with $n, k \in \mathbb{N}_0$, exact coordinates give

$$2^x = 2^n M^k, \qquad 3^x = 3^n A^k, \tag{12}$$

and the pair (n, k) is uniquely determined by x. The coordinate gcd $\gcd(n, k)$ turns out to be an intrinsic arithmetic invariant of the output pair, not merely of its coordinates.

10.1 Simultaneous perfect powers

Theorem 10.1 (Exact common-power criterion)**.** *Let $x = n + k\beta \in \mathcal{S}$ and let $r \geq 1$. The following are equivalent:*

(i) *both 2^x and 3^x are perfect r-th powers in $\mathbb{N}$;*

(ii) *$x/r \in \mathcal{S}$;*

(iii) *$r \mid n$ and $r \mid k$.*

Proof. If $2^x = U^r$ and $3^x = V^r$ with $U, V > 0$, uniqueness of positive real r-th roots gives

$$2^{x/r} = U, \qquad 3^{x/r} = V,$$

so $x/r \in \mathcal{S}$. The converse follows by raising the two integer values at x/r to the r-th power.

For the coordinate form, write $x/r = p + q\beta$ in the unique coordinates supplied by the exact primitive-generator theorem of Section 5. Then $n + k\beta = rp + rq\beta$, and uniqueness of coordinates forces $n = rp$ and $k = rq$, which is exactly the pair of divisibility conditions. Conversely, if $r \mid n$ and $r \mid k$, then $(n/r) + (k/r)\beta \in \mathcal{S}$ and equals x/r. ☐

For $x > 0$ define the *common output power degree*

$$g(x) := \max\{r \geq 1 : 2^x \text{ and } 3^x \text{ are both perfect } r\text{-th powers}\}.$$

Corollary 10.2 (Coordinate gcd)**.** *If $x = n + k\beta > 0$, then*

$$g(x) = \gcd(n, k),$$

with the convention $\gcd(n, 0) = n$*. The output pair is simultaneously power-primitive exactly when* $\gcd(n, k) = 1$*.*

Proof. By Theorem 10.1 the admissible degrees r are precisely the common divisors of n and k, and the set of common divisors of a pair that is not $(0, 0)$ has a largest element, namely $\gcd(n, k)$. ☐

Consequently every nonzero output pair has a unique decomposition

$$(2^x, 3^x) = (U^g, V^g), \qquad g = \gcd(n, k),$$

where $(U, V) = (2^{x/g}, 3^{x/g})$ is a simultaneously power-primitive output pair belonging to the solution x/g. The repository proves power-primitivity for the least pair only; this corollary computes the intrinsic power degree at every point of the solution monoid. For integral $x = n$ it returns $g(n) = n$, which is the familiar statement that 2^n and 3^n are perfect n-th powers and no better; for $x = \beta$ it returns $g(\beta) = \gcd(0, 1) = 1$, which is exactly the kernel-verified exclusion of a common perfect-power degree at the least pair.

Status: **[paper proof in this report]** — Theorem 10.1 and Corollary 10.2 are paper results of this report. Their inputs are kernel-verified: the unique-coordinate clause of the exact primitive-generator theorem (`PrimitiveGenerator`) and the least-solution corollaries of `LeastSolution`. The implication (i)⇒(ii) is the special case $r = 0$ of the Lean affine descent theorem `affine_descent_integral_powers` in `ArithmeticRigidity`; the converse and the coordinate criterion (iii) are new here.

10.2 Matched affine decompositions

A descent that strips the same distinguished-base depth and the same outer power from both outputs is the natural common generalization of the two operations above. The repository's affine descent theorem shows that such a stripping produces a solution; the following criterion says exactly when one exists.

Theorem 10.3 (Matched affine decomposition criterion). *Let $x = n + k\beta \in \mathcal{S}$, let $d \geq 1$, and let $r \in \mathbb{N}_0$. The following are equivalent:*

(i) *there exist $U, V \in \mathbb{N}_{>0}$ with*
$$2^x = 2^r U^d, \qquad 3^x = 3^r V^d;$$

(ii) $(x - r)/d \in \mathcal{S}$;

(iii) $0 \leq r \leq n$, $d \mid k$, *and* $d \mid (n - r)$.

When these conditions hold,

$$\frac{x-r}{d} = \frac{n-r}{d} + \frac{k}{d}\beta, \qquad U = 2^{(x-r)/d}, \qquad V = 3^{(x-r)/d}.$$

Proof. Conditions (i) and (ii) are equivalent by uniqueness of positive real roots, applied after removing the common factor 2^r from the first output and 3^r from the second. Expanding (ii) in unique coordinates gives
$$n + k\beta = r + d(p + q\beta)$$
for some $p, q \in \mathbb{N}_0$, so $n = r + dp$ and $k = dq$; this is exactly (iii), and it also yields the displayed formulas. Conversely, (iii) lets one define $p = (n-r)/d \in \mathbb{N}_0$ and $q = k/d \in \mathbb{N}_0$, and then $p + q\beta \in \mathcal{S}$ equals $(x - r)/d$. ☐

At the least pair the criterion collapses. For $x = \beta$ we have $n = 0$ and $k = 1$, so (iii) forces $r = 0$ and $d = 1$: no nontrivial matched decomposition of (M, A) exists. That is the kernel-verified exact affine primitivity statement of Section 5, recovered here as one instance of a criterion valid at every point.

For a nonintegral solution $x = n + k\beta$ with $k > 0$, Euclidean division $n = qk + r$ with $0 \leq r < k$ gives a canonical reduced affine presentation

$$x = r + k(q + \beta), \tag{13}$$

and hence
$$2^x = 2^r \left(2^q M\right)^k, \qquad 3^x = 3^r \left(3^q A\right)^k.$$

The normalization $0 \leq r < k$ makes the presentation unique. It separates the outer affine degree k — an invariant of the point, being its β-coordinate — from the reduced common base depth r.

Status: **[paper proof in this report]** — Theorem 10.3 and the reduced presentation (13) are paper results. The implication (i)⇒(ii) is the kernel-verified affine descent theorem; the equivalence with the coordinate condition (iii), and hence the classification of *all* matched decompositions, is new.

10.3 What is already formal, and what is new here

The table below separates the two layers. The middle column is the kernel-verified statement about the least pair (M, A); the right column is what this section adds for a general point $x = n + k\beta$ of the monoid.

Statement	Kernel-verified at β	Extension in this section
Matched base depth	$\min\{v_2(M)\,, v_3(A)\} = 0$, in `LeastSolution`	Admissible depths at x are exactly $0 \le r \le n$ with $d \mid (n-r)$
Common perfect-power degree	no $k \ge 2$ with M, A both k-th powers, in `LeastSolution`	the exact degree is $g(x) = \gcd(n, k)$ (Corollary 10.2)
Affine primitivity	$M = 2^r U^k$, $A = 3^r V^k$ force $r = 0$, $k = 1$, in `LeastSolution`	full criterion for every x, with the least pair as the case $(n, k) = (0, 1)$ (Theorem 10.3)
Affine descent	$(x-r)/k \in \mathcal{S}$ from a matched factorization, in `ArithmeticRigidity`	the descent condition is an equivalence, not merely an implication
Output normalization	$M = 2^a w^d$, $A = 3^b v^e$ with $a = 0$ or $b = 0$ and $\gcd(\gcd(d, e), \lvert b-a \rvert) = 1$, in `OutputNormalization`	unchanged; that invariant constrains the two *separate* degrees, whereas g measures only the common one

The kernel-verified identifiers behind the middle column are

`IsLeastTwoBaseNonintegerSolution.not_both_base_dvd_outputs`

`IsLeastTwoBaseNonintegerSolution.no_common_perfect_power`

`IsLeastTwoBaseNonintegerSolution.affine_primitive`

in `LeastSolution`, together with `exists_simultaneous_primitive_output_normalization` in `OutputNormalization` and `affine_descent_integral_powers` in `ArithmeticRigidity`. None of the right-hand column is machine-checked. A formalization would be light: the coordinate criteria follow from the already accepted unique-representation clause, so the missing work is bookkeeping over $\mathbb{N}_0^2$ rather than new arithmetic.

10.4 Why this does not yet descend below β

If n and k have a common divisor, Theorem 10.1 descends to $x/\gcd(n, k)$. If a matched decomposition exists, Theorem 10.3 descends to $(x-r)/d$. But exact coordinates guarantee that the descended point is again $p + q\beta$ with $p, q \in \mathbb{N}_0$, and it lies below β only when $q = 0$, in which case it is integral and carries no information. The least nonintegral point β is already primitive for every descent of this shape, which is precisely why the kernel-verified corollaries at β are exclusions rather than contradictions.

A successful descent argument therefore needs a transformation not captured by common perfect powers and common distinguished-base depths. Both operations act inside the monoid $\mathbb{N}_0^2$; anything that could close the problem must leave it.

10.5 Total and primitive counts

Corollary 10.2 converts ordinary coprimality statistics of lattice points into exact statistics of hypothetical output pairs. For $T > 0$ put

$$N_\beta(T) = \#\{(n,k) \in \mathbb{N}_0^2 \setminus \{(0,0)\} : n + k\beta < T\}$$

and

$$P_\beta(T) = \#\{(n,k) \in \mathbb{N}_0^2 : n + k\beta < T,\ \gcd(n,k) = 1\}.$$

The first counts nonzero solutions below T; by Corollary 10.2 the second counts exactly those output pairs with no nontrivial common perfect-power degree.

Counting each column separately, for every real $U > 0$,

$$N_\beta(U) = \sum_{0 \le k < U/\beta} \lceil U - k\beta \rceil - 1 = \frac{U^2}{2\beta} + O_\beta(U+1), \tag{14}$$

uniformly in U. For integer T this is the kernel-verified cumulative count $C(T)$ of Section 5 minus the point $x = 0$, so (14) agrees in leading order with `QuadraticAsymptotic`; the uniform real-variable form used below is elementary.

Theorem 10.4 (Primitive output pairs)**.** *As $T \to \infty$,*

$$P_\beta(T) = \frac{T^2}{2\beta\zeta(2)} + O_\beta(T \log T) = \frac{3T^2}{\pi^2\beta} + O_\beta(T \log T).$$

Consequently

$$\frac{P_\beta(T)}{N_\beta(T)} \longrightarrow \frac{1}{\zeta(2)} = \frac{6}{\pi^2}.$$

Proof. Every nonzero coordinate pair has the unique factorization

$$(n,k) = d\,(n_0, k_0), \qquad d = \gcd(n,k), \quad \gcd(n_0,k_0) = 1,$$

and $n + k\beta < T$ holds if and only if $n_0 + k_0\beta < T/d$. Hence

$$N_\beta(T) = \sum_{d \ge 1} P_\beta(T/d),$$

a finite sum for each T, and Möbius inversion gives

$$P_\beta(T) = \sum_{d \le T} \mu(d)\, N_\beta(T/d).$$

Inserting the uniform estimate (14),

$$\begin{aligned} P_\beta(T) &= \frac{T^2}{2\beta} \sum_{d \le T} \frac{\mu(d)}{d^2} + O_\beta\left(\sum_{d \le T} \Big(\frac{T}{d} + 1 \Big) \right) \\ &= \frac{T^2}{2\beta\zeta(2)} + O_\beta(T \log T), \end{aligned}$$

where the tail $\sum_{d>T} \mu(d)/d^2 = O(1/T)$ contributes $O_\beta(T)$. Dividing by (14) gives the stated limit. □

The constant $6/\pi^2$ is universal: it does not depend on the hypothetical generator at all. The geometry of the monoid contributes the factor $1/(2\beta)$, and the primitive-lattice density contributes $1/\zeta(2)$. In particular no choice of counterexample can make the primitive proportion small.

10.6 Distribution of the maximal common power degree

For fixed $g \geq 1$ let

$$P_{\beta,g}(T) := \#\{x = n + k\beta : 0 < x < T,\ \gcd(n,k) = g\}$$

be the number of solutions below T whose output pair has common power degree exactly g.

Corollary 10.5 (Exact-degree distribution)**.** *For fixed $g \geq 1$,*

$$P_{\beta,g}(T) = P_\beta(T/g) = \frac{3T^2}{\pi^2 \beta g^2} + O_\beta\!\left(\frac{T}{g} \log \frac{2T}{g}\right).$$

The limiting probability that an output pair below T has maximal common power degree exactly g is

$$\frac{6}{\pi^2 g^2}.$$

Proof. The map $(n_0, k_0) \mapsto g\,(n_0, k_0)$ is a bijection from the primitive pairs below T/g onto the pairs of gcd exactly g below T, so $P_{\beta,g}(T) = P_\beta(T/g)$; now apply Theorem 10.4 and divide by (14). ☐

The probabilities sum to one, since $\sum_{g\geq 1} 6/(\pi^2 g^2) = 1$. Likewise, by Theorem 10.1 both outputs are perfect r-th powers exactly when $r \mid n$ and $r \mid k$, so

$$\#\{0 < x < T : 2^x \text{ and } 3^x \text{ are both perfect } r\text{-th powers}\} = N_\beta(T/r), \tag{15}$$

and the limiting proportion is $1/r^2$.

10.7 Exact generating series

The coordinate monoid has an equally explicit Laplace series. For $s > 0$,

$$\begin{aligned} Z_\beta(s) &:= \sum_{x \in \mathcal{S}\setminus\{0\}} e^{-sx} = \sum_{n,k\geq 0} e^{-s(n+k\beta)} - 1 \\ &= \frac{1}{(1-e^{-s})(1-e^{-\beta s})} - 1, \end{aligned} \tag{16}$$

the two factors being the two free generators 1 and β. Möbius inversion in the degree gives the primitive-output series

$$Z_\beta^{\text{prim}}(s) = \sum_{d\geq 1} \mu(d)\, Z_\beta(ds).$$

At the candidate level the same factorization appears multiplicatively: since $\mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\}$ with unique coordinates,

$$\sum_{m \in \mathcal{C}} m^{-s} = \frac{1}{(1-2^{-s})(1-M^{-s})}, \qquad \Re s > 0. \tag{17}$$

These identities make the two free generators visible in analytic form and give clean starting points for weighted Tauberian refinements of Theorem 10.4.

Status: **[paper proof in this report]** — Theorem 10.4, Corollary 10.5, and the series (16)–(17) are paper results of this report. Their only nonelementary input is the kernel-verified exact monoid classification of Section 5, whose leading-order count is itself **[kernel-verified Lean]** through `ExactCounting` and `QuadraticAsymptotic`. The finite scans of Section 25 play no role here; the exponent bound in force remains the kernel-verified $x \geq 12$.

10.8 Interpretation

Roughly 60.79% of hypothetical output pairs would be simultaneously power-primitive, and the remaining ones split across degrees $g \geq 2$ with the exact frequencies $6/(\pi^2 g^2)$. This is not pressure toward a contradiction. It is precisely the primitive proportion expected of a free rank-two monoid, and it would be the same for any generator: the statistic is diagnostic, not restrictive.

Its diagnostic value is a warning about method. The kernel-verified primitivity corollaries at β, and the classification of this section at a general point, both say that perfect-power and matched-depth structure is the exception among hypothetical solutions. Any descent argument built on extracting a common root or a common base depth therefore acts on a minority of the conditional population — asymptotically the proportion $1 - 6/\pi^2 \approx 0.3921$ of pairs that admit any nontrivial common degree at all — and leaves the primitive majority untouched. A method that closes the problem must engage the power-primitive pairs directly, which is exactly where the transcendence content of Section 7 sits.

11 Candidate divisibility and ambient root saturation

The exact classification of Section 5 describes the conditional candidate set by its *generators*: under failure, $\mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\}$ with unique coordinates. Generation alone says nothing about how candidates sit inside $\mathbb{N}$ under the two operations that descent arguments and searches actually use — divisibility, and extraction of roots. This section answers both exactly. The candidate monoid has a simple valuation geometry: divisibility is coordinatewise order on a two-dimensional lattice cone, $\mathcal{C}$ is closed under gcd and lcm, and the set of integers some power of which is a candidate is again an explicit cone — one that contains $\mathcal{C}$ with an exact finite-index defect measured by the rational-power index d of Section 5, which is also the exact radical degree of Section 6.

Every statement below is **[paper proof in this report]**. Its inputs — the canonical primitive odd core, the rational-power index, and the exact primitive-generator theorem — are **[kernel-verified Lean]**; what is added here is elementary lattice combinatorics on top of them. None of it is a step toward a contradiction: as the closing discussion makes explicit, the closure properties proved here are compatible with failure by construction, and the kernel-verified exponent bound remains $x \geq 12$.

11.1 A two-dimensional valuation cone

Lemma 11.1 (Valuation coordinates and divisibility)**.** *Assume Conjecture 2.1 is false and let w, d, a, β, M be the canonical data of Section 5, so that $w > 1$ is odd and power-primitive, $d > 0$, and*

$$M = 2^\beta = 2^a w^d.$$

Put $U := w^d$, the odd part of M, and for $n, k \in \mathbb{N}_0$ write

$$m(n,k) := 2^n M^k = 2^{n+ak} U^k, \qquad s := v_2(m(n,k)) = n + ak.$$

Then:

(i) $U > 1$;

(ii) *the map $m(n,k) \mapsto (s,k)$ is a bijection from $\mathcal{C}$ onto the lattice cone*

$$\Lambda_a := \{(s,k) \in \mathbb{N}_0^2 : s \geq ak\};$$

(iii) *writing $m_i = m(n_i, k_i)$ and $s_i = n_i + ak_i$,*

$$m_1 \mid m_2 \iff s_1 \leq s_2 \text{ and } k_1 \leq k_2. \tag{18}$$

Proof. (i) If $U = 1$ then $M = 2^a$ and $\beta = a \in \mathbb{Z}$, contradicting the irrationality of the least generator.

(ii) By the exact primitive-generator theorem of Section 5, every candidate is $m(n,k)$ for a unique pair $(n,k) \in \mathbb{N}_0^2$. The reparameterization $(n,k) \mapsto (n+ak, k)$ is affine with inverse $(s,k) \mapsto (s-ak, k)$, and $n \geq 0$ holds exactly when $s \geq ak$.

(iii) Since U is odd, $v_2(m(n,k)) = s$, and for an odd prime p we have $v_p(m(n,k)) = k\, v_p(U)$, which vanishes unless $p \mid U$. Divisibility in $\mathbb{N}$ is comparison of all prime valuations. If $m_1 \mid m_2$ then comparison at 2 gives $s_1 \leq s_2$, and comparison at any prime $p \mid U$ — one exists by (i) — gives $k_1 v_p(U) \leq k_2 v_p(U)$ with $v_p(U) > 0$, hence $k_1 \leq k_2$. Conversely those two inequalities force every prime valuation of m_1 to be at most the corresponding valuation of m_2. □

Thus candidate divisibility is nothing but the coordinatewise order on Λ_a. The inequality $s \geq ak$ is the only constraint, and it records the base-adic depth a of the least generator: a candidate cannot carry the odd part U^k without carrying at least ak factors of two along with it.

11.2 The candidate gcd/lcm lattice

Theorem 11.2 (Candidate gcd/lcm lattice)**.** *Unconditionally, $\mathcal{C}$ is closed under* gcd *and* lcm*. If Conjecture 2.1 fails, then in the notation of Lemma 11.1,*

$$\begin{aligned}
\gcd(m_1, m_2) &= m(\min(s_1,s_2) - a\min(k_1,k_2),\ \min(k_1,k_2)),\\
\operatorname{lcm}(m_1, m_2) &= m(\max(s_1,s_2) - a\max(k_1,k_2),\ \max(k_1,k_2));
\end{aligned}$$

equivalently, in cone coordinates gcd *and* lcm *are coordinatewise minimum and maximum on Λ_a. Candidate divisibility is therefore a distributive lattice, and Λ_a is a sublattice of $\mathbb{N}_0^2$.*

Proof. If the conjecture holds then $\mathcal{C} = \{2^n : n \in \mathbb{N}_0\}$, whose gcds and lcms are again powers of two; this gives the unconditional half.

Assume failure. By Lemma 11.1 the candidate $m(n,k)$ has valuation vector s at the prime 2 and $k\, v_p(U)$ at each odd p. Gcd and lcm take coordinatewise minima and maxima of valuation vectors, so

$$\gcd(m_1, m_2) = 2^{\min(s_1,s_2)} U^{\min(k_1,k_2)}, \qquad \operatorname{lcm}(m_1, m_2) = 2^{\max(s_1,s_2)} U^{\max(k_1,k_2)}.$$

It remains to check that both exponent pairs lie in Λ_a. Suppose $s_1 \le s_2$; then $\min(s_1, s_2) = s_1 \ge ak_1 \ge a\min(k_1, k_2)$ because $a \ge 0$. For the maximum, suppose $k_1 \le k_2$; then $\max(s_1, s_2) \ge s_2 \ge ak_2 = a\max(k_1, k_2)$. Hence Λ_a is closed under coordinatewise min and max, that is, it is a sublattice of $\mathbb{N}_0^2$, and the translation back to (n, k) coordinates is the stated substitution $n = s - ak$. A sublattice of a product of chains is distributive. □

Status: [kernel-verified Lean] — `twoBaseNaturalCandidate_gcd` and `twoBaseNaturalCandidate_lcm` in `CandidateLattice`, resting on the coordinatewise arithmetic lemmas `gcd_two_pow_mul_odd_pow` and `lcm_two_pow_mul_odd_pow` of the same module and on the [kernel-verified Lean] exact primitive-generator theorem of Section 5. The formal statements are unconditional: the Lean proof splits on the conjecture and disposes of the affirmative branch through the powers of two. No transcendence input and no analytic estimate enters.

Corollary 11.3 (Counting candidate divisors)**.** *Under failure, the number of candidates dividing $m(n, k)$, including 1 and $m(n, k)$ itself, is*

$$(k + 1)(n + 1) + \frac{ak(k + 1)}{2}.$$

Proof. By (18) a candidate divisor corresponds to a point $(s', j) \in \Lambda_a$ with $0 \le j \le k$ and $aj \le s' \le n + ak$. For fixed j there are $n + ak - aj + 1$ admissible values of s'. Summing the arithmetic progression over $j = 0, 1, \ldots, k$ gives $(k + 1)(n + 1) + a\sum_{i=0}^{k} i$, which is the stated value. □

Remark 11.4 (Candidates are not divisor-closed). The full divisor count of $m(n, k) = 2^s U^k$ is $(s + 1)\prod_{p|U}(k\, v_p(U) + 1)$, which for $k \ge 1$ and U with more than one prime factor grows much faster than the count of Corollary 11.3. So $\mathcal{C}$ is emphatically not closed under taking divisors; it is closed only under the two lattice operations. The gap between the two counts is a concrete measure of how thin the candidate sublattice is inside the divisor lattice of a single candidate.

11.3 Fixed prime support

The lattice picture has an immediate consequence for the shape of a hypothetical counterexample branch: the supports do not vary at all.

Corollary 11.5 (Prime-support rigidity)**.** *Assume failure and use the simultaneous primitive output normalization of Section 5,*

$$M = 2^{\beta} = 2^a w^d, \qquad A = 3^{\beta} = 3^b v^e.$$

Then for $x = n + k\beta$,

$$2^x = 2^{n+ak} w^{dk}, \qquad 3^x = 3^{n+bk} v^{ek},$$

so every nonintegral solution — that is, every solution with $k \ge 1$ — has exactly the same odd prime support $\mathrm{supp}(w)$ *in its base-2 output, and exactly the same prime support away from 3, namely* $\mathrm{supp}(v)$*, in its base-3 output. The only variation across the entire nonintegral branch is whether the distinguished primes 2 and 3 occur at all.*

Proof. Substituting M and A into $(2^x, 3^x) = (2^n M^k, 3^n A^k)$ gives the displayed factorizations. For $k \geq 1$ the odd part of 2^x is w^{dk}, and $\operatorname{supp}(w^{dk}) = \operatorname{supp}(w)$ because $dk \geq 1$; similarly for v^{ek}. The exponents $n + ak$ and $n + bk$ may vanish, which is exactly the stated freedom at 2 and 3. $\square$

A counterexample therefore locks the whole nonintegral branch onto a single fixed pair of finite prime sets. Computational reconnaissance should exploit this directly, enumerating primitive cores and supports rather than raw integers scanned independently; the finite verifications reported in Section 25 are **[interval computation]** and this observation does not change their status, only the shape of the search that would extend them.

11.4 Ambient root saturation and the index d

Reparameterizing Lemma 11.1 by the exponent of the primitive core w rather than of $U = w^d$ gives the description that governs roots. With $s = v_2(m)$ and t the exponent of w in the odd part of m,

$$\mathcal{C} = \{2^s w^t : s, t \in \mathbb{N}_0,\ d \mid t,\ ds \geq at\}. \tag{19}$$

Indeed $m = 2^n M^k$ has $s = n + ak$ and $t = dk$, so $d \mid t$, and $n \geq 0$ is equivalent to $s \geq at/d$, that is, to $ds \geq at$.

Definition 11.6 (Ambient root saturation)**.** For a multiplicative submonoid $H \subseteq \mathbb{N}_{>0}$ put

$$\operatorname{Sat}(H) := \{u \in \mathbb{N}_{>0} : u^r \in H \text{ for some } r \geq 1\},$$

the set of positive integers admitting some positive power inside H.

Theorem 11.7 (Exact ambient root saturation)**.** *Assume Conjecture 2.1 is false. Then*

$$\operatorname{Sat}(\mathcal{C}) = \{2^s w^t : s, t \in \mathbb{N}_0,\ ds \geq at\},$$

and for $u = 2^s w^t$ in this set the least positive r with $u^r \in \mathcal{C}$ is

$$r_{\min}(u) = \frac{d}{\gcd(d, t)}.$$

Proof. First let $u = 2^s w^t$ with $s, t \in \mathbb{N}_0$ and $ds \geq at$, and put $r = d/\gcd(d, t)$. Then $rt = d \cdot t/\gcd(d, t)$ is divisible by d, and multiplying $ds \geq at$ by $r > 0$ gives $d(rs) \geq a(rt)$. By (19), $u^r = 2^{rs} w^{rt} \in \mathcal{C}$, so $u \in \operatorname{Sat}(\mathcal{C})$. Conversely, if $u^{r'} \in \mathcal{C}$ for some $r' \geq 1$ then (19) forces $d \mid r't$, since w is power-primitive and hence the exponent of w in the odd part of $u^{r'}$ is exactly $r't$. Now $d \mid r't$ is equivalent to $(d/\gcd(d, t)) \mid r'$, so $r' \geq r$ and r is least.

Now suppose merely that $u \in \mathbb{N}_{>0}$ satisfies $u^r \in \mathcal{C}$ for some $r \geq 1$, and write $u^r = 2^S w^{dk}$ with $dS \geq a\,dk$. A prime divides u if and only if it divides u^r, so no prime outside $\{2\} \cup \operatorname{supp}(w)$ divides u. Put $c_p := v_p(w)$ for $p \mid w$; comparing valuations in $u^r = 2^S w^{dk}$ gives

$$r\, v_p(u) = dk\, c_p \qquad \text{for every } p \mid w.$$

Because w is power-primitive, $\gcd\{c_p : p \mid w\} = 1$, so there are integers r_p, almost all zero, with $\sum_p r_p c_p = 1$; then

$$\frac{dk}{r} = \sum_p r_p \cdot \frac{dk\, c_p}{r} = \sum_p r_p\, v_p(u) \in \mathbb{Z}.$$

Call this integer $t \geq 0$. Then $v_p(u) = tc_p$ for every $p \mid w$, so the odd part of u is exactly w^t, while $S = rs$ for $s = v_2(u)$. Substituting $S = rs$ and $dk = rt$ into $dS \geq a\,dk$ and dividing by $r > 0$ yields $ds \geq at$. Hence $u = 2^s w^t$ lies in the stated cone. □

Status: [paper proof in this report] — the only nonformal ingredient is the Bézout step, which is the same primitivity mechanism already kernel-verified in `PrimitiveOddCore` through `oddPrimeValuationGCD`. Everything else is unique factorization and integer arithmetic.

Corollary 11.8 (Ambient saturation index)**.** *Under failure, $\mathcal{C} = \mathrm{Sat}(\mathcal{C})$ if and only if $d = 1$, and in general*

$$d = \min\{r \geq 1 : \exists\, s \in \mathbb{N}_0,\ (2^s w)^r \in \mathcal{C}\}.$$

Proof. Take $t = 1$ and any $s \geq a$, so that $ds \geq a = at$; by Theorem 11.7 the integer $2^s w$ lies in $\mathrm{Sat}(\mathcal{C})$ with $r_{\min} = d/\gcd(d,1) = d$. This proves the displayed formula, and it shows that $2^s w \in \mathcal{C}$ forces $d = 1$. Conversely, if $d = 1$ then the divisibility condition $d \mid t$ in (19) is vacuous and the two sets coincide. □

Corollary 11.9 (Which integer roots of a candidate are again candidates)**.** *Assume failure and let $m = 2^n M^k \in \mathcal{C}$ and $r \geq 1$.*

(a) *There is an integer u with $u^r = m$ if and only if $r \mid n + ak$ and $r \mid dk$; such a u always lies in $\mathrm{Sat}(\mathcal{C})$.*

(b) *That integer root is itself a candidate if and only if $r \mid n$ and $r \mid k$, in which case $u = 2^{n/r} M^{k/r}$.*

Proof. (a) Write $m = 2^s w^t$ with $s = n + ak$ and $t = dk$. If $u^r = m$ then, exactly as in the proof of Theorem 11.7, u is supported in $\{2\} \cup \mathrm{supp}(w)$, $r \mid s$, and $t/r \in \mathbb{Z}$ by the Bézout step; conversely $u = 2^{s/r} w^{t/r}$ is an integer r-th root when $r \mid s$ and $r \mid t$. Membership in $\mathrm{Sat}(\mathcal{C})$ is immediate from Definition 11.6.

(b) If $u \in \mathcal{C}$ then $u = 2^{n'} M^{k'}$ for unique $n', k' \in \mathbb{N}_0$, and $u^r = 2^{rn'} M^{rk'}$ together with uniqueness of coordinates gives $n = rn'$ and $k = rk'$. The converse is the direct computation $\left(2^{n/r} M^{k/r}\right)^r = m$. □

Observation 11.10 (Normal intrinsically, saturated only up to index d)**.** Let $L := \{2^s w^t : s, t \in \mathbb{Z}\}$ be the ambient multiplicative group generated by 2 and w inside $\mathbb{Q}^{\times}_{>0}$, and let $L_{\mathcal{C}}$ be the group generated by $\mathcal{C}$. Then

$$L_{\mathcal{C}} = \{2^s w^t : s \in \mathbb{Z},\ d \mid t\}, \qquad [L : L_{\mathcal{C}}] = d, \qquad \mathcal{C} = \mathrm{Sat}(\mathcal{C}) \cap L_{\mathcal{C}}.$$

In particular $\mathrm{Sat}(\mathcal{C})$ is itself a submonoid of $\mathbb{N}_{>0}$, namely the full lattice points of the rational cone $ds \geq at$, $s, t \geq 0$.

Proof. $L_{\mathcal{C}}$ is generated by 2 and $M = 2^a w^d$, so it consists of the elements $2^{i+aj} w^{dj}$ with $i, j \in \mathbb{Z}$, which is the stated set; the quotient $L/L_{\mathcal{C}}$ is isomorphic to $\mathbb{Z}/d\mathbb{Z}$ through the exponent of w. Intersecting the cone description of Theorem 11.7 with the congruence $d \mid t$ returns (19). Closure of the cone under multiplication is additivity of the inequality $ds \geq at$ in (s, t). □

A terminology caution is worth stating explicitly, because the word "normal" is used differently in affine-semigroup theory. There, normality of a semigroup is measured relative to the group that the semigroup itself generates. By Observation 11.10 that group is $L_{\mathcal{C}}$, in which the congruence $t \equiv 0 \pmod{d}$ already holds identically, and $\mathcal{C}$ is the full set of cone points of $L_{\mathcal{C}}$ for every value of d: intrinsically, $\mathcal{C}$ is normal. Theorem 11.7 measures something different, namely saturation inside the larger ambient lattice L containing every $2^s w^t$. So d is an over-lattice saturation defect of finite index, not a failure of intrinsic normality. This is precisely parallel to its two other exact meanings established elsewhere in this report: d is the least rational-power index of Section 5 and the exact radical degree $[\mathbb{Q}(w^{\vartheta}) : \mathbb{Q}]$ of Section 6. One integer invariant, three characterizations — algebraic degree, least rational power, and lattice saturation defect.

11.5 Why lattice saturation is a good Lean target

Two points must be kept apart. The gcd/lcm closure of Theorem 11.2 and the saturation formula of Theorem 11.7 are compatible with failure by construction: they describe the counterexample world rather than obstruct it, and no contradiction can be extracted from them alone. They are structural information and search normalization. With that said, they are unusually attractive as the next formalization increment.

The mathematical inputs are already kernel-verified. The canonical primitive odd core, the rational-power index, the three-power denominator normalization, and the exact primitive generator with unique coordinates are all **[kernel-verified Lean]** and live in `ExponentialIdentities/TwoBaseIntegerExponent/`. Everything added in this section is finite combinatorics over those results: unique factorization in $\mathbb{N}$, one Bézout step that already exists in `PrimitiveOddCore` as `oddPrimeValuationGCD`, and elementary min/max arithmetic on $\mathbb{N}_0^2$. There is no analysis, no transcendence input, and no numerical certificate anywhere in the chain — a rare profile in this corpus, and the reason the work should be cheap relative to its yield. A single module, sitting beside `PrimitiveGenerator` and `RationalPowerIndex`, would carry the coordinate bijection of Lemma 11.1, the divisibility criterion (18), the lattice operations, the divisor count, the cone form (19), the saturation set, the exact value of $r_{\min}$, and the index characterization of d. The natural name is

```
CandidateLattice
```

and the blueprint conventions of Section 28 apply to it unchanged.

The contribution to the search is threefold. First, a kernel-checked normal form: candidate enumeration can be phrased over lattice coordinates rather than over raw integers, so rejection tests become integer inequalities of the form $ds \geq at$ and $d \mid t$ instead of factorizations recomputed for every scanned value. This is exactly the discipline recommended in the computational priority of Section 27, and it applies equally to the certificate architecture that any extension of the **[interval computation]** finite verifications of Section 25 would need; the kernel-verified region itself would still stop at $x \geq 12$ until new certificates are replayed. Second, a single invariant with three independently proved meanings: once Corollary 11.8 is formal alongside the radical-degree proposition of Section 6, any future argument that bounds d — from Kummer theory, from ramification, or from a determinant estimate — bounds all three at once, and the Lean statements make that transfer mechanical rather than a matter of prose. Third, and most concretely, root extraction is the step where descent arguments live. Every descent so far — no matched base depth, no common perfect-power degree, exact affine primitivity in Section 5 — proceeds by extracting a root of the output pair and contradicting leastness of β. Corollary 11.9 says exactly which integer roots of a candidate survive as candidates and which merely land in the saturation, so a formalized version supplies the reusable side condition that such arguments currently re-prove by hand. That is a modest but genuine improvement in the leverage of the conditional structure, and it costs no new mathematics.

12 A sufficient condition: independence of prime-log products

The reductions of Section 6 isolate the obstruction as a statement about one radical at a time. This section isolates it differently, as a single statement about $\mathbb{Q}$-linear independence of a natural family of real numbers. The resulting criterion is weaker in appearance than the four exponentials conjecture and is of a different kind: it does not mention exponentials at all.

12.1 Failure as a vanishing linear form in log-products

Suppose the conjecture fails, and let x be a nonintegral solution with outputs

$$M = 2^x \in \mathbb{N}, \qquad A = 3^x \in \mathbb{N}.$$

Taking logarithms of $M = 2^x$ and $A = 3^x$ and eliminating x gives the defining relation already recorded in (2):

$$\log M \,\log 3 - \log A \,\log 2 = 0. \tag{20}$$

Now expand both integers over their prime factorizations,

$$M = \prod_p p^{\alpha_p}, \qquad A = \prod_p p^{\gamma_p}, \qquad \alpha_p, \gamma_p \in \mathbb{N}_0,$$

so that $\log M = \sum_p \alpha_p \log p$ and $\log A = \sum_p \gamma_p \log p$. Substituting into (20),

$$\sum_p \alpha_p(\log p \,\log 3) - \sum_p \gamma_p(\log p \,\log 2) = 0. \tag{21}$$

Every term of (21) is an integer multiple of a product of exactly two logarithms of primes. Collecting coefficients in the family

$$\{\log p \,\log q \;:\; p \le q \text{ prime}\},$$

the unordered pair $\{p, 3\}$ with $p \notin \{2, 3\}$ receives the coefficient α_p; the pair $\{p, 2\}$ with $p \notin \{2, 3\}$ receives $-\gamma_p$; the diagonal pair $\{3, 3\}$ receives α_3; the diagonal pair $\{2, 2\}$ receives $-\gamma_2$; and the mixed pair $\{2, 3\}$ receives $\alpha_2 - \gamma_3$.

Thus *failure of the conjecture is exactly the existence of a nontrivial vanishing integer-coefficient linear form in products of two prime logarithms*, with the additional positivity constraint that the coefficients arise from prime factorizations.

12.2 The independence criterion

Definition 12.1 (Prime-log-product independence)**.** Say that *prime-log products are independent* if the real numbers $\log p \,\log q$, indexed by pairs of primes $p \le q$, are linearly independent over $\mathbb{Q}$.

Equivalently, in the integer-coefficient form convenient for formalization: for every finite set S of primes and every $c : S \times S \to \mathbb{Z}$,

$$\sum_{p \in S} \sum_{q \in S} c_{pq} \log p \log q = 0 \qquad \Longrightarrow \qquad c_{pq} + c_{qp} = 0 \ \text{ for all } p, q \in S.$$

(The diagonal case $p = q$ reads $2c_{pp} = 0$, hence $c_{pp} = 0$.) Clearing denominators shows the two formulations agree.

Theorem 12.2 (Prime-log-product independence implies the conjecture)**.** *If prime-log products are independent, then Conjecture 2.1 holds.*

Proof. Let x satisfy $2^x, 3^x \in \mathbb{Z}$. Both powers are positive, so $M := 2^x$ and $A := 3^x$ are positive integers, and $x \geq 0$. Relation (20) holds, hence so does (21).

Apply the independence hypothesis to the finite set $S = \mathrm{primes}(M) \cup \mathrm{primes}(A) \cup \{2, 3\}$ and to the coefficient family

$$c_{pq} = \begin{cases} \alpha_p, & q = 3, \\ -\gamma_p, & q = 2, \\ 0, & \text{otherwise}, \end{cases}$$

which reproduces the left-hand side of (21). The conclusion $c_{pq} + c_{qp} = 0$ yields, coefficient by coefficient:

- taking $q = 3$ and $p \in S \setminus \{2, 3\}$: $\alpha_p + 0 = 0$, so $\alpha_p = 0$;
- taking $p = q = 3$: $2\alpha_3 = 0$, so $\alpha_3 = 0$;
- taking $q = 2$ and $p \in S \setminus \{2, 3\}$: $-\gamma_p + 0 = 0$, so $\gamma_p = 0$;
- taking $p = q = 2$: $-2\gamma_2 = 0$, so $\gamma_2 = 0$;
- taking $p = 2$, $q = 3$: $\alpha_2 - \gamma_3 = 0$.

Primes outside S divide neither M nor A, so their exponents vanish as well. Hence $\alpha_p = 0$ for every $p \neq 2$, that is

$$M = 2^{\alpha_2},$$

and symmetrically $A = 3^{\gamma_3} = 3^{\alpha_2}$. Writing $n = \alpha_2 \in \mathbb{N}_0$ we get $2^x = M = 2^n$, and strict monotonicity of $t \mapsto 2^t$ gives $x = n \in \mathbb{Z}$. ☐

Status: **[paper proof in this report]** — a Lean module `LogProductIndependence` implementing this argument (with the independence hypothesis recorded as an explicit `Prop`, never asserted) is in preparation; the theorem statement there is `alaogluErdosConjecture_of_primeLogProductIndependence`.

12.3 What the criterion is, and is not

Three remarks place Theorem 12.2 correctly.

It is a consequence of Schanuel's conjecture. Schanuel's conjecture implies that logarithms of multiplicatively independent positive rationals are algebraically independent over $\mathbb{Q}$; in particular the numbers $\log p$, over all primes p, would be algebraically independent, and then the products $\log p \log q$ are $\mathbb{Q}$-linearly independent because distinct monomials of degree two in algebraically independent elements are linearly independent. So Definition 12.1 is strictly weaker than Schanuel and lies well inside the standard expectations.

Its strength beyond the conjecture lies in pairs that never occur. This calibration matters and is easy to miss. Collecting the relation (21) back up,

$$\sum_p \alpha_p(\log p \log 3) - \sum_p \gamma_p(\log p \log 2) = \log 3 \, \log M - \log 2 \, \log A,$$

so the only products ever exercised are those of the form $\log 2 \log p$ and $\log 3 \log p$. Write F for that sub-family. A vanishing rational relation over F factors as $\log 2 \, \log u + \log 3 \, \log v = 0$ for positive rationals u, v, i.e. says precisely that some positive rational $\neq 1$ has a rational ϑ-th power. Equivalently, $\mathbb{Q}$-linear independence of F is the *rational-output* form of the problem:

$$2^x \in \mathbb{Q} \text{ and } 3^x \in \mathbb{Q} \qquad \Longrightarrow \qquad x \in \mathbb{Z}. \tag{22}$$

This is worth separating from Conjecture 2.1. It implies the conjecture, since integers are rational, and it is *a priori strictly stronger*: a rational counterexample such as $2^x = 3/2$ is not an integer counterexample, and nothing in the corpus reduces one to the other. Both follow from four exponentials, because rationals are algebraic.

The proof of Theorem 12.2 establishes (22) verbatim: for rational outputs the prime exponents α_p, γ_p are integers of either sign, and the argument never used their positivity. So the criterion is not a restatement of the conjecture; it delivers the stronger rational-output form. What the calibration does show is where its surplus lies: independence of pairs $\log p \log q$ with $p, q \geq 5$ is never exercised, so one should not expect the full criterion to be easier than (22). Its unconditional dividend is the two-prime case of Section 12.5.

Status: **[kernel-verified Lean]** for the integer-output form (`alaogluErdosConjecture_of_primeLogProductIndependence`); the rational-output strengthening (22) is **[paper proof in this report]** and awaits a Lean version stated over `Rat` valuations.

It is nevertheless of a different kind from four exponentials. The four exponentials conjecture is a statement about 2×2 matrices of logarithms and their exponentials; the present criterion mentions no exponentials and is purely a linear-independence assertion about a countable family of real numbers. Neither is known, and neither is known to imply the other in general; what Theorem 12.2 shows is that *either* suffices for the Alaoglu–Erdős conjecture. This is worth recording because the two hypotheses invite different techniques: four exponentials is attacked by auxiliary functions and zero estimates, while linear independence of log-products is a question about the arithmetic of the multiplicative group of $\mathbb{Q}_{>0}$ and its second symmetric power.

The smallest instances are already open. Independence of the family includes, as its very first nontrivial case, the assertion that $(\log 3)^2$, $(\log 2)(\log 3)$ and $(\log 2)^2$ are $\mathbb{Q}$-linearly independent, equivalently that $\vartheta = \log_2 3$ is not a root of a rational quadratic — a statement about the irrationality of ϑ^2 modulo $\mathbb{Q}(\vartheta)$ that is not decided by present transcendence theory. Indeed the relation exhibited in Section 6 for a 3-smooth counterexample would already contradict the single independence

$$\alpha_2(\log 2)(\log 3) + \alpha_3(\log 3)^2 - \gamma_2(\log 2)^2 - \gamma_3(\log 2)(\log 3) \neq 0$$

for the relevant nonzero integer vector. Restricted to the two primes 2 and 3, the criterion needed is thus exactly: 1*,* ϑ *and* ϑ^2 *are linearly independent over* $\mathbb{Q}$, which is the 3-smooth shadow of Conjecture 6.3.

12.4 A finite form of the criterion

Because a counterexample has a fixed finite prime support, the full strength of Definition 12.1 is never needed at once.

Corollary 12.3 (Finite prime-support form)**.** *Let S be a finite set of primes containing 2 and 3. If the products $\log p \log q$ with $p \le q$ in S are $\mathbb{Q}$-linearly independent, then no counterexample to Conjecture 2.1 has both outputs supported on S.*

Proof. The proof of Theorem 12.2 used only the pairs drawn from the prime support of MA together with 2 and 3. ☐

12.5 The two-prime case is a theorem, and it is sharp

The finite form is more than a bookkeeping device, because its smallest instance is *provable.*

Proposition 12.4 (Two-prime log-product independence)**.** *Let $p \neq q$ be primes. Then $(\log p)^2$, $\log p\, \log q$ and $(\log q)^2$ are linearly independent over $\mathbb{Q}$.*

Proof. A vanishing relation $a(\log p)^2 + b \log p \log q + c(\log q)^2 = 0$ with rational a, b, c, divided by $(\log p)^2$, reads $a + bt + ct^2 = 0$ where $t = \log q/ \log p$. If $(a, b, c) \neq 0$ this makes t a root of a nonzero rational polynomial, hence algebraic. But t is transcendental: it is irrational by unique factorization, and were it algebraic irrational then Gelfond–Schneider applied to $p^{\,t} = q$ would make the integer q transcendental. ☐

Consequently the hypothesis of Corollary 12.3 is *unconditionally true* for $S = \{2, 3\}$, and the corollary becomes an unconditional exclusion. Stated directly, and proved in Lean without any appeal to Definition 12.1:

Theorem 12.5 (Three-smooth outputs force an integral exponent)**.** *If $2^x \in \mathbb{Z}$ and $3^x \in \mathbb{Z}$ are both 3-smooth — divisible by no prime outside $\{2, 3\}$ — then $x \in \mathbb{Z}$.*

Proof. Write $2^x = 2^a 3^b$ and $3^x = 2^c 3^d$. Taking base-two logarithms gives the two relations $x = a + b\vartheta$ and $x\vartheta = c + d\vartheta$. Substituting the first into the second eliminates x and leaves

$$b\,\vartheta^2 + (a - d)\,\vartheta - c = 0.$$

If $b \neq 0$ this makes ϑ a root of a nonzero rational quadratic, contradicting its transcendence. Hence $b = 0$ and $x = a$. ☐

Corollary 12.6 (A smooth candidate has a rough partner)**.** *If x is a nonintegral solution and 2^x is 3-smooth, then 3^x has a prime factor at least 5. The two outputs of a counterexample can never both be built from $\{2, 3\}$.*

Status: [kernel-verified Lean] — `integer_of_threeSmooth_outputs` and `exists_prime_five_le_dvd_of_threeSmooth_candidate` in `SmoothOutputs`. The proofs use only transcendence of ϑ, itself kernel-verified in this corpus.

Two remarks fix the position of this result precisely.

The two-sided hypothesis is essential. Assuming only that 2^x is 3-smooth gives $x = a + b\vartheta$ and then requires $\left(3^{\vartheta}\right)^b \in \mathbb{Q}$, which is exactly the open localized-radical condition of Section 6. What the second smoothness hypothesis supplies is a *second* linear relation between x and ϑ; two such relations eliminate x and leave an algebraic equation in ϑ alone. This degree drop is the whole mechanism, and it is worth isolating as a template: *each independent multiplicative constraint on an output buys one linear relation, and two relations force algebraicity of ϑ.* The difficulty of the general problem is precisely that one never gets a second constraint for free.

Three primes is where the criterion becomes conjectural. Proposition 12.4 exhausts what transcendence of a single logarithmic ratio can give. For $S = \{2, 3, 5\}$ a vanishing relation among the six products, divided by $(\log 2)^2$, reads

$$a + b\vartheta + c\vartheta_5 + d\vartheta^2 + e\vartheta\vartheta_5 + f\vartheta_5^2 = 0, \qquad \vartheta_5 = \log_2 5,$$

that is, the point (ϑ, ϑ_5) lies on a rational conic. Both coordinates are transcendental and $1, \vartheta, \vartheta_5$ are $\mathbb{Q}$-independent by unique factorization, but no present method excludes a quadratic relation between them. So Definition 12.1 is a theorem for two primes, open from three onwards, and the Alaoglu–Erdős conjecture sits exactly on that boundary.

13 The algebraic auxiliary-base spectrum

Section 9 classified the *natural* auxiliary bases whose x-th powers remain algebraic, and had to tag the transcendence half of that classification as classical background: the repository's interpolation determinant carried integer entries, so it could prove $B^x \notin \mathbb{Q}$ but not $B^x \notin \overline{\mathbb{Q}}$. That boundary has moved. The same square determinant now accepts algebraic entries of bounded height inside a number field, and the resulting classification is exact over *all* positive real algebraic bases. A hypothetical nonintegral solution turns out to have one fixed, explicitly described set of good bases — a dense but very thin subgroup of $\mathbb{R}_{>0}$ — and every positive algebraic base outside it produces a transcendental value.

13.1 The saturated group of positive two-three units

The multiplicative group $\overline{\mathbb{Q}}_{>0}$ of positive real algebraic numbers is a vector space over $\mathbb{Q}$ once rational scalars act through the unique positive real root:

$$q \cdot \alpha = \alpha^q, \qquad q \in \mathbb{Q}, \quad \alpha \in \overline{\mathbb{Q}}_{>0}.$$

For multiplicatively independent $a, b \in \overline{\mathbb{Q}}_{>0}$ define the *saturated group*

$$\Gamma(a, b) = \{a^r b^s : r, s \in \mathbb{Q}\}.$$

The map $\mathbb{Q}^2 \to \Gamma(a,b)$, $(r,s) \mapsto a^r b^s$, is an isomorphism of $\mathbb{Q}$-vector spaces; injectivity is exactly multiplicative independence after clearing denominators. For the original problem write

$$\Gamma = \Gamma(2,3) = \{2^r 3^s : r,s \in \mathbb{Q}\}.$$

This group has a second description that is the one the Lean development actually uses. Let $\langle 2,3\rangle_{\mathbb{Z}} = \{2^i 3^j/(2^k 3^\ell) : i,j,k,\ell \in \mathbb{N}_0\}$ be the group of positive rational 2, 3-units, and let

$$\sqrt[\infty]{\langle 2,3\rangle_{\mathbb{Z}}} := \{\gamma > 0 : \exists\, n \geq 1,\ \gamma^n \in \langle 2,3\rangle_{\mathbb{Z}}\}$$

be its divisible hull inside $\mathbb{R}_{>0}$.

Observation 13.1. $\sqrt[\infty]{\langle 2,3\rangle_{\mathbb{Z}}} = \Gamma$. In particular every element of the divisible hull is automatically algebraic.

Indeed, if $\gamma^n = 2^i 3^j/(2^k 3^\ell)$ then γ is the unique positive real n-th root of a positive rational, hence $\gamma = 2^{(i-k)/n} 3^{(j-\ell)/n} \in \Gamma$; conversely $2^r 3^s$ raised to a common denominator of r,s lands in $\langle 2,3\rangle_{\mathbb{Z}}$. The Lean predicate `HasPositiveTwoThreeUnitPower` is the literal transcription of membership in the hull, so it and membership in Γ may be used interchangeably below.

13.2 The exact spectrum theorem

The kernel-verified corpus contains the determinant in fully symmetric form: three positive algebraic bases with injective nonnegative monomials cannot all have algebraic x-th powers when x is irrational. The next theorem is the coordinate-free synthesis for an arbitrary independent pair; its proof is a direct application of the same classical six-exponentials input, and it is stated here at paper level because the Lean development fixes the pair $(2,3)$.

Theorem 13.2 (Exact algebraic-base spectrum)**.** *Let $a,b \in \overline{\mathbb{Q}}_{>0}$ be multiplicatively independent, let $x \in \mathbb{R} \setminus \mathbb{Q}$, and assume $a^x, b^x \in \overline{\mathbb{Q}}$. Then, for every $\alpha \in \overline{\mathbb{Q}}_{>0}$,*

$$\alpha^x \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad \alpha \in \Gamma(a,b).$$

[paper proof in this report]

Proof. If $\alpha = a^r b^s$ with $r,s \in \mathbb{Q}$, positivity gives $\alpha^x = (a^x)^r (b^x)^s$, which is algebraic.

Conversely, suppose $\alpha \notin \Gamma(a,b)$. Then $\log a$, $\log b$, $\log \alpha$ are linearly independent over $\mathbb{Q}$: a rational relation clears to an integer multiplicative relation, and if the coefficient of $\log \alpha$ vanishes it contradicts the multiplicative independence of a,b, while otherwise it places α in the saturated group. The pair $1,x$ is linearly independent over $\mathbb{Q}$ because x is irrational. Apply the six exponentials theorem (Theorem 9.3) to the three rows $\log a, \log b, \log \alpha$ and the two columns $1,x$. Five of the six exponential values,

$$a, \quad b, \quad \alpha, \quad a^x, \quad b^x,$$

are algebraic. Therefore the sixth, α^x, is transcendental. □

Conceptually this says more than any list of good auxiliary bases: two algebraic outputs have already exhausted the maximum algebraic multiplicative rank that six exponentials permits.

Specializing to $a = 2$, $b = 3$ gives the statement the repository proves outright.

Corollary 13.3 (Complete algebraic spectrum of a counterexample). *Let $x \in \mathcal{S} \setminus \mathbb{Z}$. For every $\alpha \in \overline{\mathbb{Q}}_{>0}$,*

$$\alpha^x \in \overline{\mathbb{Q}} \quad \iff \quad \alpha \in \Gamma.$$

In particular the spectrum does not depend on which hypothetical nonintegral solution is chosen. ***[kernel-verified Lean]***

A rational solution is integral, so x is irrational and Theorem 13.2 applies; the formal route is the divisible-hull equivalence of the next subsection together with Observation 13.1.

Status: [kernel-verified Lean] — the divisible-hull equivalence lives in `AlgebraicBaseUnitPower`.

Its two forms are

`real_rpow_isAlgebraic_iff_integer_or_hasPositiveTwoThreeUnitPower`

together with its counterexample-relative form

`real_rpow_isAlgebraic_iff_hasPositiveTwoThreeUnitPower`

in the `TwoBaseNonintegerSolution` namespace, both resting on the injectivity criterion `real_two_three_gamma_monomial_injective_iff`. The natural-base and rational-base specializations are `rpow_isAlgebraic_iff_integer_or_eq_two_pow_mul_three_pow` in `AlgebraicThirdBase` and `rat_rpow_isAlgebraic_iff_isTwoThreeUnit_of_not_integer` in `AlgebraicRationalBase`; the half-plane, mixed-iterate and rank-one ratio consequences are in `AlgebraicOutputLocus`; and the simultaneous exclusion of both canonical radicals from the entire hull is in `CanonicalRadicalDivisibleHull`.

The group Γ is countable, divisible, of rational rank two, and dense in $\mathbb{R}_{>0}$ — density already follows from the subgroup $\{2^r : r \in \mathbb{Q}\}$. A counterexample would therefore possess a dense set of algebraic bases with algebraic output, while every algebraic base outside that exact dense subgroup would give a transcendental one.

Corollary 13.4 (Rank bound for any family of algebraic outputs). *Let $x \in \mathcal{S} \setminus \mathbb{Z}$ and let $\alpha_1, \ldots, \alpha_t \in \overline{\mathbb{Q}}_{>0}$ satisfy $\alpha_i^x \in \overline{\mathbb{Q}}$. Then every α_i lies in Γ and the multiplicative rank of $\alpha_1, \ldots, \alpha_t$ is at most two.* ***[paper proof in this report]***

Both ingredients are kernel-verified — membership in the hull by `AlgebraicBaseUnitPower` and the rank ceiling by `AlgebraicRankThreeEquivalence` — but the combined family statement is recorded here at paper level.

13.3 The divisible-hull formulation

Stated without reference to a fixed counterexample, the kernel-verified form is a single biconditional valid for every positive real algebraic base. For $\gamma \in \overline{\mathbb{Q}}_{>0}$, under $2^x, 3^x \in \mathbb{Z}$,

$$\gamma^x \in \overline{\mathbb{Q}} \quad \iff \quad x \in \mathbb{Z} \quad \text{or} \quad \gamma \in \sqrt[\infty]{\langle 2, 3 \rangle_{\mathbb{Z}}}. \tag{23}$$

The arithmetic core is an exact combinatorial equivalence: γ has *no* positive natural power in $\langle 2, 3 \rangle_{\mathbb{Z}}$ precisely when the monomial map $(i, j, k) \mapsto 2^i 3^j \gamma^k$ is injective on $\mathbb{N}_0^3$. The forward implication of

(23) then feeds that injectivity into the generic algebraic determinant, and the reverse implication descends algebraicity from a positive integral power. Specializing γ to a natural number recovers the 3-smooth classification of Section 9 with “rational” upgraded to “algebraic”, so for a hypothetical nonintegral solution the three properties $a^x \in \mathbb{Z}$, $a^x \in \mathbb{Q}$ and $a^x \in \overline{\mathbb{Q}}$ coincide for $a \in \mathbb{N}_{>0}$, and hold exactly at $2,3$-smooth a. Specializing to a positive rational q gives $q^x \in \overline{\mathbb{Q}} \iff q \in \langle 2,3\rangle_{\mathbb{Z}}$; there is no algebraic-irrational middle case anywhere in the picture.

13.4 The determinant engines behind the spectrum

Two generic determinant theorems, neither of which mentions the bases $2,3$, carry the whole classification. The rational one states that if $a,b,c \in \mathbb{Q}_{>0}$ have injective nonnegative monomials $(i,j,k) \mapsto a^i b^j c^k$ and $a^x, b^x, c^x \in \mathbb{Q}$, then $x \in \mathbb{Q}$. The algebraic one removes both restrictions: if a,b,c are positive real algebraic numbers with injective nonnegative monomials and a^x, b^x, c^x are all algebraic, then $x \in \mathbb{Q}$. Both allow bases below 1 and negative exponents, and both use $1+|x|$ in the size estimate.

Status: [kernel-verified Lean] — `rational_of_three_rat_rpows_rational_of_monomial_injective` in `RationalSixExponentials`, and `rational_of_three_real_rpows_isAlgebraic_of_monomial_injective` in `AlgebraicSixExponentials`.

The step from rational to algebraic outputs is exactly the replacement of the trivial bound $|D| \geq 1$ for a nonzero integer determinant by a field-norm bound for a nonzero algebraic integer determinant, which is the extension anticipated in Section 9. The six input and output values are placed in one number field K with one common nonzero natural scale s making every entry lie in $\mathcal{O}_K$. For side parameter n the square matrix has size n^6 and each entry acquires the factor s^{n^5}, so the determinant acquires $s^{n^{11}}$; at every embedding other than the distinguished one the six factor groups in an entry are bounded by T^{6n^5}, and the norm of the nonzero algebraic-integer determinant gives

$$\frac{1}{\left((n^6)!\,T^{6n^{11}}\right)^{[K:\mathbb{Q}]-1}} \;\leq\; \left|\sigma(\det A)\right|.$$

The verified estimates $(n^6)! \leq 2^{n^{11}}$ and $\left((n^6)!\,T^{6n^{11}}\right)^{[K:\mathbb{Q}]-1} \leq \left((2T^6)^{[K:\mathbb{Q}]-1}\right)^{n^{11}}$ put the entire conjugate and degree loss on the same n^{11} scale as the old rational denominator loss, where the existing explicit analytic margin already absorbs it. No new analytic estimate was required — only honest bookkeeping.

Status: [kernel-verified Lean] — `one_div_conjugate_entry_bound_le_norm_det` and `one_div_entry_bound_le_scaled_norm_det` in `AlgebraicIntegerDeterminant`, with the number-field realizations `AlgebraicOutputBridge.ofRealIsAlgebraic` and `FiniteAlgebraicOutputBridge.ofFinSixIsAlgebraic`.

13.5 The strong spectrum via Baker and Roy

Let $\mathcal{L}$ denote the $\overline{\mathbb{Q}}$-vector space generated by 1 and by all logarithms of nonzero algebraic numbers, with arbitrary complex branches allowed; for positive real algebraic numbers we use the real logarithm, which is one of those branches. Roy's strong six exponentials theorem [8, 11] states that if three complex numbers are linearly independent over $\overline{\mathbb{Q}}$ and two more are linearly independent over $\overline{\mathbb{Q}}$, then at least one of the six pairwise products lies outside $\mathcal{L}$.

Theorem 13.5 (Strong algebraic-base spectrum)**.** *Under the hypotheses of Theorem 13.2, if $\alpha \notin \Gamma(a, b)$, then*

$$x \log \alpha \notin \mathcal{L}.$$

In particular α^x is transcendental. ***[paper proof in this report]***

Proof. As in Theorem 13.2, the logarithms $\log a, \log b, \log \alpha$ are $\mathbb{Q}$-linearly independent, and Baker's homogeneous theorem [3] upgrades that to linear independence over $\overline{\mathbb{Q}}$. The number x is transcendental: were it algebraic, its irrationality together with $a \neq 0, 1$ would make a^x transcendental by Gelfond–Schneider. Hence $1, x$ are linearly independent over $\overline{\mathbb{Q}}$. Apply Roy's theorem to the rows $\log a, \log b, \log \alpha$ and the columns $1, x$. Five products lie in $\mathcal{L}$, namely $\log a$, $\log b$, $\log \alpha$, $x \log a = \log(a^x)$ and $x \log b = \log(b^x)$. Therefore the sixth product $x \log \alpha$ lies outside $\mathcal{L}$. □

Definition 13.6. Call a positive exponential value e^z *log-strongly transcendental* when $z \notin \mathcal{L}$. This implies ordinary transcendence of e^z, since the logarithm of a positive algebraic number belongs to $\mathcal{L}$.

So outside Γ the failure is not merely that α^x is transcendental: the number $x \log \alpha$ is not even a $\overline{\mathbb{Q}}$-linear combination of 1 and logarithms of algebraic numbers. This is a genuinely stronger exclusion than Corollary 13.3, and it is the form in which the spectrum interacts with linear forms in logarithms.

13.6 One base is enough: detectors and rank equivalences

Because the spectrum is the same for every hypothetical nonintegral solution, a *single* auxiliary base outside Γ already decides the whole conjecture.

Theorem 13.7 (Algebraic detector equivalence)**.** *Fix $\alpha \in \overline{\mathbb{Q}}_{>0} \setminus \Gamma$. Then the Alaoglu–Erdős conjecture is equivalent to*

$$\forall x \in \mathbb{R}, \quad 2^x, 3^x \in \mathbb{Z} \Longrightarrow \alpha^x \in \overline{\mathbb{Q}},$$

and equally to the assertion that every such x satisfies $x \log \alpha \in \mathcal{L}$. ***[paper proof in this report]***

Proof. If the conjecture holds, every solution is a nonnegative integer and α^x is algebraic, with $x \log \alpha \in \mathcal{L}$. Conversely a nonintegral solution would make α^x transcendental by Corollary 13.3 and would put $x \log \alpha$ outside $\mathcal{L}$ by Theorem 13.5. □

This is a paper-level universalization of a pointwise locus that the kernel already knows. Valid detectors include, for instance,

$$5, \qquad \sqrt{5}, \qquad 1 + \sqrt{2}, \qquad \frac{7 + 3\sqrt{5}}{2},$$

provided the chosen number does not accidentally lie in Γ; for an algebraic integer with a prime divisor outside $\{2, 3\}$ that exclusion is immediate, and in general it is a multiplicative independence check. Nothing arithmetically special about 5 is used.

Four instances of this phenomenon are themselves kernel-verified, and they are the most quotable consequences of the whole algebraic-output layer.

One-base equivalents of the conjecture. Each of the following is equivalent to the Alaoglu–Erdős conjecture, and each equivalence is accepted by the Lean kernel.

(i) Every x with $2^x, 3^x \in \mathbb{Z}$ has 5^x algebraic.

(ii) Every x with $2^x, 3^x \in \mathbb{Z}$ has 6^{x^2} algebraic.

(iii) For every x with $2^x, 3^x \in \mathbb{Z}$, the positive-algebraic output locus

$$\{\alpha \in \overline{\mathbb{Q}}_{>0} : \alpha^x \in \overline{\mathbb{Q}}\}$$

has multiplicative rank three, i.e. contains a, b, c with injective nonnegative monomials.

(iv) For every x with $2^x, 3^x \in \mathbb{Z}$, the rational squared-output ratio locus has rank two, i.e. contains algebraic members $(2^i/3^j)^{x^2}$ and $(2^k/3^\ell)^{x^2}$ with $i\ell \neq kj$.

A counterexample would collapse (iii) to rank at most two and confine the entire locus in (iv) to a single rational ray in the exponent lattice. [kernel-verified Lean]

Status: [kernel-verified Lean] — all four equivalences are accepted by the kernel; the Lean names follow.

Items (i) and (ii) are `alaogluErdosConjecture_iff_five_rpow_isAlgebraic` and `alaogluErdosConjecture_iff_six_squared_exponent_isAlgebraic` in `AlgebraicConsequences`, which also carries the two-iterate variant `alaogluErdosConjecture_iff_both_iterated_outputs_isAlgebraic`. Item (iii) is

`alaogluErdosConjecture_iff_algebraicRpowOutputLocusHasRankThree`

in `AlgebraicRankThreeEquivalence`, and item (iv) is

`alaogluErdosConjecture_iff_ratioSquaredAlgebraicLocusHasRankTwo`

in `AlgebraicOutputLocus`. The same modules also record the pointwise forms `ratioSquaredAlgebraicLocusHasRankTwo_iff_integer` and `algebraicRpowOutputLocusHasRankThree_iff_integer`, and the explicit p-adic line `exists_external_prime_algebraic_output_ratio_padic_line` on which a counterexample would force every algebraic ratio output to sit.

There is a corresponding zero-one law: assuming the conjecture fails and fixing $\alpha \in \overline{\mathbb{Q}}_{>0}$, exactly one of two things happens. Either $\alpha \in \Gamma$ and α^x is algebraic for *every* nonintegral solution x, or $\alpha \notin \Gamma$ and α^x is transcendental for every nonintegral solution. No base can behave one way for one counterexample and the other way for another.

13.7 Why the exact spectrum does not settle the conjecture

It is worth being precise about what has and has not been gained. The spectrum is an exact answer to the question "which algebraic bases stay algebraic?", and its answer is $\Gamma = \{2^r 3^s : r, s \in \mathbb{Q}\}$ — but that is precisely the set of bases whose logarithms lie in the rational span of $\log 2$ and $\log 3$. Every base the spectrum certifies is, logarithmically, a rational recombination of the two columns already present in the four-exponentials matrix of Section 7. Six exponentials needs a third row genuinely independent of $\log 2, \log 3$ together with algebraicity of its output; the spectrum theorem shows that a counterexample supplies algebraic outputs at exactly the bases that fail this requirement, and transcendental outputs at exactly the bases that would meet it. The classification is therefore self-consistent with a counterexample rather than obstructive to one: it converts every attempt to manufacture a third independent base into the very transcendence statement one was trying to prove. The kernel-verified radical work makes the same wall concrete: the simultaneous normalization of the two canonical radicals produces an explicit multiplicative relation among 2, the primitive odd core, and the swapped radical, so that the most tempting three-base application of the algebraic six-exponentials theorem fails its monomial-injectivity hypothesis for a structural, kernel-verified reason (`swappedRadical_relation_blocks_monomial_injectivity` in `CanonicalRadicalFieldNorm`). Closing the gap needs a new source of size — nonarchimedean, or from a sharper five/six-exponentials interface — not a wider supply of algebraic bases. The conjecture remains open.

14 Rational and integral outputs inside the spectrum

The algebraic spectrum identifies the divisible group

$$\Gamma = \{2^r 3^s : r, s \in \mathbb{Q}\}$$

as the exact set of positive algebraic bases whose powers at a hypothetical nonintegral solution remain algebraic: outside Γ every such power is transcendental, and inside Γ every such power is algebraic. A finer arithmetic question survives inside the spectrum. If $\alpha = 2^r 3^s \in \Gamma$, when is α^x rational, and when is it a positive integer?

For a fixed solution the answer is completely encoded by prime valuations of the output pair. Rationality becomes a congruence condition, integrality adds a cone condition, and the gap between the obvious rational bases $2^u 3^v$ with $u, v \in \mathbb{Z}$ and the full rational locus is one explicit finite abelian group whose order is a single gcd of 2×2 minors. This is a strict refinement of the divisible-hull description of the algebraic locus, and it is exact rather than asymptotic.

14.1 The valuation matrix of an output pair

Let $x \in \mathcal{S} \setminus \mathbb{Z}$ be a nonintegral solution and write

$$M = 2^x \in \mathbb{N}, \qquad A = 3^x \in \mathbb{N}.$$

Let P be the finite union of the prime supports of M and A, and form the column vectors

$$\mathbf{m} = (v_p(M))_{p \in P}, \qquad \mathbf{a} = (v_p(A))_{p \in P}, \qquad V_x = \begin{pmatrix} \mathbf{m} & \mathbf{a} \end{pmatrix} \in M_{P \times 2}(\mathbb{Z}).$$

For a column $q = (r, s)^T \in \mathbb{Q}^2$ put $\alpha(q) = 2^r 3^s$, so that Γ is exactly the image of $q \mapsto \alpha(q)$. Taking the unique positive real interpretation of the rational powers,

$$\alpha(q)^x = (2^x)^r (3^x)^s = M^r A^s = \prod_{p\in P} p^{r v_p(M) + s v_p(A)}. \tag{24}$$

The exponent vector appearing in (24) is precisely $V_x q$.

Lemma 14.1 (The valuation matrix has rank two)**.** *The two columns of V_x are linearly independent over $\mathbb{Q}$.*

Proof. Suppose $r\mathbf{m}+s\mathbf{a} = 0$ with $r, s \in \mathbb{Q}$. Then every prime exponent in (24) vanishes, so $M^r A^s = 1$. Taking real logarithms and using $\log M = x \log 2$, $\log A = x \log 3$ gives $x(r \log 2 + s \log 3) = 0$. Since a solution with $M = 2^x \in \mathbb{N}$, $M > 1$ has $x > 0$, and since $\vartheta = \log_2 3$ is irrational, $r = s = 0$. □

Proposition 14.2 (Exact rational and integral auxiliary-output spectrum)**.** *Let $x \in \mathcal{S} \setminus \mathbb{Z}$, let V_x be as above, and let $q \in \mathbb{Q}^2$. Then*

$$\alpha(q)^x \in \mathbb{Q}_{>0} \iff V_x q \in \mathbb{Z}^P, \qquad \alpha(q)^x \in \mathbb{N} \iff V_x q \in \mathbb{N}_0^P.$$

Together with the algebraic spectrum, this classifies algebraic, rational and integral outputs for every positive algebraic auxiliary base. ***[paper proof in this report]***

Proof. Equation (24) exhibits $\alpha(q)^x$ as a finite product of primes with rational exponents. Choose a common denominator d for the entries of $V_x q$. If $\alpha(q)^x$ is rational, then raising to the d-th power gives a rational number whose ordinary p-adic valuation equals d times the p-th entry of $V_x q$; comparing valuations of numerator and denominator shows every entry of $V_x q$ is an integer. Conversely, integer exponents produce a positive rational number. Such a rational number is a positive integer exactly when all of its prime valuations are nonnegative, which is the second equivalence. □

Three nested spectra therefore occur, all determined by V_x:

$$\begin{aligned} \mathcal{B}_{\text{alg}}(x) &= \{\alpha(q) : q \in \mathbb{Q}^2\} = \Gamma, \\ \mathcal{B}_{\text{rat}}(x) &= \{\alpha(q) : q \in \Lambda_x\}, \\ \mathcal{B}_{\text{int}}(x) &= \{\alpha(q) : q \in \Lambda_x^+\}, \end{aligned} \qquad \begin{aligned} \Lambda_x &= \{q \in \mathbb{Q}^2 : V_x q \in \mathbb{Z}^P\}, \\ \Lambda_x^+ &= \{q \in \Lambda_x : V_x q \geq 0\}. \end{aligned}$$

The outer spectrum does not depend on x; the two inner ones may.

14.2 Smith normal form and the finite rational-output defect

By Lemma 14.1 the matrix V_x has rank two, so Λ_x is a rank-two lattice in $\mathbb{Q}^2$ containing $\mathbb{Z}^2$ with finite index. Write

$$\delta_1(V_x) = \gcd\{\text{all entries of } V_x\}, \qquad \delta_2(V_x) = \gcd\left\{ \left| \det \begin{pmatrix} v_p(M) & v_p(A) \\ v_{p'}(M) & v_{p'}(A) \end{pmatrix} \right| : p, p' \in P \right\}$$

for the two determinantal divisors. The second is positive precisely because the rank is two.

Proposition 14.3 (Smith classification of the rational locus)**.** *Let $d_1 \mid d_2$ be the Smith invariant factors of V_x, so that $d_1 = \delta_1(V_x)$ and $d_1 d_2 = \delta_2(V_x)$. After a unimodular change of coordinates in $\mathbb{Q}^2$,*

$$\Lambda_x = \tfrac{1}{d_1}\mathbb{Z} \oplus \tfrac{1}{d_2}\mathbb{Z},$$

and consequently

$$\Lambda_x/\mathbb{Z}^2 \cong \mathbb{Z}/d_1\mathbb{Z} \oplus \mathbb{Z}/d_2\mathbb{Z}, \qquad \left|\Lambda_x/\mathbb{Z}^2\right| = \delta_2(V_x). \qquad \textit{[paper proof in this report]}$$

Proof. Choose unimodular matrices $U \in \mathrm{GL}_P(\mathbb{Z})$ and $W \in \mathrm{GL}_2(\mathbb{Z})$ with

$$UV_xW = \begin{pmatrix} d_1 & 0 \\ 0 & d_2 \\ 0 & 0 \\ \vdots & \vdots \end{pmatrix}.$$

Since U preserves $\mathbb{Z}^P$, the condition $V_x q \in \mathbb{Z}^P$ is equivalent to $UV_x q \in \mathbb{Z}^P$, and substituting $q = Wq'$ turns it into $d_1 q'_1 \in \mathbb{Z}$ and $d_2 q'_2 \in \mathbb{Z}$. Since W preserves $\mathbb{Z}^2$, it identifies $\Lambda_x/\mathbb{Z}^2$ with the quotient computed in the primed coordinates, which is $\mathbb{Z}/d_1\mathbb{Z} \oplus \mathbb{Z}/d_2\mathbb{Z}$. The identifications $d_1 = \delta_1(V_x)$ and $d_1 d_2 = \delta_2(V_x)$ are the standard determinantal-divisor formulae for the Smith normal form. □

The first invariant factor has a direct output interpretation: d_1 is the largest integer d such that M and A are both perfect d-th powers, because $d \mid \delta_1(V_x)$ says exactly that d divides every prime valuation of M and of A. For the canonical least nonintegral solution β, primitivity of the least output pair (Section 5) gives $d_1 = 1$, and then

$$\Lambda_\beta/\mathbb{Z}^2 \cong \mathbb{Z}/\delta_2(V_\beta)\mathbb{Z}.$$

Thus a single integer — the gcd of all 2×2 valuation minors of the least output pair — measures the entire finite defect between the rational β-outputs at algebraic $2,3$-radical bases and the obvious rational $2,3$-unit bases. We call $\delta_2(V_\beta)$ the *rational-output defect*. It is a finite, in principle computable, invariant of a hypothetical counterexample, and it equals 1 exactly when no auxiliary base outside $\{2^u 3^v : u, v \in \mathbb{Z}\}$ produces a rational output.

The integral spectrum is the intersection of this lattice with the rational polyhedral cone $V_x q \geq 0$. By Gordan's lemma Λ_x^+ is a finitely generated normal affine semigroup, so $\mathcal{B}_{\mathrm{int}}(x)$ has a finite generating set of auxiliary bases. This is an exact structural replacement for attempting to enumerate algebraic auxiliary bases one at a time.

Status: [kernel-verified Lean] and [paper proof in this report]

The lattice statements themselves — Lemma 14.1, Proposition 14.2, Proposition 14.3 and the Gordan consequence — are [paper proof in this report]: they are proved on paper in this report and are not yet formalized in that form. What *is* kernel-verified is the arithmetic core on which they rest, in three modules.

`CanonicalRadicalValuation` proves the divisibility dictionary between perfect powers and valuations that underlies the meaning of d_1: a divisor of the odd-prime valuation gcd of a positive integer is an honest exponent of an exact power (`exists_pow_eq_of_dvd_oddPrimeValuationGCD`), the least rational-power index is coprime to the joint gcd of the denominator exponent and the numerator's valuation gcd (`rationalPowerIndex_gcd_denominator_valuationGCD_eq_one`), and every prime dividing that index leaves some prime valuation of the normalized numerator undivided, giving the exact radical degree together with the valuation restriction (`exists_exact_radical_degree_and_valuation_restriction`).

`SimultaneousRadicalDegrees` kernel-verifies the exactness statements that the $d_1 = 1$ normalization buys: under the simultaneous primitive normalization the displayed exponents are the least rational-power indices and hence the exact degrees over $\mathbb{Q}$ (`oddCoreRpow_exact_rationalPowerIndex_of_simultaneous_normalization` and `threeFreeCoreRpow_exact_radical_degree_of_simultaneous_normalization`), and when both base-adic depths vanish the two indices are coprime and the mixed product has exact degree equal to their product (`mixed_coreRpow_exact_radical_degree_of_zero_baseDepths`). The descent used there is precisely the one behind $d_1 = 1$: a proper index divisor would make both outputs of the least nonintegral solution common perfect powers, which is excluded by `IsLeastTwoBaseNonintegerSolution.no_common_perfect_power` in `LeastSolution` (Section 5). That exclusion is exactly the assertion $d_1 = 1$ for V_β.

`CanonicalRadicalMixedOutput` kernel-verifies the transcendence side of the mixed case: a genuinely mixed positive monomial in the two canonical radicals retains an external prime in every rational power, so all of its outputs at the least nonintegral solution are transcendental, and an algebraic mixed output is therefore forced onto a coordinate face. The theorem is

`TwoBaseNonintegerSolution.transcendental_mixed_canonical_radical_output`,

packaged under failure of the conjecture as

`exists_simultaneous_canonical_radical_mixed_output_transcendence`.

These are statements about the canonical radicals rather than about Λ_x itself; they confirm the geometry of the lattice picture without yet formalizing it.

14.3 Two translated solutions remove the fractional defect

The algebraic spectrum is solution-independent, but the rational locus Λ_x need not be. Translation of the exponent by one gives a sharp intersection identity, and it shows that the defect cannot persist across the whole solution monoid.

Proposition 14.4 (Translation intersection)**.** *Let $x \in \mathcal{S} \setminus \mathbb{Z}$. Then*

$$\Lambda_x \cap \Lambda_{x+1} = \mathbb{Z}^2.$$

Equivalently, for $\alpha \in \overline{\mathbb{Q}}_{>0}$, if α^x and α^{x+1} are both rational, then $\alpha = 2^u 3^v$ with $u, v \in \mathbb{Z}$. ***[paper proof in this report]***

Proof. One inclusion is immediate: for $q \in \mathbb{Z}^2$ the base $\alpha(q) = 2^u 3^v$ has $\alpha(q)^y = M^u A^v \in \mathbb{Q}_{>0}$ for every solution y, in particular for $y = x$ and $y = x + 1$. Conversely, suppose α^x and α^{x+1} are rational. Both are algebraic, so the algebraic spectrum gives $\alpha = 2^r 3^s$ with $r, s \in \mathbb{Q}$, and

$$\alpha = \frac{\alpha^{x+1}}{\alpha^x} \in \mathbb{Q}_{>0}.$$

A positive rational of the form $2^r 3^s$ with $r, s \in \mathbb{Q}$ has $r, s \in \mathbb{Z}$, by comparing ordinary 2-adic and 3-adic valuations after clearing a common denominator. □

Corollary 14.5 (Universal rational and integral spectra under failure)**.** *Assume Conjecture 2.1 fails and let $\alpha \in \overline{\mathbb{Q}}_{>0}$.* ***[paper proof in this report]***

1. *α^y is rational for every nonintegral solution y if and only if $\alpha = 2^u 3^v$ with $u, v \in \mathbb{Z}$.*
2. *α^y is a positive integer for every solution y if and only if $\alpha = 2^u 3^v$ with $u, v \in \mathbb{N}_0$.*

Proof. For the first part, apply Proposition 14.4 to a nonintegral solution x and to $x + 1$, which is again a nonintegral solution. Conversely, rational 2, 3-units have rational outputs $M^u A^v$ at every solution.

For the second part, assume $\alpha^y \in \mathbb{N}$ for every $y \in \mathcal{S}$. Taking $y = x$ and $y = x+1$ makes $\alpha = \alpha^{x+1}/\alpha^x$ a positive rational, say a/b in lowest terms. Put $z = \alpha^x \in \mathbb{N}$. Every $x + n$ with $n \geq 0$ is again a solution, so the number $z(a/b)^n = \alpha^{x+n}$ is an integer for every n, whence b^n divides the fixed integer z for every n, forcing $b = 1$ and $\alpha \in \mathbb{N}$. Since α^x is an integer, the algebraic spectrum gives $\alpha \in \Gamma$, and an integer in Γ is exactly $2^u 3^v$ with $u, v \in \mathbb{N}_0$. The converse is immediate. □

Corollary 14.5 says that the rational-output defect is a genuinely per-solution phenomenon. A base can enjoy a rational output at one nonintegral solution only by failing at the next one, unless it is a rational 2, 3-unit already.

14.4 The two smooth branches are mutually exclusive

The number of rows of V_x is another exact invariant of a counterexample, and the smallest conceivable case is excluded outright. Call a positive integer $\{2, 3\}$-smooth (equivalently 3-smooth) when it is divisible by no prime outside $\{2, 3\}$. In the notation above, M is $\{2, 3\}$-smooth exactly when the column $\mathbf{m}$ is supported on $\{2, 3\}$, and both outputs are $\{2, 3\}$-smooth exactly when $P \subseteq \{2, 3\}$, that is, when V_x is a 2×2 integer matrix.

Proposition 14.6 (The smooth cases are mutually exclusive)**.** *For a nonintegral solution x, the outputs $M = 2^x$ and $A = 3^x$ cannot both be $\{2, 3\}$-smooth. Equivalently, the valuation matrix V_x of a counterexample always has a row indexed by a prime $p \geq 5$.* ***[kernel-verified Lean]***

Proof. Suppose $M = 2^a 3^b$ and $A = 2^c 3^e$ with $a, b, c, e \in \mathbb{N}_0$. Taking base-two logarithms of $2^x = M$ and of $3^x = A$ gives the two relations

$$x = a + b\vartheta, \qquad x\vartheta = c + e\vartheta, \qquad \vartheta = \log_2 3.$$

Substituting the first into the second eliminates x and leaves

$$b\,\vartheta^2 + (a-e)\,\vartheta - c = 0.$$

Were any of b, $a-e$, c nonzero, this would exhibit ϑ as a root of a nonzero rational polynomial of degree at most two, contradicting its transcendence. Hence $b = c = 0$ and $a = e$, so $x = a \in \mathbb{Z}$, contrary to hypothesis. ☐

Corollary 14.7 (A smooth output has a rough partner)**.** *Let x be a nonintegral solution with $M = 2^x$ $\{2,3\}$-smooth. Then $A = 3^x$ has a prime divisor $p \geq 5$.* ***[kernel-verified Lean]***

Proof. If every prime divisor of A were 2 or 3, then A would be $\{2,3\}$-smooth, and Proposition 14.6 would force $x \in \mathbb{Z}$. ☐

Status: [kernel-verified Lean]

`integer_of_threeSmooth_outputs` and `exists_prime_five_le_dvd_of_threeSmooth_candidate` in `SmoothOutputs`, with the integral restatement `integer_of_two_three_rpow_integer_of_threeSmooth_outputs`. The proofs use only the transcendence of ϑ, itself kernel-verified in this corpus from Gelfond–Schneider. The same statement is also derived independently at paper level in the research report by the identical degree-drop elimination, so the formalized and unformalized routes confirm each other.

Two remarks fix the position of this exclusion.

The two-sided hypothesis is essential. Assuming only that M is $\{2,3\}$-smooth gives $x = a + b\vartheta$ and then requires $\left(3^{\vartheta}\right)^b$ to be rational up to a power of 3, which is exactly the open localized-radical condition of Section 6. What the second smoothness hypothesis supplies is a *second* linear relation between x and ϑ; two such relations eliminate x and leave an algebraic equation in ϑ alone. This degree drop is the entire mechanism, and it is worth isolating as a template: each independent multiplicative constraint on an output buys one linear relation, and two relations force algebraicity of ϑ. The difficulty of the general problem is precisely that a second constraint is never free.

The exclusion feeds back into the lattice. Because $P \not\subseteq \{2,3\}$, the valuation matrix of a counterexample is never square, so $\delta_2(V_x)$ is a genuine gcd over at least three rows rather than a single determinant, and the rational locus Λ_x is cut out by at least three congruences. In particular the rational-output defect is bounded by the absolute value of any one 2×2 minor of V_x, and any two minors that are coprime force $\delta_2(V_x) = 1$, in which case no auxiliary base outside $\{2^u 3^v : u, v \in \mathbb{Z}\}$ has a rational output at x at all.

15 The second-iterate kernel and multiplicative dependence

The algebraic spectrum of Section 13 controls one application of the exponent x. Asking which bases of Γ *return* to Γ after that application turns second iterates into finite-dimensional linear algebra, and — as this section records — into an entirely elementary statement about four integers.

15.1 The kernel and its invariance

For a nonintegral solution x with outputs $M = 2^x$ and $A = 3^x$, set

$$K_x = \{(r,s) \in \mathbb{Q}^2 \ : \ M^r A^s \in \Gamma\}, \qquad \Gamma = \{2^u 3^v : u, v \in \mathbb{Q}\}. \tag{25}$$

Writing $\mathbf{m}_{\text{ext}} = (v_p(M))_{p\neq 2,3}$ and $\mathbf{a}_{\text{ext}} = (v_p(A))_{p\neq 2,3}$ for the external valuation vectors, membership of $M^r A^s$ in Γ is exactly the vanishing of every prime valuation outside $\{2,3\}$, so

$$K_x = \ker\bigl((r,s) \mapsto r\,\mathbf{m}_{\text{ext}} + s\,\mathbf{a}_{\text{ext}}\bigr). \tag{26}$$

Proposition 15.1 (Kernel dimension)**.** $\dim_{\mathbb{Q}} K_x \le 1$*; the kernel contains no vector with* $rs > 0$*; it equals* $\mathbb{Q}(1,0)$ *exactly when* M *is* 3*-smooth and* $\mathbb{Q}(0,1)$ *exactly when* A *is* 3*-smooth.*

Proof. At least one external vector is nonzero: otherwise both M and A would be 3-smooth, which Theorem 12.5 excludes. Hence the map in (26) has rank at least one and its kernel has dimension at most one. If $r, s > 0$ then any external prime dividing M or A has positive valuation in $M^r A^s$, and if $r, s < 0$ it has negative valuation, so no same-sign vector lies in the kernel. The two smooth cases are immediate from (26) by testing $(1,0)$ and $(0,1)$. □

Status: [kernel-verified Lean] — `SecondIterateKernel` carries the kernel as an integral subgroup `kernelAddSubgroup` of $\mathbb{Z}^2$, with `kernelPairs_mul_nonpos` for the absence of same-sign vectors, `kernelPairs_det_eq_zero` and `kernelPairs_prop_dim` for the rank bound, `mem_kernelPairs_one_zero_iff` for the two smooth cases, and `saturatedKernel_invariant` for $K_x = K_y$. The input that M and A cannot both be 3-smooth is `integer_of_threeSmooth_outputs` in `SmoothOutputs`. Clearing denominators lets the formal statement use integer rather than rational pairs, which is equivalent and lighter.

The kernel is not an artifact of the chosen counterexample. If x, y are two nonintegral solutions and $\alpha = 2^r 3^s$, then $B = \alpha^x$ is algebraic, and applying the spectrum theorem to y gives $\alpha^{xy} = B^y \in \overline{\mathbb{Q}}$ if and only if $B \in \Gamma$, that is, if and only if $(r,s) \in K_x$. The left-hand condition is symmetric in x and y, so $K_x = K_y$: the kernel is an invariant of the problem, not of the solution.

15.2 Multiplicative dependence of four integers

The kernel condition has a formulation with no logarithms, no projective linear group, and no transcendence vocabulary at all. It is the observation of this section.

Theorem 15.2 (Multiplicative form of the kernel dichotomy)**.** *Let* x *be a nonintegral solution with outputs* $M = 2^x$ *and* $A = 3^x$*. The following are equivalent:*

(i) $K_x \neq \{0\}$*;*

(ii) *the four integers* M*,* A*,* 2*,* 3 *are multiplicatively dependent, that is,*

$$M^a A^b 2^c 3^d = 1 \qquad \text{for some } (a,b,c,d) \in \mathbb{Z}^4 \setminus \{0\};$$

(iii) 1*,* ϑ*,* x*,* $x\vartheta$ *are linearly dependent over* $\mathbb{Q}$*;*

(iv) *x lies in the* $\mathrm{PGL}_2(\mathbb{Q})$*-orbit of* ϑ*, that is,* $x = (u + v\vartheta)/(r + s\vartheta)$ *with rational* u, v, r, s *and* $us - vr \neq 0$.

Proof. (i) implies (ii). If $(r, s) \in K_x$ is nonzero then $M^r A^s = 2^u 3^v$ with $u, v \in \mathbb{Q}$; clearing a common denominator of r, s, u, v produces integers with $M^a A^b 2^c 3^d = 1$ and $(a, b) \neq (0, 0)$.

(ii) implies (i). Given such a relation, note first that $(a, b) \neq (0, 0)$: otherwise $2^c 3^d = 1$, and unique factorization forces $c = d = 0$, contradicting nontriviality. Then $M^a A^b = 2^{-c} 3^{-d} \in \Gamma$, so $(a, b) \in K_x \setminus \{0\}$.

(ii) is equivalent to (iii). Taking logarithms of $M^a A^b 2^c 3^d = 1$ and using $\log M = x \log 2$ and $\log A = x \log 3$ gives

$$(ax + c) \log 2 + (bx + d) \log 3 = 0,$$

and division by $\log 2$ turns this into

$$x\,(a + b\vartheta) + (c + d\vartheta) = 0, \tag{27}$$

which is precisely a nontrivial rational linear relation among $1, \vartheta, x, x\vartheta$. Each step reverses.

(iii) is equivalent to (iv). In (27) the coefficient $a + b\vartheta$ cannot vanish, since ϑ is irrational and $(a, b) \neq (0, 0)$; hence $x = -(c + d\vartheta)/(a + b\vartheta)$. If the determinant $ad - bc$ vanished, numerator and denominator would be rationally proportional and x would be rational, hence integral by Lemma 2.3, contrary to hypothesis. Conversely, any such expression cross-multiplies to (27). □

Status: [kernel-verified Lean] — `KernelDichotomy` formalizes the equivalence of (ii), (iii) and (iv), as `multiplicativelyDependentOutputs_iff_secondIterateKernelRelation` together with `exists_moebius_of_multiplicativelyDependentOutputs` and `multiplicativelyDependentOutputs_of_moebius`; the packaged branching is `twoBaseNonintegerSolution_kernel_dichotomy`. Clause (i) is `SecondIterateKernel`. The formalization is slightly stronger than the statement above: (ii) $\iff$ (iii) needs only irrationality of ϑ, and nonintegrality of x enters just for the nonvanishing of the Möbius determinant.

15.3 Why the multiplicative form is worth having

Two consequences follow from stating the dichotomy multiplicatively.

It removes the transcendence vocabulary from the branching. Every counterexample falls into exactly one of two cases, and the case distinction is a question about four ordinary integers: either they satisfy a multiplicative relation or they do not. In the dependent branch Proposition 15.1 says the relation is essentially unique up to scaling, and its exponents on M and A have opposite signs unless one of the outputs is 3-smooth.

It makes Baker's theorem applicable to the independent branch. This is the point that the Diophantine formulations obscure. In the independent branch $M, A, 2, 3$ are multiplicatively independent, so $\log M, \log A, \log 2, \log 3$ are $\mathbb{Q}$-linearly independent *logarithms of algebraic numbers* — all four arguments are positive integers. Baker's homogeneous theorem then gives more than $\mathbb{Q}$-linear independence: these four logarithms are linearly independent over $\overline{\mathbb{Q}}$. The corpus has not previously used this input.

Corollary 15.3 (Algebraic-coefficient rigidity in the independent branch)**.** *Suppose x is a nonintegral solution for which $M, A, 2, 3$ are multiplicatively independent. Then no relation*

$$\gamma_1 \log M + \gamma_2 \log A + \gamma_3 \log 2 + \gamma_4 \log 3 = 0$$

holds with algebraic $\gamma_1, \gamma_2, \gamma_3, \gamma_4$ not all zero.

Status: [classical/open background] — immediate from Baker's homogeneous theorem on $\overline{\mathbb{Q}}$-linear independence of logarithms of algebraic numbers, applied to the four positive integers $M, A, 2, 3$.

It must be said plainly what this does and does not do. It does *not* close the independent branch: the defining relation $\log M \log 3 = \log A \log 2$ has coefficients $\log 3$ and $-\log 2$, which are transcendental by Lindemann, so it is not a $\overline{\mathbb{Q}}$-linear relation and Corollary 15.3 does not apply to it. That is exactly the "Baker is linear, not bilinear" obstruction recorded in Appendix L. What changes is that the independent branch is no longer structureless: it carries a genuine unconditional rigidity statement, and any future argument that manages to produce an algebraic-coefficient relation among these four logarithms — for instance by clearing the transcendental coefficients against a second, independent relation — would close it immediately.

16 Depth-three tower rigidity

A hypothetical counterexample makes the first level of the tower $2 \mapsto 2^x$ integral by assumption. This section answers the exact question that the corpus keeps circling: starting from an arbitrary positive algebraic base, how many times can one apply the same nonintegral exponent x before transcendence is forced? The answer is at most three, the bound is attained, and the depth at which the first transcendental value appears is determined by a single rational-linear invariant.

For positive real bases there is no branch ambiguity between iterated powers and powers of powers, so the tower is unambiguous:

$$((\gamma^x)^x)^x = \gamma^{x^3}.$$

Throughout this section x denotes a hypothetical nonintegral solution, that is $x \in \mathcal{S} \setminus \mathbb{N}_0$, and we write

$$m = 2^x \in \mathbb{N}, \qquad n = 3^x \in \mathbb{N}.$$

The letter γ is reserved for a positive algebraic base, matching the Lean naming in `AlgebraicBaseUnitPower`.

16.1 The saturated spectrum and its strong form

Give $\overline{\mathbb{Q}}_{>0}$ its structure of $\mathbb{Q}$-vector space, with rational scalars acting through the unique positive real root, $q \cdot \gamma = \gamma^q$, and set

$$\Gamma = \{2^r 3^s : r, s \in \mathbb{Q}\} \subset \mathbb{R}_{>0}.$$

This is the divisible hull of the group generated by 2 and 3: a countable, divisible subgroup of rational rank two, dense in $\mathbb{R}_{>0}$ already through $\{2^r : r \in \mathbb{Q}\}$, and isomorphic to $\mathbb{Q}^2$ as a $\mathbb{Q}$-vector space via

$(r, s) \mapsto 2^r 3^s$. An *integer* lies in Γ exactly when it is three-smooth, since clearing denominators in $N = 2^r 3^s$ gives $N^k = 2^p 3^q$ with $p, q \in \mathbb{Z}$ and unique factorization then confines the primes of N to $\{2, 3\}$.

Theorem 16.1 (Exact algebraic-base spectrum)**.** *Let $x \in \mathcal{S} \setminus \mathbb{N}_0$ and let $\gamma \in \overline{\mathbb{Q}}_{>0}$. Then*

$$\gamma^x \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad \gamma \in \Gamma.$$

In particular the algebraic spectrum of a counterexample does not depend on which nonintegral solution is used.

Status: [kernel-verified Lean] — in `AlgebraicBaseUnitPower`, the pair

`TwoBaseNonintegerSolution.real_rpow_isAlgebraic_iff_hasPositiveTwoThreeUnitPower`

and

`TwoBaseNonintegerSolution.transcendental_real_rpow_of_no_positiveTwoThreeUnitPower`

The Lean predicate is `HasPositiveTwoThreeUnitPower`: some positive natural power of γ is a positive rational $2, 3$-unit. For positive algebraic γ this is exactly membership in Γ — one direction raises $2^r 3^s$ to a common denominator, the other takes the unique positive real root — so the kernel-verified biconditional is the displayed spectrum verbatim.

The spectrum is already a strong statement: two algebraic outputs exhaust the maximal algebraic multiplicative rank that the two-base configuration can support, and every positive algebraic base outside one explicit dense subgroup produces a transcendental value. For the tower we need the sharper conclusion that the offending *logarithm* is not merely irrational but escapes the algebraic span of all logarithms.

Let $\mathcal{L}$ denote the $\overline{\mathbb{Q}}$-vector space spanned by 1 and by all logarithms of nonzero algebraic numbers, arbitrary complex branches allowed; for a positive real algebraic number the real logarithm is one of those branches.

Definition 16.2. A positive real value e^z is *log-strongly transcendental* when $z \notin \mathcal{L}$. This implies ordinary transcendence, because the logarithm of a positive algebraic number lies in $\mathcal{L}$ by definition.

Theorem 16.3 (Strong algebraic-base spectrum)**.** *Let $x \in \mathcal{S} \setminus \mathbb{N}_0$ and let $\gamma \in \overline{\mathbb{Q}}_{>0} \setminus \Gamma$. Then*

$$x \log \gamma \notin \mathcal{L},$$

so γ^x is log-strongly transcendental. The same conclusion holds with $2, 3$ replaced by any multiplicatively independent pair $a, b \in \overline{\mathbb{Q}}_{>0}$ with $a^x, b^x \in \overline{\mathbb{Q}}$, and Γ replaced by $\Gamma(a, b) = \{a^r b^s : r, s \in \mathbb{Q}\}$.

Proof. The logarithms $\log 2, \log 3, \log \gamma$ are linearly independent over $\mathbb{Q}$: a rational relation clears to an integral multiplicative relation, which either contradicts the multiplicative independence of 2 and 3 or places γ in Γ. Baker's homogeneous theorem [3] upgrades this to linear independence over $\overline{\mathbb{Q}}$.

The exponent x is transcendental: it is irrational by the kernel-verified rational-exponent lemma of Section 4, and an algebraic irrational x would make 2^x transcendental by Gelfond–Schneider [9]. Hence 1 and x are linearly independent over $\overline{\mathbb{Q}}$.

Apply Roy's strong six exponentials theorem [8, 11] to the three rows $\log 2, \log 3, \log\gamma$ and the two columns $1, x$. Five of the six pairwise products lie in $\mathcal{L}$, namely

$$\log 2, \quad \log 3, \quad \log\gamma, \quad x\log 2 = \log m, \quad x\log 3 = \log n.$$

Therefore the sixth product $x\log\gamma$ lies outside $\mathcal{L}$. □

Status: [paper proof in this report] — paper proof in this report. It rests on Baker's theorem and on Roy's strong form of the six exponentials theorem, neither of which is formalized in the repository; the repository's own interpolation determinant is specialized to integer entries. Theorem 16.1, the ordinary-transcendence shadow of this statement, is **[kernel-verified Lean]**.

16.2 The return kernel

The spectrum controls one application of x. The tower needs to know which bases *inside* Γ come back to Γ after that application. On prime valuations this is finite-dimensional linear algebra.

Proposition 16.4 (The two outputs are not both three-smooth)**.** *For $x \in \mathcal{S} \setminus \mathbb{N}_0$, at most one of $m = 2^x$ and $n = 3^x$ is three-smooth.*

Proof. Suppose $m = 2^a 3^b$ and $n = 2^c 3^d$ with $a, b, c, d \in \mathbb{N}_0$. Taking logarithms in base 2 and in base 3 gives $x = a + b\vartheta$ and $x = d + c/\vartheta$; eliminating x and clearing the denominator yields

$$b\,\vartheta^2 + (a-d)\,\vartheta - c = 0. \tag{28}$$

The kernel-verified transcendence of ϑ forces $b = c = 0$ and $a = d$, hence $x = a \in \mathbb{N}_0$, contrary to hypothesis. □

Status: [paper proof in this report] — the only nonelementary input is the kernel-verified transcendence of ϑ,

`transcendental_logThreeDivLogTwo`

in `Transcendence`. This is the same computation as the three-smooth output test of Section 9, restated here in the form the kernel below consumes.

For $x \in \mathcal{S} \setminus \mathbb{N}_0$ define the *return kernel*

$$K_x = \{\gamma \in \Gamma : \gamma^x \in \Gamma\} = \{2^r 3^s : (r, s) \in \mathbb{Q}^2,\ m^r n^s \in \Gamma\}. \tag{29}$$

Under the coordinate isomorphism $\Gamma \cong \mathbb{Q}^2$ this is a $\mathbb{Q}$-subspace, because $(2^r 3^s)^x = m^r n^s$ is multiplicative in the coordinates. Write

$$\mathbf{m}_{\text{ext}} = \big(v_p(m)\big)_{p \neq 2,3}, \qquad \mathbf{n}_{\text{ext}} = \big(v_p(n)\big)_{p \neq 2,3},$$

two finitely supported nonnegative integer vectors.

Proposition 16.5 (Kernel formula and dimension)**.** *In the coordinates* (r, s)*,*

$$K_x = \ker\Big(\mathbb{Q}^2 \longrightarrow \mathbb{Q}^{(\mathbb{P}\setminus\{2,3\})}, \quad (r, s) \longmapsto r\,\mathbf{m}_{\mathrm{ext}} + s\,\mathbf{n}_{\mathrm{ext}}\Big). \tag{30}$$

Moreover:

(i) $\dim_{\mathbb{Q}} K_x \le 1$*;*

(ii) K_x *contains no* $2^r 3^s$ *with* $rs > 0$*;*

(iii) $K_x = \{2^r : r \in \mathbb{Q}\}$ *exactly when* m *is three-smooth;*

(iv) $K_x = \{3^s : s \in \mathbb{Q}\}$ *exactly when* n *is three-smooth.*

Proof. Membership of $m^r n^s$ in Γ is precisely the vanishing of every prime valuation outside $\{2, 3\}$, which is (30). By Proposition 16.4 at least one of the two external vectors is nonzero, so the displayed linear map has rank at least one and its kernel has dimension at most one.

If $r, s > 0$, then any external prime p occurring in m or in n has $rv_p(m) + sv_p(n) > 0$, since both valuations are nonnegative and at least one is positive; the case $r, s < 0$ is symmetric. Hence no same-sign direction survives.

Finally, if m is three-smooth then $\mathbf{m}_{\mathrm{ext}} = 0$ while $\mathbf{n}_{\mathrm{ext}} \ne 0$, so the kernel is exactly the first coordinate axis; the converse follows by testing the direction $(1, 0)$, which lies in K_x exactly when $m \in \Gamma$, that is, exactly when the integer m is three-smooth. The statement for n is the mirror image. □

Status: [paper proof in this report] — elementary given Proposition 16.4. No transcendence input beyond the transcendence of ϑ enters the proof; the six exponentials theorem is used only to interpret K_x, not to compute it.

The dynamical content of the kernel is that it admits no nontrivial internal orbit of length two.

Lemma 16.6 (No two-step internal orbit)**.** *Let* $x \in \mathcal{S} \setminus \mathbb{N}_0$ *and let* $\gamma \in K_x \setminus \{1\}$*. Then* $\gamma^x \notin K_x$*; equivalently,*

$$\gamma \in \Gamma \text{ and } \gamma^x \in \Gamma \quad \Longrightarrow \quad \gamma^{x^2} \notin \Gamma.$$

Proof. Suppose γ and $\gamma_1 = \gamma^x$ both lie in $K_x \setminus \{1\}$. By Proposition 16.5(i) the space K_x is at most one-dimensional over $\mathbb{Q}$, so two of its nonzero elements are rational scalar multiples of one another in the additive notation: $\gamma_1 = \gamma^q$ for some $q \in \mathbb{Q}$. Since also $\gamma_1 = \gamma^x$ and $\gamma \ne 1$, comparing logarithms gives $x = q$, contradicting the irrationality of x. Hence $\gamma_1 \notin K_x$, which by (29) says exactly that $\gamma_1^x = \gamma^{x^2} \notin \Gamma$. □

There are therefore no cycles, no fixed rays, and no length-two paths lying entirely inside Γ: a rational direction may survive one return to the algebraic spectrum, but the returned direction cannot itself survive.

16.3 Exact survival depth

Theorem 16.7 (Depth-three algebraic-tower theorem)**.** *Let $x \in \mathcal{S} \setminus \mathbb{N}_0$ and let $\gamma \in \overline{\mathbb{Q}}_{>0} \setminus \{1\}$. Then at least one of*

$$\gamma^x, \qquad \gamma^{x^2}, \qquad \gamma^{x^3}$$

is transcendental. More precisely, the first transcendental term occurs at exactly one of the following three depths.

(1) ***Depth one.*** *If $\gamma \notin \Gamma$, then γ^x is log-strongly transcendental.*

(2) ***Depth two.*** *If $\gamma \in \Gamma \setminus K_x$, then $\gamma^x \in \overline{\mathbb{Q}} \setminus \Gamma$ and γ^{x^2} is log-strongly transcendental.*

(3) ***Depth three.*** *If $\gamma \in K_x \setminus \{1\}$, then $\gamma^x \in \Gamma$, $\gamma^{x^2} \in \overline{\mathbb{Q}} \setminus \Gamma$, and γ^{x^3} is log-strongly transcendental.*

Equivalently, for some $j \in \{1, 2, 3\}$ one has $x^j \log \gamma \notin \mathcal{L}$.

Proof. Depth one is Theorem 16.3.

Assume $\gamma \in \Gamma$, so that $\gamma_1 = \gamma^x$ is algebraic by Theorem 16.1. If $\gamma \notin K_x$, then $\gamma_1 \notin \Gamma$ by (29), and γ_1 is a positive algebraic base outside Γ; the strong spectrum applied to γ_1 and the same exponent x makes $\gamma_1^x = \gamma^{x^2}$ log-strongly transcendental.

If instead $\gamma \in K_x \setminus \{1\}$, then $\gamma_1 \in \Gamma$, so $\gamma_2 = \gamma_1^x = \gamma^{x^2}$ is algebraic. By Lemma 16.6 we have $\gamma_1 \notin K_x$, hence $\gamma_2 \notin \Gamma$. One last application of the strong spectrum, now to the base γ_2, gives $\gamma_2^x = \gamma^{x^3}$ log-strongly transcendental.

The three cases are mutually exclusive and exhaustive, and in each case all earlier terms of the tower are algebraic, so the stated depth is exact rather than an upper bound. ☐

The classification is compact enough to tabulate. For a nonintegral solution x and a positive algebraic base $\gamma \neq 1$, the first level of the tower whose logarithm escapes $\mathcal{L}$ is:

Location of γ	First log outside $\mathcal{L}$	Structural reason
$\gamma \notin \Gamma$	$x \log \gamma$	exact algebraic spectrum
$\gamma \in \Gamma \setminus K_x$	$x^2 \log \gamma$	the first image leaves Γ
$\gamma \in K_x \setminus \{1\}$	$x^3 \log \gamma$	no two-step internal orbit

Status: [paper proof in this report] — paper proof in this report, resting on Theorem 16.3 and hence on Baker's theorem and Roy's strong six exponentials theorem. The ordinary-transcendence version of depth one is **[kernel-verified Lean]**; the ordinary-transcendence version of depth two for the mixed integer bases $2^i 3^j$ with $i, j > 0$ is also **[kernel-verified Lean]**, as recorded below.

The theorem places a strict ceiling on how far an algebraic power tower can imitate the integral first level of a counterexample. A counterexample buys two algebraic outputs; it never buys a third independent one, and the tower makes the deficit visible after at most three steps regardless of the base chosen.

16.4 Exact behaviour of the bases two, three, and six

Corollary 16.8 (Base-two tower dichotomy). *Let $x \in \mathcal{S} \setminus \mathbb{N}_0$ and $m = 2^x$.*

(a) *If m is not three-smooth, then 2^{x^2} is log-strongly transcendental.*

(b) *If m is three-smooth, say $m = 2^a 3^b$ with $a, b \in \mathbb{N}_0$, then*

$$2^{x^2} = m^x = (2^x)^a (3^x)^b = m^a n^b \in \mathbb{N}, \tag{31}$$

and 2^{x^3} is log-strongly transcendental.

The mirror statement holds with 2 replaced by 3 and m by n.

Proof. The base 2 lies in Γ, and its coordinate vector is $(1, 0)$, which lies in K_x exactly when $m \in \Gamma$, that is, by Proposition 16.5(iii), exactly when the integer m is three-smooth. Now apply Theorem 16.7: the non-smooth case is depth two and the smooth case is depth three. In the smooth case the intermediate value is computed by (31), where the middle equality is the identity $(2^a 3^b)^x = (2^x)^a (3^x)^b$ for positive real bases and the last membership holds because $m, n \in \mathbb{N}$. $\square$

Equation (31) is worth pausing on. In the smooth branch the second level of the tower is not merely algebraic but an explicit positive integer, expressed in the two original outputs with the same exponents that describe the first output. The tower does not become transcendental because the arithmetic runs out immediately; it becomes transcendental because the third level has nowhere left to land.

Corollary 16.9 (Base six always fails at depth two). *For every $x \in \mathcal{S} \setminus \mathbb{N}_0$ and all $r, s \in \mathbb{Q}$ of the same nonzero sign,*

$$x^2 (r \log 2 + s \log 3) \notin \mathcal{L}.$$

In particular $x^2 \log 6 \notin \mathcal{L}$, so 6^{x^2} is log-strongly transcendental.

Proof. The base $\gamma = 2^r 3^s$ lies in Γ, and its coordinate vector (r, s) has positive coordinate product, so $\gamma \notin K_x$ by Proposition 16.5(ii). Its exact survival depth is therefore two, by Theorem 16.7(2). The case $r = s = 1$ is $\gamma = 6$. $\square$

This recovers and strengthens a kernel-verified family. The Lean theorem

```
TwoBaseNonintegerSolution.transcendental_mixed_two_three_squared_exponents
```

in `AlgebraicOutputLocus` proves that $\left(2^i 3^j\right)^{x^2}$ is transcendental for all natural $i, j > 0$, which is exactly the statement that no genuinely mixed three-smooth integer base survives to the second level. Corollary 16.9 extends that to all rational same-sign directions, including negative ones, and upgrades the conclusion from transcendence to escape from $\mathcal{L}$; more importantly it explains *why* the mixed bases fail, namely that the exceptional directions of K_x are confined to the two coordinate axes, which the mixed monomials avoid by construction. The kernel-verified case is the positive-integer corner of a statement that is really about a line in $\mathbb{Q}^2$.

Status: [paper proof in this report] for Corollaries 16.8 and 16.9 as stated. [kernel-verified Lean] for the ordinary transcendence of $(2^i 3^j)^{x^2}$ with integer $i, j > 0$, in `AlgebraicOutputLocus`. The gap between the two is exactly the general six exponentials input discussed in Section 9.

16.5 One-base tower equivalences

The depth theorem converts into a reformulation of the conjecture that is striking in its indifference to the base.

Universal three-level detector (Theorem 16.10). Fix *any* positive algebraic $\gamma \neq 1$ — $\gamma = 5$, $\gamma = \sqrt{2}$, $\gamma = 2$, $\gamma = 3$, $\gamma = \frac{1}{2}(7 + 3\sqrt{5})$, anything at all. Conjecture 2.1 is equivalent to

$$\forall x \in \mathbb{R}, \quad 2^x, 3^x \in \mathbb{Z} \implies \gamma^x,\ \gamma^{x^2},\ \gamma^{x^3} \in \overline{\mathbb{Q}},$$

and equally to the assertion that all three numbers $x \log \gamma$, $x^2 \log \gamma$, $x^3 \log \gamma$ lie in $\mathcal{L}$. No arithmetic feature of the base is used: three levels of the tower detect a counterexample for every positive algebraic base simultaneously. [paper proof in this report]

Theorem 16.10 (Universal three-level detector)**.** *Fix $\gamma \in \overline{\mathbb{Q}}_{>0} \setminus \{1\}$. Then Conjecture 2.1 is equivalent to*

$$\forall x \in \mathbb{R}, \quad 2^x, 3^x \in \mathbb{Z} \implies \gamma^x, \gamma^{x^2}, \gamma^{x^3} \in \overline{\mathbb{Q}}, \tag{32}$$

and also to the same implication with $\overline{\mathbb{Q}}$-algebraicity replaced by the three membership conditions $x^j \log \gamma \in \mathcal{L}$, $j = 1, 2, 3$.

Proof. If the conjecture holds, then $\mathcal{S} = \mathbb{N}_0$, every exponent x, x^2, x^3 appearing in (32) is a nonnegative integer, and all three powers of the algebraic number γ are algebraic, with logarithms in $\mathcal{L}$. If the conjecture fails, apply Theorem 16.7 to a nonintegral solution: one of the three values is log-strongly transcendental, so (32) fails and so does its $\mathcal{L}$-version. □

Bases already inside Γ have their first level algebraic for free, which removes one of the three conditions and yields the shortest form of the equivalence.

Corollary 16.11 (Base-two cubic-tower equivalence)**.** *Conjecture 2.1 is equivalent to*

$$\forall x \in \mathbb{R}, \quad 2^x, 3^x \in \mathbb{Z} \implies 2^{x^2} \in \overline{\mathbb{Q}} \ \textit{ and } \ 2^{x^3} \in \overline{\mathbb{Q}}. \tag{33}$$

The same statement holds with the base 3 *in place of the base* 2.

Proof. Take $\gamma = 2$ in Theorem 16.10 and drop the first condition, which is automatic because 2^x is an integer under the hypothesis. □

This is a genuinely different equivalence from the second-level algebraicity tests of Section 9, which require *both* 2^{x^2} and 3^{x^2} to be algebraic. One original base suffices if one is willing to climb a single extra level. The trade is informative: a second base and a third tower level carry exactly the same amount of information, which is another way of seeing that the two-base configuration supplies only two independent logarithmic directions no matter how the exponent is iterated.

Status: [paper proof in this report] — paper proof in this report. Both equivalences inherit the classical inputs of Theorem 16.3; the ordinary-algebraicity forms (32) and (33) need only the six exponentials theorem, since Theorem 16.1 is **[kernel-verified Lean]** and only the depth-two and depth-three steps require the general form. Neither is a proof of Conjecture 2.1, which remains open: the tower reformulations relocate the difficulty to the second and third levels without supplying a mechanism that produces algebraicity there.

17 Symmetric tower constants

The exponent $\vartheta = \log_2 3$ and its reciprocal

$$\eta = \frac{1}{\vartheta} = \frac{\log 2}{\log 3}$$

generate two symmetric "next outputs",

$$E_3 = 3^\vartheta = 5.704522494691117635\ldots, \qquad E_2 = 2^\eta = 1.548562652630242907\ldots$$

The first is the canonical radical of Section 6 in the special branch where the primitive odd core equals 3; the second is its base-swapped mirror. Neither constant is known to be transcendental, and for E_3 even irrationality is open. What is available unconditionally is a statement about the *pair*, and what a counterexample would add is an exact classification of which of the two becomes algebraic.

17.1 An unconditional strong dichotomy

Theorem 17.1 (At least one symmetric tower constant is strongly transcendental)**.** *At least one of the two numbers*

$$\vartheta \log 3 = \log E_3, \qquad \eta \log 2 = \log E_2$$

lies outside $\mathcal{L}$. In particular, at least one of E_3 and E_2 is transcendental.

Proof. The three numbers $1, \vartheta, \eta$ are linearly independent over $\overline{\mathbb{Q}}$: a relation $a + b\vartheta + c\eta = 0$ with algebraic coefficients, multiplied by ϑ, becomes $b\vartheta^2 + a\vartheta + c = 0$, which is impossible for a transcendental ϑ unless $a = b = c = 0$. The two numbers $\log 2, \log 3$ are linearly independent over $\mathbb{Q}$ by unique factorization, hence over $\overline{\mathbb{Q}}$ by Baker's theorem [3].

Apply Roy's strong six exponentials theorem [8, 11] to the three rows $1, \vartheta, \eta$ and the two columns $\log 2, \log 3$. Four of the six pairwise products are already logarithms of algebraic numbers,

$$\log 2, \qquad \log 3, \qquad \vartheta \log 2 = \log 3, \qquad \eta \log 3 = \log 2,$$

and therefore lie in $\mathcal{L}$. Hence at least one of the two remaining products, $\vartheta \log 3$ and $\eta \log 2$, lies outside $\mathcal{L}$. □

Status: [paper proof in this report] — paper proof in this report, using Baker's theorem and Roy's strong six exponentials theorem. The ordinary six exponentials theorem gives the weaker dichotomy that one of E_3, E_2 is transcendental. The strong form is worth the extra input because it survives verbatim into the conditional branches below.

Note what the theorem does not say. It does not identify which constant is transcendental, and no known method identifies either one individually; that is precisely the wall described in Section 6. The value of the dichotomy is that a counterexample would resolve it, and would resolve it in a completely explicit way.

17.2 Exact conditional classification

Throughout this subsection $x \in \mathcal{S} \setminus \mathbb{N}_0$, $m = 2^x$, and $n = 3^x$.

Theorem 17.2 (The E_3 equivalences)**.** *The following are equivalent.*

(i) $E_3 = 3^\vartheta$ *is algebraic;*

(ii) $\vartheta \log 3 \in \mathcal{L}$*;*

(iii) m *is three-smooth;*

(iv) *there are* $a, b \in \mathbb{N}_0$ *with* $b > 0$ *such that* $m = 2^a 3^b$ *and* $x = a + b\vartheta$*;*

(v) $x \in \mathbb{Q} + \mathbb{Q}\vartheta$*.*

If these conditions fail, then $\vartheta \log 3 \notin \mathcal{L}$ *and* E_3 *is log-strongly transcendental. If they hold, then*

$$E_3^b = \frac{n}{3^a} \in \mathbb{Q}_{>0}, \tag{34}$$

so E_3 *is an explicit algebraic radical.*

Proof. (iv)⇒(iii)⇒(v): the first is trivial, and taking logarithms of $m = 2^a 3^b$ gives $x = a + b\vartheta$, which lies in $\mathbb{Q} + \mathbb{Q}\vartheta$.

(v)⇒(iv): write $x = u + v\vartheta$ with $u, v \in \mathbb{Q}$ and clear a common positive denominator d, so that $m^d = 2^p 3^q$ with $p, q \in \mathbb{Z}$. The left side is a positive integer, so prime valuations force $p, q \geq 0$ and $d \mid p$, $d \mid q$. Hence $u = a$ and $v = b$ with $a, b \in \mathbb{N}_0$, and $b > 0$ because $x \notin \mathbb{N}_0$.

(i)⇒(iv): apply the six exponentials theorem to the rows $1, x, \vartheta$ and the columns $\log 2, \log 3$. The six exponential values are

$$2, \quad 3, \quad m, \quad n, \quad 3, \quad E_3,$$

all algebraic by hypothesis. The theorem therefore forbids both independence conditions simultaneously; the columns $\log 2, \log 3$ are $\mathbb{Q}$-independent, so the rows are $\mathbb{Q}$-dependent and $x = u + v\vartheta$ for some $u, v \in \mathbb{Q}$. Now (v)⇒(iv) applies.

(iv)⇒(i): from $x = a + b\vartheta$ we get $n = 3^x = 3^a E_3^b$, which is (34) and exhibits E_3 as a positive real radical of a positive rational, hence algebraic.

Finally the strong statement. Suppose m is not three-smooth. Then $\log 2, \log m, \log 3$ are linearly independent over $\mathbb{Q}$: a rational relation with nonzero coefficient on $\log m$ would place m in Γ, and an integer in Γ is three-smooth, while a relation with zero coefficient on $\log m$ contradicts the independence of $\log 2, \log 3$. Baker's theorem upgrades this to independence over $\overline{\mathbb{Q}}$, so after division by $\log 2$ the three numbers $1, x, \vartheta$ are $\overline{\mathbb{Q}}$-linearly independent. Apply Roy's theorem to the rows $1, x, \vartheta$ and the columns $\log 2, \log 3$: five of the six products, namely $\log 2, \log 3, \log m, \log n$, and $\vartheta \log 2 = \log 3$, lie in $\mathcal{L}$, so $\vartheta \log 3 \notin \mathcal{L}$. Thus failure of (iii) implies failure of (ii); and (i) implies (ii) because $\log E_3 = \vartheta \log 3$ is then the logarithm of an algebraic number. ∎

Theorem 17.3 (The E_2 equivalences)**.** *The following are equivalent.*

(i) $E_2 = 2^\eta$ *is algebraic;*

(ii) $\eta \log 2 \in \mathcal{L}$*;*

(iii) n *is three-smooth;*

(iv) *there are* $c, d \in \mathbb{N}_0$ *with* $c > 0$ *such that* $n = 2^c 3^d$ *and* $x = d + c\eta$*;*

(v) $x \in \mathbb{Q} + \mathbb{Q}\eta$*.*

If these conditions fail, then $\eta \log 2 \notin \mathcal{L}$ *and* E_2 *is log-strongly transcendental. If they hold, then*

$$E_2^c = \frac{m}{2^d} \in \mathbb{Q}_{>0}. \tag{35}$$

Proof. Interchange the roles of 2 and 3, of m and n, and of ϑ and η in the proof of Theorem 17.2. For the strong step, use the $\mathbb{Q}$-linear independence of $\log 3, \log n, \log 2$ when n is not three-smooth. ☐

Corollary 17.4 (The two branches are mutually exclusive)**.** *For a nonintegral solution,* E_3 *and* E_2 *cannot both be algebraic. Equivalently, at least one of* $\vartheta \log 3$ *and* $\eta \log 2$ *lies outside* $\mathcal{L}$ *under a counterexample as well as unconditionally.*

Proof. By Theorems 17.2 and 17.3, algebraicity of E_3 is equivalent to three-smoothness of m and algebraicity of E_2 to three-smoothness of n. Proposition 16.4 forbids both, since eliminating x from $x = a + b\vartheta$ and $x = d + c\eta$ produces the quadratic (28) for the transcendental ϑ. ☐

So under a counterexample the unconditional dichotomy of Theorem 17.1 becomes a genuine classification rather than an alternative: at most one of the two constants is algebraic, and each is algebraic exactly in its own smooth-output branch, with an explicit radical description.

Smooth branch	Return kernel	Symmetric constants
$m = 2^a 3^b$, $b > 0$	$\{2^r : r \in \mathbb{Q}\}$	$E_3^b = n/3^a \in \mathbb{Q}_{>0}$; E_2 log-strongly transcendental
$n = 2^c 3^d$, $c > 0$	$\{3^s : s \in \mathbb{Q}\}$	$E_2^c = m/2^d \in \mathbb{Q}_{>0}$; E_3 log-strongly transcendental
neither output smooth	no coordinate axis	both E_3 and E_2 log-strongly transcendental

The three rows of the table are exactly the three surviving rows of the primitive-core trichotomy of Section 9: the first is the case $w = 3$, in which the canonical radical $E = w^\vartheta$ of Section 6 degenerates to E_3 itself with index $d = b$; the second is its mirror $v = 2$; and the third is the generic case with two independent external prime supports. The missing fourth pattern, both outputs three-smooth, is the one excluded by Proposition 16.4.

Two remarks are worth making about the direction of the implications. First, nothing here proves that E_3 or E_2 is transcendental: a counterexample would make one of them an explicit algebraic radical, and Corollary 17.4 only says that it cannot make both algebraic. The conjecture remains open, and the smallest instance — irrationality of $E_3 = 3^{\log_2 3}$ — is exactly the wall named in Section 6. Second, the implication that is usable in the other direction is the strong one: proving

$\vartheta \log 3 \notin \mathcal{L}$ would kill the entire $w = 3$ branch, and proving $\eta \log 2 \notin \mathcal{L}$ would kill the $v = 2$ branch, leaving only the generic case, in which both outputs carry prime support outside $\{2, 3\}$ and the local determinant methods have two independent external primes to work with.

Status: [paper proof in this report] — paper proofs in this report. Theorems 17.2 and 17.3 use the six exponentials theorem for the algebraicity equivalences and Baker's theorem together with Roy's strong six exponentials theorem for the $\mathcal{L}$-statements; none of those classical inputs is formalized in the repository. Corollary 17.4 adds only Proposition 16.4, whose sole nonelementary ingredient is the kernel-verified transcendence of ϑ, so the mutual exclusion of the two smooth branches — as opposed to the transcendence conclusions attached to them — is within immediate reach of the Lean development.

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18 Finite-difference obstructions

The first local obstruction recorded in the corpus forbids three candidates in arithmetic progression under the convenient hypothesis $h^5 \le n$. That hypothesis is an artifact of the estimate, not of the mechanism: the sharp arbitrary-order curvature theorem is now kernel-verified for every $r \ge 2$, and it contains the three- and four-term cases as specializations.

For a function f and a step $h > 0$ write $\Delta_h f(t) = f(t+h) - f(t)$ and $\Delta_h^r = \Delta_h \circ \cdots \circ \Delta_h$ (r factors). Iterating the fundamental theorem of calculus gives the mean-value identity in integral form,

$$\Delta_h^r f(n) = \int_{[0,h]^r} f^{(r)}(n + t_1 + \cdots + t_r)\, dt_1 \cdots dt_r. \tag{36}$$

For $f(t) = t^\vartheta$ and $r \ge 2$ the derivative is

$$f^{(r)}(t) = (\vartheta)_{\underline{r}}\, t^{\vartheta - r}, \qquad (\vartheta)_{\underline{r}} := \prod_{j=0}^{r-1} (\vartheta - j) = \vartheta(\vartheta - 1)\cdots(\vartheta - r + 1).$$

Since $1 < \vartheta < 2$ the falling factorial never vanishes, so $f^{(r)}$ has a fixed nonzero sign on $(0, \infty)$, and $|f^{(r)}|$ decreases in t. These two facts — nonvanishing and decay — are the whole content of the obstruction.

Theorem 18.1 (Higher-difference progression obstruction)**.** *Let $r \ge 2$, let $n, h \in \mathbb{N}$ be positive, and set $c_r = |\,(\vartheta)_{\underline{r}}\,|$. If all of $n, n+h, \ldots, n+rh$ are candidates, then*

$$1 \le c_r\, h^r n^{\vartheta - r}.$$

Such a progression is therefore impossible whenever

$$c_r\, h^r < n^{\,r - \vartheta}. \tag{37}$$

Proof. If all $n + jh$ are candidates, every $(n + jh)^\vartheta$ is an integer, so the integer combination $\Delta_h^r f(n) = \sum_{j=0}^{r} (-1)^{r-j} \binom{r}{j} f(n + jh)$ is an integer. By (36) and the fixed sign of $f^{(r)}$ it is nonzero, hence of absolute value at least one. Bounding $|f^{(r)}|$ by its value at the left endpoint n,

$$1 \le |\Delta_h^r f(n)| \le c_r\, h^r n^{\vartheta - r}.$$ □

Status: [kernel-verified Lean] — `HigherDifference` proves the generic iterated mean-value identity `exists_iterated_fwdDiff_eq_pow_mul`, its real-power specialization `exists_iterated_fwdDiff_rpow_point`, integrality of the binomial forward difference, and the obstruction `not_arithmetic_progression_twoBaseNaturalCandidates_of_curvature` uniformly for every $r \geq 2$. The displayed iterated-integral identity (36) is not used by the Lean proof; the formal argument iterates the mean-value theorem instead, which avoids any integration theory.

18.1 The sharp three-term criterion

For $r = 2$ the criterion (37) reads

$$\vartheta(\vartheta - 1)h^2 < n^{2-\vartheta}, \qquad \text{i.e.} \qquad h < 1.038547802\ldots \cdot n^{0.2075187\ldots}, \tag{38}$$

strictly weaker as a hypothesis than $h \leq n^{1/5}$, whose exponent is only 0.2. The real form is `second_difference_logThreeDivLogTwo_ap_of_curvature` in `SharperProgressions`, built on the curvature estimate `second_difference_logThreeDivLogTwo_ap_bound`.

Almost all of the gain survives in a purely rational hypothesis, which is what a kernel can check without any real-number decision procedure. The step exponent $83/400 = 0.2075$ is admissible:

Theorem 18.2 (Sharp rational three-term criterion). *Let $n, h \geq 1$ be integers with $h^{400} \leq n^{83}$. Then n, $n + h$, $n + 2h$ are not all candidates.*

Status: [kernel-verified Lean] — `SharpProgression` proves `second_difference_logThreeDivLogTwo_ap_sharp` and `not_three_arithmetic_progression_twoBaseNaturalCandidates_sharp`. The two numerical ingredients are the enclosure $\vartheta < 317/200$, replayed by the kernel from the exact power inequality $3^{200} < 2^{317}$ through `logThreeDivLogTwo_lt_317_div_200`, and the consequence $\vartheta(\vartheta - 1) < 37089/40000 < 1$. For $r = 3$, `ThirdDifference` kernel-checks the exact third-difference identity and the corresponding four-candidate obstruction.

18.2 Thresholds for higher order

Writing (37) as a bound on the step, a progression of length $r + 1$ starting at n is impossible once

$$h < C_r\, n^{1-\vartheta/r}, \qquad C_r = c_r^{-1/r}.$$

Table 2 lists the first thresholds.

18.3 Why longer progressions do not close the problem

As r grows the admissible step exponent $1 - \vartheta/r$ approaches one, which looks tantalizing: for large r the theorem forbids progressions with steps almost as large as the starting point itself. It does not contradict the exact candidate monoid, and it cannot be bootstrapped into a proof, for a purely combinatorial reason.

r	length $r+1$	exponent $1-\vartheta/r$	C_r
2	3	0.207518750	1.038547802
3	4	0.471679166	1.374849673
4	5	0.603759375	1.164124485
5	6	0.683007500	0.946705202
6	7	0.735839583	0.778537835
7	8	0.773576786	0.652646038
8	9	0.801879687	0.557368999

Table 2: Step thresholds for the higher-difference obstruction. A progression of length $r+1$ with step $h < C_r n^{1-\vartheta/r}$ cannot consist of candidates.

Observation 18.3 (Combinatorial limitation)**.** Conditionally on the structure described in Section 5, a dyadic block near X contains $O(\log X)$ candidates, so the average additive gap inside the block is of order $X/\log X$. For each fixed r this exceeds $X^{1-\vartheta/r}$ once X is large. A rank-two multiplicative monoid carries no reason whatsoever to contain long *additive* progressions; the hypothesis of Theorem 18.1 is therefore vacuous for the configurations that actually occur.

Extracting leverage from long progressions would require r growing with the height, control of the rapidly growing constant $c_r = |(\vartheta)_{\underline{r}}|$, and — decisively — a combinatorial source guaranteeing such progressions exist. None of the three is available. The productive response is to drop the additive-structure hypothesis altogether, which is what Section 19 does.

18.4 The optimized variable-order scale

Theorem 18.1 is uniform in r, so nothing forces the difference order to stay fixed while the starting point grows. Table 2 records the first fixed-order thresholds, but each row is only a snapshot of a one-parameter family that may be re-optimized at every height. Carrying out that optimization strengthens every fixed-order corollary at once and, more valuably, replaces the vague complaint that the candidate density "feels too small" by a single explicit constant. Throughout this subsection log denotes the natural logarithm.

The exact derivative coefficient. Fix $r \geq 2$ and recall the constant $c_r = |(\vartheta)_{\underline{r}}|$ of Theorem 18.1. Since $1 < \vartheta < 2$, the first two factors ϑ and $\vartheta - 1$ are positive while every later factor $\vartheta - j$ with $j \geq 2$ is negative, so taking absolute values termwise gives

$$c_r = \vartheta(\vartheta-1)\prod_{j=2}^{r-1}(j-\vartheta),$$

and the recursion $\Gamma(z+1) = z\Gamma(z)$ telescopes the product into closed form:

$$c_r = \frac{\vartheta(\vartheta-1)}{\Gamma(2-\vartheta)}\,\Gamma(r-\vartheta). \tag{39}$$

So c_r grows at factorial speed, which is exactly why no fixed order can be optimal for all n: the constant eventually defeats the improving exponent. Stirling's formula applied to (39) gives the root asymptotic that the optimization needs,

$$c_r^{1/r} = \frac{r}{e}\left(1 + O\left(\frac{\log r}{r}\right)\right). \tag{40}$$

Optimizing the step threshold over the order. The formalized condition (37) forbids a progression of length $r+1$ as soon as $h < c_r^{-1/r} n^{1-\vartheta/r}$, and substituting (40) rewrites that threshold as

$$c_r^{-1/r} n^{1-\vartheta/r} = \frac{en}{r} \exp\Big(-\frac{\vartheta \log n}{r}\Big) \Big(1 + O\Big(\frac{\log r}{r}\Big)\Big). \tag{41}$$

The two r-dependent factors pull in opposite directions: $c_r^{-1/r} \sim e/r$ decays in r, while $n^{-\vartheta/r}$ increases to 1. Discarding the lower-order factor and differentiating the logarithm of the leading term,

$$\frac{d}{dr}\Big(1 + \log n - \log r - \frac{\vartheta \log n}{r}\Big) = -\frac{1}{r} + \frac{\vartheta \log n}{r^2},$$

which is positive for $r < \vartheta \log n$, vanishes at $r = \vartheta \log n$, and is negative afterwards. The leading term is therefore unimodal in real $r > 0$ with a unique maximum at $r = \vartheta \log n$, and at that point the explicit factor e cancels against $\exp(-\vartheta \log n/r) = e^{-1}$, leaving the clean scale $n/(\vartheta \log n)$.

Theorem 18.4 (Optimized variable-order obstruction)**.** *Let $0 < \varepsilon < 1$. For all sufficiently large n choose an integer*

$$r_n = \vartheta \log n + O(1), \qquad r_n \geq 2.$$

If

$$1 \leq h \leq (1-\varepsilon)\,\frac{n}{\vartheta \log n},$$

then $n, n+h, \ldots, n+r_n h$ are not all candidates. Moreover, the largest leading-order step scale obtainable by optimizing the curvature condition (37) over r is

$$h_{\max}(n) \sim \frac{n}{\vartheta \log n} = \frac{\log 2}{\log 3} \cdot \frac{n}{\log n} = 0.6309297535\ldots \cdot \frac{n}{\log n}.$$

Proof. Take r_n to be either nearest integer to $\vartheta \log n$. Then $r_n \to \infty$, so (40) and hence (41) apply, and the quotient of $c_{r_n}^{-1/r_n} n^{1-\vartheta/r_n}$ by $n/(\vartheta \log n)$ tends to 1. The fixed margin $1-\varepsilon$ absorbs both the Stirling error $1 + O(\log r/r)$ and the $O(1)$ rounding of $\vartheta \log n$ to an integer, so (37) holds for every h in the stated range, and Theorem 18.1 excludes the progression.

For the optimality assertion, the leading approximation in (41) is unimodal in real $r > 0$ with its unique maximum at $r = \vartheta \log n$, where its value is $n/(\vartheta \log n)$. Any regime in which r stays bounded, or grows outside a constant multiple of $\log n$, returns a strictly smaller order or a strictly smaller leading constant. □

Status: **[paper proof in this report]** — the obstruction being optimized is **[kernel-verified Lean]** and holds uniformly for every $r \geq 2$ with no upper bound on the order, so substituting $r = r_n$ inside the formalized statement is legitimate; the relevant module is `HigherDifference`. The optimization itself, the Gamma evaluation (39), and the Stirling asymptotic (40) are paper arguments and have not been submitted to the kernel.

Comparison with the conditional candidate spacing. A dyadic block $[X, 2X)$ is an interval of length one in exponent space, so the exact block count of Section 5 applies verbatim: conditional on a counterexample with least nonintegral generator β,

$$\#(\mathcal{C} \cap [X, 2X)) = \frac{\log_2 X}{\beta} + O(1) = \frac{\log X}{\beta \log 2} + O_\beta(1). \tag{42}$$

The average additive spacing between consecutive candidates inside that block is therefore

$$\sim \beta \log 2 \cdot \frac{X}{\log X},$$

whereas Theorem 18.4 taken at $n \asymp X$ reaches only

$$h_{\max} \sim \frac{1}{\vartheta} \cdot \frac{X}{\log X}.$$

Both are of order $X/\log X$, so the comparison is a pure constant, and in it the transcendental ϑ cancels against the $\log 2$:

$$\frac{\beta \log 2}{1/\vartheta} = \beta\,\vartheta \log 2 = \beta \log 3. \tag{43}$$

The kernel-verified zero-free region $2^x < 4096$ forces $\beta > 12$ (Section 5), which already makes the ratio exceed $12 \log 3 = 13.1833\ldots$; the interval scan of Section 25 raises it past $20 \log 3 = 21.9722\ldots$, and the extended directed-rounding scan through 2^{26} raises it past $26 \log 3 = 28.5639\ldots$. The last two improvements are **[interval computation]** and do not enter any kernel-verified statement; the exponent bound the kernel actually certifies remains $x \geq 12$.

The same constant governs a second, purely cardinal comparison, which is the more telling of the two. The configuration forbidden by Theorem 18.4 consists of

$$r_n + 1 \sim \vartheta \log X$$

candidates arranged in one structured pattern, while the whole dyadic block contains only $\sim \log X/(\beta \log 2)$ candidates by (42). The ratio is again exactly $\beta \log 3$: the optimized theorem asks a block to supply about $\beta \log 3$ times as many candidates as it can possibly hold. Concretely, at $X = 2^{100}$ the block $[X, 2X)$ holds $100/\beta + O(1)$ candidates — about eight if β were as small as its kernel-certified floor, and about four under the 2^{26} scan — while the shortest progression the theorem forbids at that height already needs $r_n + 1 \approx 111$ terms.

Remark 18.5 (Quantified barrier). Optimizing over r materially strengthens the local theorem: the admissible step scale rises from $n^{1-\vartheta/r}$ at fixed r to $n^{1-o(1)}$, missing n itself only by the single logarithmic factor $\vartheta \log n$. It still cannot contradict the conditional monoid, because the conditional candidate density in a dyadic block is too small by the fixed factor $\beta \log 3 > 13.18$. No argument using only the number of candidates in a dyadic block can force the progression that the current finite-difference functional requires. A successful continuation must exploit additional arithmetic correlation — for example valuation congruences that force a progression of a special shape — or replace Δ_h^r by a different integer-valued functional with a better curvature-to-spacing constant.

This is a limitation of the method, not evidence against Conjecture 2.1. Its value is to replace "the density seems too weak" by a sharp asymptotic constant and a concrete numerical target that any improved functional must beat.

Agreement with the one-logarithm deficit. Section 22 audits the semigroup determinant in the same currency and identifies the same bottleneck. In both analyses the binding quantity is the conditional dyadic population $\asymp \log X$ of (42), and in both the archimedean inequality turns out to be comfortably satisfied by the exact monoid rather than violated by it. To that extent the two diagnoses agree. They differ, instructively, on the size of the shortfall. The semigroup determinant needs $r \asymp (\log X)^2$ nodes and is supplied $\asymp \log X$: a deficit of a full power of $\log X$. The optimized finite difference needs $\vartheta \log X$ nodes and is supplied $\log X/(\beta \log 2)$: a deficit of the bounded factor $\beta \log 3$. The finite difference is thus quantitatively much closer — but only because it buys that closeness with a hypothesis it cannot discharge, namely that the candidates lie in exact arithmetic progression, which Observation 18.3 says a rank-two multiplicative monoid has no reason to provide. Discarding that hypothesis is exactly what the determinants of Sections 19 and 20 do, and the one-logarithm deficit is the precise price charged for it. Read together, the two sections say that the present archimedean toolkit is short by one factor of $\log X$ in generality or by a factor of $\beta \log 3$ in strength, and that tuning constants converts neither shortfall into the other.

19 New result I: arbitrary-node lattice repulsion

The higher-difference theorem of Section 18 uses equally spaced nodes, because the forward-difference operator is built from an arithmetic progression. The same integer-versus-small-real principle extends to *arbitrary* nodes through interpolation determinants. The resulting statement constrains every configuration of candidates, not only the additively structured ones, which is exactly the gap identified in Observation 18.3.

19.1 The determinant identity

For distinct real numbers $x_0 < \cdots < x_r$ and a function f, write $f[x_0, \ldots, x_r]$ for the divided difference. The standard determinant identity is

$$\det \begin{pmatrix} 1 & x_0 & \cdots & x_0^{r-1} & f(x_0) \\ 1 & x_1 & \cdots & x_1^{r-1} & f(x_1) \\ \vdots & \vdots & & \vdots & \vdots \\ 1 & x_r & \cdots & x_r^{r-1} & f(x_r) \end{pmatrix} = V(x_0, \ldots, x_r)\, f[x_0, \ldots, x_r], \tag{44}$$

where

$$V(x_0, \ldots, x_r) = \prod_{0 \le i < j \le r} (x_j - x_i).$$

If $f \in C^r$ on the interval spanned by the nodes, the generalized mean-value theorem for divided differences supplies some $\xi \in (x_0, x_r)$ with

$$f[x_0, \ldots, x_r] = \frac{f^{(r)}(\xi)}{r!}. \tag{45}$$

For $f(t) = t^\vartheta$ this is again $f^{(r)}(t) = (\vartheta)_{\underline{r}}\, t^{\vartheta - r}$, and since ϑ is irrational — in particular not an integer — the coefficient $(\vartheta)_{\underline{r}}$ never vanishes.

19.2 The arbitrary-node theorem

Theorem 19.1 (Vandermonde repulsion for candidates)**.** *Let $r \ge 2$ and let*

$$1 \le m_0 < m_1 < \cdots < m_r$$

be natural candidates, so that $m_i^\vartheta \in \mathbb{N}$ for every i. Then

$$\prod_{0 \le i < j \le r} (m_j - m_i) \ge \frac{r!}{|\,(\vartheta)_{\underline{r}}\,|}\, m_0^{r-\vartheta}. \tag{46}$$

Proof. Apply (44) to $f(t) = t^\vartheta$ with the nodes m_i. Every entry of the matrix is an integer: the ordinary powers m_i^k are integral by construction, and the last column is integral precisely because the m_i are candidates. By (45) the determinant equals

$$D = V(m_0, \ldots, m_r)\,\frac{(\vartheta)_{\underline{r}}}{r!}\,\xi^{\vartheta-r} \qquad (m_0 < \xi < m_r).$$

Each factor is nonzero, so $D \in \mathbb{Z}\setminus\{0\}$ and therefore $|D| \geq 1$. Because $r \geq 2 > \vartheta$ the exponent $\vartheta - r$ is negative, so $\xi^{\vartheta-r} \leq m_0^{\vartheta-r}$. Therefore

$$1 \leq |D| \leq V(m_0, \ldots, m_r)\,\frac{|\,(\vartheta)_{\underline{r}}\,|}{r!}\,m_0^{\vartheta-r},$$

which rearranges to (46). □

Status: [paper proof in this report] — the argument is unconditional and elementary once the divided-difference mean-value theorem is available. It is not an instance of any committed Lean theorem: `HigherDifference` assumes arithmetic progressions. A formalization blueprint appears in Section 28.

Remark 19.2 (Formalization state). A Lean draft `ArbitraryNodeDeterminant` exists in the working tree. It supplies the algebraic half deliberately free of analysis — the node matrix, an explicit integral model, integrality of the determinant with the resulting bound $1 \leq |\det|$, the Vandermonde product and its positivity on strictly increasing nodes. **[Lean draft, not yet accepted]**: it has not yet been accepted by the kernel, and the analytic half (the divided-difference mean-value theorem for $t \mapsto t^\vartheta$) is still an input rather than a proved ingredient. Nothing in this section may be cited as machine-checked.

19.3 Three points: a lattice-triangle inequality

For $r = 2$ the determinant is

$$D = \det\begin{pmatrix} 1 & m_0 & m_0^\vartheta \\ 1 & m_1 & m_1^\vartheta \\ 1 & m_2 & m_2^\vartheta \end{pmatrix},$$

and $|D|$ is twice the area of the lattice triangle with vertices (m_i, m_i^ϑ). Strict convexity of $t \mapsto t^\vartheta$ makes the area positive; lattice integrality makes it at least $1/2$. The bound below is the quantitative form of that picture.

Corollary 19.3 (Three-candidate span)**.** *If three candidates lie in $[N, N+H]$ with $N \geq 1$, then*

$$H \geq \left(\frac{8}{\vartheta(\vartheta-1)}\right)^{1/3} N^{(2-\vartheta)/3}. \tag{47}$$

Rounded to ten decimal places the constant is 2.0510723957 *and the exponent* 0.1383458331*; in particular* $H \geq 2.0510723956\,N^{0.1383458330}$.

Proof. Write $a = m_1 - m_0$ and $b = m_2 - m_1$. Then $a, b > 0$ and $a + b \leq H$, so the Vandermonde product satisfies

$$V = ab(a+b) \leq \frac{H^3}{4},$$

the continuous maximum being attained at $a = b = H/2$. Theorem 19.1 at $r = 2$ gives

$$V \geq \frac{2}{\vartheta(\vartheta-1)} N^{2-\vartheta}.$$

Combining the two inequalities yields (47). □

19.4 General local packing bound

Put

$$R_r := \binom{r+1}{2} = \frac{r(r+1)}{2}, \qquad C_r := \left(\frac{r!}{|\,(\vartheta)_{\underline{r}}\,|}\right)^{1/R_r}, \qquad \alpha_r := \frac{r-\vartheta}{R_r} = \frac{2(r-\vartheta)}{r(r+1)}.$$

Every pairwise gap among points of $[N, N+H]$ is at most H, so the Vandermonde product is at most H^{R_r}, and Theorem 19.1 immediately gives the following.

Corollary 19.4 (At most r points below the repulsion scale)**.** *If $r+1$ candidates lie in $[N, N+H]$, then*

$$H \geq C_r N^{\alpha_r}. \tag{48}$$

Equivalently, every interval of length strictly less than $C_r N^{\alpha_r}$ beginning at or after N contains at most r candidates.

The exponent can be optimized in r. As a function of a real variable $r > \vartheta$,

$$\alpha(r) = \frac{2(r-\vartheta)}{r(r+1)}$$

attains its maximum at

$$r_* = \vartheta + \sqrt{\vartheta^2 + \vartheta} = 3.6090841941\ldots,$$

so the best integer order in this ordinary-power family is $r = 4$ — not the three-point case that geometry makes most vivid. Table 3 lists the constants.

r	R_r	$\lvert(\vartheta)_{\underline{r}}\rvert$	α_r	C_r
2	3	0.9271436280	0.1383458331	1.2920946431
3	6	0.3847993728	0.2358395832	1.5805909191
4	10	0.5445055422	0.2415037499	1.4602287461
5	15	1.3150013031	0.2276691666	1.3510881062
6	21	4.4907787617	0.2102398809	1.2735045942
7	28	19.8269566337	0.1933941964	1.2187063909
8	36	107.3637136680	0.1781954861	1.1790125192

Table 3: Ordinary-power arbitrary-node repulsion constants. For $r = 2$ the sharper three-point constant of Corollary 19.3 uses the exact maximum of the three-gap product rather than the crude bound H^{R_2}.

Corollary 19.5 (Best ordinary-power local statement)**.** *Every interval*

$$[N, N+H], \qquad H < 1.4602287460\, N^{0.2415037499},$$

contains at most four natural candidates. The exact constant is $C_4 = (4!/|\,(\vartheta)_{\underline{4}}\,|)^{1/10} = 1.4602287461$ *to ten decimal places, and the exact exponent is* $\alpha_4 = (4-\vartheta)/10 = 0.2415037499\ldots$

19.5 A bound for any longer interval

The span estimate converts into a counting inequality. Let $m_1 < \cdots < m_s$ be all candidates in $[N, N+H]$. For each $i \le s-r$ the consecutive window $m_i, \dots, m_{i+r}$ has span at least $C_r N^{\alpha_r}$ by Corollary 19.4. Summing these spans, each adjacent gap $m_{j+1} - m_j$ is counted at most r times, so

$$(s-r)C_r N^{\alpha_r} \le r(m_s - m_1) \le rH.$$

Corollary 19.6 (Arbitrary-node counting inequality). *For every $r \ge 2$,*

$$\#(\mathcal{C} \cap [N, N+H]) \le r + \frac{r}{C_r} H N^{-\alpha_r}. \tag{49}$$

This is a genuine strengthening of progression avoidance: it constrains every possible configuration of candidates in an interval, with no additive hypothesis at all. It remains far from the conditional benchmark of Section 5. At the optimal $r = 4$ and $H = N$, the right-hand side of (49) is $O(N^{0.7585\cdots})$, whereas the exact population of a failed conjecture would be only $O(\log N)$. The gap is enormous, and closing it by ordinary powers alone is hopeless — which motivates enlarging the column space, the subject of Section 20.

19.6 Relation to higher differences

For equally spaced nodes $m_i = n + ih$ the Vandermonde factor is explicit:

$$V = h^{R_r} \prod_{j=1}^{r} j!.$$

Substituting in (44) recovers an integer interpolation remainder closely related to the rth forward difference of Section 18. The committed repository theorem is sharper for this special configuration, because it uses the exact binomial combination rather than bounding all pairwise gaps by a common span. Conversely, Theorem 19.1 applies when the nodes carry no additive structure whatsoever. The two results are complementary, and only the arbitrary-node form escapes the vacuity diagnosed in Observation 18.3.

20 New result II: semigroup generalized Vandermonde determinants

The determinant of Section 19 used only the integral columns $1, m, m^2, \dots$ and one extra integral column m^ϑ. But a candidate carries much more integral data. If $m^\vartheta = n \in \mathbb{N}$, then for every $a, b \in \mathbb{N}_0$,

$$m^{a+b\vartheta} = m^a n^b \in \mathbb{N}. \tag{50}$$

The full set of available exponents is therefore

$$\Lambda_\vartheta := \{a + b\vartheta : a, b \in \mathbb{N}_0\}.$$

Since ϑ is irrational, all pairs (a, b) give distinct exponents.

20.1 Positivity of generalized Vandermonde determinants

Let
$$0 \le \lambda_0 < \lambda_1 < \cdots < \lambda_r \qquad \text{and} \quad 0 < x_0 < x_1 < \cdots < x_r.$$
The functions x^{λ_j} form an extended complete Chebyshev system on $(0,\infty)$. Their Wronskian is classical in the Karlin–Studden theory of Tchebycheff systems:
$$W(x) = x^{\sum_j \lambda_j - R_r} \prod_{0\le i<j\le r} (\lambda_j - \lambda_i) > 0. \tag{51}$$
Indeed, after factoring x^{λ_j} from column j and x^{-i} from derivative row i, the remaining determinant is the Vandermonde determinant of the λ_j, expressed in the falling-factorial basis. Positivity of all initial Wronskians implies strict positivity of every ordered evaluation determinant.

For completeness, there is also an elementary zero-count proof. Put $x = e^t$. A nonzero linear combination of x^{λ_j} becomes an exponential polynomial $\sum c_j e^{\lambda_j t}$. After factoring the lowest exponential, differentiation removes the constant term and leaves one fewer exponential. Induction with Rolle's theorem shows that a combination of $r+1$ such functions has at most r real zeros counting multiplicity. Consequently an evaluation determinant at $r+1$ ordered nodes cannot vanish. Its sign is constant on the connected chamber $x_0 < \cdots < x_r$ and is positive in the coalescing-node limit by (51).

Lemma 20.1 (Generalized Vandermonde positivity)**.** *For strictly increasing nonnegative exponents λ_j and strictly increasing positive nodes x_i,*
$$\det(x_i^{\lambda_j})_{0\le i,j\le r} > 0. \tag{52}$$

The two arguments just sketched are carried out in full, with all induction bookkeeping made explicit, in Appendix D.

20.2 Integer determinants from candidates

Theorem 20.2 (Semigroup integer determinant)**.** *Let $m_0 < \cdots < m_r$ be natural candidates. Choose distinct pairs $(a_j, b_j) \in \mathbb{N}_0^2$ and order*
$$\lambda_j = a_j + b_j\vartheta$$
so that $\lambda_0 < \cdots < \lambda_r$. Then
$$D_{\boldsymbol{\lambda}}(\boldsymbol{m}) := \det(m_i^{\lambda_j})_{0\le i,j\le r} \tag{53}$$
is a positive integer. In particular, $D_{\boldsymbol{\lambda}}(\boldsymbol{m}) \ge 1$.

Proof. By (50), each entry equals
$$m_i^{a_j}(m_i^{\vartheta})^{b_j} \in \mathbb{N},$$
so the determinant is an integer. It is strictly positive by Lemma 20.1. ☐

Status: [paper proof in this report]

This theorem packages all mixed integrality consequences into one reusable object. It is not the same as the repository's exponential interpolation determinant in `MultiplicativeRank.lean`: here the rows are arbitrary natural candidates and the columns are chosen from the two-dimensional exponent semigroup.

20.3 Divided-difference factorization

Let $f_j(x) = x^{\lambda_j}$ and let

$$V(\boldsymbol{m}) = \prod_{i<j}(m_j - m_i).$$

Applying the Newton divided-difference row transformation gives

$$D_{\boldsymbol{\lambda}}(\boldsymbol{m}) = V(\boldsymbol{m}) \det\big(f_j[m_0, \dots, m_i]\big)_{0\le i,j\le r}. \tag{54}$$

The transformation is exact; its determinant is $V(\boldsymbol{m})^{-1}$.

Suppose

$$N \le m_0 < \cdots < m_r \le N + H \le 2N.$$

For each divided difference, the mean-value theorem gives a point in $[N, 2N]$ and

$$|f_j[m_0, \dots, m_i]| \le \frac{|\,(\lambda_j)_{\underline{i}}\,|}{i!} \sup_{N\le x\le 2N} x^{\lambda_j - i}. \tag{55}$$

We need a uniform elementary coefficient estimate.

Lemma 20.3 (Falling-factorial coefficient bound)**.** *For $\lambda \ge 0$ and $i \in \mathbb{N}_0$,*

$$\frac{|\,(\lambda)_{\underline{i}}\,|}{i!} \le 2^{\lambda+i}. \tag{56}$$

Proof. If $0 \le \lambda \le 1$, each factor satisfies $|\lambda - k| \le k + 1$, so the quotient is at most 1. If $\lambda \ge 1$, let $n = \lceil \lambda \rceil$. Then

$$\frac{|\,(\lambda)_{\underline{i}}\,|}{i!} \le \prod_{k=0}^{i-1} \frac{\lambda + k}{k+1} \le \prod_{k=0}^{i-1} \frac{n+k}{k+1} = \binom{n+i-1}{i}.$$

The last binomial coefficient is at most the sum of its row, 2^{n+i-1}, and $n - 1 \le \lambda$. Hence it is at most $2^{\lambda+i}$. □

For $x \in [N, 2N]$,

$$x^{\lambda-i} \le 2^{\lambda} N^{\lambda-i}. \tag{57}$$

If $\lambda - i \ge 0$, use $x \le 2N$; if it is negative, use $x \ge N$. Combining (55), (56), and (57),

$$|f_j[m_0, \dots, m_i]| \le 2^{2\lambda_j+i} N^{\lambda_j - i}. \tag{58}$$

20.4 Quantitative semigroup gap theorem

Set

$$R := R_r = \frac{r(r+1)}{2}, \qquad \Sigma := \sum_{j=0}^{r} \lambda_j.$$

By the Leibniz formula and (58), every permutation term in the small determinant in (54) is at most

$$2^{2\Sigma+R} N^{\Sigma-R},$$

and there are $(r+1)!$ terms. Since every pairwise node difference is at most H, $V(\boldsymbol{m}) \le H^R$. The positive integer lower bound from Theorem 20.2 now gives the following.

Theorem 20.4 (Semigroup determinant gap)**.** *Let $r \ge 1$. Let $\lambda_0 < \cdots < \lambda_r$ be any $r+1$ exponents in Λ_ϑ, and put $R = r(r+1)/2$ and $\Sigma = \sum_j \lambda_j$. If $r+1$ natural candidates lie in*

$$[N, N+H] \subseteq [N, 2N], \qquad N \ge 1,$$

then

$$H \ge ((r+1)!)^{-1/R} 2^{-1-2\Sigma/R} N^{1-\Sigma/R}. \tag{59}$$

Proof. Equations (54) and (58) imply

$$1 \le D_{\boldsymbol{\lambda}}(\boldsymbol{m}) \le H^R (r+1)! 2^{R+2\Sigma} N^{\Sigma-R}.$$

Taking Rth roots and rearranging proves (59). □

The exponent $1-\Sigma/R$ is the key. We should choose the $r+1$ smallest available semigroup exponents to minimize Σ.

20.5 The ordered exponent semigroup

Write the elements of Λ_ϑ in increasing order:

$$0 = \lambda_0 < \lambda_1 < \lambda_2 < \cdots .$$

The beginning is

$$0,\ 1,\ \vartheta,\ 2,\ 1+\vartheta,\ 3,\ 2\vartheta,\ 2+\vartheta,\ 4,\ 1+2\vartheta,\ 3+\vartheta,\ 3\vartheta,\ 5, \ldots$$

The counting function is elementary:

$$\begin{aligned} L(T) &:= \#\{\lambda \in \Lambda_\vartheta : \lambda \le T\} \\ &= \sum_{0 \le b \le T/\vartheta} (\lfloor T - b\vartheta \rfloor + 1) = \frac{T^2}{2\vartheta} + O(T). \end{aligned} \tag{60}$$

Irrationality of ϑ ensures that different pairs (a,b) are counted distinctly. Inverting (60) and summing the order statistics gives

$$\lambda_r = \sqrt{2\vartheta r} + O(1), \tag{61}$$

$$\Sigma_r := \sum_{j=0}^{r} \lambda_j = \frac{2}{3}\sqrt{2\vartheta}\, r^{3/2} + O(r), \tag{62}$$

$$\frac{\Sigma_r}{R_r} = \frac{4}{3}\sqrt{\frac{2\vartheta}{r}} + O(r^{-1}). \tag{63}$$

Consequently the power of N in (59) tends to 1:

$$1 - \frac{\Sigma_r}{R_r} = 1 - \frac{4}{3}\sqrt{\frac{2\vartheta}{r}} + O(r^{-1}). \tag{64}$$

r	Σ_r	$1-\Sigma_r/R_r$
2	2.5849625007	0.1383458331
3	4.5849625007	0.2358395832
4	7.1699250014	0.2830074999
5	10.1699250014	0.3220049999
6	13.3398500029	0.3647690475
7	16.9248125036	0.3955424106
8	20.9248125036	0.4187552082
9	25.0947375050	0.4423391666
10	29.6797000058	0.4603690908
12	39.4345875079	0.4944283653
20	87.7939875195	0.5819333928

Table 4: Exponent gain from using the first $r+1$ mixed exponents in $\mathbb{N}_0+\mathbb{N}_0\vartheta$. The ordinary-power family peaks and then decays; the semigroup family improves monotonically in these data and tends to exponent 1.

20.6 An explicit coarse form

For the global counting corollary we need no delicate lattice-point error term. A simple bound suffices.

Lemma 20.5 (Crude semigroup weight bound)**.** *For $r \geq 2$, the first $r+1$ semigroup exponents satisfy*

$$\lambda_r \leq 6\sqrt{r}, \qquad \frac{\Sigma_r}{R_r} \leq \frac{12}{\sqrt{r}}. \tag{65}$$

Proof. Let $q = \lceil \sqrt{r+1} \rceil$. The $(q+1)^2$ pairs $0 \leq a, b \leq q$ give distinct exponents and each is at most $q(1+\vartheta) < 3q$. Since $(q+1)^2 > r+1$, the rth ordered exponent is below $3q$. For $r \geq 2$, $q \leq \sqrt{r+1}+1 \leq 2\sqrt{r}$, hence $\lambda_r \leq 6\sqrt{r}$. Then

$$\Sigma_r \leq (r+1)\lambda_r \leq 6(r+1)\sqrt{r},$$

and division by $R_r = r(r+1)/2$ gives the second inequality. □

For $r \geq 144$, (65) gives $\Sigma_r/R_r \leq 1$. Also

$$((r+1)!)^{1/R_r} \leq (r+1)^{(r+1)/R_r} = (r+1)^{2/r} \leq 3$$

for $r \geq 2$. Therefore Theorem 20.4 yields a particularly clean form.

Corollary 20.6 (Explicit near-linear span)**.** *If $r \geq 144$ and $r+1$ natural candidates lie in $[N, N+H] \subseteq [N, 2N]$, then*

$$H \geq \frac{1}{24} N^{1-12/\sqrt{r}}. \tag{66}$$

Proof. In (59), use $((r+1)!)^{-1/R} \geq 1/3$, $2^{-1-2\Sigma/R} \geq 2^{-3} = 1/8$, and, since $N \geq 1$, $N^{1-\Sigma/R} \geq N^{1-12/\sqrt{r}}$. □

20.7 Counting candidates in a general interval

As in the ordinary-power count of Section 19, sum the spans of consecutive windows of $r + 1$ candidates. Every adjacent gap is counted at most r times.

Corollary 20.7 (Semigroup interval count)**.** *For $r \geq 144$ and $1 \leq H \leq N$,*

$$\#(\mathcal{C} \cap [N, N + H]) \leq r + 24rHN^{-1+12/\sqrt{r}}. \tag{67}$$

Proof. If s is the number of candidates and $s \leq r$, there is nothing to prove. Otherwise, Corollary 20.6 gives a lower bound $L = N^{1-12/\sqrt{r}}/24$ for each span $m_{i+r} - m_i$. Thus

$$(s - r)L \leq rH,$$

which is equivalent to (67). □

20.8 An independent dyadic logarithmic-square bound

Corollary 20.8 (Explicit $O((\log X)^2)$ dyadic bound)**.** *For every real $X \geq 1$,*

$$\#(\mathcal{C} \cap [X, 2X]) \leq 10000 \max\{1, \log X\}^2. \tag{68}$$

Proof. Put $L = \max\{1, \log X\}$ and

$$r = \left\lceil 144L^2 \right\rceil.$$

Then $r \geq 144$, $\sqrt{r} \geq 12L$, and

$$X^{12/\sqrt{r}} = \exp\left(\frac{12 \log X}{\sqrt{r}}\right) \leq e.$$

Apply (67) with $N = H = X$:

$$\#(\mathcal{C} \cap [X, 2X]) \leq r + 24rX^{12/\sqrt{r}} \leq (1 + 24e)r.$$

Since $L \geq 1$, $r \leq 145L^2$, and $(1 + 24e)145 < 10000$. □

Status: [paper proof in this report]

The bound is numerically coarse by design. Its value is conceptual independence: it comes from positivity and integrality of generalized Vandermonde determinants, not from the prime-factor valuation rank of candidates. Refining constants is much less important than finding an additional source of determinant lower bound.

Status: [Lean draft, not yet accepted]

Lean drafts of this hierarchy exist in the repository but have not yet been accepted by the kernel: `MixedExponentSemigroup` for (50) and the ordered exponent semigroup, `SemigroupDeterminant` for Theorem 20.2 and Theorem 20.4, `FallingFactorialBound` for Lemma 20.3, and `SemigroupWeightBound` for Lemma 20.5 and its explicit consequences. Until those files compile, every statement in this section rests on the paper proofs given above.

21 The min-plus ceiling: why term-wise p-adic amplification cannot close a determinant argument

Every interpolation-determinant argument in this report squeezes a nonzero integer determinant D between a lower and an upper bound. A recurring proposal — registered in the deficit analysis of Section 22 as the most natural way to supply the missing logarithm — is to manufacture the lower bound from p-adic content: choose the nodes and the column exponents so that every monomial in the Leibniz expansion of D is divisible by a high power of 2 and of 3, and conclude

$$v_p(D) \;\geq\; V_p^* := \min_\sigma \sum_i v_p\Big(E_{i\sigma(i)}\Big), \qquad |D| \;\geq\; 2^{V_2^*}3^{V_3^*}. \tag{69}$$

This section records why the proposal cannot work, and — as importantly — corrects the quantitative form in which the obstruction has been stated.

21.1 The threshold

Fix a column set $S = \{(a_j, b_j)\}_{j=0}^r$ of distinct mixed exponents, put $\lambda_j = a_j + b_j\vartheta$ and $W = \sum_j \lambda_j$, and let D be the semigroup determinant $\det\big(c_i^{a_j}(c_i^\vartheta)^{b_j}\big)$ built on the candidates $c_0 < \cdots < c_r$ of a single dyadic block $[2^T, 2^{T+1})$. Every entry satisfies $\log_2 E_{ij} \leq (T+1)\lambda_j$, so the Leibniz bound gives

$$\log_2 |D| \;\leq\; (T+1)W + \log_2(r+1)! \;=\; TW\,(1 + o(1)). \tag{70}$$

An amplification scheme yielding a guaranteed lower bound $|D| \geq 2^V$ therefore needs $V > (T+1)W + \log_2(r+1)!$ for a contradiction. We call the right-hand side of (70) the *bar*.

21.2 The ceiling

The obstruction is not quantitative at all. It is the observation that the $\{2,3\}$-part of an integer divides the integer.

Theorem 21.1 (Min-plus ceiling)**.** *Let $n \geq 1$ be a natural number, let $V_2, V_3 \geq 0$ satisfy $2^{V_2} \mid n$ and $3^{V_3} \mid n$, and suppose $n \leq 2^B$. Then*

$$V_2 + V_3\,\vartheta \;\leq\; B.$$

More generally, for any finite set S of primes, $\sum_{p\in S} v_p(n)\log p \leq \log n$.

Proof. 2^{V_2} and 3^{V_3} are coprime and both divide n, so $2^{V_2}3^{V_3} \mid n$ and hence $2^{V_2}3^{V_3} \leq n \leq 2^B$. Taking $\log_2$ gives the claim. For general S the same argument applies to $\prod_{p\in S} p^{v_p(n)}$, which divides n by pairwise coprimality. □

Status: [kernel-verified Lean] — `minPlusContent_le_of_le` and `minPlusContent_le_of_natAbs_le` in `MinPlusCeiling`, with the general prime-set form `primeContent_le_log` and the divisibility input `prod_primePow_dvd`.

Corollary 21.2 (Term-wise amplification is inert)**.** *Let $D \neq 0$ be the determinant of an interpolation system, let V_2^*, V_3^* be any certified min-plus lower bounds as in (69), and let B be any correct archimedean upper bound $\log_2 |D| \leq B$. Then $V_2^* + V_3^*\vartheta \leq B$.*

Consequently no choice of node system, no choice of column set, and no enlargement of the prime set can make a term-wise divisibility bound contradict a correct archimedean bound.

The corollary is unconditional and needs no equidistribution input, no averaging, and no property of the candidates beyond $D \in \mathbb{Z} \setminus \{0\}$. Its content is that term-wise divisibility contributes nothing beyond what $|D| \geq 1$ already gives: it is a lower bound for $|D|$ derived from $|D|$ itself.

21.3 A correction to the averaging form of the obstruction

The obstruction has previously been argued by averaging: the minimum over permutations is at most the mean over permutations, the mean of $\sum_i v_p(E_{i\sigma(i)})$ is $\sum_j \big(a_j\bar{\nu} + b_j\bar{\mu}\big)$ with $\bar{\nu}, \bar{\mu}$ the row-averaged valuations, and the depths n_k equidistribute over $[0, T]$, so $\bar{\nu} = T/2$. That computation is correct, and its conclusion — that the ceiling is governed by the *same* block equidistribution that defeats the shrinking-target route of Section 23 — is a genuine structural insight worth keeping: at the structural places the two walls of this problem are one wall.

The numerical conclusion drawn from it, however, is too strong. Bounding the 2-adic place and the ϑ-weighted 3-adic place separately gives $TW/2$ *each*; their sum is TW, which is the bar, not half of it. With $M = 2^\beta$ and $A = 3^\beta$ and the normalization $s_M = 2^{v_2(M)}3^{v_3(M)}$, $s_A = 2^{v_2(A)}3^{v_3(A)}$ for the 3-smooth parts, the mean bound reads

$$V^* \;\leq\; \frac{T}{2}\left[W + \frac{1}{\beta}\sum_j \Big(a_j \log_2 s_M + b_j \log_2 s_A\Big)\right], \tag{71}$$

and the bracket is at most $2W$, with equality exactly when M and A are both 3-smooth. Theorem 12.5 excludes that case, so the mean bound is strictly below the bar; but the deficiency it provides is proportional to the roughness of the outputs and degenerates as the outputs approach smoothness.

The factor $1/2$ requires $3 \nmid M$ *and* $2 \nmid A$. The matched-depth theorem of Section 5 excludes only the pair $2 \mid M$ and $3 \mid A$; the crossed pair $3 \mid M$ and $2 \mid A$ is not excluded, and it is exactly the configuration in which the ceiling climbs.

21.4 Calibration: the exact ceilings

The mean bound is not the true ceiling, because the minimum is genuinely lower than the mean. The exact constants were obtained in the continuation report by an optimal-transport computation, and they replace every heuristic figure that has circulated for this obstruction.

Write $\mathfrak{G}$ for the Gini norm of a normalized prime-valuation slope vector, and c_p for the slope vector at p. The termwise all-prime divisor fraction of the balanced mixed-semigroup determinant is exactly $1 - \frac{3}{8}\sum_p \mathfrak{G}(c_p) + o(1)$, and the primitive condition $\min\big(v_2(M)\,, v_3(A)\big) = 0$ forces $\sum_p \mathfrak{G}(c_p) \geq 8/15$.

System	Termwise-divisor ceiling
Ordinary input/output Vandermonde product	< 0.8710491
Mixed single block, arbitrary branch	$4/5$
Mixed single block, cross-coprime	$7/10$
Mixed single block, places 2 and 3 only, cross-coprime	$3/10$
Complete triangular row grid	$\pi/4 = 0.7853982\ldots$
First tied fibers at 2 and 3	$o(1)$ of the block budget

Two figures that have been proposed for this obstruction are not valid in the stated generality. A structural ceiling of $1/2$ requires dropping the depths $k\,u\,v_2(M)$ and $k\,v\,v_3(A)$ simultaneously, which assumes $v_2(M) = v_3(A) = 0$; the primitive theorem of Section 5 supplies only $\min(v_2(M)\,, v_3(A)) = 0$. An all-prime ceiling of $2/3$ does not follow from "minimum $\leq$ average": for an external positive weight proportional to $X_1 + X_2$ on the standard triangle one has $\mathbb{E}Y = 2/3$ and $\Phi(1,1) = 4/15$, so the minimum assignment is $4/5$ of half the mean, not $2/3$.

An independent numerical check, computing exact assignment minima for the block node system over the places 2 and 3 only, in the cross-coprime branch and with a square-box column family, gives a ratio settling near 0.33 across block heights $T = 400, 1200, 3600, 10000$, consistent with the exact continuum value $3/10$ for those places. The same computation shows what happens outside the balanced primitive normalization: with a degenerate column family such as the pure-b columns $(0, j)$, and with A within $\log_2 3$ of being a power of three, the ratio rises to 0.9964. That configuration violates none of the kernel-verified constraints, but it saturates *tautologically* — the entries are then nearly powers of three, their 3-adic content is nearly their size, and (69) degenerates. The same degeneration occurs on the a-side at $b = 0$, where $D = \prod_{i<i'}(c_{i'} - c_i)$ and $v_2(D) = \sum_{i<i'} \min(n_i, n_{i'})$ exactly. This is why the balanced and primitive normalizations are not cosmetic: they are what make the $4/5$ and $7/10$ ceilings meaningful.

Status: **[paper proof in this report]** for the exact ceilings, from the continuation report's transport computation — they follow from the closed-form Gini identity, not from experiment. **[interval computation]** for the independent $3/10$ cross-check and the degenerate-family observation: exact rational node data, exact Jonker–Volgenant assignment minima, double-precision ϑ. Neither is needed for Corollary 21.2, which is **[kernel-verified Lean]** and unconditional.

21.5 Where cancellation can live

The ceilings bound only the *tropical* divisor. They say nothing about the true valuation $v_p(D)$, which can be larger when several permutations attain the minimum. The following observation localizes that possibility exactly.

Proposition 21.3 (A unique minimizer forbids cancellation)**.** *Let $D = \det(E_{ij})$ be a nonzero integer determinant and let p be a prime. If a unique permutation σ_0 minimizes $\sum_i v_p(E_{i\sigma(i)})$, then $v_p(D) = \sum_i v_p(E_{i\sigma_0(i)})$.*

Proof. Divide the Leibniz expansion by p^{τ_p} with τ_p the minimum. The term indexed by σ_0 becomes a p-adic unit and every other term stays divisible by p, so the sum is a unit modulo p. □

Status: [kernel-verified Lean] — `det_dvd_and_not_dvd_of_unique_min` in `MinPlusCeiling`, stated in the divisibility form '$q^\tau \mid \det$ and $q^{\tau+1} \nmid \det$' so that no p-adic valuation machinery is needed.

Consequently the missing fifth, or more, can arise only at primes where the assignment problem ties, and the tie geometry is explicit: for $p \neq 2, 3$ columns tie along the rational lines orthogonal to the external valuation vector, and in the cross-coprime branch the 2-adic cost depends only on u and the 3-adic cost only on v, so every vertical or horizontal column fiber creates equal-cost permutations. The continuation report evaluates the first such tied layer and finds its valuation to be $O\big(T^{3/2}(\log T)^2\big)$ against a block budget $H_T \asymp T^{5/2}$, hence $o(1)$ of the budget.

21.6 What this leaves

Three statements survive every normalization.

First, term-wise amplification is retired unconditionally against the crude entry bound. Corollary 21.2 is a structural identity, not a numerical near-miss, and it holds for every prime set, every node system and every column set.

Second, the quantitative ceilings above say how much of the archimedean budget automatic divisibility can supply in each concrete construction, and none reaches 1. Adding every prime in the fixed support does not close any determinant; it improves the diagnosis from a heuristic "about one half" to a proved 4/5.

Third — and this is the redirection — a successful p-adic proof has to be a *cancellation* theorem, not a divisibility-by-every-term theorem. Proposition 21.3 says exactly where such a theorem could live: inside the equal-cost fibers, and it would have to repeat through many tropical layers or be combined with a sharper archimedean free energy than the crude bar. Quantifying the gain of the refined divided-difference bounds of Sections 19 and 20 over the crude bar is the concrete companion question, and it concerns the archimedean side alone.

Status: [paper proof in this report] for the reading of the obstruction; [kernel-verified Lean] for Theorem 21.1, Corollary 21.2 and Proposition 21.3. The structural observation that the ceiling and the shrinking-target obstruction are governed by the same block equidistribution is recorded as an interpretive claim, not a theorem.

22 Why the new hierarchy still stops short

The semigroup determinant is much stronger locally than the ordinary-power determinant: its repulsion exponent approaches 1. It is therefore important to identify exactly why it does not prove the conjecture.

22.1 How many nodes are needed to approach scale X?

The asymptotic exponent in (64) is

$$\delta_r := 1 - \frac{\Sigma_r}{R_r} = 1 - \frac{c_\vartheta}{\sqrt{r}} + O(r^{-1}), \qquad c_\vartheta = \frac{4}{3}\sqrt{2\vartheta} = 2.3739044262\ldots.$$

For nodes near X, the lower span has the shape

$$X^{\delta_r} = X \exp\Bigl(-\frac{c_\vartheta \log X}{\sqrt{r}} + O\Bigl(\frac{\log X}{r}\Bigr)\Bigr). \tag{72}$$

To make the exponential loss in (72) bounded independently of X, one needs

$$\sqrt{r} \asymp \log X, \qquad \text{that is,} \qquad r \asymp (\log X)^2. \tag{73}$$

This is precisely why Corollary 20.8 has a logarithmic square.

22.2 How many nodes a counterexample supplies

Conditional on a counterexample, the exact dyadic block count established in Section 5 gives

$$\#(\mathcal{C} \cap [X, 2X]) = \frac{\log_2 X}{\beta} + O(1) \asymp \log X. \tag{74}$$

Thus the actual conditional population is smaller than the determinant threshold (73) by one factor of $\log X$.

Taking $r \asymp \log X$ in (72) yields only

$$X \exp(-C\sqrt{\log X}),$$

which is much smaller than the average spacing $X/\log X$ among the $\asymp \log X$ conditional candidates. The inequality is therefore compatible with the exact monoid even at the level of expected spacing.

22.3 Broader multiplicative intervals do not help

The cumulative conditional count below X is $\asymp (\log X)^2$, so one might try to use all those points at once. But their nodes range from 1 to X, and the gap theorem depends on the smallest node. Restricting to $[X^\gamma, X]$ with fixed $0 < \gamma < 1$ does retain $\asymp (\log X)^2$ points, but its additive length is $\asymp X$, while the lower scale is at most $X^{\gamma+o(1)}$. Again there is no contradiction.

This is not a mere defect of coarse constants. It reflects a mismatch between additive node geometry and the logarithmic triangular lattice of exponents (n, k) in $m = 2^n M^k$.

22.4 The determinant needs one more source of size

The archimedean argument uses only

$$D \in \mathbb{Z}_{>0} \implies D \geq 1.$$

If the exact monoid coordinates forced

$$D \equiv 0 \pmod{Q} \qquad \text{with} \quad \log Q \gg R \log X/\sqrt{r},$$

then the lower bound $D \geq Q$ could remove the one-logarithm deficit. Alternatively, one could seek a determinant with total exponent weight $\Sigma = O(r)$ rather than $\Sigma \asymp r^{3/2}$, but the two-dimensional semigroup counting law $L(T) \asymp T^2$ makes that impossible using only distinct nonnegative mixed monomials.

This gives a concrete design specification:

> A successful determinant proof must add nonarchimedean divisibility, introduce lower-weight functions outside the mixed-monomial semigroup, or exploit the exact two-dimensional row grid so efficiently that $O(\log X)$ nodes suffice.

Status: [paper proof in this report]

The deficit analysis of this section is a paper argument about the limits of the method, not a formalized theorem. It does not weaken any statement proved above; it explains why the explicit $O((\log X)^2)$ bound of Corollary 20.8 cannot be pushed to the conditional $\asymp \log X$ population by tuning constants alone. The conjecture remains open.

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23 Further equivalent reformulations

Several equivalent forms are already formalized in the repository and were catalogued in Section 4. The exact conditional structure of Section 5 also yields useful paper-level reformulations that expose different possible proof technologies. Each restatement below is an exact equivalence, not a heuristic analogy: a proof of any one of them is a proof of Conjecture 2.1.

23.1 Integer points on a transcendental lattice curve

The candidate formulation of Section 2 says that

$$\{(m, n) \in \mathbb{N}^2 : n = m^{\vartheta}\} = \{(2^k, 3^k) : k \in \mathbb{N}_0\}. \tag{75}$$

The curve $n = m^{\vartheta}$ is strictly convex and transcendental, and (75) asserts that its only lattice points are the trivial exponential ones. The arbitrary-node repulsion theorem of Section 19 can be read as a lower bound for the volumes of lattice simplices cut from this curve; the deficit analysis of Section 22 measures exactly how far the current bound is from excluding the nontrivial points.

23.2 Vanishing determinant of four logarithms

Writing $M = 2^x$ and $A = 3^x$, a solution is a positive integer pair with

$$\log M \, \log 3 - \log A \, \log 2 = 0, \quad M, A \in \mathbb{N}, \quad \Longrightarrow \quad (M, A) = (2^k, 3^k). \tag{76}$$

The hypothesis of (76) is a vanishing 2×2 determinant of logarithms of positive integers. This is the most direct bridge to the four-exponentials circle of ideas and to algebraic independence of logarithms, and it is the form in which the kernel-verified four-exponentials reduction is stated.

23.3 Primitive localized-radical exclusion

The kernel-verified radical form of Section 6 is

$$\forall\, w > 1 \text{ odd and power-primitive}, \quad \forall\, d > 0,\ a \geq 0, \qquad 3^a \left(w^{\vartheta}\right)^d \notin \mathbb{N}. \tag{77}$$

If (77) fails with least rational-power index d, then $X^d - c/3^a$ is the exact minimal polynomial of w^{ϑ} over $\mathbb{Q}$ for the corresponding rational c.

This formulation is well suited to Kummer-theoretic, norm, and ramification investigations, because all redundant proper powers have been removed by primitivity and the algebraic degree of the hypothetical radical is known exactly rather than merely bounded.

23.4 Exact growth dichotomy

Write

$$B(T) := \#(\mathcal{C} \cap [2^T, 2^{T+1}))$$

for the Tth dyadic block count. The repository formalizes the cumulative subquadratic equivalence of Section 5; the exact block formula there, namely

$$B(T) = 1 + \left\lfloor \frac{T+1}{\beta} \right\rfloor \qquad \text{under failure with least generator } \beta, \tag{78}$$

gives several immediate variants.

Proposition 23.1 (Block-growth reformulations)**.** *Conjecture 2.1 is equivalent to each of the following:*

1. *the dyadic block counts $B(T)$ are bounded;*
2. $B(T) = o(T)$*;*
3. $\liminf_{T\to\infty} B(T)/T = 0$*;*
4. $B(T) = 1$ *for every* $T \in \mathbb{N}_0$*.*

Proof. If the conjecture holds, every block contains only 2^T, so all four properties hold. If it fails, (78) gives $B(T) = 1 + \lfloor (T+1)/\beta \rfloor$, so $B(T)/T \to 1/\beta > 0$ and none of the first three properties holds. The fourth is visibly equivalent to the candidate statement. □

This is useful because a future analytic theorem need not classify candidates individually. Any uniform $o(T)$ bound for the number of candidates in the Tth dyadic block would prove the conjecture.

The exactness of (78) permits a sharpening of item (4) that is, at first sight, dramatically weaker than the conjecture and yet still equivalent to it.

Proposition 23.2 (Infinitely many singleton blocks suffice)**.** *Conjecture 2.1 is equivalent to the statement*

$$B(T) = 1 \quad \textit{for infinitely many } T \in \mathbb{N}_0.$$

Proof. If the conjecture holds then $B(T) = 1$ for every T, so certainly for infinitely many. Conversely, suppose the conjecture fails, and let $\beta > 0$ be the irrational least generator of the exact solution monoid. By (78),

$$B(T) = 1 \iff \left\lfloor \frac{T+1}{\beta} \right\rfloor = 0 \iff T + 1 < \beta,$$

which holds for at most the finitely many indices $T < \beta - 1$. Hence failure forces $B(T) = 1$ for only finitely many T, and the contrapositive gives the equivalence. □

Remark 23.3. Proposition 23.2 cannot be weakened further to "$B(T) = 1$ for some T". That statement is already known unconditionally and carries no information: the kernel-verified zero-free region $2^x < 4096$ gives $\beta \geq 12$, hence $B(T) = 1$ for $T = 0, \ldots, 10$, and the interval computation of Section 25 pushes the same conclusion to $T \leq 19$ at the weaker **[interval computation]** trust level. The gap between "for some T" and "for infinitely many T" is exactly the gap between a finite check and the conjecture, because none of the present results bounds β from above: a counterexample with a very large least generator would have $B(T) = 1$ for every $T < \beta - 1$, that is, on any range a fixed finite computation could ever reach. Only the infinitude of singleton blocks is decisive.

Status: **[paper proof in this report]** — Proposition 23.1 and Proposition 23.2 are paper proofs in this report, resting on the kernel-verified exact dyadic block counts. A Lean draft `BlockGrowthReformulations` exists but has not yet been accepted by the kernel. **[Lean draft, not yet accepted]**

23.5 Compact synchronized S-integral orbit

The S-integral orbit of Section 5 yields another equivalence, this time one that isolates the synchronization of the two denominators.

Proposition 23.4 (Compact-orbit reformulation)**.** *Conjecture 2.1 fails if and only if there exist integers $M, A > 1$ and an irrational $\beta > 0$ such that, for every $k \geq 1$, the point*

$$\left(\frac{M^k}{2^{\lfloor k\beta \rfloor}}, \frac{A^k}{3^{\lfloor k\beta \rfloor}} \right) \tag{79}$$

lies on the compact arc

$$\mathcal{A} = \{(2^t, 3^t) : 0 \leq t < 1\}.$$

Under these conditions the orbit is dense in $\mathcal{A}$.

Proof. Failure supplies the least generator β, with $M = 2^\beta$ and $A = 3^\beta$; then (79) is exactly $(2^{\{k\beta\}}, 3^{\{k\beta\}})$, and density follows from irrational rotation.

Conversely, arc membership gives $t_k \in [0, 1)$ such that

$$k \log_2 M - \lfloor k\beta \rfloor = t_k = k \log_3 A - \lfloor k\beta \rfloor.$$

Thus $\log_2 M = \log_3 A =: \gamma$, and $0 \leq k\gamma - \lfloor k\beta \rfloor < 1$ for every k. Hence $\lfloor k\gamma \rfloor = \lfloor k\beta \rfloor$, so $|k(\gamma - \beta)| < 1$ for every k; therefore $\gamma = \beta$. Consequently $M = 2^\beta$ and $A = 3^\beta$ are integers while β is nonintegral, so the conjecture fails. The same identity now shows that the orbit equals the irrational-rotation orbit and is dense. □

Status: **[paper proof in this report]** — paper proof in this report; the Lean draft `CompactOrbit` has not yet been accepted by the kernel. **[Lean draft, not yet accepted]**

This formulation isolates the synchronized-denominator feature that a specialized S-integral counting theorem would need to exploit: the two coordinates are S-integral for disjoint sets of primes, yet their denominator exponents are forced to agree exactly at every height.

23.6 Two-dimensional exponential grid

Under failure, every solution coordinate (n, k) gives

$$m_{n,k} = 2^n M^k, \qquad m^{\vartheta}_{n,k} = 3^n A^k,$$

and for every mixed exponent $a + b\vartheta$ with $a, b \in \mathbb{N}_0$,

$$m^{a+b\vartheta}_{n,k} = (2^a 3^b)^n (M^a A^b)^k. \tag{80}$$

Thus the candidate evaluation matrix factors through a two-dimensional row lattice and a two-dimensional column lattice. The conjecture can be viewed as the assertion that this integer exponential grid cannot exist unless (M, A) is a common integral power of $(2, 3)$. This is the most natural starting point for a bespoke 2×2 interpolation determinant that uses more than the arbitrary-node geometry of Section 19, and it is the structure the semigroup hierarchy of Section 20 exploits only partially.

23.7 An operator formulation on the prime-logarithm space

All the reformulations above are Diophantine. There is also a purely linear-algebraic one, which recasts the problem as a statement about multiplication by ϑ acting on a single $\mathbb{Q}$-vector space.

Let

$$W := \operatorname{span}_{\mathbb{Q}}\{\log p \ : \ p \text{ prime}\} \subset \mathbb{R},$$

a $\mathbb{Q}$-vector space of countably infinite dimension, with the $\log p$ as a basis by unique factorization. Multiplication by ϑ does not preserve W, so consider the largest part of W that it does return to W:

$$U := \{\, w \in W \ : \ \vartheta\, w \in W \,\}.$$

This is a $\mathbb{Q}$-subspace, and it is never zero: $\vartheta \log 2 = \log 3 \in W$, so $\mathbb{Q} \log 2 \subseteq U$.

Proposition 23.5 (Operator form)**.** $\dim_{\mathbb{Q}} U = \operatorname{rank} K$*, where* $K = \{v \in \mathbb{Q}_{>0} : v^{\vartheta} \in \mathbb{Q}\}$ *is the multiplicative group of Section 12. Consequently the rational-output form* (22) *— and hence Conjecture 2.1 — is equivalent to*

$$\boxed{\dim_{\mathbb{Q}}\{\, w \in W : \vartheta\, w \in W \,\} = 1.}$$

Proof. An element of W is $w = \sum_p c_p \log p$ with rational c_p, almost all zero. Clearing a common denominator N gives $Nw = \log v$ for a positive rational v. Now $\vartheta w \in W$ says $\vartheta \log v = \sum_p d_p \log p$ with rational d_p; clearing a denominator M turns this into $\vartheta \log(v^M) = \log u$ with $u \in \mathbb{Q}_{>0}$, that is $(v^M)^{\vartheta} = u$, i.e. $v^M \in K$. So U is exactly the $\mathbb{Q}$-span of $\log K$ inside W, whence $\dim_{\mathbb{Q}} U = \operatorname{rank} K$. The conjecture in its rational-output form says $K = \langle 2 \rangle$, of rank one. □

Status: **[paper proof in this report]** — an elementary translation, but a genuinely different framing: it replaces a Diophantine assertion by the statement that the operator "multiply by ϑ" carries only a single line of W back into W.

Two remarks on what this framing does and does not give.

It explains why finite-dimensional arguments cannot apply directly. If some nonzero finite-dimensional subspace $V \subseteq W$ were *ϑ-stable*, then ϑ would be an eigenvalue of the rational matrix representing multiplication by ϑ on V, hence algebraic, contradicting transcendence. So a proof cannot proceed by exhibiting an invariant finite-dimensional subspace: none exists, unconditionally. The subspace U above is not ϑ-stable — $\vartheta U \subseteq W$ but generally $\vartheta U \not\subseteq U$ — and this failure of stability is precisely what blocks the eigenvalue argument.

The natural stable candidate is empty for the same reason. Setting $U_\infty = \{w \in W : \vartheta^k w \in W \text{ for all } k \geq 0\}$ does produce a ϑ-stable space, but it is either zero or infinite-dimensional, and infinite dimension is automatic whenever it is nonzero: if $0 \neq w \in U_\infty$, the powers $\vartheta^k w$ are $\mathbb{Q}$-linearly independent, since a relation $\sum_k c_k \vartheta^k w = 0$ with $w \neq 0$ would make ϑ algebraic. So the operator picture reproduces the transcendence barrier exactly, rather than circumventing it. Its value is diagnostic: it shows the obstruction is not a missing estimate but the absence of any finite-dimensional invariant structure to work with.

24 A consolidated catalogue of equivalent formulations

The preceding sections restate Conjecture 2.1 many times over: as a candidate-base identity, as a localized radical exclusion, as a counting dichotomy, as a statement about fractional parts, and as a closure property of the solution monoid. Because these restatements were introduced where they were needed rather than where they belong together, this section collects them in one place.

Throughout, $\vartheta = \log_2 3$,

$$\mathcal{S} = \{x \in \mathbb{R} : 2^x, 3^x \in \mathbb{Z}\}, \qquad \mathcal{C} = \{m \in \mathbb{N} : m > 0,\ m^\vartheta \in \mathbb{N}\},$$

which correspond under $x \mapsto 2^x$ by the exponent–candidate bijection of Section 2, and

$$B(T) := \#(\mathcal{C} \cap [2^T, 2^{T+1}))$$

is the Tth dyadic block count.

24.1 How to read the catalogue

Every item (E1)–(E26) below is an *exact biconditional*: a proof of any one of them is a proof of Conjecture 2.1, and a disproof of any one of them is a counterexample. None of them is proved. The status tag on each item records the strength of the evidence for the *equivalence*, never for the conjecture: **[kernel-verified Lean]** means the biconditional itself has been accepted by the Lean kernel, **[paper proof in this report]** means it is a paper proof in this report, and **[classical/open background]** means the equivalence is obtained by combining a paper proof here with a classical transcendence theorem.

Two adjacent notions are deliberately excluded from the catalogue and treated separately in Section 24.8 below: *sufficient conditions*, which are one-way implications from hypotheses that are themselves unproved, and *unconditional partial results*, which settle a bounded region rather than the statement. Conflating the three is the most common way to overstate what the corpus establishes.

24.2 Cluster A: normalized restatements

These three items are the conjecture rewritten in the coordinates in which the rest of the corpus works. They carry no arithmetic content beyond that bijection, and their value is that each suggests a different toolkit.

(E1) **[kernel-verified Lean]** *Candidate form.* $\mathcal{C} = \{2^n : n \in \mathbb{N}_0\}$; equivalently, the only positive integers m with $m^\vartheta \in \mathbb{N}$ are the powers of two (Section 2).

(E2) **[kernel-verified Lean]** *Vanishing four-log determinant.* The only positive integer pairs (M, A) with $\log M \, \log 3 - \log A \, \log 2 = 0$ are $(M, A) = (2^n, 3^n)$, $n \in \mathbb{N}_0$ (Section 2); this is the 2×2 logarithmic determinant whose entries include two logarithms of unknown integers, and the form in which the four-exponentials reduction of Section 7 is stated.

(E3) **[kernel-verified Lean]** *One-parameter subgroup intersection.* The real one-parameter subgroup $\Gamma = \{(2^t, 3^t) : t \in \mathbb{R}\}$ of the multiplicative torus meets $\mathbb{N}^2$ exactly in $\{(2^n, 3^n) : n \in \mathbb{N}_0\}$ (Section 2); a geometric restatement of (E2), with the difficulty that the direction vector $(\log 2, \log 3)$ is transcendental rather than algebraic.

Status: **[kernel-verified Lean]** — the exponent–candidate bijection is `exists_twoBaseNaturalCandidate_of_two_three_rpow_integer` together with `TwoBaseNaturalCandidate.integer_powers`, and the equivalence (E1) is

`alaogluErdosConjecture_iff_candidates_are_powers_of_two`.

Items (E2) and (E3) are the same content transported along that bijection and along $m \mapsto (m, m^\vartheta)$; they carry no separate Lean identifier.

24.3 Cluster B: localized radicals

This cluster is the most compressed form of the problem known: a statement quantified over all real x becomes a statement about one integer radical at a time, with all 2-adic and all but one 3-adic degree of freedom eliminated. It is also the cluster closest to established transcendence theory.

(E4) **[kernel-verified Lean]** *Localized radical exclusion.* There is no odd integer $u > 1$ with $u^\vartheta \in \mathbb{Z}[1/3]$ (Section 6).

(E5) **[kernel-verified Lean]** *Integral localized form.* There is no pair (u, a) with $u > 1$ odd, $a \in \mathbb{N}_0$ and $3^a u^\vartheta \in \mathbb{N}$; equivalently, no candidate has odd part greater than one (Section 6).

(E6) **[kernel-verified Lean]** *Primitive localized radical exclusion.* There is no power-primitive odd $w > 1$ and $d \geq 1$ with $\left(w^\vartheta\right)^d \in \mathbb{Z}[1/3]$ (Section 6); primitivity removes every proper-power redundancy, and under failure the minimal polynomial of w^ϑ is known exactly.

(E7) [**kernel-verified Lean**] *Symmetric base-swapped form.* With $\eta = \log_3 2 = 1/\vartheta$, there is no integer $v > 1$ with $3 \nmid v$ and $v^\eta \in \mathbb{Z}[1/2]$ (Appendix C); the two normalizations produce the same least exponent, so the reduction is not an artifact of privileging the base 2.

Status: [**kernel-verified Lean**] — (E4) is

`alaogluErdosConjecture_iff_odd_rpow_not_localized`

in `Localization`, restated as

`alaogluErdosConjecture_iff_oddLocalizedRadicalExclusion`

in `RadicalReformulation`; (E5) is the independently proved candidate-level variant

`alaogluErdosConjecture_iff_no_odd_multiple_candidate`;

(E6) is

`alaogluErdosConjecture_iff_primitiveOddLocalizedRadicalExclusion`

in `PrimitiveRadical`; and (E7) is

`alaogluErdosConjecture_iff_threeFreeLocalizedRadicalExclusion`

in `RadicalReformulation`. The equivalences are kernel-verified; the exclusions on their right-hand sides are the open conjecture.

24.4 Cluster C: counting and block growth

The counting cluster replaces an integrality statement about individual candidates by an upper bound on how many of them there can be. It is attractive because an analytic theorem need never identify a candidate, and unattractive because the target quantities are extremely far from the arithmetic conclusion; Section 22 measures that distance precisely.

(E8) [**kernel-verified Lean**] *Subquadratic cumulative growth.* $\#(\mathcal{C} \cap [1, 2^T])$ is $o(T^2)$ as $T \to \infty$ (Section 5).

(E9) [**paper proof in this report**] *Bounded blocks.* The dyadic block counts $B(T)$ are bounded (Section 23).

(E10) [**paper proof in this report**] *Sublinear blocks.* $B(T) = o(T)$ (Section 23).

(E11) [**paper proof in this report**] *Sublinear along a subsequence.* $\liminf_{T\to\infty} B(T)/T = 0$ (Section 23).

(E12) [**paper proof in this report**] *Singleton blocks.* $B(T) = 1$ for every $T \in \mathbb{N}_0$ (Section 23); an immediate restatement of (E1) blockwise.

(E13) [**paper proof in this report**] *Infinitely many singleton blocks.* $B(T) = 1$ for infinitely many $T \in \mathbb{N}_0$ (Section 23).

Remark 24.1. Item (E13) is the sharpest illustration of how exact the conditional structure theory is. It looks dramatically weaker than (E12), yet the exact block formula $B(T) = 1 + \lfloor (T+1)/\beta \rfloor$ under failure with least generator β forces $B(T) = 1$ to hold only for the finitely many $T < \beta - 1$. It cannot be weakened further to "$B(T) = 1$ for some T": that statement is already known unconditionally and carries no information, since the kernel-verified zero-free region gives $\beta \geq 12$. The gap between "for some" and "for infinitely many" is exactly the gap between a finite check and the conjecture.

Status: [kernel-verified Lean] for (E8) —

`alaogluErdosConjecture_iff_candidate_count_below_two_pow_subquadratic`.

[paper proof in this report] for (E9)–(E13) — paper proofs in this report, resting on the kernel-verified exact dyadic block counts of `ExactCounting` and the quadratic asymptotic of `QuadraticAsymptotic`. A Lean draft `BlockGrowthReformulations` exists but has not been accepted by the kernel. **[Lean draft, not yet accepted]**

24.5 Cluster D: orbits and fractional parts

Under failure the solution set is an irrational-rotation orbit in disguise. This cluster converts the conjecture into the assertion that no such orbit can be arithmetically synchronized, and into three gap statements on the unit interval.

(E14) **[paper proof in this report]** *Compact synchronized S-integral orbit.* There are no integers $M, A > 1$ and irrational $\beta > 0$ such that $(M^k/2^{\lfloor k\beta \rfloor}, A^k/3^{\lfloor k\beta \rfloor})$ lies on the compact arc $\{(2^t, 3^t) : 0 \leq t < 1\}$ for every $k \geq 1$ (Section 23). The two coordinates are S-integral for disjoint prime sets, yet their denominator exponents must agree exactly at every height.

(E15) **[kernel-verified Lean]** *Avoided interval.* Some nonempty open interval $(u, v) \subseteq (0, 1)$ contains no fractional part $\{y\}$ of a nonintegral solution y (Section 4).

(E16) **[kernel-verified Lean]** *Gap at the top.* The fractional parts of nonintegral solutions are bounded away from 1 (Section 4).

(E17) **[kernel-verified Lean]** *Gap at the bottom.* The fractional parts of nonintegral solutions are bounded away from 0 (Section 4).

Items (E15)–(E17) are unconditional biconditionals, and they say something quantitative about the elementary gap bounds of Section 4: those bounds place $\{x\}$ away from 0 and 1 by margins that decay like $2^{-\lfloor x \rfloor}$, and the equivalences show that *any* height-independent margin, however small and wherever located, is already the full conjecture. The decay is therefore essential rather than an artifact of the estimate.

Status: [paper proof in this report] for (E14) — paper proof in this report; the Lean draft `CompactOrbit` has not been accepted by the kernel ([Lean draft, not yet accepted]). [kernel-verified Lean] for (E15)–(E17) —

`alaogluErdosConjecture_iff_fract_avoids_interval`,

`alaogluErdosConjecture_iff_fract_bounded_away_one`,

`alaogluErdosConjecture_iff_fract_bounded_away_zero`

in `FractionalPartDensity`.

24.6 Cluster E: closure, smoothness, and algebraicity

The last cluster is the least classical and, for that reason, the most interesting as a source of new attacks. It says that the conjecture is equivalent to the failure of a single nonlinear closure property: the solution monoid, which is trivially closed under addition, must not also be closed under multiplication.

(E18) [paper proof in this report] *Multiplicative closure.* $\mathcal{S}$ is closed under multiplication.

(E19) [paper proof in this report] *Square closure.* $\mathcal{S}$ is closed under squaring.

(E20) [paper proof in this report] *Star closure.* $\mathcal{C}$ is closed under the operation $m \star n = 2^{\log_2 m \log_2 n}$.

(E21) [paper proof in this report] *Star idempotence.* Every $m \in \mathcal{C}$ satisfies $m \star m \in \mathcal{C}$.

(E22) [paper proof in this report] *Smooth outputs.* Every $x \in \mathcal{S}$ has both 2^x and 3^x 3-smooth.

(E23) [paper proof in this report] *Smooth product output.* Every $x \in \mathcal{S}$ has 6^x 3-smooth.

(E24) [classical/open background] *Square-exponent algebraicity.* Every $x \in \mathcal{S}$ has $6^{x^2} \in \overline{\mathbb{Q}}$.

(E25) [classical/open background] *Split square-exponent algebraicity.* Every $x \in \mathcal{S}$ has both $2^{x^2} \in \overline{\mathbb{Q}}$ and $3^{x^2} \in \overline{\mathbb{Q}}$.

(E26) [classical/open background] *Mixed-product algebraicity.* All $x, y \in \mathcal{S}$ satisfy $6^{xy} \in \overline{\mathbb{Q}}$.

Remark 24.2 (Manufacturing a third base). Items (E22)–(E26) admit a concrete reading. By the kernel-verified 3-smooth classification of Section 4, a hypothetical nonintegral solution β admits an integer base B with $B^\beta \in \mathbb{Z}$ only when B is 3-smooth. It would therefore suffice to manufacture *one* non-3-smooth positive integer B with $B^\beta \in \mathbb{Z}$. The obvious candidates $B = 2^\beta$, $B = 3^\beta$, $B = 6^\beta$ have β-th powers 2^{β^2}, 3^{β^2}, 6^{β^2}, so "manufacture a third base" is exactly the square-closure problem of (E19) in different language. Whenever the chosen base is not 3-smooth its β-th power is already known to be transcendental by six exponentials, so any proof of algebraicity would finish the conjecture at once. What is missing is a single bridge:

$$\boxed{\text{derive one nonlinear algebraic value from the two linear integer values.}}$$

Neither unique factorization, nor the ordinary exponent laws, nor linear forms in logarithms supplies it.

Status: [**paper proof in this report**] for (E18)–(E23) — paper proofs in this report, from the closure equivalences and the output-smoothness theorem; no Lean formalization yet. [**classical/open background**] for (E24)–(E26) — the equivalences combine those paper proofs with the classical six exponentials theorem, which supplies transcendence of a non-3-smooth base raised to the hypothetical exponent. None of the nine items has been accepted by the kernel.

24.7 Which formulations are worth attacking

The twenty-six items are not equally promising, and it is worth saying so plainly. Cluster C is exact but analytically distant: the deficit analysis of Section 22 shows the determinant hierarchy reaching $O((\log X)^2)$ while a counterexample supplies $\asymp \log X$ candidates, a full logarithm short of the $o(T)$ that (E10) would need. Cluster A is a change of coordinates, useful for orientation and for locating the four-exponentials barrier, but not itself a method. Cluster B is closest to established transcendence theory and is where a Kummer-theoretic, norm, or ramification argument would land; its smallest open instance is the irrationality of $3^{\log_2 3}$. Cluster D isolates a synchronization phenomenon that a bespoke S-integral counting theorem might exploit. Cluster E is the only one whose hypotheses are elementary and nonlinear at the same time, and it is the natural home of any attempt to manufacture a third base.

24.8 Equivalences versus sufficient conditions

An equivalence and a sufficient condition play very different roles, and the distinction is easy to lose in a corpus this size. Items (E1)–(E26) are biconditionals: each is exactly as strong as Conjecture 2.1, so proving any of them proves the conjecture and no weakening of any of them is safe without checking that the reverse implication survives. The statements below are strictly one-way. Each is a hypothesis that is itself open, each implies the conjecture, and none is known to follow from it — so none belongs in the catalogue.

(S1) *The real four exponentials conjecture.* If for all $\mathbb{Q}$-linearly independent x_1, x_2 and $\mathbb{Q}$-linearly independent y_1, y_2 some $e^{x_i y_j}$ is transcendental, then Conjecture 2.1 holds. The implication is kernel-verified (Section 7); the hypothesis is open, and is the reason no routine combination of present methods closes the problem.

(S2) *Schanuel's conjecture.* It implies algebraic independence of the logarithms of the primes, hence (S3), hence Conjecture 2.1 (Section 12).

(S3) *Prime-log-product independence.* If the real numbers $\log p \log q$, indexed by pairs of primes $p \le q$, are linearly independent over $\mathbb{Q}$, then Conjecture 2.1 holds (Section 12). This criterion mentions no exponentials at all, is strictly weaker than Schanuel, and is not known to imply or to follow from (S1).

Status: [kernel-verified Lean] for (S1) —

`alaogluErdosConjecture_of_realFourExponentialsConjecture`

in `FourExponentialsReduction`, which formalizes the implication without assuming its hypothesis. **[paper proof in this report]** for (S2) and (S3) — paper proofs in this report; a Lean module `LogProductIndependence` recording the independence hypothesis as an explicit `Prop`, never asserted, is in preparation, with headline statement

`alaogluErdosConjecture_of_primeLogProductIndependence`.

The finite prime-support form of (S3) is weaker still and correspondingly more approachable: if the products $\log p \log q$ over a fixed finite prime set $S \ni 2, 3$ are $\mathbb{Q}$-linearly independent, then no counterexample has both outputs supported on S. For $S = \{2, 3\}$ this reduces to $\mathbb{Q}$-linear independence of 1, ϑ and ϑ^2, which would exclude every 3-smooth counterexample. That is a partial exclusion, not an equivalence.

A third category deserves the same care. Unconditional partial results settle a bounded region and are neither equivalences nor sufficient conditions: the kernel-verified zero-free region gives $2^x \geq 4096$, hence $x \geq 12$, for every nonintegral solution (Section 25), **[kernel-verified Lean]**; the directed-rounding scan through 2^{27} extends the same conclusion at the weaker trust level of a software computation, **[interval computation]**, and is not a kernel proof. The kernel-verified exponent bound remains $x \geq 12$. No item in this section, and no combination of them, proves Conjecture 2.1; the catalogue records where a proof could enter, not that one has.

25 Independent finite verification

Exactly one zero-free region in this report is a kernel certificate. Lean proves that a solution with $2^x < 4096$ forces $x \in \mathbb{Z}$, so that a nonintegral solution satisfies $x \geq 12$; the statement lives in `FiniteCheck4096` as

`twelve_le_of_not_integer_of_two_three_rpow_integer`

and it is the only finite exclusion that any other statement here may use without adding trust assumptions. **[kernel-verified Lean]**

One further finite computation pushes the excluded range far past 4096: a directed-rounding exclusion through 2^{27}, described in Section 40. It supersedes the earlier interval scans, which are not restated here. It is a software result: it does not replay in the Lean kernel, it is not interchangeable with `FiniteCheck4096`, and it does not raise the kernel-verified exponent bound above $x \geq 12$. They are recorded here because they strengthen the explicit conditional picture, test the conditional models of Section 5 against data, and supply the extremal cases that a future generated certificate must clear.

25.1 What each scan must certify

Write $\vartheta = \log_2 3$, and let m be a positive integer that is not a power of two. Excluding m means proving $m^\vartheta \notin \mathbb{N}$, and it suffices to exhibit an integer n with $n < m^\vartheta < n + 1$. Exponentiation is never performed: the pair of strict inequalities is equivalent, after taking logarithms and clearing the positive factor $\log 2$, to

$$\log m \,\log 3 - \log n \,\log 2 > 0, \tag{81}$$

$$\log(n+1) \,\log 2 - \log m \,\log 3 > 0. \tag{82}$$

Every logarithm and every product occurring in the scanned ranges is positive, so the required rounding directions are monotone and unambiguous.

In both scans an ordinary floating-point evaluation merely *proposes* $n \approx \lfloor m^\vartheta \rfloor$. Once certified lower bounds for (81) and (82) are both strictly positive, the floating proposal has no remaining logical role: it is a witness generator, not a step of the argument. A locator error can therefore create a spurious failure report, but it cannot create a spurious exclusion. Powers of two are skipped rather than tested, since 2^q has the integral value $(2^q)^\vartheta = 3^q$; an exactly integral value at a non-power of two would satisfy neither strict inequality for any n and would be reported as a failure.

25.2 Superseded scans

Two earlier software exclusions — an interval-arithmetic scan through 2^{20} and a directed-rounding scan through 2^{26} — have been absorbed by the extension through 2^{27} of Section 40, which uses the same directed-rounding method, the same margin conventions and the same manifest discipline over the wider range. Only the strongest exclusion is stated in this report; the superseded ranges and their chunk tables are not reproduced.

25.3 The trust boundary, stated plainly

Kernel. $2^x < 4096$ and $2^x, 3^x \in \mathbb{Z}$ imply $x \in \mathbb{Z}$; a nonintegral solution has $x \geq 12$. **[kernel-verified Lean]** `FiniteCheck4096`.
Interval scan. Under CPython and `mpmath` interval arithmetic, a nonintegral solution has $x > 20$. **[interval computation]**
Directed-rounding scan. Under the C source, GCC code generation, OpenMP scheduling, the MPFR and GMP shared libraries, the x86-64 arithmetic environment, and the operating system, a nonintegral solution has $x > 26$. **[interval computation]**

MPFR's correct-rounding contract makes the numerical reasoning crisp, and the interval scan is reproducible from a short script, but neither removes the software assumptions listed above. Nothing in Section 5 or Section 27 that is tagged **[kernel-verified Lean]** depends on either scan; wherever a numeric threshold is used inside a kernel-verified statement, that threshold is 12.

A drafted extension of the kernel region. An attempt to raise the kernel-checked region from $2^x < 4096$ to $2^x < 16384$ exists in draft. It follows the `FiniteCheck4096` pattern, splitting the enlarged table into four chunks so that no single `decide` call carries the whole certificate. The certificate rows themselves verify: each chunk's Boolean recursion evaluates and each row's exact integer comparison is accepted. What has not been accepted is the assembly — the per-chunk length lemmas that glue the four tables into a single covering statement exceeded the elaborator's recursion limit, so the module does not compile end to end and no theorem is available for use. Until that is resolved the kernel bound is unchanged at $x \geq 12$. **[Lean draft, not yet accepted]**

From scan to kernel certificate. The repository's own certificate architecture (Section 4) shows the stronger trust model: generate rational certificates externally, revalidate them independently, and replay only exact integer comparisons in Lean by `decide`. The next step is certificate extraction, not a larger blind scan. For each block one records rational lower and upper approximants to ϑ together with integer inequalities sufficient to replay every exclusion, exploiting monotonicity and convexity so that a single certificate covers a run of consecutive m rather than storing one record per input. A scalable formal certificate should share convergent enclosures across ranges, chunk the data modules to bound elaboration memory — the failure mode of the 16384 draft is precisely that the chunk bookkeeping, not the arithmetic, is the expensive part — and rely on verified binary powering rather than giant literals. The extremal pairs identified by both scans predict the required tier depth. A practical intermediate target is 2^{16}, followed by a generated hierarchical certificate through 2^{26}.

26 Attempts at a full proof and their exact failure points

This section records the principal avenues that were pushed until they broke, together with the exact place where each one breaks. The purpose is not merely negative. Every failure point names a quantitative or structural feature that a successful proof must improve, and several of them turn out to be the same feature seen from different sides; the last subsection gives that common feature its exact name.

26.1 Collapsing rank two to rank one

The multiplicative-rank bound is kernel-verified: the exponent vectors of the candidate set span a subgroup of rank at most two. Under failure the rank-two possibility is not vague but known exactly, $\mathcal{C} = \langle 2, M \rangle$, with M the primitive generator of Section 5. Nothing in the rank machinery distinguishes this free rank-two monoid from the allowed monoid of powers of two.

Failure point. A rank bound is an upper bound on dimension; it supplies no mechanism for forcing the dimension down to one. Excluding the second generator *is* the original transcendence problem, restated.

26.2 Rational approximation to a least generator

Assume failure and let β be the least nonintegral solution, with $M = 2^{\beta}$ and $A = 3^{\beta}$. Let p/q approximate β and put $\varepsilon = q\beta - p$. Then

$$M^q - 2^p = 2^p\big(e^{\varepsilon \log 2} - 1\big), \tag{83}$$

$$A^q - 3^p = 3^p\big(e^{\varepsilon \log 3} - 1\big). \tag{84}$$

Both left-hand sides are integers, so a nonzero value has absolute value at least 1, which for small ε gives bounds of the shape

$$|\varepsilon| \gtrsim 2^{-p} \qquad \text{and} \qquad |\varepsilon| \gtrsim 3^{-p}.$$

These are exponentially tiny in q. Continued fractions only guarantee infinitely many approximants with $|\varepsilon| \ll q^{-1}$, and Baker-type lower bounds for ordinary linear forms in logarithms are polynomial in the relevant heights [3, 10]. Neither scale conflicts with a lower bound as weak as M^{-q} or A^{-q}. Multiplying (83) by (84), or eliminating ε between them, only recovers $\log_2 3$ to first order; the higher terms are far too small to create an integer contradiction.

Failure point. The two integer gaps are correlated rather than independent, so the two exponential inequalities carry essentially one piece of information, not two.

Observation 26.1. Any rational-approximation proof needs more than "a nonzero integer has absolute value at least one." It needs a divisibility lower bound growing exponentially with q, or an approximation to β exponentially better than continued fractions supply in general.

26.3 Equidistribution against exponentially shrinking targets

At paper level the orbit $\{k\beta\}$ is equidistributed modulo 1 (Section 5); the finite-Fourier convergence needed for Weyl's criterion, covering every fixed mode and every finite trigonometric polynomial, is kernel-verified. Meanwhile, integrality of 2^{n+t} between adjacent integral powers pushes the fractional part t away from both endpoints by an amount comparable with 2^{-n}, and the base-3 condition imposes a comparable 3^{-n} restriction; at height k the admissible target width is of exponential scale M^{-k}, a bound also kernel-verified in the shrinking-target module.

Equidistribution controls visits to fixed intervals and, with effective discrepancy, to polynomially shrinking ones along subsequences. It does not force visits to a summable family of targets of length e^{-ck}: the Borel–Cantelli heuristic predicts only finitely many such visits for a typical rotation, and numbers with bounded partial quotients show that equidistributed multiples can avoid exponentially small targets forever. The exact conditional monoid is perfectly compatible with that behaviour.

Failure point. The blockwise Weyl theorems are structurally correct but point the wrong way: they describe how already existing solutions distribute inside each block, not how likely arbitrary exponents are to hit the integer lattice. Closing the route needs Diophantine information specific to the transcendental β with $2^\beta, 3^\beta \in \mathbb{N}$, not generic metric theory.

26.4 Finite differences and additive progressions

For $f(t) = t^\vartheta$ the repository proves the exact mean-value formula for every forward difference order r,

$$\Delta_h^r f(n) = (\vartheta)_{\underline{r}}\; h^r \xi^{\vartheta - r} \qquad (n < \xi < n + rh),$$

so that if all $r+1$ values $f(n), f(n+h), \ldots, f(n+rh)$ are integers, the left side is an integer; under $|\,(\vartheta)_{\underline{r}}\,| h^r < n^{r-\vartheta}$ its absolute value lies strictly between 0 and 1. This is a sharp uniform obstruction to candidate arithmetic progressions, and its rational effective three-term shadow $h^{400} \le n^{83}$ is kernel-verified together with the arbitrary-order form.

The difficulty is combinatorial rather than analytic. Conditional failure produces only $O(\log X)$ candidates per multiplicative block, far below the positive-density regime of Roth-type theorems, and a rank-two multiplicative monoid need not contain long additive progressions at all: many points in a short interval need not be equally spaced. A three-term progression of candidates would solve the S-unit equation

$$2^{n_1}M^{k_1} + 2^{n_3}M^{k_3} = 2^{n_2+1}M^{k_2},$$

and general S-unit theory yields finiteness statements, not the required nonexistence.

Failure point. Sparse exact structure and sharp local curvature are mutually consistent. The arbitrary-node determinant of Section 19 removes the equal-spacing hypothesis, but its packing exponent is still too weak to contradict the exact conditional count.

26.5 Radical fields and conjugates

The normalized radical $E = w^{\vartheta}$ is algebraic with exact binomial minimal polynomial $X^d - c/3^a$, and its conjugates are $\zeta_d^j E$; degree, norm, and denominator support are all kernel-verified (Section 6). What this does not supply is a way to transport the real analytic identity $\log E = \vartheta \log w$ to the other conjugates.

Failure point. The chosen positive real logarithm is not Galois-equivariant: the remaining roots differ by complex logarithmic periods, and taking norms merely reproduces $E^d = c/3^a$. Ramification is equally inconclusive, since the equation relates the constrained primes through a quotient of logarithms rather than through a field embedding; standard Kummer theory sees the radical equation but not the special value $\vartheta = \log_2 3$.

Observation 26.2. A radical-field proof needs a genuinely mixed invariant: something simultaneously sensitive to the algebraic conjugates of E and to the logarithmic relation among $2, 3, w, E$. At present the algebraic and the real-analytic arguments meet only at the four-logarithm determinant.

26.6 The compact S-integral orbit

From the least generator define

$$U_k := \frac{M^k}{2^{\lfloor k\beta \rfloor}} = 2^{\{k\beta\}} \in \mathbb{Z}[1/2], \qquad V_k := \frac{A^k}{3^{\lfloor k\beta \rfloor}} = 3^{\{k\beta\}} \in \mathbb{Z}[1/3].$$

The points (U_k, V_k) lie on the compact analytic arc $\mathcal{A} = \{(2^t, 3^t) : 0 \le t < 1\}$ and are dense there because β is irrational. This is the compact-orbit reformulation of Section 23, and it invites o-minimal and S-integral point counting; Butler relates special cases of Wilkie-type counting to real three- and four-exponentials statements [5].

Failure point. The counting is not strong enough by an enormous margin. The logarithmic heights of the orbit points grow linearly in k, so the arc carries only about $\log H$ orbit points of height at most H, far below the subpolynomial count H^{ε} that Pila–Wilkie style theorems already tolerate outside algebraic parts. What would suffice is a denominator-sensitive refinement counting only points whose coordinate denominators are supported exactly on $\{2\}$ and on $\{3\}$ with synchronized exponents; no theorem of that strength appears in the inspected corpus.

26.7 Forcing an auxiliary base

If one could exhibit an integer $b > 1$ multiplicatively independent of 2 and 3 with b^x algebraic — or rational with suitably controlled denominator — then the six exponentials theorem, or the repository's rational-third-base theorem, would force $x \in \mathbb{Q}$ and hence $x \in \mathbb{Z}$. The five exponentials theorem [10, 8] similarly makes $e^{\gamma x}$ transcendental for every nonzero algebraic γ, given the four algebraic corner exponentials, and the kernel-verified 3-smooth classification is the formalized integer-level shadow of the six exponentials theorem.

Failure point. The hypotheses provide no such b. The natural auxiliary quantities $p^x = e^{x \log p}$ have transcendental coefficient $\log p$, so the five exponentials theorem does not convert a divisor of M or A into a contradiction; and the six exponentials theorem controls b^x, not the distinct question of whether b divides the integer 2^x. The obvious constructions $b = 2^u 3^v$, and any construction from M and A, collapse to the same rank-two logarithmic span. An auxiliary base must come from an arithmetic operation whose exponential behaviour stays controlled while its logarithm leaves that span — an algebraic independence problem in disguise.

26.8 Exploiting the primitive output pair

The primitive pair (M, A) is canonical and rigid: kernel-verified theorems show that it admits no matched depth, no common perfect power, and no nontrivial affine decomposition, and that the odd core and its coordinates are unique. Every structural handle one might hope to grip has therefore been shown to be absent by design rather than by accident.

Failure point. After all of that normalization, what remains is exactly the single relation $\log M \, \log 3 = \log A \, \log 2$, a product of logarithms lying outside the scope of linear-forms machinery.

26.9 The linear-form-in-log-products formulation of the wall

The obstruction can be given one further exact name. A counterexample is a pair of integers $m, A \geq 2$ with $\log m \, \log 3 = \log A \, \log 2$. Expanding both integers over their prime factorizations $m = \prod_p p^{m_p}$ and $A = \prod_p p^{a_p}$, failure of Conjecture 2.1 is equivalent to the nontrivial vanishing of

$$D \;=\; \sum_p \bigl(m_p \log 3 - a_p \log 2\bigr) \log p, \tag{85}$$

an *integer-coefficient linear form in the products of two prime logarithms* $\{\log p \cdot \log 2,\ \log p \cdot \log 3\}$. Baker's theory bounds linear forms in logarithms with *algebraic* coefficients [3]; here every coefficient of $\log p$ is itself a transcendental logarithm, and no lower-bound technology exists for such forms. Indeed even the $\mathbb{Q}$-linear independence of the family $\{\log p \cdot \log q : p < q \text{ prime}\}$ is open — it follows from, and is essentially a fragment of, the four exponentials circle of conjectures. Section 12 turns exactly that independence statement into a sufficient condition for the conjecture.

Failure point. Every route examined above — determinants, product formulas in the radical field, auxiliary exponentials, equidistribution against shrinking targets, orbit counting — terminates at a special case of this one independence statement. A genuine proof therefore requires either a fundamentally new lower bound for linear forms in log-products, or a mechanism exploiting the positivity and integrality of the coefficients (m_p, a_p) in (85) that has no analogue in the general four-exponentials setting.

27 Consolidated research program

The two source programs overlapped substantially; the exact structure theory of Section 5 settles some directions outright and re-prioritizes the rest. The list below is ordered by how directly each item attacks the quantitative deficit of Section 22 — the one missing logarithm between what the determinant hierarchy delivers and what a contradiction requires.

27.1 Priority A: p-adic amplification of semigroup determinants

The semigroup determinant $D = \det\big(m_i^{a_j}(m_i^\vartheta)^{b_j}\big)$ is a positive integer, and under the exact monoid every entry has explicit valuations obtained from $m_i = 2^{n_i}M^{k_i}$ and $m_i^\vartheta = 3^{n_i}A^{k_i}$:

$$v_2\Big(\text{entry}_{ij}\Big) = a_j\,(n_i + v_2(M)\,k_i) + b_j v_2(A)\,k_i,$$
$$v_3\Big(\text{entry}_{ij}\Big) = a_j v_3(M)\,k_i + b_j\,(n_i + v_3(A)\,k_i).$$

What this would buy is decisive: archimedean reasoning offers only the universal lower bound $|D| \geq 1$, whereas a systematic p-adic factor supplies $|D| \geq p^L$ and could erase the missing logarithm outright.

Question 27.1. *Can one choose $O(\log X)$ candidate rows and mixed-monomial columns so that the resulting nonzero determinant satisfies $\log|D|_p^{-1} \gg (\log X)^2$ for $p = 2$ or $p = 3$, while the archimedean estimate remains $\log|D|_\infty = o\big((\log X)^2\big)$?*

The difficulty is that a naive row or column factor is lost because the exponent set contains $(0, 0)$; only more subtle min-plus determinant estimates, applied after suitable row and column finite-difference operations, have a chance. The exact affine primitivity and the gcd condition on the normalized degrees should be fed into the estimate rather than discarded.

27.2 Priority B: exploit the full (n, k) grid, not sorted magnitudes

The arbitrary-node argument forgets the canonical coordinates and retains only the sorted integers m_i. Restoring them exposes rectangular, often Kronecker-like, bivariate exponential matrices, which is exactly the balanced 2×2 setting where ordinary six exponentials estimates stop; integrality, positivity, and the one-depth-zero normalization may create a margin absent from the general four-exponentials problem. The right experiment is not another large generic determinant but an optimized minor selected from triangular coordinate regions $n + k\beta \leq T$ and $a + b\vartheta \leq L$, with exact accounting of archimedean size, zero estimates, and 2- and 3-adic valuations. The hard part is that the optimization is genuinely two-parameter and the useful shapes are not predicted by any current heuristic, so a search must precede a theorem.

27.3 Priority C: denominator-sensitive counting on the compact arc

The compact-orbit reformulation asks for infinitely many points on $y = x^\vartheta$ in $[1, 2) \times [1, 3)$ whose first denominator is a power of 2, whose second denominator is a power of 3, and whose denominator exponents are synchronized by the same integer $\lfloor k\beta \rfloor$. A theorem asserting that a compact transcendental arc with nonzero curvature carries $o(H)$ points of logarithmic height at most H with denominators supported on two prescribed disjoint prime sets and a common exponent parameter would close the case, since the failed-conjecture orbit has $\asymp H$ such points. What makes it hard is that standard rational-point counting is sensitive to denominator *size* but not to denominator *support* or synchronization; Butler's connection between Wilkie-type statements and real exponential conjectures [5] suggests this direction sits on the transcendence frontier rather than being a routine application.

27.4 Priority D: mixed archimedean–radical invariants

The normalized radical $E = w^{\vartheta}$ has exact degree d and binomial minimal polynomial, so the Kummer field $\mathbb{Q}(E, \zeta_d)$ is completely explicit. The questions worth pursuing are whether the logarithmic relation $\log E \log 2 - \log w \log 3 = 0$ forces an impossible regulator or height relation there; whether simultaneous normalization at both outputs constrains the two normalized degrees through field intersections or discriminants beyond the known gcd condition; whether all complex branches $\log(\zeta_d^j E) = \log E + 2\pi i j/d$ can populate a multi-logarithm determinant with algebraic entries of controlled height; and whether a p-adic logarithm branch is compatible with the real identity after raising to the normalized degree. A warning is essential, and it is the same one that closed Section 26.5: merely taking norms or conjugates of $X^d - c/3^a$ reproduces known relations, so the new invariant must mix conjugation with the special logarithmic direction.

27.5 Priority E: formalize the new determinant results

The arbitrary-node repulsion theorem of Section 19 and the semigroup generalized-Vandermonde hierarchy of Section 20 are currently **[paper proof in this report]**. Formalizing them converts the strongest new tools in this report into reusable kernel-checked objects and, just as importantly, forces the constants in the gap theorem and the explicit dyadic bound to be stated uniformly. Lean drafts of `ArbitraryNodeDeterminant`, `MixedExponentSemigroup`, `SemigroupDeterminant`, and `FallingFactorialBound` already exist but are **[Lean draft, not yet accepted]**; the module layout is described in Section 28. The hard part is analytic bookkeeping rather than mathematics: divided differences at arbitrary nodes, falling-factorial growth, and generalized-Vandermonde positivity all need uniform statements before the main theorems can be assembled cheaply.

27.6 Priority F: force a third algebraic exponential

The denominator-controlled theorem in `RationalThirdBase` is a precise gateway: for an auxiliary integer q containing a prime outside $\{2, 3\}$ it suffices, in several forms, to prove that q^x is rational with denominator supported by the known outputs, after which monomial injectivity and the three-base determinant force $x \in \mathbb{Q}$ and hence $x \in \mathbb{Z}$. What this would buy is a complete proof. The obstacle is the one identified in Section 26.7: straight monomials cannot introduce a new prime direction, so candidates must come from norms, resultants, or algebraic combinations arising in the primitive radical field, whose xth powers would then have to simplify.

27.7 Priority G: five and sharp six exponentials

Formalizing the five exponentials theorem, or Roy's sharp six exponentials theorem [8], would add genuinely new restrictions — for instance transcendence of $e^{\gamma x}$ for algebraic $\gamma \neq 0$ — and would extend the rational-third-base result from a controlled denominator to an unrestricted rational output. The effort is comparable to the completed Gelfond–Schneider port, which is encouraging; the risk is that it be undertaken without a specific downstream lemma in mind, since as Section 26.7 shows these theorems constrain b^x and not the divisibility facts the conjecture actually needs.

27.8 Priority H: larger kernel-replayed regions and effective measures

The zero-free region $2^x < 4096$ is kernel-verified; the 2^{14} extension is in progress and 2^{20} is the natural next target, matching the region already covered by interval computation (**[interval computation]**) in Section 25. Every descent constant improves by bookkeeping once the region grows, so the payoff is broad but shallow. The bottleneck is a compact kernel-replayable proof object: adaptive convergent enclosures, chunked certificate modules, and tiers shared along ranges. In the same vein, effective irrationality measures for ϑ from Páde or hypergeometric methods would replace per-value certificate tiers with a uniform polynomial criterion and likely extend verified regions by orders of magnitude. Their limitation is sharp: they say nothing directly about the generator $\beta = \log_2 M$ for unknown M, which would require an effective measure derived from the log-product relation (85) and therefore lies beyond ordinary linear forms in logarithms.

27.9 Priority I: computational reconnaissance with exact certificates

Finite scanning is evidence, not a path to infinity, but computation remains valuable in three focused roles: searching candidate determinant shapes for large systematic 2- or 3-adic valuations, feeding Priority A and Priority B; enumerating small primitive radical patterns (w, d, a, c) and recording local obstructions with exact rational inequalities rather than floating-point rejection; and rejecting candidates early using the primitive-pair conditions — no matched depth, no common perfect power, no matched affine decomposition — while recording near misses keyed by odd core, denominator, and rational-power index. The goal must be pattern discovery and theorem generation; the difficulty is the discipline not to spend the budget on extending a decimal barrier that proves nothing new.

27.10 Priority J: the colossally abundant interface

Formalizing the original application — that the rational-power form of the conjecture implies that quotients of consecutive colossally abundant numbers are prime [1] — would connect the corpus to its historical source and give the whole development a visible consequence. It is logically downstream of everything above, and the work is combinatorial and definitional rather than transcendental, so it should follow the core structure modules rather than compete with them.

28 Lean formalization blueprint

The new arguments of Sections 19, 20, 22, 23 and 12 were written so that each analytic step is a reusable lemma rather than a step inside one monolithic proof. The intended module layout sits under `ExponentialIdentities/TwoBaseIntegerExponent/` alongside the existing kernel-verified corpus of Section 4.

Since the first draft of this blueprint was written, Lean sources have been committed for most of the modules below. They are drafts in the precise sense that matters here: the files exist, contain no `sorry`, and are written against the current corpus, but they have *not* yet been accepted by the Lean kernel in a completed build, and all but `FallingFactorialBound` are not yet on the aggregate import surface `ExponentialIdentities.lean`. Every such module is therefore marked **[Lean draft, not yet accepted]** below, and the underlying mathematics keeps whatever status Section 19 or Section 20 assigned it. Nothing in this section may be read as a claim of kernel verification.

28.1 `ArbitraryNodeDeterminant`

Status: **[Lean draft, not yet accepted]** — `ArbitraryNodeDeterminant.lean`; the underlying mathematics is **[paper proof in this report]** (Section 19).

This module formalizes the ordinary-power determinant and the arbitrary-node repulsion theorem of Section 19. Its dependencies are:

1. a determinant/divided-difference identity for the matrix whose columns are $1, x, \dots, x^{r-1}, f(x)$ evaluated at the nodes, so that the determinant factors as the Vandermonde product times the rth divided difference of f;
2. a generalized mean value theorem for divided differences at distinct ordered nodes, which converts that divided difference into a derivative value at an interior point;
3. the existing derivative tower `rpowDerivativeFamily` and the falling-power coefficient `fallingRpowCoeff`, both already available in `HigherDifference`;
4. the integer-entry determinant lower bound: a matrix all of whose entries are integers has integer determinant, so a nonzero determinant has absolute value at least one.

The draft splits exactly along those four lines. It defines `nodeMatrix` and its integer shadow `nodeIntMatrix`, proves that candidate nodes make every entry an integer, derives `one_le_abs_det_nodeMatrix` from that, and then isolates the algebra of the final step in a hypothesis-carrying form: the analytic factorization $\det = V \cdot dd$ and the divided-difference bound $|dd| \le B$ are inputs, and the conclusion is the repulsion inequality. A target statement for the fully analytic version is:

```
theorem vandermonde_lower_bound_of_twoBaseNaturalCandidates
    {r : N} (hr : 2 <= r)
    (m : Fin (r + 1) -> N)
    (hstrict : StrictMono m)
    (hcand : forall i, TwoBaseNaturalCandidate (m i)) :
    (r.factorial : R) / |fallingRpowCoeff logThreeDivLogTwo r| *
        (m 0 : R) ^ ((r : R) - logThreeDivLogTwo)
      <= vandermondeProduct r m
```

Listing 2: Proposed arbitrary-node theorem signature.

The intermediate form actually proved in the draft is the hypothesis-carrying one, which is worth stating separately because it is reusable for any f whose divided differences can be bounded:

```
theorem inv_le_vandermondeProduct_of_dividedDifference (r : N)
    (m : Fin (r + 1) -> N) (hstrict : StrictMono m)
    (hcand : forall i, TwoBaseNaturalCandidate (m i)) {dd B : R}
    (hB : 0 < B)
    (hdet : (nodeMatrix r m).det = vandermondeProduct r m * dd)
    (hdd_bound : |dd| <= B) (hne : (nodeMatrix r m).det <> 0) :
    1 / B <= vandermondeProduct r m
```

Listing 3: Packaged repulsion step (draft form).

Working with a `Fin`-indexed strictly monotone family and an explicit pair product is easier than working with a `Finset` of nodes; the `Finset` form is recovered at the end through the monotone enumeration `Finset.orderEmbOfFin`.

28.2 `GeneralizedVandermonde`

Status: Design note; the analytic content is **[paper proof in this report]** (Appendix D), and the nonvanishing half is drafted in `ExponentialPolynomialZeros`.

The positivity theorem for generalized Vandermonde determinants with arbitrary real exponents is useful well beyond this conjecture, and two formalization routes were available.

Wronskian route. Prove the Wronskian determinant formula of Appendix D directly, then invoke or develop the criterion that positive initial Wronskians produce an extended complete Chebyshev system and hence positive ordered evaluation determinants. This needs a small Chebyshev-system API that mathlib does not yet have.

Zero-count route. After the substitution $x = e^t$, prove by induction that a nonzero exponential polynomial with $r + 1$ distinct real exponents has at most r real zeros. Deduce nonvanishing of the evaluation determinant, and if the sign is wanted, obtain it by continuity and a coalescing-node limit.

The decisive observation for the present application is that only *nonvanishing* is needed downstream. The determinant in question has integer entries, and a nonzero integer already has absolute value at least one; the sign never enters the repulsion inequality. That removes the entire coalescing-limit argument from the critical path, and it is why the draft took the zero-count route and left strict positivity unproved.

28.3 `ExponentialPolynomialZeros`

Status: **[Lean draft, not yet accepted]** — `ExponentialPolynomialZeros.lean`.

This module carries the zero-count induction. Write $g(t) = \sum_{j \le n} c_j e^{\lambda_j t}$ with $\lambda_0 < \cdots < \lambda_n$. Normalizing by $e^{-\lambda_0 t}$ preserves the zero set; differentiating kills the constant term and lowers the number of exponents by one; Rolle's theorem between consecutive zeros supplies the zeros of the derivative. Induction on n then forces every coefficient to vanish.

Two Lean engineering difficulties dominate the effort, and neither is visible in the paper proof.

First, the Rolle step must produce n *distinct and ordered* interior points from $n+1$ ordered zeros. Mathlib's `exists_hasDerivAt_eq_zero` yields one interior point per gap, but nothing about how the points from different gaps compare. The draft chooses them with `choose`, then proves strict monotonicity by chaining $y_i < x_{i+1} \le x_j < y_j$ through the monotonicity of the node family. When the hypothesis arrives as a `Finset` of at least $n+1$ reals rather than an indexed family, the monotone enumeration `Finset.orderEmbOfFin` converts it, after cutting the set down to exactly $n+1$ elements with `Finset.exists_subset_card_eq`.

Second, the induction reindexes. The derivative of an $(m+1)$-term exponential polynomial is an m-term one whose exponents are $\lambda_{j+1} - \lambda_0$ and whose coefficients are $c_{j+1}(\lambda_{j+1} - \lambda_0)$, so every statement has to be transported along `Fin.succ`, and the induction has to be set up with the order quantified first so that `induction` may generalize over exponents, coefficients and nodes simultaneously. In the draft this is done through a private auxiliary theorem of the shape `forall (n : N) (lam c : Fin (n + 1) -> R), ...`, from which the user-facing statements are one-line consequences.

```
theorem card_le_of_expPoly_eq_zero {n : N} (lam c : Fin (n + 1) -> R)
    (hlam : StrictMono lam) (hc : exists j, c j <> 0) (S : Finset R)
    (hzero : forall t, t in S -> expPoly lam c t = 0) :
    S.card <= n

theorem det_rpowVandermonde_ne_zero {n : N} (x lam : Fin (n + 1) -> R)
    (hx : StrictMono x) (hx0 : 0 < x 0) (hlam : StrictMono lam) :
    (rpowVandermonde x lam).det <> 0
```

Listing 4: Zero-count theorem and its determinant form.

A vanishing determinant yields a nonzero kernel vector, via `Matrix.exists_mulVec_eq_zero_iff`, and it is a real combination of the functions $x \mapsto x^{\lambda_j}$ vanishing at $n+1$ distinct positive nodes. So the determinant statement follows from the zero-count statement.

28.4 `MixedExponentSemigroup` and `SemigroupDeterminant`

Status: **[Lean draft, not yet accepted]** — `MixedExponentSemigroup.lean` and `SemigroupDeterminant.lean`; the mathematics is **[paper proof in this report]** (Section 20).

The pair-indexed exponent is the whole of the definitional layer:

```
noncomputable def mixedExponent (ab : N x N) : R :=
  (ab.1 : R) + (ab.2 : R) * logThreeDivLogTwo
```

and the module proves, in this order:

- injectivity of `mixedExponent`, from irrationality of ϑ;

- integrality of $m^{a+b\vartheta} = m^a\,(m^\vartheta)^b$ for every candidate m, packaged as a natural-number witness `candidateRpowNat`;
- closure under addition, so that the image is an `AddSubmonoid R`;
- the elementary comparisons $a+b \le a+b\vartheta \le a+2b$ and $a+b\vartheta \le 3q$ for $a,b \le q$, which are what the counting arguments actually consume.

`SemigroupDeterminant` then builds the matrix $m_i^{\lambda_j}$ for candidate nodes and mixed exponents, shows every entry is a natural number, and concludes integrality and the lower bound $|\det| \ge 1$ whenever the determinant is nonzero — the nonvanishing being imported from `ExponentialPolynomialZeros`.

```
theorem one_le_abs_det_semigroupMatrix (r : N) (m : Fin (r + 1) -> N)
    (hcand : forall i, TwoBaseNaturalCandidate (m i))
    (ab : Fin (r + 1) -> N x N)
    (hne : (semigroupMatrix r m ab).det <> 0) :
    1 <= |(semigroupMatrix r m ab).det|
```

Listing 5: Semigroup determinant integrality.

It is deliberately unnecessary to define the globally sorted semigroup here. The determinant estimate accepts an arbitrary finite family of pairs together with a proof that their mixed exponents are strictly increasing; the sorted enumeration is a convenience of the paper proof, not a requirement of the formal one.

28.5 `FallingFactorialBound`

Status: [Lean draft, not yet accepted] — `FallingFactorialBound.lean`; already on the aggregate import surface.

This module carries the two elementary estimates behind every Newton-row bound of Section 20: the falling-coefficient bound $|(\lambda)_i|/i! \le 2^{\lambda+i}$ for $\lambda \ge 0$, and the strip bound $x^{\lambda-i} \le 2^\lambda N^{\lambda-i}$ for $N \le x \le 2N$, $N \ge 1$. The proof of the first compares each factor $|\lambda - k|$ with $m+1+k$ where $m = \lfloor\lambda\rfloor$, identifies $\prod_{k<i}(m+1+k)$ with $i!\binom{m+i}{i}$, and bounds the binomial coefficient by the row sum 2^{m+i}. Their product is the single inequality consumed downstream:

```
theorem abs_fallingRpowCoeff_div_factorial_mul_rpow_le
    {N x lam : R} (hN : 1 <= N) (h1 : N <= x) (h2 : x <= 2 * N)
    (hlam : 0 <= lam) (i : N) :
    |fallingRpowCoeff lam i| / (i.factorial : R) * x ^ (lam - (i : R))
      <= (2 : R) ^ (2 * lam + (i : R)) * N ^ (lam - (i : R))
```

Listing 6: Newton-row entry bound.

The determinant estimate then follows from `Matrix.det_apply` or, better, from a generic Leibniz-bound lemma worth adding once:

> if every entry in row i, column j is bounded by a separable quantity A_iB_j, then the determinant is at most $(r+1)!\prod_i A_i \prod_j B_j$.

28.6 SemigroupWeightBound

Status: [Lean draft, not yet accepted] — `SemigroupWeightBound.lean`.

For the explicit logarithmic-square corollary the full asymptotic count of the ordered semigroup is optional. What is needed is the square-box construction of Section 20 in *existence form*, which sidesteps real-valued order statistics entirely:

- the $(q+1)^2$ pairs in $[0,q]^2$ have pairwise distinct mixed exponents;
- every such exponent is at most $3q$, using $1 < \vartheta < 2$;
- taking $q = \lfloor \sqrt{r} \rfloor$ makes $r+1$ pairs fit, with total weight at most $6(r+1)\sqrt{r}$;
- dividing by $R_r = r(r+1)/2$ gives the deficit ratio $\Sigma_r/R_r \le 12/\sqrt{r}$.

The draft realizes the injective family explicitly by base-$(q+1)$ digits, $j \mapsto (j \bmod (q+1),\ j \div (q+1))$, so no sorting is performed and no canonical ordered semigroup is ever constructed. The statements are:

```
theorem exists_mixedExponent_family (r : N) :
    exists ab : Fin (r + 1) -> N x N, Function.Injective ab and
      (forall j, mixedExponent (ab j) <= 6 * Real.sqrt (r : R)) and
      (sum j, mixedExponent (ab j)) <= 6 * ((r : R) + 1) * Real.sqrt (r : R)
```

Listing 7: Square-box weight bound, existence form.

The module also contains the abstract counting bridge that turns any span lower bound into a cardinality bound, and which serves the ordinary-power corollary of Section 19 equally well: if $S \subseteq [N, N+H]$ and every $r+1$ elements of S contain two points at distance at least $L > 0$, then $\#S \le r + rH/L$. The proof partitions $[N, N+H]$ into $\lfloor H/L \rfloor + 1$ half-open blocks of length L, observes that each block meets S in at most r points, and applies pigeonhole.

28.7 SemigroupDyadicBound

Status: Planned; no draft yet. The mathematics is [paper proof in this report] (Section 20).

This is the module that combines the span theorem with the consecutive-window summation and produces the explicit $O((\log X)^2)$ dyadic bound. The final theorem could state:

```
theorem card_twoBaseNaturalCandidates_Icc_le_log_sq (X : N) (hX : 1 <= X) :
  ((Finset.Icc X (2 * X)).filter TwoBaseNaturalCandidate).card
    <= Nat.ceil (10000 * max 1 (Real.log X) ^ 2)
```

Listing 8: Proposed explicit dyadic bound.

A cleaner all-natural statement may replace the real logarithm by the binary ceiling logarithm, obtaining a larger but much easier constant times `Nat.clog 2 X ^ 2`. Since the repository already has logarithmic-square counting infrastructure in `MultiplicativeRank`, its finite-set definitions and cardinality lemmas can be reused while keeping the proof mechanism entirely independent — which is the point of proving the same bound twice.

28.8 `BlockGrowthReformulations`

Status: **[Lean draft, not yet accepted]** — `BlockGrowthReformulations.lean`; the mathematics is **[paper proof in this report]** (Section 23).

The exact conditional block formula proved in the corpus makes the dyadic block count $B(T) = \#\{\text{candidates in } [2^T, 2^{T+1})\}$ either identically 1 or bounded below by a positive multiple of T. That dichotomy collapses a whole family of apparently weaker growth hypotheses onto the conjecture itself, and the module states all four equivalences together so that a future argument may pick whichever is most convenient:

```
theorem alaogluErdosConjecture_block_growth_reformulations :
    (AlaogluErdosConjecture <-> exists C : N, forall T : N, dyadicBlockCount T <= C) and
    (AlaogluErdosConjecture <->
       (fun T : N => (dyadicBlockCount T : R)) =o[atTop] fun T : N => (T : R)) and
    (AlaogluErdosConjecture <->
       forall eps : R, 0 < eps ->
         frequently T in atTop, (dyadicBlockCount T : R) < eps * (T : R)) and
    (AlaogluErdosConjecture <-> forall T : N, dyadicBlockCount T = 1)
```

Listing 9: Block-growth reformulations.

The practical value is negative and worth stating plainly: proving mere boundedness, or even vanishing lower density along a subsequence, is exactly as hard as the conjecture. No softening of the counting target buys anything.

28.9 `CompactOrbit`

Status: **[Lean draft, not yet accepted]** — `CompactOrbit.lean`; the mathematics is **[paper proof in this report]** (Section 23).

This module states the conjecture as the nonexistence of a synchronized integral orbit on the compact arc $\{(2^t, 3^t) : 0 \le t < 1\}$. The orbit point is

$$(M^k/2^{\lfloor k\beta \rfloor},\ A^k/3^{\lfloor k\beta \rfloor}),$$

both coordinates divided by a power of their own base with the *same* integer exponent $\lfloor k\beta \rfloor$; that shared exponent is what makes the reformulation nontrivial.

```
theorem alaogluErdosConjecture_iff_no_compactOrbit :
    AlaogluErdosConjecture <->
      forall M A : N, forall beta : R,
        1 < M -> 1 < A -> Irrational beta -> 0 < beta ->
          exists k : N, 1 <= k and
            synchronizedOrbitPoint M A beta k not_in compactExponentialArc
```

Listing 10: Compact-orbit reformulation.

The forward direction routes through the canonical primitive generator, so the produced β is the least noninteger solution and M, A are its normalized outputs. The converse needs neither $M, A > 1$ nor $\beta > 0$: arc membership by itself already pins $M = 2^\beta$ and $A = 3^\beta$. Formalizing that asymmetry was the only delicate part.

28.10 `LogProductIndependence`

Status: **[Lean draft, not yet accepted]** — `LogProductIndependence.lean`; the mathematics is **[paper proof in this report]** (Section 12).

This module records the prime-log-product criterion of Section 12. The open hypothesis is never asserted; it is a named `Prop` that appears only as an explicit argument, so the exported theorem is unconditional Lean about a conditional mathematical statement.

```
def PrimeLogProductIndependence : Prop :=
  forall (S : Finset N) (c : N -> N -> Z),
    (forall p, p in S -> Nat.Prime p) ->
    (sum p in S, sum q in S, (c p q : R) * (Real.log p * Real.log q)) = 0 ->
    forall p, p in S -> forall q, q in S -> c p q + c q p = 0

theorem alaogluErdosConjecture_of_primeLogProductIndependence
    (hind : PrimeLogProductIndependence) : AlaogluErdosConjecture
```

Listing 11: Prime-log-product criterion.

The supporting lemma is the prime-factorization expansion $\log n = \sum_{p|n} v_p(n) \log p$, after which the relation $M^{\log 3} = A^{\log 2}$ becomes a vanishing integer combination of the products $\log p \log q$; symmetrized coefficientwise vanishing then forces M to be a power of 2 and the exponent to be an integer.

28.11 Proof-audit discipline

The discipline that governs every status label in this report is mechanical, and it is worth restating because several modules above are new. Each headline theorem must enter the aggregate import surface `ExponentialIdentities.lean` and be checked with `#print axioms`, so that the trust boundary of each result is visible rather than inferred. A result stays **[Lean draft, not yet accepted]** or **[paper proof in this report]** until a full build succeeds; the existence of a `sorry`-free source file is not verification, and neither is a successful build of an earlier revision. Numerical constants appearing in the tables of this report are explanatory only — the formal statements use exact expressions or safe rational constants such as 1/24 and 10000, chosen so that the arithmetic side conditions close by `norm_num` rather than by a tuned estimate.

29 Comparison of methods

29.1 Prime-valuation rank versus interpolation geometry

The module `MultiplicativeRank` proves that any three candidates have linearly dependent prime-exponent vectors, of rank at most two. It does so by a three-base, six-exponentials determinant: if three candidate bases were multiplicatively independent, integrality of all three ϑth powers would force ϑ rational, contradicting its irrationality. A finite candidate set therefore lies in a rational plane inside the free vector space on primes. Choosing two prime coordinates injectively encodes the set and yields an explicit logarithmic-square count.

The semigroup determinant of Section 20 proves a logarithmic-square dyadic bound by an entirely different mechanism.

Prime-valuation method	Semigroup determinant method
Uses prime factorization of candidate bases	Uses only integer values on $y = x^{\vartheta}$
Global multiplicative rank at most two	Local total positivity and interpolation
Invokes a six-exponentials consequence	Uses elementary determinant integrality once ϑ is fixed
Produces coordinates in two prime directions	Produces repulsion for arbitrary ordered nodes
Naturally global in magnitude	Naturally local in additive intervals
Kernel-verified **[kernel-verified Lean]**	Paper proof **[paper proof in this report]**, Lean draft **[Lean draft, not yet accepted]**

Their agreement at logarithmic-square scale is suggestive rather than coincidental. Both methods see a two-dimensional structure, but they see different two-dimensional structures: prime-exponent rank two on one side, mixed-exponent lattice dimension two on the other. Neither refinement alone has closed the gap, and a future proof may well need to combine the two dual lattices rather than push either one further in isolation.

29.2 Exact conditional rank one over two generators

Conditional on failure of the conjecture, the candidate monoid is not merely of rank at most two — it is exactly free on 2 and M. In prime-valuation space its points form the nonnegative lattice generated by the prime-exponent vectors of 2 and M. In mixed-exponent evaluation space, the values factor across a two-dimensional grid indexed by the pair (n, k) with value $2^n M^k$. This exact grid is far stronger than the unconditional rank theorem, and yet every presently known consequence remains compatible with it. That compatibility is the honest measure of how much room is left.

The strongest likely synthesis is a determinant whose rows use primitive-generator coordinates and whose columns use mixed exponents. Such a matrix would be simultaneously:

- integral, hence subject to the $|\det| \geq 1$ lower bound;
- totally positive after ordering by magnitude, hence nonvanishing for free;

- explicitly factored into powers of $2, 3, M, A$;
- constrained by affine primitivity and by simultaneous output normalization.

No current module combines all four properties. `MultiplicativeRank` has the first and third; the semigroup modules have the first and second; `PrimitiveGenerator` and `OutputNormalization` have the fourth. Building the module that has all four is the most concrete open formalization task suggested by this report.

29.3 Finite differences as confluent determinants

Forward differences are the equally spaced specialization of interpolation, and Wronskians are its confluent limit as nodes coalesce. The determinant hierarchy of Sections 19 and 20 therefore provides a single conceptual envelope for four things that the corpus currently treats separately:

- second-, third-, and arbitrary-order finite differences;
- arbitrary-node ordinary-power determinants;
- mixed-monomial generalized Vandermonde determinants;
- confluent determinants using derivatives or repeated coordinate directions.

The corresponding formalization strategy is to build a general interpolation-determinant layer once and recover `HigherDifference` as a specialization of it, rather than to keep the two developments parallel. That would remove duplicated real analysis and, more usefully, make experimental determinant choices cheap to certify: a new choice of nodes and exponents would need only a divided-difference bound, with integrality and nonvanishing supplied by the shared layer.

30 Conclusions

The Alaoglu–Erdős conjecture (Conjecture 2.1) remains open. Nothing in this report proves it, and no combination of the results collected here proves it. What has changed is that the reason can now be stated exactly rather than impressionistically.

The kernel-verified core is broad and its themes are complementary. Interpolation determinants settle three independent bases; Gelfond–Schneider excludes algebraic irrational exponents; exact certificates create a growing zero-free region, currently $2^x < 4096$; unique factorization yields a canonical primitive odd core and, conditionally, an exact solution monoid; arbitrary-order forward differences force sharp local additive sparsity, with the rational three-term criterion $h^{400} \leq n^{83}$ as its effective form; leastness gives exact affine primitivity; and unconditional equivalences with fractional-part gaps and localized radicals identify the entire remaining distance to the conjecture.

Conditional on a single counterexample, the corpus determines the whole picture. A counterexample would not produce a messy exceptional set that density or semigroup arguments might squeeze; it would produce the perfectly rigid free monoid

$$\mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0\beta, \qquad \mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\},$$

whose block counts, cumulative asymptotics, fractional parts, rotation statistics, rational-power index, radical degree, and simultaneous output normalization are all explicit and internally consistent with every verified gap and density bound. The surviving obstruction is correspondingly concentrated:

$$\text{exclude } w^{\log_2 3} \in \mathbb{Z}[1/3] \text{ for every primitive odd } w > 1.$$

That single exclusion is equivalent to, and exactly as open as, the original conjecture.

The new determinant hierarchy sharpens the picture quantitatively. The arbitrary-node theorem of Section 19 shows that candidate lattice points cannot cluster freely: every Vandermonde product over candidate nodes obeys a power-law lower bound. The mixed-exponent semigroup of Section 20 upgrades this to arbitrarily high-order positive integer determinants with a repulsion exponent tending to one, and yields a second, conceptually independent logarithmic-square counting theorem.

The most valuable outcome, however, is the quantified failure point of Section 22. Pure archimedean interpolation needs about $(\log X)^2$ nodes to operate at scale X, whereas the exact conditional monoid supplies only about $\log X$ of them. The method misses by exactly one factor of $\log X$ — not by a constant, and not by an epsilon. This makes future work sharper by exclusion: improving constants, tightening the falling-factorial bound, or extending finite searches cannot close a gap of that shape. The plausible routes are precisely those capable of supplying one additional logarithm of determinant strength: 2- or 3-adic divisibility used simultaneously with the archimedean estimate, a synchronized-denominator counting theorem on the compact exponential arc of Section 23, or a mechanism forcing a third algebraically independent exponential direction.

Section 12 isolates a different sufficient condition, and it is worth separating from the rest because it does not live at the four-exponentials boundary at all. If the products $\log p \log q$ of prime logarithms are linearly independent over $\mathbb{Q}$, the conjecture follows by an argument that is elementary once the hypothesis is granted. That independence is a consequence of Schanuel's conjecture and is not known unconditionally — indeed even the irrationality of $(\log 3)^2/(\log 2)^2$ is open — but it is a different classical conjecture, with a different literature and different partial results. Having two inequivalent sufficient conditions, one transcendence-theoretic and one arithmetic, is more informative than having one.

The near-term program is the one consolidated in Section 27: finish the exact conditional consequences, in particular the functional-analytic extension of blockwise equidistribution to arbitrary continuous tests and interval indicators; exploit the uniform higher-difference theorem combinatorially, allowing the order to grow with height; kernel-replay the 2^{20} region as a compact certificate; use the explicit radical field for a genuinely number-field argument; and pursue auxiliary-function methods tailored to integrality, where all corner values are positive integers and the primitive pair is canonical. Alongside those, the immediate formalization work is the module list of Section 28: bring the nine existing drafts through a full kernel build and onto the aggregate import surface, then attempt the combined determinant identified in Section 29.

A complete proof still requires a new idea. But the target is now narrow, the conditional structure is exact, the failure mode of the best available elementary method is measured rather than guessed, and the surrounding formal infrastructure is strong enough to test the next idea precisely.

31 Continuation: executive answer, scope, and trust boundaries

The sections that follow were produced as a continuation of the preceding material and are merged here unchanged in substance. They supply a symbolic-dynamical model of a hypothetical counterexample, an effective replacement for the elementary continued-fraction estimate, exact all-support valuations of the block determinants, and the sharp constants that govern how much of the archimedean budget term-wise divisibility can supply. Section 21 states the unconditional ceiling that underlies all of them and records the two constants this part corrects.

31.1 The requested goal and the outcome

The requested primary goal was to continue the attached research program and prove the following statement.

Conjecture 31.1 (Alaoglu–Erdős). *If $x \in \mathbb{R}$ and $2^x, 3^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$.*

I did not find an unconditional proof. In particular, none of the arguments below converts

$$\log(2^x)\log 3 - \log(3^x)\log 2 = 0$$

into a forbidden linear form in logarithms of algebraic numbers. The equality is bilinear in logarithms, and normalizing it gives exactly the unresolved 2×2 determinant behind the four exponentials conjecture. Baker-type estimates control every *nonzero* linear form that occurs after fixing integer coefficients; they do not prove that this special bilinear form cannot vanish.

The continuation nevertheless changes the technical map in useful ways. It obtains an exact symbolic-dynamical formulation, replaces an elementary continued-fraction estimate by an effective polynomial one, computes the full leading p-adic content of ordinary block Vandermondes, and gives sharp continuum formulas for the termwise divisibility of two much larger determinant constructions. These calculations do more than say that a method "seems too weak": they quantify the leading constant that is available and isolate the precise kind of extra cancellation a successful proof would have to produce.

31.2 The principal new results

Result	Content	Status
Synchronized Sturmian orbit	A counterexample is equivalent to one irrational mechanical word simultaneously controlling the dyadic and triadic denominator jumps of two compact rational recurrences.	**[paper proof in this report]**
Effective irrationality of β	Baker–Wüstholz/Matveev gives $\lvert\beta - p/q\rvert \geq c(M)q^{-\mu(M)}$ and hence polynomial, rather than exponential, recurrence bounds for continued-fraction denominators.	**[classical/open background]** plus **[paper proof in this report]**

Result	Content	Status
Ordinary Vandermonde audit	The valuations at every prime in the fixed supports of M and A are computed to leading order. The product construction has universal ceiling $1-\log 2/(3\log 6)=0.871049\ldots$.	[paper proof in this report]
Gini assignment formula	For the balanced mixed-semigroup determinant, the exact termwise all-prime divisor fraction is $1-\frac{3}{8}\sum_p \mathfrak{G}(c_p)+o(1)$ for an explicit norm $\mathfrak{G}$.	[paper proof in this report]
Universal 4/5 ceiling	The primitive condition forces $\sum_p \mathfrak{G}(c_p)\geq 8/15$. Thus entrywise/min-plus divisibility can supply at most 80% of the leading block determinant height.	[paper proof in this report]
Cross-coprime constants	If $\gcd(MA,6)=1$, the all-prime ceiling is $7/10$, and the places 2 and 3 alone contribute exactly $3/10$ in the continuum limit.	[paper proof in this report]
Full-grid transport ceiling	On the complete triangular (n,k) grid, any termwise all-prime divisor is bounded by the countermonotone energy $\pi/8$, while the largest determinant term has energy $1/2$; the ratio is $\pi/4$.	[paper proof in this report]
HCIZ reformulation	The full-grid archimedean problem is expressed by the Harish–Chandra–Itzykson–Zuber integral, identifying a concrete free-energy inequality as the remaining analytic target.	[classical/open back-ground] plus [paper proof in this report]

31.3 Evidence labels used in this part

This part uses the report-wide convention of Section 3.1, with two additions. **[repository build reported]** marks a statement for which a repository commit or build log reports success but which was not independently replayed by the author of the continuation; it is repository metadata, not a kernel audit. **[proposed target]** marks a statement proposed as a precise target rather than asserted. Where the continuation wrote **[classical/open background]** for standard literature the report-wide label is used unchanged.

31.4 What the determinant ceilings do and do not prove

A ceiling such as 4/5 has a specific meaning. Given an integer determinant $D=\det(E_{ij})$, define at each prime p the minimum valuation of a Leibniz term,

$$\tau_p=\min_\sigma \sum_i v_p(E_{i,\sigma(i)}).$$

Every term, and therefore D, is divisible by p^{τ_p}. The product $\prod_p p^{\tau_p}$ is the *termwise* or *tropical* divisor. It ignores extra valuation caused by cancellation among several terms attaining the same minimum. The 4/5 theorem says that this automatic divisor cannot occupy more than four-fifths of the leading archimedean height for the balanced block construction. It does *not* say that the true valuation $v_p(D)$ is at most the tropical minimum; the opposite inequality holds, and extra cancellation may be large. Thus the result retires only the naïve entrywise-divisibility version of the method. It leaves a sharply defined cancellation problem.

32 Baseline inherited from the unified report

32.1 Notation

Set

$$\vartheta = \log_2 3.$$

The solution and candidate sets are

$$\mathcal{S} = \{x \geq 0 : 2^x, 3^x \in \mathbb{N}\}, \qquad \mathcal{C} = \{m \in \mathbb{N} : m^\vartheta \in \mathbb{N}\}.$$

The map $x \mapsto 2^x$ is a bijection $\mathcal{S} \to \mathcal{C}$. Integer exponents give the obvious solutions $\mathbb{N}_0 \subseteq \mathcal{S}$ and candidates $2^{\mathbb{N}_0} \subseteq \mathcal{C}$.

Throughout the conditional sections, suppose the conjecture fails. The exact structure proved in the source report then supplies a least nonintegral solution $\beta > 0$ and primitive outputs

$$M = 2^\beta \in \mathbb{N}, \qquad A = 3^\beta \in \mathbb{N}.$$

Every solution and candidate has a unique coordinate representation

$$\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\}, \qquad \mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\}. \tag{86}$$

The associated output is

$$(2^n M^k)^\vartheta = 3^n A^k.$$

We write

$$a = v_2(M), \qquad b = v_3(A), \qquad c = v_2(A), \qquad d = v_3(M).$$

The inherited primitive-output theorem includes the crucial condition

$$\min(a, b) = 0. \tag{87}$$

It is important not to silently strengthen (87) to $a = b = 0$.

32.2 Inherited results used here

The following facts are taken from the attached unified report and its referenced Lean modules.

- **[kernel-verified Lean]** A nonintegral solution, if it exists, is transcendental. The constant ϑ is transcendental, by Gelfond–Schneider.
- **[kernel-verified Lean]** Failure gives the exact free rank-two monoid structure (86), including uniqueness of (n, k).

- [kernel-verified Lean] In a dyadic exponent block $[T, T+1)$, the solutions are exactly
$$T \quad \text{and} \quad T - \lfloor k\beta \rfloor + k\beta, \qquad 1 \le k \le \left\lfloor \frac{T+1}{\beta} \right\rfloor.$$
Thus the block count is $1 + \lfloor (T+1)/\beta \rfloor$.
- [kernel-verified Lean] The primitive pair obeys (87); both outputs cannot simultaneously be $\{2,3\}$-smooth.
- [kernel-verified Lean] Mixed monomials $m^u(m^\vartheta)^v$ are integers for $(u,v) \in \mathbb{N}_0^2$, and generalized Vandermonde determinants built from distinct candidate nodes and distinct exponents $u + \vartheta v$ are nonzero integers.
- [kernel-verified Lean] The kernel-checked finite exclusion in the source report gives $x \ge 12$ for a nonintegral solution. The report separately records a directed-rounding software exclusion through $x \le 26$.

The continuation does not reprove this large inherited corpus. It restates every inherited fact at the point where it enters a new argument.

32.3 Repository reconciliation as of August 21, 2026

The attached source is internally newer than one bibliographic pin printed near its end: its body and status matrix already discuss the enlarged algebraic-output and determinant module surface, although a repository item still names commit `5d3982ceb12644fdd3499569d9db5f62f9a31dc6`. During this continuation I checked the current public repository history. Commit `78f80c5d3f6f9afd64ecb72ddb154c85e2036bc1` reports a successful aggregate build with 67 imported modules and merges the later algebraic-output work. Subsequent commits add a literature compass and a short p-adic priority note; the latest snapshot consulted while writing was `ff9a9f5aa3a3f02599de93fc053e60918f269855`.

Status: [repository build reported]

The build claims in the preceding paragraph are taken from repository commit messages and files. I did not independently run all Lean jobs. The mathematical arguments newly stated below are therefore labelled [paper proof in this report], not [kernel-verified Lean].

32.4 The exact remaining logarithmic wall

Writing $M = 2^\beta$ and $A = 3^\beta$ gives
$$\beta = \frac{\log M}{\log 2} = \frac{\log A}{\log 3}$$
and hence
$$\log M \, \log 3 - \log A \, \log 2 = 0. \tag{88}$$
The four numbers $\log 2, \log 3, \log M, \log A$ are logarithms of algebraic numbers. Equation (88) is a vanishing 2×2 determinant. Four exponentials would exclude it because $2, 3$ are multiplicatively independent and $1, \beta$ are linearly independent over $\mathbb{Q}$, but four exponentials remains open. The new work below therefore pursues information that is invisible in the bare real logarithmic determinant: exact floor coding, effective separation, and valuations of large integer determinants.

33 A synchronized Sturmian model of a counterexample

33.1 Fractional-part normalization

Write

$$\beta = B + \alpha, \qquad B = \lfloor \beta \rfloor \in \mathbb{N}_0, \qquad 0 < \alpha < 1.$$

Since a nonintegral solution is transcendental, α is irrational. Define the rational localized multipliers

$$U = \frac{M}{2^B} = 2^\alpha \in \mathbb{Q} \cap (1,2), \qquad V = \frac{A}{3^B} = 3^\alpha \in \mathbb{Q} \cap (1,3). \tag{89}$$

For $k \geq 0$, put

$$e_k = \lfloor k\alpha \rfloor, \qquad s_k = e_{k+1} - e_k \in \{0,1\}.$$

The binary sequence $s = s_0 s_1 s_2 \cdots$ is the lower mechanical word of slope α. Define compact normalized orbits

$$R_k = \frac{U^k}{2^{e_k}} = 2^{\{k\alpha\}} \in [1,2), \qquad S_k = \frac{V^k}{3^{e_k}} = 3^{\{k\alpha\}} \in [1,3). \tag{90}$$

Both orbits use the *same* fractional part and hence the same symbolic itinerary.

Theorem 33.1 (Synchronized dyadic–triadic Sturmian recurrence)**.** *Assume a primitive nonintegral solution β exists. Then the rational sequences R_k, S_k in (90) satisfy*

$$R_0 = S_0 = 1, \qquad R_{k+1} = \frac{UR_k}{2^{s_k}}, \qquad S_{k+1} = \frac{VS_k}{3^{s_k}}, \tag{91}$$

where

$$s_k = 1 \iff \{k\alpha\} \geq 1-\alpha \iff UR_k \geq 2 \iff VS_k \geq 3. \tag{92}$$

Moreover,

$$\sum_{j=0}^{N-1} s_j = \lfloor N\alpha \rfloor \qquad (N \geq 1). \tag{93}$$

Proof. The recurrence follows by subtracting consecutive floor exponents:

$$R_{k+1} = \frac{U^{k+1}}{2^{e_{k+1}}} = \frac{UU^k}{2^{e_k+s_k}} = \frac{UR_k}{2^{s_k}},$$

and the same calculation with base 3 gives the formula for S_k. Since $e_{k+1} = e_k + 1$ exactly when $\{k\alpha\} + \alpha \geq 1$, the first equivalence in (92) holds. The identities $UR_k = 2^{\alpha+\{k\alpha\}}$ and $VS_k = 3^{\alpha+\{k\alpha\}}$ give the other two. Finally, (93) telescopes: $\sum_{j<N} s_j = e_N - e_0 = \lfloor N\alpha \rfloor$. □

33.2 Sturmian consequences

The following facts are classical consequences of irrational mechanical coding [25, 26].

Corollary 33.2 (Aperiodicity, balance, and minimal complexity)**.** *The word s in Theorem 33.1 is aperiodic. Any two factors of equal length contain numbers of 1's differing by at most one, and the number of distinct factors of length n is exactly $n+1$.*

Aperiodicity has an elementary proof in this setting. If s were eventually periodic, then its limiting density of 1's would be rational. Equation (93) shows that this density is α, contradicting irrationality. Balance and factor complexity are the standard Sturmian characterization.

The point is not merely terminological. A hypothetical counterexample is no longer described only by two rational numbers U, V satisfying a logarithmic equality. It determines one minimal-complexity aperiodic word, and that same word must be realized by two rational piecewise-multiplicative systems with multiplicatively independent reset bases 2 and 3. This is a much more rigid synchronized object.

33.3 Exact reduced denominators

The primitive factor depths determine how much cancellation occurs in (90). Write

$$M = 2^a W, \quad W \text{ odd}, \qquad A = 3^b Z, \quad 3 \nmid Z.$$

Because M is not a power of 2 and A is not a power of 3, one has $W, Z > 1$.

Proposition 33.3 (Denominator exponents and common jump word)**.** *In lowest terms,*

$$R_k = \frac{W^k}{2^{q_k^{(2)}}}, \qquad q_k^{(2)} = \lfloor k\beta \rfloor - ak = k(B-a) + \lfloor k\alpha \rfloor, \tag{94}$$

$$S_k = \frac{Z^k}{3^{q_k^{(3)}}}, \qquad q_k^{(3)} = \lfloor k\beta \rfloor - bk = k(B-b) + \lfloor k\alpha \rfloor. \tag{95}$$

Consequently

$$q_{k+1}^{(2)} - q_k^{(2)} = B - a + s_k, \qquad q_{k+1}^{(3)} - q_k^{(3)} = B - b + s_k.$$

At least one of the two sequences has baseline jump B, because $\min(a, b) = 0$.

Proof. Using $\lfloor k\beta \rfloor = kB + \lfloor k\alpha \rfloor$,

$$R_k = \frac{M^k}{2^{\lfloor k\beta \rfloor}} = \frac{2^{ak} W^k}{2^{\lfloor k\beta \rfloor}} = \frac{W^k}{2^{\lfloor k\beta \rfloor - ak}}.$$

The numerator is odd, so this is reduced. The base-3 calculation is identical. Taking successive differences gives the jump formulas, and the last claim is the inherited primitive condition (87). ☐

Thus the same Sturmian bit s_k records whether *both* reduced denominators take their large or small permitted jump at time k. One denominator can have a shallower constant baseline, but the aperiodic fluctuation is synchronized exactly.

33.4 An exact equivalent exclusion principle

The source report already contains localization reformulations such as $2^\alpha \in \mathbb{Z}[1/2]$ and $3^\alpha \in \mathbb{Z}[1/3]$. The recurrence packages them into a symbolic statement.

Theorem 33.4 (Synchronized Sturmian exclusion is equivalent to Alaoglu–Erdős)**.** *The Alaoglu–Erdős conjecture is equivalent to the nonexistence of data*

$$\alpha \in (0,1) \setminus \mathbb{Q}, \qquad U \in \mathbb{Q} \cap (1,2), \qquad V \in \mathbb{Q} \cap (1,3)$$

such that, for the lower mechanical word $s_k = \lfloor (k+1)\alpha \rfloor - \lfloor k\alpha \rfloor$, the recurrences

$$R_0 = S_0 = 1, \qquad R_{k+1} = UR_k/2^{s_k}, \qquad S_{k+1} = VS_k/3^{s_k}$$

remain in $[1,2) \cap \mathbb{Q}$ and $[1,3) \cap \mathbb{Q}$, respectively, for every k, with $U = 2^\alpha$ and $V = 3^\alpha$. Equivalently, it suffices to rule out irrational α for which

$$2^\alpha \in \mathbb{Z}[1/2], \qquad 3^\alpha \in \mathbb{Z}[1/3].$$

Proof. A counterexample $\beta = B + \alpha$ gives the data by Theorem 33.1. Conversely, if such α, U, V exist, choose an integer B at least as large as the powers of 2 and 3 occurring in the respective reduced denominators of U and V. Then

$$2^{B+\alpha} = 2^B U \in \mathbb{N}, \qquad 3^{B+\alpha} = 3^B V \in \mathbb{N},$$

while $B + \alpha \notin \mathbb{Z}$. The recurrence identities are equivalent to $R_k = 2^{\{k\alpha\}}$ and $S_k = 3^{\{k\alpha\}}$, so they add an exact dynamic coding but no hidden hypothesis. ☐

33.5 Cobham-type and Mahler-type diagnostics

Cobham's theorem says that a sequence automatic in two multiplicatively independent integer bases is ultimately periodic [27, 28]. Sturmian words of irrational slope are not automatic in any integer base. Theorem 33.4 therefore suggests a tempting strategy: show that the dyadic rational recurrence makes s 2-automatic and that the triadic recurrence makes it 3-automatic.

That implication is *not* presently available. Rationality of the multiplier does not turn the compact rational state R_k or S_k into a finite-state object; the exact reduced denominators grow without bound. Cobham's theorem becomes applicable only after proving a finite-kernel or finite-substitution property, and that is essentially a new arithmetic statement. The synchronized formulation is useful because it identifies this missing lemma cleanly rather than treating "automaticity" as a metaphor.

Research target 33.5 (Finite-kernel extraction)**.** Show that simultaneous rational realizability of the same irrational mechanical word by (91) forces a finite 2-kernel and a finite 3-kernel, or derive a weaker pair of incompatible substitutive structures.

A Mahler-function variant is also conceivable: encode the jump word by $F(z) = \sum_{k \geq 0} s_k z^k$. Aperiodicity and the finite coefficient set already imply, by classical results of Fatou type, that F is transcendental over $\mathbb{Q}(z)$. A proof would have to derive incompatible functional equations from the two rational recurrences. No such equations were found here; the recurrence is multiplicative in the state, not self-similar in the index.

34 Effective logarithmic separation

34.1 From linear forms in two logarithms to an irrationality measure

The source report proves an elementary denominator-growth statement: a very good rational approximation to a hypothetical solution β would force a correspondingly large output, and consecutive continued-fraction denominators obey a single-exponential recurrence. Since β is itself a ratio of logarithms of fixed integers, Baker's theory gives a substantially stronger conclusion.

Theorem 34.1 (Effective finite irrationality measure for the primitive generator)**.** *Assume a primitive counterexample β exists, and put $M = 2^{\beta}$. There are effectively computable constants $c_M > 0$ and $\mu_M < \infty$ such that, for every $p \in \mathbb{Z}$ and $q \geq 2$,*

$$\left|\beta - \frac{p}{q}\right| \geq c_M q^{-\mu_M}. \tag{96}$$

One may choose μ_M bounded by an effectively computable absolute constant times $1 + \log M$, hence by an absolute constant times $1 + \beta$. The same conclusion follows from $\beta = \log A / \log 3$.

Proof. Consider the linear form

$$\Lambda = q \log M - p \log 2.$$

It is nonzero: otherwise $M^q = 2^p$, which would make $\beta = p/q$ rational; a rational solution is integral by the elementary rational-exponent classification already formalized in the source corpus. The Baker–Wüstholz theorem, or Matveev's explicit lower bound for a homogeneous linear form in logarithms of algebraic numbers, gives

$$\log |\Lambda| \geq -C\,(1 + \log M) \log H - C'(1 + \log M),$$

where $H \geq \max(3, |p|, q)$ and C, C' are effective absolute constants after the fixed number 2 is absorbed. For approximations with $|p| \leq (\beta + 1)q$, one has $H \ll_M q$; for all other p the desired inequality is trivial after decreasing c_M. Since

$$\left|\beta - \frac{p}{q}\right| = \frac{|\Lambda|}{q \log 2},$$

absorbing fixed factors and one additional power of q gives (96). Matveev's dependence on the logarithmic heights is linear in $\log M = \beta \log 2$ here, which gives the final scale assertion. ∎

Status: [classical/open background] plus **[paper proof in this report]**

The lower-bound theorem for logarithmic forms is classical [3, 19, 20]. Its application to the primitive generator, the continued-fraction consequences below, and the comparison with the source report are new deductions in this continuation. No useful small numerical value of μ_M is claimed; standard explicit constants are very large.

34.2 Polynomial recurrence for convergent denominators

Let p_n/q_n be the continued-fraction convergents of β. The standard upper bound

$$\left|\beta - \frac{p_n}{q_n}\right| < \frac{1}{q_n q_{n+1}}$$

combined with (96) gives the following.

Corollary 34.2 (Polynomial denominator growth)**.** *There is an effective constant $C_M > 0$ such that*

$$q_{n+1} \le C_M q_n^{\mu_M - 1} \tag{97}$$

for every n with $q_n \ge 2$. Consequently the next partial quotient satisfies $a_{n+1} = O_M(q_n^{\mu_M-2})$.

This is qualitatively stronger than the inherited elementary estimate $q_{n+1} < 2M^{q_n}$. It rules out Liouville behavior and says that the exceptionally long near-repetitions of the Sturmian word are polynomially controlled in the preceding standard word length. The exponent depends on the unknown integer M, however, and may be enormous.

34.3 Polynomial spacing in conditional blocks

The irrationality measure also upgrades the separation of the block nodes. From (96), for $d \ge 1$ and the nearest integer r to $d\beta$,

$$\|d\beta\|_{\mathbb{Z}} = |d\beta - r| \ge c_M d^{1-\mu_M}. \tag{98}$$

For an integer $T \ge 0$, put

$$K_T = \left\lfloor \frac{T+1}{\beta} \right\rfloor, \qquad x_{T,k} = T - \lfloor k\beta \rfloor + k\beta = T + \{k\beta\}, \quad 0 \le k \le K_T.$$

The associated candidate and output nodes are

$$m_{T,k} = 2^{x_{T,k}} = 2^{T-\lfloor k\beta \rfloor} M^k, \qquad a_{T,k} = 3^{x_{T,k}} = 3^{T-\lfloor k\beta \rfloor} A^k. \tag{99}$$

Corollary 34.3 (Effective block separation)**.** *For distinct $i, j \in \{0, \dots, K_T\}$,*

$$|x_{T,j} - x_{T,i}| \ge c_M K_T^{1-\mu_M}, \tag{100}$$

$$|m_{T,j} - m_{T,i}| \ge (\log 2) c_M 2^T K_T^{1-\mu_M}, \tag{101}$$

$$|a_{T,j} - a_{T,i}| \ge (\log 3) c_M 3^T K_T^{1-\mu_M}. \tag{102}$$

Proof. The ordinary distance between two fractional parts is at least their circular distance, so

$$|\{j\beta\} - \{i\beta\}| \ge \|(j-i)\beta\|_{\mathbb{Z}} \ge c_M |j-i|^{1-\mu_M} \ge c_M K_T^{1-\mu_M},$$

because $1 - \mu_M < 0$. This proves (100). The mean value theorem applied to 2^x and 3^x on $[T, T+1]$ gives the remaining inequalities. □

34.4 Separation on the full (n, k) lattice

The same estimate applies to the two-dimensional conditional exponent grid.

Corollary 34.4 (Polynomial separation of conditional exponents)**.** *There are effective $c_M, \mu_M > 0$ such that, for distinct integer pairs $(n, k), (n', k')$ with $0 \le k, k' \le H$,*

$$|(n + k\beta) - (n' + k'\beta)| \ge \min\{1, c_M H^{1-\mu_M}\}.$$

Proof. If $k = k'$, the difference is a nonzero integer. Otherwise divide the linear form $(n-n')+(k-k')\beta$ by $|k - k'|$ and apply (96). □

The column exponent lattice $u + \vartheta v$ has an analogous effective polynomial spacing with absolute constants, because $\vartheta = \log_2 3$ is a fixed ratio of two logarithms. More refined explicit estimates are known for linear forms in $\log 2$ and $\log 3$; for example Wu's simultaneous linear-independence measure leads, in the usual normalization, to an irrationality exponent for ϑ below approximately 8.62 [24]. The exact convention-dependent numerical conversion is not needed below.

34.5 Why the Baker upgrade does not close the conjecture

The effective spacing is useful, but its scale is still compatible with the exact conditional population. A dyadic exponent block contains only $K_T + 1 \asymp T$ nodes. A polynomial lower bound on their minimum spacing imposes no contradiction on T points in a fixed interval. Likewise, the full triangular grid has $\asymp T^2$ points in an interval of exponent length T, and polynomially small gaps are expected.

Baker's theorem therefore strengthens compactness and determinant estimates but does not supply the missing logarithm by itself. Its most promising roles are indirect:

- controlling the smallest row and column gaps in an exact Harish–Chandra–Itzykson–Zuber factorization;
- bounding how often a p-adic assignment problem can be nearly degenerate;
- limiting the lengths of near-periodic Sturmian blocks that a congruence argument must handle;
- making computational searches over continued-fraction branches provably finite once a candidate output M is fixed.

35 Exact all-support valuations of ordinary block Vandermondes

35.1 The block determinants

Fix an integer T and abbreviate $K = K_T$. Let $m_k = m_{T,k}$ and $a_k = a_{T,k}$ be the $K+1$ input–output pairs in (99). Their ordinary Vandermonde determinants are

$$V_T^{(2)} = \det(m_i^j)_{0\le i,j\le K} = \prod_{0\le i<j\le K} (m_j - m_i), \tag{103}$$

$$V_T^{(3)} = \det(a_i^j)_{0\le i,j\le K} = \prod_{0\le i<j\le K} (a_j - a_i). \tag{104}$$

They are nonzero integers. Since $m_k \in [2^T, 2^{T+1})$ and $a_k \in [3^T, 3^{T+1})$,

$$\log |V_T^{(2)}| \le R_T T \log 2, \tag{105}$$

$$\log |V_T^{(3)}| \le R_T (T \log 3 + \log 2), \tag{106}$$

where

$$R_T = \binom{K+1}{2} = \frac{K(K+1)}{2}.$$

The additive $R_T \log 2$ in (106) is lower order.

35.2 The input determinant: an exact formula at every support prime

Factor

$$M = 2^a W, \qquad W \text{ odd}, \qquad \delta_2 = \beta - a = \log_2 W.$$

As noted above, $W > 1$, so $W \geq 3$ and

$$\delta_2 \geq \log_2 3 > 1. \tag{107}$$

Because a is an integer,

$$m_k = 2^{T - \lfloor k\delta_2 \rfloor} W^k. \tag{108}$$

The 2-adic valuations in (108) strictly decrease with k, while every odd support-prime valuation strictly increases.

Proposition 35.1 (Exact support valuations for $V_T^{(2)}$)**.** *For $0 \leq i < j \leq K$,*

$$v_2(m_j - m_i) = T - \lfloor j\delta_2 \rfloor, \tag{109}$$

$$v_p(m_j - m_i) = i\, v_p(W) \qquad (p \mid W). \tag{110}$$

Consequently

$$v_2(V_T^{(2)}) = \sum_{j=1}^{K} j\big(T - \lfloor j\delta_2 \rfloor\big), \tag{111}$$

$$v_p(V_T^{(2)}) = v_p(W) \sum_{i=0}^{K-1} i(K - i) \qquad (p \mid W). \tag{112}$$

Proof. Put $\rho_k = T - \lfloor k\delta_2 \rfloor$. By (107), $\rho_i > \rho_j$ when $i < j$. From (108),

$$m_j - m_i = 2^{\rho_j} W^i \left(W^{j-i} - 2^{\rho_i - \rho_j}\right).$$

The parenthesis is odd, proving (109). For an odd prime $p \mid W$, the two terms m_i and m_j have distinct valuations $iv_p(W) < jv_p(W)$, so the ultrametric equality gives (110). Summing over pairs gives (111)–(112). □

Let $Q_T^{(2)}$ be the part of $V_T^{(2)}$ supported on the primes dividing $2M$. Then Proposition 35.1 gives the exact identity

$$\log Q_T^{(2)} = \log 2 \sum_{j=1}^{K} j\big(T - \lfloor j\delta_2 \rfloor\big) + \log W \sum_{i=0}^{K-1} i(K - i). \tag{113}$$

The two elementary sums have asymptotics

$$\sum_{j=1}^{K} j\big(T - \lfloor j\delta_2 \rfloor\big) = \frac{TK^2}{2} - \frac{\delta_2 K^3}{3} + O_\beta(T^2),$$

$$\sum_{i=0}^{K-1} i(K - i) = \frac{K^3}{6} + O(K^2).$$

Using $\log W = \delta_2 \log 2$ and $K/T \to 1/\beta$ yields

$$\frac{\log Q_T^{(2)}}{R_T T \log 2} \longrightarrow 1 - \frac{\delta_2}{3\beta}. \tag{114}$$

This includes every prime appearing in the entries. Any further prime divisor of the Vandermonde comes from subtraction and hence from genuine cancellation beyond termwise entry valuations.

35.3 The output determinant and bounded tie corrections

Factor

$$A = 3^b Z, \qquad 3 \nmid Z, \qquad \delta_3 = \beta - b = \log_3 Z > 0.$$

Then

$$a_k = 3^{T - \lfloor k\delta_3 \rfloor} Z^k. \tag{115}$$

Unlike δ_2, the number δ_3 can be smaller than 1; for example the least possible base-free factor is 2. Equal consecutive 3-adic valuations can therefore occur. They contribute only a lower-order correction.

Proposition 35.2 (Output support valuations)**.** *For every prime $p \mid Z$ and $0 \le i < j \le K$,*

$$v_p(a_j - a_i) = i\, v_p(Z).$$

At $p = 3$,

$$v_3(a_j - a_i) = T - \lfloor j\delta_3 \rfloor$$

after adding a nonnegative correction that is zero whenever $\lfloor i\delta_3 \rfloor < \lfloor j\delta_3 \rfloor$. The total correction over all pairs is $O_{A,\beta}(K^2)$. Consequently, if $Q_T^{(3)}$ denotes the part of $V_T^{(3)}$ supported on primes dividing $3A$, then

$$\frac{\log Q_T^{(3)}}{R_T T \log 3} \longrightarrow 1 - \frac{\delta_3}{3\beta}. \tag{116}$$

Proof. The assertion for $p \mid Z$ is the same distinct-valuation argument as before. Put $\rho_k = T - \lfloor k\delta_3 \rfloor$. If $\rho_i > \rho_j$, then

$$a_j - a_i = 3^{\rho_j} Z^i \left(Z^{j-i} - 3^{\rho_i - \rho_j} \right)$$

and the parenthesis is not divisible by 3, so the valuation is exactly ρ_j. If $\rho_i = \rho_j$, then

$$v_3(a_j - a_i) = \rho_j + v_3(Z^{j-i} - 1).$$

Equality of the floors implies $(j - i)\delta_3 < 1$, so $j - i$ belongs to a fixed finite set depending only on δ_3. Hence the extra valuation is bounded by a constant depending on A and β. There are $O(K^2)$ pairs. The asymptotic calculation is then identical to (113). ☐

35.4 The depth-sensitive product ceiling

Combining (105), (106), (114), and (116) gives the principal conclusion of the ordinary Vandermonde audit.

Theorem 35.3 (All-support ordinary Vandermonde fraction)**.** *For a hypothetical primitive counterexample,*

$$\frac{\log(Q_T^{(2)} Q_T^{(3)})}{R_T T \log 6} \longrightarrow 1 - \frac{\delta_2 \log 2 + \delta_3 \log 3}{3\beta \log 6}, \qquad \delta_2 = \beta - a, \quad \delta_3 = \beta - b. \tag{117}$$

Because $\min(a, b) = 0$*, the limit is at most*

$$1 - \frac{\log 2}{3 \log 6} = 0.871049\,064\ldots. \tag{118}$$

If $b = 0$, the sharper ceiling is

$$1 - \frac{\log 3}{3 \log 6} = 0.795599\,921\ldots.$$

If $a = b = 0$, the limit is exactly $2/3$.

Proof. Only the numerical ceiling remains to be shown. If $a = 0$, then $\delta_2 = \beta$, so the deficit in (117) is at least $\log 2/(3 \log 6)$. If $b = 0$, it is at least $\log 3/(3 \log 6)$. Taking the larger possible fraction gives (118). If both vanish, the numerator of the deficit is $\beta \log 6$. □

Table 6: Leading termwise-divisor ceilings for ordinary block Vandermondes.

Branch information	Exact/upper fraction	Decimal
$a = b = 0$	$2/3$	$0.666666\ldots$
$b = 0$	$\leq 1 - \log 3/(3 \log 6)$	$0.795599\ldots$
No matched depth only	$\leq 1 - \log 2/(3 \log 6)$	$0.871049\ldots$

35.5 Why even 87% is not enough

The deficit in Theorem 35.3 is a fixed positive fraction once the hypothetical primitive pair is fixed. The archimedean height is $\asymp T^3$, whereas all factorial, spacing, and tie corrections are $O_\beta(T^2 \log T)$. Thus no lower-order refinement of the ordinary Vandermonde can close the remaining gap.

This conclusion is stronger than the earlier observation that the 2-adic valuation alone captures one third in the base-free case. It includes every prime already present in the entries and every possible base depth. A successful ordinary-Vandermonde proof would need a new leading-order source of divisibility in the prime factors of the *differences*; the fixed-support valuations are completely accounted for by Theorem 35.3.

36 The mixed-semigroup determinant and its Gini norm

36.1 Balanced low-weight columns

Let

$$\Lambda = \{u + \vartheta v : (u, v) \in \mathbb{N}_0^2\}.$$

Because ϑ is irrational, every element has a unique pair (u, v). List the pairs in increasing order of $u + \vartheta v$:

$$(u_0, v_0), (u_1, v_1), \ldots.$$

For $N \geq 1$, let

$$L_N = \max_{0 \leq j < N} (u_j + \vartheta v_j).$$

The elementary lattice count

$$\#\{(u, v) \in \mathbb{N}_0^2 : u + \vartheta v \leq L\} = \frac{L^2}{2\vartheta} + O_\vartheta(L) \tag{119}$$

shows that

$$L_N \sim \sqrt{2\vartheta N}. \tag{120}$$

Moreover, the empirical distribution of

$$X_j = \left(\frac{u_j}{L_N}, \frac{\vartheta v_j}{L_N}\right) \tag{121}$$

converges to uniform area measure on the standard triangle

$$\mathcal{T} = \{(x,z) \in \mathbb{R}^2 : x \geq 0, z \geq 0, x+z \leq 1\}. \tag{122}$$

Here and below "uniform" means probability measure with density 2 on this triangle.

For the block at height T, set $K = K_T$ and $N = K+1$. Define

$$D_T = \det\left(m_{T,k}^{u_j} a_{T,k}^{v_j}\right)_{0 \leq k,j < N}. \tag{123}$$

The generalized-Vandermonde theorem inherited from the source report implies that D_T is a nonzero integer. Put

$$B_j = 2^{u_j} 3^{v_j}.$$

Since $m_{T,k} = 2^{x_{T,k}}$ and $a_{T,k} = 3^{x_{T,k}}$,

$$m_{T,k}^{u_j} a_{T,k}^{v_j} = B_j^{x_{T,k}} = B_j^{T+\{k\beta\}}. \tag{124}$$

Thus the Leibniz expansion gives the clean archimedean bound

$$\log|D_T| \leq (T+1) \sum_{j<N} \log B_j + \log N!. \tag{125}$$

The leading height is

$$H_T = T \sum_{j<N} \log B_j. \tag{126}$$

By (120) and the mean $\mathbb{E}(X_1 + X_2) = 2/3$ on $\mathcal{T}$,

$$H_T \sim \frac{2}{3} T N L_N \log 2 \asymp_\beta T^{5/2}. \tag{127}$$

Both $\sum_j \log B_j$ and $\log N!$ in the error of (125) are $o(H_T)$.

36.2 The all-prime tropical divisor

For a prime p, define the assignment minimum

$$\tau_{p,T} = \min_{\sigma \in S_N} \sum_{k=0}^{N-1} v_p\left(m_{T,k}^{u_{\sigma(k)}} a_{T,k}^{v_{\sigma(k)}}\right). \tag{128}$$

Every Leibniz term is divisible by $p^{\tau_{p,T}}$, so

$$Q_T^{\mathrm{trop}} = \prod_p p^{\tau_{p,T}} \quad \text{divides} \quad D_T. \tag{129}$$

Only primes dividing $6MA$ occur in the product.

The problem is now to compute the leading asymptotic of $\log Q_T^{\mathrm{trop}}/H_T$. A crude estimate "minimum $\leq$ average" loses the precise branch dependence. The exact answer is controlled by a simple probability norm.

36.3 Antimonotone assignment and the Gini norm

Let $X = (X_1, X_2)$ be uniform on $\mathcal{T}$, and let X' be an independent copy. For $c = (r, s) \in \mathbb{R}^2$, define

$$\mathfrak{G}(r, s) = \mathbb{E}\left|r(X_1 - X_1') + s(X_2 - X_2')\right|. \tag{130}$$

This is a norm on $\mathbb{R}^2$: homogeneity and the triangle inequality are immediate, and the support of $X - X'$ spans $\mathbb{R}^2$, so only $(r, s) = (0, 0)$ has norm zero.

Let $Y = rX_1 + sX_2$, with ascending quantile $Q_Y(t)$. Pairing an increasing uniform row parameter with the decreasing ordering of Y gives the continuum assignment functional

$$\Phi(r, s) = \int_0^1 t\, Q_Y(1 - t)\, dt. \tag{131}$$

Lemma 36.1 (Gini formula for the minimum assignment)**.** *For every* $(r, s) \in \mathbb{R}^2$*,*

$$\Phi(r, s) = \frac{1}{2}\,\mathbb{E}Y - \frac{1}{4}\,\mathfrak{G}(r, s) = \frac{r + s}{6} - \frac{1}{4}\,\mathfrak{G}(r, s). \tag{132}$$

Proof. The rearrangement inequality says that the minimum scalar product pairs the increasing row parameter with the decreasing list of column values. Changing variables $u = 1 - t$ in (131) gives

$$\Phi = \int_0^1 (1 - u) Q_Y(u)\, du.$$

If Y' is an independent copy, the minimum of Y, Y' has quantile density $2(1 - u)$, so

$$\Phi = \frac{1}{2}\mathbb{E}\min(Y, Y').$$

Since $\min(Y, Y') = (Y + Y' - |Y - Y'|)/2$, this becomes $\frac{1}{2}\mathbb{E}Y - \frac{1}{4}\mathbb{E}|Y - Y'|$. Finally $\mathbb{E}X_1 = \mathbb{E}X_2 = 1/3$. □

36.4 Closed form of the Gini norm

The norm (130) is elementary. Sort the three vertex values

$$0, \qquad r, \qquad s$$

as $\xi_0 \le \xi_1 \le \xi_2$, and put

$$A_0 = \xi_1 - \xi_0, \qquad B_0 = \xi_2 - \xi_1.$$

Proposition 36.2 (Explicit triangular Gini norm)**.** *If* $A_0 + B_0 > 0$*, then*

$$\mathfrak{G}(r, s) = \frac{2\left(2A_0^2 + 3A_0B_0 + 2B_0^2\right)}{15(A_0 + B_0)}. \tag{133}$$

At $(r, s) = (0, 0)$ *the value is* 0*. In particular,*

$$\mathfrak{G}(1, 0) = \mathfrak{G}(0, 1) = \mathfrak{G}(1, 1) = \frac{4}{15}, \qquad \mathfrak{G}(1, -1) = \frac{7}{15}. \tag{134}$$

Proof. Translation of the three vertex values does not change a difference $Y - Y'$, so take $\xi_0 = 0$, $\xi_1 = A_0$, and $\xi_2 = A_0 + B_0 = L$. Projection of uniform area measure on a triangle by a linear function has density

$$f(y) = \begin{cases} \dfrac{2y}{A_0 L}, & 0 \le y \le A_0, \\ \dfrac{2(L-y)}{B_0 L}, & A_0 \le y \le L. \end{cases}$$

Thus its distribution function is $F(y) = y^2/(A_0 L)$ on the first interval and $F(y) = 1-(L-y)^2/(B_0 L)$ on the second. The standard identity

$$\mathbb{E}|Y - Y'| = 2\int_{-\infty}^{\infty} F(y)(1 - F(y))\, dy$$

gives (133) after two polynomial integrations. Cases with one gap zero follow by continuity. □

For later use, if $\gamma \ge 0$, then

$$\mathfrak{G}(-1,\gamma) = \frac{2(2+3\gamma+2\gamma^2)}{15(1+\gamma)} = \frac{4}{15} + \frac{2\gamma(1+2\gamma)}{15(1+\gamma)} \ge \frac{4}{15}. \tag{135}$$

36.5 Normalized prime-valuation slope vectors

Recall

$$a = v_2(M), \quad b = v_3(A), \quad c = v_2(A), \quad d = v_3(M).$$

For each prime, define a normalized slope vector $c_p \in \mathbb{R}^2$ by

$$c_2 = \left(-1 + \frac{a}{\beta},\ \frac{c}{\beta\vartheta}\right), \tag{136}$$

$$c_3 = \left(\frac{\vartheta d}{\beta},\ -1 + \frac{b}{\beta}\right), \tag{137}$$

$$c_p = \left(\frac{v_p(M)\log p}{\beta \log 2}, \frac{v_p(A)\log p}{\beta\log 3}\right), \qquad p \ne 2, 3. \tag{138}$$

The logarithmic factorizations of M and A give the conservation law

$$\sum_p c_p = (0,0). \tag{139}$$

Indeed, the first coordinate sum is

$$-1 + \frac{a\log 2 + d\log 3 + \sum_{p\ne 2,3} v_p(M)\log p}{\beta\log 2} = 0,$$

and similarly for the second coordinate.

The meaning of these vectors can be read directly from the valuation matrices. Put $s_k = \beta k/T$. Up to $o(L_N)$ after division by T, the logarithmic p-adic cost of the entry indexed by a scaled column (x, z) is

Prime	Cost divided by $TL_N \log 2$
2	$x + s_k\, c_2 \cdot (x, z)$
3	$z + s_k\, c_3 \cdot (x, z)$
$p \neq 2, 3$	$s_k\, c_p \cdot (x, z)$

For example,

$$v_2(m_k^u a_k^v) = u(T - \lfloor k\beta \rfloor + ak) + vck,$$

and substituting $u = L_N x$, $v = L_N z/\vartheta$, $k \sim Ts_k/\beta$ gives the first row. The $O(u)$ errors from the floors sum to $O(NL_N) = o(H_T)$.

36.6 The exact continuum limit

Theorem 36.3 (All-prime tropical fraction for the mixed block determinant)**.** *Under a hypothetical primitive counterexample, the determinant* (123) *satisfies*

$$\frac{\log Q_T^{\mathrm{trop}}}{H_T} \longrightarrow 1 - \frac{3}{8} \sum_p \mathfrak{G}(c_p), \tag{140}$$

where the finitely many slope vectors c_p are given by (136)*–*(138)*.*

Proof. The row parameters $s_k = \beta k/T$ become uniformly distributed on $[0, 1]$, because $K_T \sim T/\beta$. The scaled columns become uniformly distributed on $\mathcal{T}$ by (119). The finite rearrangement inequality and Riemann-sum convergence therefore show that the assignment minimum at prime p, after normalization by $TL_N N \log 2$, tends to its constant-column mean plus $\Phi(c_p)$.

Only $p = 2$ has constant part x, and only $p = 3$ has constant part z. Their combined mean is $2/3$. By Lemma 36.1, the sum of the slope contributions is

$$\sum_p \Phi(c_p) = \frac{1}{6} \sum_p (c_{p,1} + c_{p,2}) - \frac{1}{4} \sum_p \mathfrak{G}(c_p).$$

The first sum vanishes by (139). Hence

$$\frac{\log Q_T^{\mathrm{trop}}}{TL_N N \log 2} \longrightarrow \frac{2}{3} - \frac{1}{4} \sum_p \mathfrak{G}(c_p).$$

On the other hand, (127) gives $H_T/(TL_N N \log 2) \to 2/3$. Dividing proves (140). □

This theorem is exact at the level of minimum Leibniz-term valuations. It replaces a family of separate p-adic estimates by one finite norm sum satisfying a conservation law.

The four base and cross depths already give a useful branch-sensitive bound without knowing any external prime factorization.

Corollary 36.4 (Depth-sensitive three-vector bound)**.** *For every hypothetical primitive counterexample,*

$$\limsup_{T\to\infty} \frac{\log Q_T^{\mathrm{trop}}}{H_T} \leq 1 - \frac{3}{8}\Big(\mathfrak{G}(c_2) + \mathfrak{G}(c_3) + \mathfrak{G}(-c_2 - c_3)\Big). \tag{141}$$

Equality in this upper bound can occur only when all nonzero external slope vectors are positively collinear.

Proof. The external vectors satisfy $\sum_{p\neq 2,3} c_p = -c_2 - c_3$. Hence the triangle inequality gives

$$\sum_{p\neq 2,3} \mathfrak{G}(c_p) \geq \mathfrak{G}(-c_2 - c_3).$$

Substitute this in Theorem 36.3. Equality in a norm triangle inequality requires all summands to lie on one nonnegative ray. □

The equality condition detects exactly the second-iterate kernel branch reviewed in the source report. Indeed the external vectors c_p are rescaled versions of $(v_p(M), v_p(A))$. They are all collinear if and only if the two external valuation vectors of M and A are rationally dependent, equivalently if the kernel K_β is nonzero. Thus:

Corollary 36.5 (The tropical norm sees the kernel dichotomy)**.** *In the multiplicatively independent branch $K_\beta = 0$, the inequality in (141) is strict. In the dependent branch $K_\beta \neq 0$, the external triangle inequality is an equality (including the axis cases in which one output is $\{2,3\}$-smooth). For fixed M, A, the strict gap in the independent branch is explicit:*

$$\Delta_{\text{ext}} = \sum_{p\neq 2,3} \mathfrak{G}(c_p) - \mathfrak{G}\left(\sum_{p\neq 2,3} c_p\right) > 0. \tag{142}$$

There is no uniform positive lower bound for Δ_{ext}: distinct rational rays can approach one another as the valuations grow. Nevertheless, the quantity is a computable branch invariant and gives a sharper determinant threshold for every fixed hypothetical primitive pair.

36.7 The universal $4/5$ ceiling

Theorem 36.6 (Sharp universal min-plus ceiling)**.** *For every hypothetical primitive counterexample,*

$$\sum_p \mathfrak{G}(c_p) \geq \frac{8}{15}. \tag{143}$$

Consequently

$$\limsup_{T\to\infty} \frac{\log Q_T^{\text{trop}}}{H_T} \leq \frac{4}{5}. \tag{144}$$

The constant $4/5$ is the sharp supremum allowed by the normalized valuation constraints, although equality would require a degenerate boundary incompatible with a genuine nonintegral solution.

Proof. By the primitive theorem, either $a = 0$ or $b = 0$. Suppose first that $a = 0$. Then $c_2 = (-1, \gamma)$ with $\gamma = c/(\beta\vartheta) \geq 0$. By (135), $\mathfrak{G}(c_2) \geq 4/15$. Conservation and the triangle inequality for the norm give

$$\sum_{p\neq 2} \mathfrak{G}(c_p) \geq \mathfrak{G}\left(\sum_{p\neq 2} c_p\right) = \mathfrak{G}(-c_2) = \mathfrak{G}(c_2).$$

Thus the total is at least $8/15$. If $b = 0$, apply the same argument to $c_3 = (\eta, -1)$, where $\eta = \vartheta d/\beta \geq 0$. Substitution in (140) gives $1 - (3/8)(8/15) = 4/5$. □

The proof is short because the difficult arithmetic has been compressed into two facts: the slope vectors sum to zero, and one structural vector reaches from -1 to a nonnegative vertex. No averaging estimate alone reveals this geometry.

36.8 Cross-coprime constants: $7/10$ and $3/10$

Assume the stronger branch condition

$$\gcd(MA, 6) = 1. \tag{145}$$

Then $a = b = c = d = 0$,

$$c_2 = (-1, 0), \qquad c_3 = (0, -1), \qquad \sum_{p \neq 2,3} c_p = (1, 1).$$

Corollary 36.7 (All-prime cross-coprime ceiling)**.** *Under* (145),

$$\limsup_{T \to \infty} \frac{\log Q_T^{\mathrm{trop}}}{H_T} \leq \frac{7}{10}. \tag{146}$$

If the external normalized valuation vectors are not all positively collinear, the inequality is strict by a branch-dependent positive amount.

Proof. By (134), the structural vectors each contribute $4/15$. The triangle inequality gives

$$\sum_{p \neq 2,3} \mathfrak{G}(c_p) \geq \mathfrak{G}(1, 1) = \frac{4}{15}.$$

Thus the total norm is at least $4/5$, and $1 - (3/8)(4/5) = 7/10$. Equality in the triangle inequality requires positive collinearity. ☐

The places 2 and 3 alone have an even smaller exact continuum contribution.

Corollary 36.8 (The structural-place 3/10 constant)**.** *Under* (145)*, let* $Q_T^{(2,3)} = 2^{\tau_{2,T}} 3^{\tau_{3,T}}$*. Then*

$$\frac{\log Q_T^{(2,3)}}{H_T} \longrightarrow \frac{3}{10}. \tag{147}$$

Proof. For $c = (-1, 0)$,

$$\Phi(c) = \frac{-1}{6} - \frac{1}{4} \cdot \frac{4}{15} = -\frac{7}{30},$$

and the same holds for $(0, -1)$. Adding the constant mean $2/3$ gives $2/3 - 7/15 = 1/5$. Dividing by the height mean $2/3$ gives $3/10$. ☐

Equivalently, one can compute this constant from quantiles. The descending quantile of X_1 on $\mathcal{T}$ is $1 - \sqrt{t}$, so

$$\int_0^1 t(1 - \sqrt{t})\, dt = \frac{1}{10}, \qquad \mathbb{E} X_1 = \frac{1}{3}.$$

The ratio is $(1/10)/(1/3) = 3/10$ in each structural coordinate.

36.9 Correction and calibration of a later repository note

A short p-adic priority note added to the repository after the attached unified report makes the correct strategic observation that minimum Leibniz-term valuations leave a constant fraction of the archimedean height uncovered. Two details require correction.

First, the displayed structural valuation formulas in that note simultaneously omit the terms $kuv_2(M)$ and $kvv_3(A)$; they therefore assume both $a = 0$ and $b = 0$. The inherited primitive theorem gives only $\min(a, b) = 0$. Equations (136)–(137) retain all four cross/base depths.

Second, the proposed all-prime 2/3 ceiling is not a valid universal consequence of "minimum $\leq$ average." For an external positive weight proportional to $X_1 + X_2$, the minimum assignment is 4/5 of its average assignment, not at most 2/3. The exact Gini formula shows that the universal normalized ceiling is 4/5, while the cross-coprime all-prime ceiling is 7/10. The note's qualitative conclusion survives—entrywise valuation is insufficient—but the constants and branch hypotheses are materially different.

Status: [paper proof in this report]

The correction is not based on a numerical counterexample to the Alaoglu–Erdős conjecture. It is an exact audit of the assignment inequality used in the method. Rational external valuation directions can approximate the equality configurations arbitrarily well, so no stronger uniform constant follows merely from integrality of the valuation vectors.

36.10 Where extra divisibility can occur

The tropical minimum equals the true determinant valuation whenever its minimizing permutation is unique.

Proposition 36.9 (Unique tropical term forbids cancellation)**.** *Let $D = \det(E_{ij})$ be a nonzero integer determinant. Fix a prime p, and suppose a unique permutation σ_0 minimizes $\sum_i v_p(E_{i,\sigma(i)})$. Then*

$$v_p(D) = \sum_i v_p(E_{i,\sigma_0(i)}).$$

Proof. Divide the Leibniz expansion by the corresponding power of p. The term indexed by σ_0 is a p-adic unit, while every other term is divisible by p. The sum is therefore a unit modulo p. □

Consequently, the missing 20% or more can arise only from primes at which the assignment problem has several minimizers. The tie geometry is explicit.

- For $p \neq 2, 3$, columns tie when $v_p(M)(u - u') + v_p(A)(v - v') = 0$, so ties lie on rational lines orthogonal to the external valuation vector.
- In the cross-coprime branch, the 2-adic cost depends only on u and the 3-adic cost only on v. Every vertical or horizontal column fiber creates many equal-cost permutations. Any leading extra valuation must be produced inside these fibers.
- Cross valuations $v_2(A)$ and $v_3(M)$ tilt the fibers. The floor term in the row profile perturbs the exact affine ordering but does not change the continuum slope vector.

This turns "look for p-adic cancellation" into a concrete algebra problem: factor or control the residual determinants inside equal-slope column fibers, and show that their valuation supplies at least the missing Gini deficit.

37 The complete triangular grid and a $\pi/4$ transport ceiling

The one-block determinant deliberately uses the exact $O(T)$ population in a unit interval of exponent space. The source report also suggests exploiting the full two-dimensional coordinate grid rather than sorting only by magnitude. Conditional on a counterexample, that proposal gives $\asymp T^2$ genuine integer rows below exponent height T, exactly the number that the mixed-semigroup interpolation argument seemed to be missing. It is therefore important to determine whether entrywise p-adic divisibility becomes strong enough on this larger node set.

The answer is again negative, but for a different and pleasantly universal reason. Once all places are summed, the prime-by-prime assignment minima are bounded by one antimonotone transport problem for the logarithmic entry matrix. Both the row heights and the balanced column weights have the same triangular distribution, and the ratio between antimonotone and monotone transport is exactly $\pi/4$.

37.1 Rows and columns

For $T > 0$ define the complete conditional row set

$$\mathcal{R}_T = \{(n,k) \in \mathbb{N}_0^2 : n + k\beta \le T\}, \qquad N_T = \#\mathcal{R}_T. \tag{148}$$

For $i = (n_i, k_i) \in \mathcal{R}_T$ put

$$x_i = n_i + k_i\beta, \qquad m_i = 2^{n_i} M^{k_i} = 2^{x_i}, \qquad a_i = 3^{n_i} A^{k_i} = 3^{x_i}. \tag{149}$$

Because β is irrational, the x_i are pairwise distinct.

Let $\mathcal{C}_T$ be the first N_T points $(u,v) \in \mathbb{N}_0^2$ in increasing order of

$$\lambda = u + \vartheta v, \qquad y = \log(2^u 3^v) = \lambda \log 2.$$

Write L_T for the largest selected λ. Standard lattice-point estimates for the two triangles give

$$N_T = \frac{T^2}{2\beta} + O_\beta(T), \qquad L_T = \sqrt{2\vartheta N_T} + O_\vartheta(1) = T\sqrt{\frac{\vartheta}{\beta}} + O_\beta(1). \tag{150}$$

The associated square determinant is

$$D_T^\triangle = \det\Big(m_i^{u_j} a_i^{v_j}\Big)_{i,j=1}^{N_T} = \det(e^{x_i y_j})_{i,j=1}^{N_T}. \tag{151}$$

Every entry is an integer. Ordering the x_i and y_j increasingly, strict total positivity of the exponential kernel makes $D_T^\triangle > 0$; equivalently, this is the same exponential-polynomial zero theorem used by the generalized-Vandermonde part of the source report.

Status: [paper proof in this report] for the asymptotic transport statements below. Integrality and nonvanishing are inherited from the mixed-semigroup determinant architecture [kernel-verified Lean].

37.2 The common triangular distribution

The normalized row heights $h_i = x_i/T$ and column weights $\ell_j = y_j/(L_T \log 2)$ both live in $[0,1]$. Their empirical measures have the same limit.

Lemma 37.1 (Triangular height law)**.** *For every continuous $f : [0,1] \to \mathbb{R}$,*

$$\frac{1}{N_T}\sum_{i=1}^{N_T} f(h_i) \longrightarrow \int_0^1 2t f(t)\,dt, \tag{152}$$

$$\frac{1}{N_T}\sum_{j=1}^{N_T} f(\ell_j) \longrightarrow \int_0^1 2t f(t)\,dt. \tag{153}$$

Thus the common limiting distribution has cumulative distribution function $F(t) = t^2$ and quantile function

$$Q(s) = \sqrt{s}, \qquad 0 \le s \le 1. \tag{154}$$

Proof. After scaling (n,k) to $(n/T, \beta k/T)$, the row lattice becomes equidistributed in the unit right triangle

$$\{(r,s) : r,s \ge 0,\ r+s \le 1\}.$$

The subtriangle on which $r+s \le t$ has area $t^2/2$, while the whole triangle has area $1/2$. Hence the height $r+s$ has distribution function t^2. Boundary discrepancies are $O(T)$ against $N_T \asymp T^2$, proving (152).

For the columns, scale (u,v) to $(u/L_T, \vartheta v/L_T)$. The same unit triangle appears, and $\ell = (u+\vartheta v)/L_T$ is again the sum of the scaled coordinates. The boundary shell created by taking the first exactly N_T points has $O(L_T)$ points against $N_T \asymp L_T^2$, which proves (153). □

37.3 Summing all primewise assignment minima

For each prime p, let

$$\tau_{p,T} = \min_{\sigma \in S_{N_T}} \sum_{i=1}^{N_T} v_p\Big(m_i^{u_{\sigma(i)}} a_i^{v_{\sigma(i)}}\Big), \qquad Q_T^{\triangle} = \prod_p p^{\tau_{p,T}}. \tag{155}$$

As before, $Q_T^{\triangle}$ divides every Leibniz term and hence divides $D_T^{\triangle}$. The key observation is that taking a separate minimizing permutation at each prime can only decrease the sum relative to using one permutation at all primes.

Lemma 37.2 (All-place transport domination)**.** *For every T,*

$$\log Q_T^{\triangle} \le \min_{\sigma \in S_{N_T}} \sum_{i=1}^{N_T} x_i y_{\sigma(i)}. \tag{156}$$

Proof. For a permutation σ, put

$$C_p(\sigma) = \sum_i v_p\Big(m_i^{u_{\sigma(i)}} a_i^{v_{\sigma(i)}}\Big) \log p.$$

Then

$$\sum_p \min_\sigma C_p(\sigma) \le \min_\sigma \sum_p C_p(\sigma).$$

Unique factorization and (149) give, entry by entry,

$$\sum_p v_p(m_i^{u_j} a_i^{v_j}) \log p = \log(m_i^{u_j} a_i^{v_j}) = x_i y_j.$$

Substitution proves (156). $\square$

Notice how little arithmetic remains in the last inequality: it is valid in every valuation branch, includes all primes, and does not use the primitive-depth normalization. The price is that it is only an upper bound for the common termwise divisor. That is exactly what is needed for a no-go theorem.

37.4 Optimal transport and the $\pi/4$ ratio

Let

$$E_T^- = \min_{\sigma \in S_{N_T}} \sum_i x_i y_{\sigma(i)}, \tag{157}$$

$$E_T^+ = \max_{\sigma \in S_{N_T}} \sum_i x_i y_{\sigma(i)}. \tag{158}$$

The rearrangement inequality pairs the two ordered lists oppositely for E_T^- and in the same order for E_T^+.

Theorem 37.3 (Triangular-grid transport constants)**.** *As $T \to \infty$,*

$$\frac{E_T^-}{N_T T L_T \log 2} \longrightarrow \int_0^1 \sqrt{t(1-t)}\, dt = \frac{\pi}{8}, \tag{159}$$

$$\frac{E_T^+}{N_T T L_T \log 2} \longrightarrow \int_0^1 t\, dt = \frac{1}{2}. \tag{160}$$

Consequently

$$\frac{E_T^-}{E_T^+} \longrightarrow \frac{\pi}{4} = 0.7853981633\ldots. \tag{161}$$

Proof. By Lemma 37.1, the increasing empirical quantile functions of the two normalized lists converge in L^1 to $Q(t) = \sqrt{t}$. The discrete rearrangement sums are Riemann sums for the monotone and antimonotone couplings. Thus

$$\frac{E_T^-}{N_T T L_T \log 2} \longrightarrow \int_0^1 Q(t)Q(1-t)\, dt, \qquad \frac{E_T^+}{N_T T L_T \log 2} \longrightarrow \int_0^1 Q(t)^2\, dt.$$

The first integral is the beta integral $B(3/2, 3/2) = \Gamma(3/2)^2/\Gamma(3) = \pi/8$; the second is $1/2$. $\square$

The determinant has $N_T!$ Leibniz terms, each at most $e^{E_T^+}$. Since $N_T \asymp T^2$ while $T L_T N_T \asymp T^4$,

$$\log(N_T!) = O(T^2 \log T) = o(N_T T L_T). \tag{162}$$

Combining the preceding statements gives the promised ceiling.

Corollary 37.4 (Full-grid termwise-divisor ceiling)**.** *With the notation above,*

$$\limsup_{T\to\infty} \frac{\log Q_T^{\triangle}}{E_T^+ + \log(N_T!)} \le \frac{\pi}{4}. \tag{163}$$

Thus the complete $\asymp T^2$ row grid does not make the all-prime common divisor exceed the sharp largest-Leibniz-term archimedean bound. Entrywise valuation alone leaves at least the fraction $1-\pi/4 = 0.2146018\ldots$ uncovered at leading order.

Proof. Lemma 37.2 gives $\log Q_T^{\triangle} \le E_T^-$. Divide by the standard determinant upper bound $E_T^+ + \log(N_T!)$ and apply Theorem 37.3 and (162). □

Caution 37.5 (What the $\pi/4$ theorem does not exclude)**.** The denominator in (163) is the largest-term upper bound, not the true size of $D_T^{\triangle}$. Generalized Vandermonde determinants exhibit substantial alternating cancellation. A sufficiently sharp archimedean estimate could lower the relevant budget below E_T^-, in which case the termwise divisor might become decisive. The theorem excludes the naive “more rows plus the same Hadamard/Leibniz estimate” route; it does not exclude a joint p-adic/free-energy argument.

37.5 The Harish–Chandra–Itzykson–Zuber identity

The exact analytic object behind that warning is classical. For diagonal Hermitian matrices $X = \operatorname{diag}(x_1, \ldots, x_N)$ and $Y = \operatorname{diag}(y_1, \ldots, y_N)$ with distinct increasing entries, normalized Haar measure on the unitary group satisfies

$$\int_{U(N)} \exp\bigl(\operatorname{Tr}(XUYU^*)\bigr)\, dU = \frac{\prod_{r=1}^{N-1} r!}{\Delta(x)\Delta(y)} \det(e^{x_i y_j})_{i,j=1}^{N}, \tag{164}$$

where $\Delta(x) = \prod_{i<j}(x_j - x_i)$ and similarly for $\Delta(y)$ [29, 30, 31]. Hence

$$\log D_T^{\triangle} = \log \Delta(x) + \log \Delta(y) - \sum_{r=1}^{N_T-1} \log(r!) + \log \mathcal{I}_T, \tag{165}$$

with $\mathcal{I}_T$ the unitary integral in (164).

The large powers of T hidden separately in the two Vandermonde factors and the factorial product cancel at leading logarithmic scale. What remains is a genuine free-energy problem for two triangular empirical measures. The Baker separation theorem of Section 34 gives a polynomial lower bound for the spacings in $\Delta(x)$, so the row Vandermonde cannot collapse through anomalously close conditional exponents. What is not known is a bound on the full combination (165) strong enough to beat the primewise divisor.

Research target 37.6 (HCIZ–valuation comparison)**.** Determine the T^4-scale limit, or a sufficiently sharp upper bound, of

$$\log D_T^{\triangle} - \log Q_T^{\triangle}$$

for the conditional triangular row lattice and balanced mixed columns. It would suffice to prove that this quantity is negative for all sufficiently large T. Corollary 37.4 shows that such a result must use either the HCIZ/Vandermonde cancellation in (165), extra p-adic cancellation beyond the assignment minima, or both.

This formulation has a useful advantage over a generic interpolation determinant: both limiting measures and all lattice densities are explicit. It is therefore suitable for numerical free-energy reconnaissance before attempting a proof.

38 The first tied p-adic layer is lower order

Theorem 36.6 says that extra divisibility must come from ties among minimum-valuation Leibniz terms. Ties are especially abundant in the cross-coprime branch (145): at $p = 2$ all columns with the same u have the same valuation profile, and at $p = 3$ all columns with the same v do. It is natural to hope that cancellation inside those large fibers supplies the missing leading-order fraction.

The present section computes that first cancellation layer exactly. It factors into ordinary Vandermondes of the odd row parts. Standard upper bounds for p-adic linear forms in two logarithms then show that the whole first layer is only $O(T^{3/2}(\log T)^C)$, whereas the mixed determinant has height $\asymp T^{5/2}$. This does not bound the final determinant valuation—higher tropical layers may cancel again—but it rules out a one-step fiber explanation of the missing fraction.

38.1 Tropical initial form at two

Assume throughout this section that $\gcd(MA, 6) = 1$. For the block rows, write

$$q_k = a_{T,k} = 3^{n_k} A^k, \qquad n_k = T - \lfloor k\beta \rfloor. \tag{166}$$

Both M and every q_k are odd, and

$$m^u_{T,k} a^v_{T,k} = 2^{un_k}\, M^{ku} q^v_k. \tag{167}$$

For each integer $u \geq 0$, let

$$V_u = \{v : (u, v) \text{ is one of the selected } N \text{ columns}\}, \qquad s_u = \#V_u.$$

Because columns are selected by the inequality $u + \vartheta v \leq L_N$ up to one boundary shell, each nonempty V_u is an initial interval of integers. Sort the distinct u values in increasing order. Since $n_0 > n_1 > \cdots > n_{N-1}$, every minimizing assignment for the cost un_k sends one consecutive row block R_u of size s_u to the columns in the fiber $\{u\} \times V_u$.

Let $\tau_{2,T}$ be the minimum from (128). The sum of all Leibniz terms having exactly this valuation is the 2-adic tropical initial form. Up to an overall sign and an odd factor, it is

$$I_{2,T} = \prod_u \det(q^v_k)_{k\in R_u,\ v\in V_u} = \prod_u \prod_{\substack{i<j \\ i,j\in R_u}} (q_j - q_i). \tag{168}$$

The first equality follows by grouping the optimal permutations by fibers and factoring the odd number M^{ku} from each row in a fixed u block; the second is the ordinary Vandermonde identity because V_u is consecutive.

Proposition 38.1 (Exact first 2-adic initial form)**.** *There is an integer $Z_{2,T}$ and an integer gap $g_{2,T} \geq \lfloor \beta \rfloor$ such that*

$$2^{-\tau_{2,T}} D_T = \varepsilon_{2,T} I_{2,T} + 2^{g_{2,T}} Z_{2,T}, \tag{169}$$

where $\varepsilon_{2,T}$ is odd. Consequently

$$v_2(D_T) - \tau_{2,T} \geq \min\{v_2(I_{2,T}), g_{2,T}\}, \tag{170}$$

and equality holds whenever the two numbers inside the minimum are unequal.

Proof. The rearrangement inequality with repeated column weights identifies the minimizing permutations exactly as above, and summing their signed unit parts gives (168). Any nonminimizing assignment must move at least one column across two adjacent u fibers. An exchange across the corresponding boundary changes the cost by

$$(u' - u)(n_i - n_j).$$

Here $u' - u \geq 1$, while consecutive row depths differ by $\lfloor (k+1)\beta \rfloor - \lfloor k\beta \rfloor \geq \lfloor \beta \rfloor$. Hence every nonminimal Leibniz term has valuation at least $\tau_{2,T} + \lfloor \beta \rfloor$, proving the expansion. The valuation statement follows from the ultrametric inequality. □

38.2 The symmetric initial form at three

Put

$$r_k = m_{T,k} = 2^{n_k} M^k. \tag{171}$$

At $p = 3$ the entry factors as

$$m_{T,k}^u a_{T,k}^v = 3^{vn_k}\, r_k^u A^{kv}.$$

For a selected value of v, let U_v be the consecutive set of selected u values and let R'_v be the corresponding consecutive block of rows in the minimum assignment. The same argument gives

$$I_{3,T} = \prod_v \prod_{\substack{i<j \\ i,j \in R'_v}} (r_j - r_i) \tag{172}$$

and

$$3^{-\tau_{3,T}} D_T = \varepsilon_{3,T} I_{3,T} + 3^{g_{3,T}} Z_{3,T}, \qquad g_{3,T} \geq \lfloor \beta \rfloor, \tag{173}$$

with $\varepsilon_{3,T}$ a 3-adic unit.

38.3 Two-logarithm bounds inside the fibers

For $i < j$, set $d = j - i$ and $h = n_i - n_j = \lfloor j\beta \rfloor - \lfloor i\beta \rfloor$. The common factors in the row values are units at the relevant place, so

$$v_2(q_j - q_i) = v_2(A^d - 3^h), \tag{174}$$

$$v_3(r_j - r_i) = v_3(M^d - 2^h). \tag{175}$$

Neither difference vanishes: $A^d = 3^h$ or $M^d = 2^h$ would give $d\beta = h$, contrary to the irrationality of β.

A standard consequence of p-adic linear-form theory is that for fixed multiplicatively independent positive rationals α_1, α_2 and a fixed prime p, there are effectively computable constants C_0, C_1 such that

$$v_p(\alpha_1^d - \alpha_2^h) \leq C_0 + C_1 \big(\log(2 + \max\{d, h\})\big)^2. \tag{176}$$

Much sharper versions, with linear dependence on $\log B$, are known in important two-logarithm regimes; the square is more than sufficient here [21, 22, 23]. Apply this to $(A, 3, p = 2)$ and $(M, 2, p = 3)$. Since $d, h = O_\beta(N)$, each factor in (168)–(172) has valuation $O_{M,A}((\log N)^2)$.

The number of pairs within the vertical fibers is

$$\sum_u \binom{s_u}{2} = \frac{L_N^3}{6\vartheta^2} + O_\vartheta(L_N^2) = O_\vartheta(N^{3/2}), \tag{177}$$

while the number within the horizontal fibers is

$$\sum_v \binom{\#U_v}{2} = \frac{L_N^3}{6\vartheta} + O_\vartheta(L_N^2) = O_\vartheta(N^{3/2}). \tag{178}$$

We obtain the following rigorous scale separation.

Theorem 38.2 (First tied layer is lower order)**.** *In the cross-coprime branch,*

$$v_2(I_{2,T}) + v_3(I_{3,T}) = O_{M,A}\Big(N^{3/2}(\log N)^2\Big) = O_{M,A}\Big(T^{3/2}(\log T)^2\Big). \tag{179}$$

In particular,

$$\frac{v_2(I_{2,T})\log 2 + v_3(I_{3,T})\log 3}{H_T} \longrightarrow 0. \tag{180}$$

Proof. Sum (176) over the pairs in (177) and (178), using (174)–(175). Finally use $N \asymp_\beta T$ and $H_T \asymp_\beta T^{5/2}$. ☐

Status: [paper proof in this report] modulo the quoted classical p-adic linear-form estimate (176). The factorization of the tropical initial forms is elementary and is a particularly attractive Lean target.

38.4 What a leading excess would have to do

Theorem 38.2 does not prove $v_2(D_T) - \tau_{2,T} = o(H_T)$ or its 3-adic analogue. If the initial form is divisible by a power exceeding the first tropical gap, terms from the next valuation layer enter; those can cancel, exposing another layer, and so on. A leading-order excess is therefore possible only through a *cascade* of cancellations across a number of tropical layers growing with T. Cancellation confined to the minimum-permutation fibers is quantitatively negligible.

This yields a more precise version of the p-adic research target.

Research target 38.3 (No shallow-layer proof)**.** Either prove that the tropical cancellation cascade has total depth $o(H_T)$—which would retire the single-block p-adic strategy—or identify a structural identity forcing $\gg H_T$ cumulative cancellation across successively nonminimal assignments. Pairwise congruences such as (174) and (175) cannot by themselves supply the latter, because p-adic logarithm bounds make each congruence only polylogarithmic in the row separation.

39 Synthesis of the determinant audits

The preceding calculations put several superficially different determinant proposals into the same currency: the fraction of the leading archimedean budget supplied automatically by valuations of the entries. The numerical constants should not be compared as though the underlying determinants were identical—the ordinary, mixed-block, and complete-grid systems have different sizes and different upper bounds—but they do show which mechanism is closest to a contradiction before any genuine cancellation is used.

Table 7: Leading termwise-divisor ceilings obtained in this continuation.

System	Ceiling or exact limit	Meaning
Ordinary input/output Vandermonde product	$1 - \dfrac{\delta_2 \log 2 + \delta_3 \log 3}{3\beta \log 6} < 0.8710491$	Exact fixed-support divisor fraction; all primes already occurring in M and A included.
Mixed single block, arbitrary branch	$1 - \dfrac{3}{8} \sum_p \mathfrak{G}(c_p) \leq 4/5$	Exact all-prime tropical limit and universal primitive-depth ceiling.
Mixed single block, cross-coprime	$\leq 7/10$	Uses $c_2 = (-1, 0)$, $c_3 = (0, -1)$ and the external sum $(1, 1)$.
Mixed single block, only structural primes in the cross-coprime branch	$3/10$	Exact contribution of $p = 2, 3$ before external primes or cancellation.
Complete triangular row grid	$\leq \pi/4 = 0.7853982\ldots$	All-prime common divisor versus the sharp largest-Leibniz-term transport upper bound.
First tied fibers at 2 and 3	$o(1)$ of the mixed-block budget	Their valuations are $O(T^{3/2}(\log T)^2)$ against $H_T \asymp T^{5/2}$.

Three conclusions survive every normalization.

1. Merely adding every prime in the fixed support does not close any determinant. It does, however, improve the diagnosis substantially: the universal mixed-block ceiling is 4/5, not a heuristic "about one half".

2. The exact primitive theorem is quantitatively useful. The condition $\min(v_2(M), v_3(A)) = 0$ creates one structural slope vector of norm at least 4/15; the conservation law forces a second norm contribution of at least the same size.

3. A successful p-adic proof must be a cancellation theorem, not a divisibility-by-every- term theorem. In the easiest branch, even the first large tied family contributes only a lower-order amount. The required phenomenon would have to repeat through many tropical layers or interact with a much sharper archimedean free energy.

39.1 Correction of the post-attachment "one half / two thirds" note

After the attached unified report was produced, the repository acquired a short exploratory note (merged into Section 21 of this report) at commit `ff9a9f5aa3a3f02599de93fc053e60918f269855`.

That note correctly identified the distinction between termwise min-plus divisibility and cancellation beyond the tropical minimum, and its real/p-adic compatibility diagnosis remains valuable. It proposed, however, a structural $1/2$ ceiling and an all-prime $2/3$ ceiling. Neither constant is valid in the stated generality.

The structural averaging argument omitted the base and cross depths that appear in (136)–(137). Even under $v_2(M) = 0$, the terms $v_2(A)kv$ and $v_3(M)ku + v_3(A)kv$ do not disappear. The exact structural contribution is obtained by retaining c_2 and c_3 in Theorem 36.3; it can exceed one half of H_T.

The all-prime step used the assertion that an oppositely sorted nonnegative triangular weight has assignment minimum at most two thirds of its mean. For the weight $Y = X + Z$ on the standard triangle, the exact values from Lemma 36.2 are

$$\mathbb{E}Y = \frac{2}{3}, \qquad \Phi(1,1) = \frac{4}{15}, \qquad \frac{\Phi(1,1)}{\mathbb{E}Y/2} = \frac{4}{5}.$$

Thus $4/5$, not $2/3$, is already attained by the relaxed external rank-one grading. The Gini-norm identity explains the general case and leads to the rigorous ceiling in Theorem 36.6. This is a correction of a calculation in an exploratory note, not a criticism of the source report itself; the attached unified report did not contain the claim.

Status: **[paper proof in this report]** — the correction follows from the exact transport formula, not from numerical experimentation.

40 Directed-rounding extension through 2^{27}

The attached report supplied a complete MPFR scan through 2^{26}. Because a finite extension is independent of the unresolved transcendence barrier and gives a useful stronger floor for every hypothetical generator, I extracted the literal C listing from its Appendix "Directed-rounding verifier", compiled it against MPFR 4.2.2, replayed an old audit chunk, and scanned the next full dyadic range.

40.1 Manifest audit

The literal extracted source has SHA-256 digest

`f888c0d89be9366f01e2d26f339dcbe446556d39d4c5dbcf8f579ec0d0052e1d`.

This does *not* equal the digest

`ec709623f812478fbb7fb5567aa59fbfe3df0ae639ca366bc7c72e278bcfee2f`

printed in the source of that section. None of the ordinary line-ending, final-newline, or tab normalizations explains the discrepancy. The report therefore contains a stale or externally generated source-identity hash.

The semantic reproducibility check succeeds. Replaying the old interval $[29{,}360{,}128, 33{,}554{,}432)$ at 192 bits gives exactly the report's records

$$\begin{aligned}&\text{lower: } 1.4439124872349507 \times 10^{-19} \text{ at } (30{,}103{,}832, 713{,}403{,}245{,}915),\\ &\text{upper: } 5.4132120119800561 \times 10^{-21} \text{ at } (29{,}411{,}704, 687{,}581{,}893{,}443),\end{aligned}$$

with zero failures. Thus the listing used here reproduces the old executable's mathematical behavior on an audited four-million-input overlap even though the printed file digest does not identify it.

Status: [interval computation] — this section is a directed-rounding software result. It is stronger numerically than the Lean theorem but is not a kernel proof. The manifest hash mismatch is explicitly part of the trust boundary.

40.2 The new range

The half-open interval $[2^{26}, 2^{27})$ was divided into sixteen chunks of width 2^{22}. The executable used 192-bit MPFR endpoints and outward rounding exactly as in the source report. There are 2^{26} integers in the range; the first, 2^{26}, is the sole power of two and is intentionally skipped. Hence the required nonpower count is $2^{26} - 1 = 67{,}108{,}863$.

The concatenated canonical extension log has SHA-256 digest

`b11843448cd8946442aba13758b5b666e4ac6eae50d2243fbdc32d83274cea8d`.

Every chunk reports zero failures, and the checked counts sum to the required total.

Table 8: Directed-rounding extension on $[2^{26}, 2^{27})$.

Chunk	Integer range	Checked	Smallest lower gap	Smallest upper gap
0	67,108,864–71,303,167	4,194,303	4.2837×10^{-20}	1.7723×10^{-19}
1	71,303,168–75,497,471	4,194,304	2.2961×10^{-20}	1.1433×10^{-20}
2	75,497,472–79,691,775	4,194,304	1.2813×10^{-19}	1.6063×10^{-20}
3	79,691,776–83,886,079	4,194,304	2.5420×10^{-20}	5.2012×10^{-20}
4	83,886,080–88,080,383	4,194,304	2.6166×10^{-20}	3.8496×10^{-20}
5	88,080,384–92,274,687	4,194,304	1.0923×10^{-19}	5.7144×10^{-20}
6	92,274,688–96,468,991	4,194,304	1.1017×10^{-20}	2.3227×10^{-20}
7	96,468,992–100,663,295	4,194,304	3.0759×10^{-20}	7.4810×10^{-20}
8	100,663,296–104,857,599	4,194,304	5.6465×10^{-20}	6.4744×10^{-20}
9	104,857,600–109,051,903	4,194,304	8.8144×10^{-20}	8.3662×10^{-21}
10	109,051,904–113,246,207	4,194,304	3.8892×10^{-20}	7.0327×10^{-21}
11	113,246,208–117,440,511	4,194,304	9.9365×10^{-21}	2.1814×10^{-20}
12	117,440,512–121,634,815	4,194,304	4.4964×10^{-20}	5.4132×10^{-21}
13	121,634,816–125,829,119	4,194,304	1.0774×10^{-20}	3.4093×10^{-20}
14	125,829,120–130,023,423	4,194,304	4.3609×10^{-20}	8.3073×10^{-22}
15	130,023,424–134,217,727	4,194,304	5.9082×10^{-21}	5.1326×10^{-20}
	total	**67,108,863**		

40.3 New extremal records and high-precision replay

The smallest lower logarithmic margin in the extension is

$$5.9082259992879571 \times 10^{-21} \quad \text{at} \quad (m, n) = (132{,}833{,}765, 7{,}501{,}348{,}219{,}569). \tag{181}$$

A 120-decimal-digit reevaluation gives

$$m^{\vartheta} - n = 6.39397545334704098376801255 10\ldots \times 10^{-8}.$$

The smallest upper logarithmic margin is the new global record

$$8.3073231820333117 \times 10^{-22} \quad \text{at} \quad (m, n) = (129{,}799{,}345, 7{,}231{,}571{,}292{,}851), \tag{182}$$

where

$$n + 1 - m^{\vartheta} = 8.66699043558189568532979522 88\ldots \times 10^{-9}.$$

Both one-input cases were rerun independently at 320 and 512 bits. The directed margins and nearest integer were unchanged in every displayed digit, and both replays reported zero failures.

Theorem 40.1 (Software-level finite exclusion through 2^{27})**.** *At the stated MPFR software-trust level, for every integer* $1 \le m < 2^{27}$*,*

$$m^{\vartheta} \in \mathbb{N} \quad \Longrightarrow \quad m \textit{ is a power of two.}$$

Consequently every nonintegral real x *satisfying* $2^x, 3^x \in \mathbb{Z}$ *obeys*

$$x > 27. \tag{183}$$

Proof. The attached report's complete scan covers $1 \le m < 2^{26}$, and the sixteen new chunks cover $2^{26} \le m < 2^{27}$. In each case the directed logarithmic interval for $\vartheta \log m$ lies strictly between the intervals for the logarithms of two consecutive integers, except when m is a power of two. If $m = 2^x$ came from a nonintegral solution and $m < 2^{27}$, then $m^{\vartheta} = 3^x$ would be an integer at a non-power input, contradicting the scan. The endpoint $m = 2^{27}$ corresponds to the integral exponent 27, so a nonintegral solution must have $m > 2^{27}$ and hence $x > 27$. □

This raises the software-level spacing ratio in the optimized finite-difference comparison from $26 \log 3$ to $27 \log 3 = 29.662531\ldots$. It does not change any kernel-verified theorem; the formal floor remains $x \ge 12$ until the finite certificates are replayed in Lean.

41 Additional proof routes pushed to their failure points

The determinant calculations were the main new line of attack, but they were not the only one. This section records four other routes in enough detail to prevent the same hidden loss from being rediscovered under different notation.

41.1 Transferring convergents of β to ϑ

Let p/q be a continued-fraction convergent of a hypothetical primitive generator β and put

$$\varepsilon = q\beta - p.$$

For all sufficiently large q, $|\varepsilon| < 1/2$. Define the two nonzero integer gaps

$$D_2 = M^q - 2^p = 2^p(2^\varepsilon - 1), \qquad D_3 = A^q - 3^p = 3^p(3^\varepsilon - 1). \tag{184}$$

They manufacture an exact rational approximation to ϑ:

$$R_{p,q} := \frac{2^p D_3}{3^p D_2} = \frac{3^\varepsilon - 1}{2^\varepsilon - 1} \in \mathbb{Q}. \tag{185}$$

Taylor expansion at zero gives

$$R_{p,q} = \vartheta + \frac{\log 3\,(\log 3 - \log 2)}{2\log 2}\,\varepsilon + O(\varepsilon^2), \tag{186}$$

so $|R_{p,q} - \vartheta| \ll |\varepsilon|$.

This looks promising because ϑ has an explicit irrationality measure. Wu's simultaneous linear-independence estimate for $1, \log 2, \log 3, \log 5$ gives, after setting the unused coefficients to zero, a conservative exponent

$$\left|\vartheta - \frac{r}{s}\right| \geq c_\vartheta s^{-\mu_\vartheta}, \qquad \mu_\vartheta = 8.616 \tag{187}$$

for all sufficiently large denominators s [24]. The obstacle is the denominator created by (185). Before reduction it is at most

$$3^p|D_2| \ll 3^p 2^p |\varepsilon| = 6^p|\varepsilon|. \tag{188}$$

Combining (186)–(188) with (187) gives only

$$|\varepsilon| \gg 6^{-p\mu_\vartheta/(1+\mu_\vartheta)}. \tag{189}$$

Since

$$6^{-\mu_\vartheta/(1+\mu_\vartheta)} = 2^{-2.316\ldots},$$

this is substantially weaker than the elementary bound $|\varepsilon| \gg 2^{-p}$ obtained directly from $D_2 \neq 0$. Sharper real irrationality measures cannot repair the architecture: every irrationality exponent is at least two, and the unreduced denominator already contains both 2^p and 3^p.

Proposition 41.1 (Exact denominator-loss diagnosis)**.** *The convergent-transfer construction can improve the elementary gap only if one proves an exponential lower bound for*

$$G_{p,q} = \gcd(2^p D_3, 3^p D_2) \tag{190}$$

that survives reduction of $R_{p,q}$. In the absence of such a bound, every known irrationality measure for ϑ is defeated by the factor 6^p in (188).

No mechanism forcing $G_{p,q}$ to be exponentially large was found. The two gaps share the same small real parameter ε, but real closeness does not impose a common prime divisor.

41.2 Two-row determinants and fixed-support S-units

For two block rows $i < j$, put $d = j - i$ and $h = n_i - n_j = \lfloor j\beta \rfloor - \lfloor i\beta \rfloor$. Their 2×2 determinant has the exact factorization

$$m_{T,i}a_{T,j} - m_{T,j}a_{T,i} = 6^{n_j}(MA)^i(2^hA^d - 3^hM^d). \tag{191}$$

Moreover

$$\frac{2^hA^d}{3^hM^d} = \left(\frac{3}{2}\right)^{\beta d - h}. \tag{192}$$

Thus a good rational approximation h/d to β makes the bracket in (191) a small *relative* difference of two integers supported on the fixed prime set dividing $6MA$.

Let

$$U = 2^hA^d, \qquad V = 3^hM^d, \qquad G = \gcd(U, V).$$

The reduced nonzero integer $(U - V)/G$ gives

$$|\beta d - h| \gg_{M,A} \frac{G}{\max\{U, V\}}. \tag{193}$$

The exponent of the right-hand side can be written explicitly as the positive part of a valuation-vector difference:

$$\log \frac{U}{G} = \sum_p \max\{h\mathbf{1}_{p=2} + dv_p(A) - h\mathbf{1}_{p=3} - dv_p(M), 0\} \log p. \tag{194}$$

As $h/d \to \beta$, this is $\kappa_{M,A}d + o(d)$ for a positive constant $\kappa_{M,A}$. Hence (193) is merely exponential in d, much weaker than the Baker–Matveev polynomial separation of Section 34.

The *abc* conjecture applied to the reduced equation $U/G - V/G = (U - V)/G$ would make the difference nearly as large as a fixed positive power of U/G, but it would still allow a relative gap $e^{-\epsilon d}$. Continued-fraction gaps of order $1/d$ are much larger. Thus this direct S-unit equation does not become a proof even after inserting *abc*; one would need an additional theorem controlling the prime support or radical of the difference.

41.3 Automaticity and the synchronized Sturmian word

The coding $s_k = \lfloor (k + 1)\alpha \rfloor - \lfloor k\alpha \rfloor$ from Section 33 is aperiodic Sturmian. A standard theorem in automatic-sequence theory says that no aperiodic Sturmian word is automatic in any integer base [27, 28]. One quick obstruction is frequency: the frequency of 1 in s is the irrational number α, whereas an automatic sequence having a letter frequency has rational frequency.

This observation blocks a tempting but invalid shortcut. The rationality of $U = 2^\alpha$ does *not* make the threshold itinerary of $R \mapsto UR/2^{\mathbf{1}_{UR \geq 2}}$ finite-state; its exact rational denominators grow with time. The same is true for V in base three. Therefore Cobham's theorem cannot be invoked merely because the same word is produced by a dyadic and a triadic rational system. A viable automata proof would first have to manufacture genuine finite kernels in two multiplicatively independent bases, and doing so is already a new arithmetic theorem about U and V.

41.4 Real and p-adic logarithms

At the real place the counterexample is exactly

$$\log M \ \log 3 - \log A \ \log 2 = 0. \tag{195}$$

At $p = 2$ or 3, after stripping valuation and torsion parts, the units among $M, A, 2, 3$ admit p-adic logarithms. The determinant entries then have unit parts whose p-adic logarithms are bilinear forms in the row and column coordinates. This is the natural language for the cancellation cascade isolated in Section 38.

What is missing is a transfer principle from (195) to a congruence or rank drop among those p-adic logarithms. Real exponentiation by the transcendental number β does not define a compatible p-adic exponent, and the equality $M = 2^\beta$ has no p-adic meaning. Known p-adic linear-form theorems control a nonzero form once its integer coefficients and algebraic bases are specified; they do not create a p-adic form from a real one.

Research target 41.2 (Real/p-adic compatibility)**.** Find an algebraic construction, valid for the integer pair (M, A) alone, that converts (195) into either

1. a nontrivial congruence among p-adic unit logarithms with precision $\gg T$, repeated over $\gg T^{3/2}$ fiber pairs; or
2. a rank drop in the tropical initial matrices persisting through $\gg T$ valuation layers.

Without such a construction, ordinary p-adic logarithm bounds point in the opposite direction: they show that individual congruences are only polylogarithmically deep.

41.5 Transcendental-curve point counting

The graph $y = x^\vartheta$ is definable in the real exponential field, so Pila–Wilkie type results [32] give subpolynomial bounds for rational points outside algebraic pieces. Those generic bounds are far too weak here. The curve already contains the infinite sequence $(2^n, 3^n)$, giving $\asymp \log H$ integer points of height at most H, while a hypothetical second generator produces $\asymp (\log H)^2$ points globally. A proof by point counting would therefore need a sharp $O(\log H)$ theorem with a structural classification of the extremal sequence, not merely $O(H^\epsilon)$. Such a theorem would essentially be another form of the rank-one conclusion sought by the conjecture.

42 Exact sufficient conditions extracted from the new analysis

A useful continuation should not stop at saying that the present determinant fails. The calculations above isolate numerical thresholds which, if crossed by a future theorem, would settle the conjecture. This section records those thresholds as precise implications.

42.1 The mixed-block cancellation-excess criterion

Let

$$\mathcal{P} = \{p : p \mid 6MA\}$$

be the fixed support of all determinant entries, and retain the mixed block determinant D_T, its tropical minima $\tau_{p,T}$, and the archimedean scale H_T. Define the support-prime cancellation excess

$$\mathcal{E}_T = \sum_{p\in\mathcal{P}} (v_p(D_T) - \tau_{p,T}) \log p \geq 0 \tag{196}$$

and the exact tropical deficit

$$\delta(M, A) = \frac{3}{8} \sum_{p\in\mathcal{P}} \mathfrak{G}(c_p). \tag{197}$$

Theorem 36.3 says $\log Q_T^{\mathrm{trop}} = (1 - \delta(M, A) + o(1))H_T$, while the elementary Leibniz upper bound is $\log D_T \leq H_T + o(H_T)$.

Theorem 42.1 (Cancellation-excess criterion)**.** *If a hypothetical primitive counterexample satisfies*

$$\liminf_{T\to\infty} \frac{\mathcal{E}_T}{H_T} > \delta(M, A), \tag{198}$$

then no such counterexample exists. In particular it is enough to prove an excess greater than $H_T/5$ uniformly in all primitive branches; in the cross-coprime branch the exact threshold is at least $3H_T/10$.

Proof. The selected support primes alone give

$$\log D_T \geq \sum_{p\in\mathcal{P}} v_p(D_T) \log p = \log Q_T^{\mathrm{trop}} + \mathcal{E}_T.$$

Under (198), the right-hand side exceeds H_T by a fixed positive multiple for all large T, contradicting the Leibniz upper bound. The numerical thresholds follow from Theorem 36.6 and Corollary 36.7. □

This criterion is not tautological: it involves only valuations at the fixed finite set of primes already present in $6MA$, whereas D_T can acquire arbitrary new prime factors. It states exactly how much same-support cancellation is required.

42.2 A denominator-cancellation criterion

For the transferred approximant $R_{p,q}$ of Section 41.1, let $s_{p,q}$ be its reduced denominator. If one could prove along infinitely many convergents that

$$s_{p,q} \ll q^{\eta} \qquad \text{for some } \eta < \frac{1 + \mu_\vartheta}{\mu_\vartheta}, \tag{199}$$

then Wu's measure would give $|R_{p,q} - \vartheta| \gg q^{-\eta\mu_\vartheta}$, whereas (186) and the convergent bound give $|R_{p,q} - \vartheta| \ll q^{-1}$. Since $\eta\mu_\vartheta < 1 + \mu_\vartheta$ is not by itself enough, one tracks $|\varepsilon| < 1/q_+$ more precisely; a contradiction follows whenever $q_+ \gg q^{\eta\mu_\vartheta+\epsilon}$ along the same subsequence. A simpler sufficient condition is

$$s_{p,q} = q^{o(1)} \tag{200}$$

for infinitely many convergents with $q_+ \geq q^{1+\epsilon}$.

Equivalently, the gcd $G_{p,q}$ in (190) would have to cancel essentially the entire exponential factor $6^p|\varepsilon|$, leaving a subpolynomial denominator. This is a sharp formulation of the otherwise vague request for “correlation between the two integer gaps.”

42.3 An automaticity criterion

The synchronized coding gives another exact, if presently inaccessible, implication.

Proposition 42.2 (Finite-kernel criterion)**.** *If under a hypothetical counterexample the common word $s_k = \lfloor (k+1)\alpha \rfloor - \lfloor k\alpha \rfloor$ were automatic in any integer base $q \geq 2$, then the Alaoglu–Erdős conjecture would follow.*

Proof. An automatic Sturmian word is ultimately periodic. Its slope, equal to the frequency of the letter 1, is therefore rational. But $\alpha = \beta - \lfloor \beta \rfloor$ would then be rational, and the rational-exponent theorem would make β integral. □

The missing step is not symbolic dynamics; it is proving automaticity from the two rational multipliers U and V. The growing denominators in (90) show exactly why the obvious state space is infinite.

43 Branch-sensitive research program

The source report already classifies a hypothetical pair much more sharply than "two integers on a curve." The next attack should exploit that classification rather than seek a single estimate that treats every branch identically.

43.1 Branch I: independent external valuation directions

Here the second-iterate kernel K_β is zero, equivalently the external prime-valuation vectors of M and A are independent. Corollary 36.5 gives a strict angular defect $\Delta_{\text{ext}} > 0$ in the all-prime tropical fraction. That makes the termwise divisor farther from the archimedean upper bound, but the same independence also supplies at least two genuinely different external primes at which cancellation can be studied.

The concrete target is a simultaneous p-adic rank theorem for the tropical initial matrices: show that if cancellation persists through L layers at two non-collinear external primes, then an integer relation appears among the two external valuation vectors. Such a theorem would contradict $K_\beta = 0$. Existing one-prime linear-form bounds do not compare the residue kernels at different primes; this is the genuinely new ingredient required.

43.2 Branch II: a common external valuation ray

When $K_\beta \neq 0$, all external vectors are collinear and the source report gives the exact Möbius-orbit condition

$$\beta = \frac{u + v\vartheta}{r + s\vartheta} \qquad (u, v, r, s \in \mathbb{Q}, us - vr \neq 0). \tag{201}$$

This is the branch in which the external triangle inequality can be an equality. It is also arithmetically lower-dimensional: after clearing denominators, the external parts of M and A are powers of one common primitive core.

The best next step is therefore not a generic all-prime determinant. It is a mixed real/radical invariant built from the canonical core and the two binomial fields already formalized in the repository. Useful intermediate targets are:

1. compare the field norms of the canonical radicals on the two sides of (201);
2. show that simultaneous radical degrees force an additional common perfect-power degree, contradicting affine primitivity; or
3. prove that the Möbius coefficients in (201) cannot have the sign pattern imposed by nonnegative prime valuations unless one output is $\{2, 3\}$-smooth.

The repository modules

`CanonicalRadicalFieldNorm`, `CanonicalRadicalValuation`,
`CanonicalRadicalSharpness`, `CanonicalRadicalDivisibleHull`,
`SimultaneousRadicalDegrees`

provide much of the algebraic plumbing for this branch.

43.3 Branch III: one smooth output

If M is $\{2, 3\}$-smooth, then $\beta = a + d\vartheta$ with $d > 0$; if A is smooth, there is a symmetric affine formula. The source corpus proves that both outputs cannot be smooth and that the other output contains a prime at least five. In the extreme case $M = 3^d$, the problem is the localized radical assertion

$$(3^d)^\vartheta \notin \mathbb{Z}[1/3].$$

This branch is the smallest transcendence-theoretic wall and deserves separate treatment. Potentially useful objects are the irreducible binomial for the least localized radical, its field norm, and reduction at primes over two and three. A successful local argument must remain sensitive to the real identity defining ϑ; ordinary ideal factorization alone is already exhausted by the canonical-radical modules.

43.4 Branch IV: cross-coprime primitive outputs

The cross-coprime branch has the strongest determinant constants and the cleanest residue matrices. Here the exact first tropical forms are the fiber Vandermondes of Section 38. This is the best laboratory for determining whether the cancellation cascade is real or illusory.

The next computational experiment should not search for counterexamples. For synthetic odd pairs (M, A) and Beatty depths generated by $\beta = \log_2 M$, compute

$$v_2(D_T) - \tau_{2,T}, \qquad v_3(D_T) - \tau_{3,T}$$

for increasing T, together with the histogram of tropical layers contributing at the final valuation. Small exact experiments performed during this work show an excess compatible with $N^{3/2}$ rather than $N^{5/2}$, matching Theorem 38.2; they are reconnaissance only, since a synthetic A does not satisfy $A = 3^\beta$. A proof should seek a uniform $o(H_T)$ bound, probably through p-adic exponential-polynomial zero estimates.

43.5 Archimedean priority: solve the triangular HCIZ limit

The complete grid finally supplies the missing T^2 rows, and its empirical measures are explicit. This makes Target 37.6 the highest-priority analytic problem. The required calculation is a large-N asymptotic for the HCIZ integral with both spectra having density $2t$ on $[0, 1]$, under the coupled scaling $x_i y_j \asymp T^2$ and $N \asymp T^2$. Even a nonsharp upper bound improving the largest-term energy by more than $1 - \pi/4$ would materially change the p-adic comparison.

A practical sequence is:

1. numerically evaluate $\log\det(e^{x_i y_j})$ using high-precision total-positive algorithms for moderate T;
2. subtract the exact Vandermonde/factorial terms in (165) to estimate the free-energy limit;
3. derive the corresponding variational problem using known HCIZ large-deviation methods;
4. only then reinsert the primewise valuation lower bounds.

This route is attractive because the analytic limit does not depend on the unknown prime factorization of M and A; only β enters the overall scaling.

43.6 Finite verification priority

The software floor is now $x > 27$. The immediate formal target is not another raw doubling of the MPFR scan but a replayable certificate architecture. A realistic hierarchy is:

1. a fast untrusted generator records only near-boundary pairs and rational interval data;
2. a small verified checker proves each interval enclosure and the monotone coverage between records;
3. chunks are combined by a theorem whose statement mentions only integer endpoints;
4. the final theorem is imported into the aggregate surface without `native\_decide` or an opaque external axiom.

The source-hash discrepancy found in Section 40 makes this especially worthwhile: mathematical replay is more robust than a manifest that identifies an external binary only by prose.

44 Lean formalization blueprint

The source corpus is already unusually strong on exact conditional structure. The most useful continuation is therefore not a second informal reimplementation of those results, but a small collection of sharply separated modules whose statements expose the genuinely new invariants. This section gives a dependency order designed to keep arithmetic, asymptotics, and transcendence inputs from leaking into one another.

44.1 Proposed module graph

Proposed module	Principal statement	Main dependencies
`SturmianNormalization`	Exact recurrences for R_k, S_k, common jump word s_k, and the exponent/output correspondence.	Existing primitive-generator and denominator-normalization modules; elementary floor arithmetic.
`ReducedOrbitDenominators`	Lowest-term denominators $q_k^{(2)}, q_k^{(3)}$ and the full-depth alternative from $\min(v_2(M), v_3(A)) = 0$.	`SturmianNormalization`; unique factorization and valuations.
`EffectiveGeneratorSeparation`	An abstract interface turning a two-logarithm lower bound into $\lvert\beta - p/q\rvert \geq cq^{-\mu}$ and a polynomial recurrence for continued-fraction denominators.	A stated external Baker–Matveev hypothesis, plus existing continued-fraction lemmas.
`BlockVandermondeValuation`	Exact v_2 and fixed-support v_p formulas for the conditional block Vandermonde; output analogue.	Existing exact block parametrization; valuation of a difference with unequal valuations.
`OrdinaryAdelicFraction`	The asymptotic fraction (117) and the universal $1 - \log 2/(3\log 6)$ ceiling.	`BlockVandermondeValuation`; polynomial floor sums; primitive-output depth theorem.
`TriangleGini`	$\mathfrak{G}$ is a norm and satisfies the explicit formula (133).	Elementary measure integration, or a finite polygonal integration library.
`FiniteTropicalAssignment`	Exact rearrangement formula for a finite row list and a finite column-weight list.	Sorting/permutations; finite rearrangement inequality.
`MixedColumnRiemannSums`	Lattice count (119) and convergence of scaled low-weight columns to uniform measure on $\mathcal{T}$.	Two-dimensional lattice-point estimates and Riemann sums.
`MixedAdelicGini`	The limit (140), conservation $\sum_p c_p = 0$, and branch-sensitive three-vector bound.	Previous three modules; finite prime support.
`MixedTropicalCeilings`	Universal $4/5$, cross-coprime $7/10$, and structural $3/10$ constants.	`MixedAdelicGini`; primitive-output depth theorem; explicit values from `TriangleGini`.
`FullGridTransport`	Row/column empirical distributions, countermonotone energy $\pi/8$, maximum energy $1/2$, and the ratio $\pi/4$.	Triangular lattice sums, quantile integration, rearrangement.
`TropicalInitialFibers`	Factorization of the first 2- and 3-adic initial forms into ordinary Vandermondes on column fibers.	Determinants over valued rings; stable sorting by valuations.
`InitialFiberValuationBound`	Under a standard p-adic logarithm hypothesis, the first tied layer contributes $o(H_T)$.	`TropicalInitialFibers`; an abstract two-logarithm valuation bound.
`FiniteCheck2Pow27`	A kernel-replayable certificate excluding non-powers of two below 2^{27}.	Existing finite-check certificate architecture; generated chunk certificates.

The first two modules are almost entirely combinatorial. The determinant modules should not import transcendence theory. Conversely, the Baker and p-adic logarithm interfaces should be abstract propositions whose classical realization is documented outside the kernel unless and until Mathlib contains a suitable theorem. This makes the trust boundary explicit.

44.2 Finite statements before asymptotics

The continuum notation in Sections 36 and 37 is compact, but Lean should first see finite inequalities. For rows $0 \le k < N$ and columns $z_j \in \mathbb{R}$, the finite rearrangement lemma should assert

$$\min_{\sigma \in S_N} \sum_{k=0}^{N-1} k z_{\sigma(k)} = \sum_{k=0}^{N-1} k z^{\uparrow}_{N-1-k}, \tag{202}$$

where $z^{\uparrow}$ is the nondecreasing rearrangement. Every tropical assignment in this report is an affine rescaling of (202). The limiting Gini identity can then be obtained by three auditable steps:

1. prove convergence of empirical distributions for the triangular lattice;
2. prove convergence of the first moments and pairwise absolute-difference moments;
3. invoke the finite identity

$$\frac{1}{N^2} \sum_{i,j} |z_i - z_j| = \frac{2}{N^2} \sum_{j=0}^{N-1} (2j - N + 1) z^{\uparrow}_j.$$

This route avoids formalizing optimal transport as a general theory.

44.3 Exact theorem sketches

The intended public statements can be phrased close to the mathematics.

```
/-- A primitive counterexample determines one common lower mechanical word
    driving both normalized rational orbits. -/
theorem exists_synchronized_sturmian_data
    (hfail : ExistsNonintegralSolution) :
    Exists fun B : Nat => Exists fun alpha U V : Real =>
      0 < alpha /\ alpha < 1 /\ Irrational alpha /\
      U = 2 ^ alpha /\ V = 3 ^ alpha /\
      (forall k : Nat,
        let e := Int.floor (k * alpha)
        normalizedTwo k = U ^ k / 2 ^ e /\
        normalizedThree k = V ^ k / 3 ^ e) := by
  ...

/-- The automatic all-prime divisor of the balanced mixed determinant
    cannot fill more than four fifths of its leading height. -/
theorem mixed_tropical_divisor_limsup_le_four_fifths
    (hprimitive : min (vTwo M) (vThree A) = 0) :
    limsup (fun T => log (tropicalDivisor T) / leadingHeight T)
      <= (4 : Real) / 5 := by
  ...

/-- The complete conditional grid has countermonotone/maximal transport
    ratio pi/4. -/
theorem full_grid_transport_ratio :
    countermonotoneEnergy / maximalEnergy = Real.pi / 4 := by
  ...
```

The actual APIs will need natural-valued denominator exponents rather than casts of negative integers, and the asymptotic statements will likely use `Tendsto` and `IsLittleO` rather than a custom `limsup`. The sketches are included to fix the mathematical interfaces, not Lean syntax.

44.4 Formalizing the first tied layer

A useful reusable notion is the initial form of a determinant at a prime. For a matrix over $\mathbb{Z}$ and an assignment minimum τ_p, reduce

$$p^{-\tau_p}\det(E_{ij}) \pmod p.$$

Only minimizing permutations survive. In the cross-coprime branch the minimizing permutations preserve equal-u fibers at $p=2$ and equal-v fibers at $p=3$; the surviving sum is a product of smaller determinants. This should be proved as a purely finite combinatorial determinant identity. The later valuation estimate can then be stated separately:

$$v_2(I_{2,T})+v_3(I_{3,T})=O\Big(N^{3/2}(\log N)^2\Big)\,. \tag{203}$$

The big-O exponent is not sacred; any $o(T^{5/2})$ theorem is sufficient for the qualitative conclusion.

44.5 Certificate architecture for the new finite bound

The existing verified bound through 4096 should remain the trusted baseline until a new certificate is accepted. A practical route to 2^{27} is hierarchical:

1. partition $[1,2^{27})$ into fixed-width chunks;
2. for each non-power-of-two m, store an integer nearest-value locator n_m and rational lower/upper enclosures proving either $m^\vartheta<n_m$ or $m^\vartheta>n_m$;
3. hash each chunk and prove a small checker correct in Lean;
4. combine the chunk theorems into one interval theorem.

The MPFR run in Section 40 is reconnaissance and a source of extremal cases; it is not the certificate itself.

45 Numerical sanity checks and reproducibility

None of the limiting constants in the determinant theorems depends on numerical evidence. The computations below were used to catch normalization errors, check convergence rates, and stress-test the closed forms before writing the proofs.

45.1 Balanced-block assignment constants

For each N, take the first N pairs (u,v) ordered by $u+\vartheta v$, pair them against N equally spaced row parameters, and solve the one-dimensional assignment problems by sorting. In the cross-coprime model, the structural places should tend to $3/10$ and the minimal all-prime three-vector model to $7/10$.

N	structural $2,3$ fraction	all-prime three-vector fraction
25	0.3224571184	0.7258906037
50	0.3135103379	0.7141968856
100	0.3050363664	0.7053583986
250	0.3024422068	0.7025311250
500	0.3012614628	0.7013136826
1000	0.3006780555	0.7006854844
2500	0.3002789586	0.7002894348
5000	0.3001354539	0.7001384826
Limit	3/10	7/10

Table 10: Finite assignment sums converging to the exact mixed-block constants.

The convergence from above is consistent with the $O(L^{-1}) = O(N^{-1/2})$ boundary error in triangular lattice counting. The table also exposes the error in the proposed 2/3 ceiling: the finite all-prime ratio is already near 0.70, and the exact limit is 7/10.

45.2 Complete-grid transport ratio

For a representative hypothetical height parameter $\beta = 27.314159$, form the row lattice $n + \beta k \leq T$ and the matching low-weight column lattice. Normalize the countermonotone assignment by the largest-term assignment. The finite ratios are:

T	number of rows	transport ratio
50	73	0.7415113
100	237	0.7559915
200	841	0.7671191
400	3157	0.7750371
800	12210	0.7796781
Limit	—	$\pi/4 = 0.7853981634\ldots$

Table 11: Finite full-grid countermonotone/maximal transport ratios.

The slow boundary convergence is expected: both empirical triangles have $O(T)$ boundary points against $O(T^2)$ bulk points, and the matching weight scale also grows with T.

45.3 Directed-rounding scan manifest

The scan in Section 40 used the literal C listing extracted from the attached LaTeX source. Its SHA-256 digest is

`f888c0d89be9366f01e2d26f339dcbe446556d39d4c5dbcf8f579ec0d0052e1d`.

This differs from the digest printed in the old report. To ensure the discrepancy was only a manifest error, the extracted source was first rerun on the old overlap interval $[29{,}360{,}128, 33{,}554{,}432)$; it reproduced the old extremal records and reported zero failures.

The extension interval $[2^{26}, 2^{27})$ was split into sixteen chunks of width 2^{22} and run at 192 bits. The concatenated chunk log has SHA-256

`b11843448cd8946442aba13758b5b666e4ac6eae50d2243fbdc32d83274cea8d`.

The two new extremal records were replayed at 320 and 512 bits. A representative build and invocation sequence was:

```
cc -O3 -std=c17 mpfr_scan.c \
  /lib/x86_64-linux-gnu/libmpfr.so.6 -lgmp -lm -o mpfr_scan

./mpfr_scan 67108864 71303168 192
# repeat consecutive width-2^22 chunks through 134217728
sha256sum extension-through-2pow27.log
```

The scanner excludes powers of two from the tested population because they are the known solutions. Across the extension it checked 67,108,863 non-powers of two and reported zero failures.

45.4 Extremal replay records

The narrowest new lower-directed separation was

$$m = 132{,}833{,}765, \qquad n = 7{,}501{,}348{,}219{,}569, \qquad \text{directed margin } 5.9082259992879571 \times 10^{-21}.$$

The corresponding high-precision ordinary difference is approximately

$$m^{\vartheta} - n = 6.39397545334704 \times 10^{-8}.$$

The narrowest new upper-directed separation was

$$m = 129{,}799{,}345, \qquad n = 7{,}231{,}571{,}292{,}851, \qquad \text{directed margin } 8.3073231820333117 \times 10^{-22},$$

with

$$n + 1 - m^{\vartheta} = 8.666990435581896 \times 10^{-9}.$$

These decimal differences are explanatory only. The certification decision uses the MPFR outward-rounded interval endpoints and the scanner's locator checks.

46 Status matrix and dependency audit

Statement	Status	Dependency or caveat
Exact primitive monoid $\mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0\beta$ and candidate monoid	**[kernel-verified Lean]**	Inherited from the source report.
Primitive-depth condition $\min(v_2(M), v_3(A)) = 0$	**[kernel-verified Lean]**	Inherited; central to both universal ceilings.
Synchronized dyadic–triadic Sturmian recurrence	**[paper proof in this report]**	Elementary from the exact monoid and floor normalization.

Statement	Status	Dependency or caveat
Lowest-term denominator formulas and full-depth alternative	**[paper proof in this report]**	Unique factorization plus primitive depth.
Effective polynomial irrationality measure for fixed M	**[classical/open background]** plus **[paper proof in this report]**	Baker–W"ustholz/Matveev; constants depend effectively on M.
Polynomial recurrence of continued-fraction denominators	**[paper proof in this report]**	Standard convergent inequality applied to the previous row.
Exact ordinary block valuations at all input-support primes	**[paper proof in this report]**	Unequal valuation of the two monomials in every difference.
Ordinary product fraction (117)	**[paper proof in this report]**	Exact floor sums; output endpoint errors are $O(T^2)$.
Universal ordinary ceiling $0.871049\ldots$	**[paper proof in this report]**	Primitive depth only; not a bound on cancellation from new primes in differences.
Explicit triangular Gini norm	**[paper proof in this report]**	Direct integration.
Mixed all-prime tropical limit (140)	**[paper proof in this report]**	Rearrangement plus triangular Riemann sums.
Universal 4/5 mixed ceiling	**[paper proof in this report]**	Conservation, Gini triangle inequality, primitive depth.
Cross-coprime 7/10 and structural 3/10	**[paper proof in this report]**	Specialization $a = b = c = d = 0$.
Full-grid $\pi/4$ ceiling	**[paper proof in this report]**	Countermonotone and comonotone transport for the common quantile $\sqrt{t}$.
HCIZ determinant identity	**[classical/open background]**	Exact finite identity; the needed large-N inequality remains open.
First tied-layer fiber factorization	**[paper proof in this report]**	Exact initial-form determinant algebra in the cross-coprime branch.
First tied layer is $o(H_T)$	**[classical/open background]** plus **[paper proof in this report]**	Standard p-adic two-logarithm estimates; no claim about higher valuation layers.
Mixed cancellation-excess criterion	**[paper proof in this report]**	Immediate sufficient condition from integrality and the Gini deficit.

Statement	Status	Dependency or caveat
Software exclusion through 2^{27}	[**interval computation**]	Complete MPFR directed-rounding scan; not a Lean proof.
Kernel exclusion through 4096	[**kernel-verified Lean**]	Remains the strongest formal finite theorem audited in the source report.
The Alaoglu–Erdős conjecture itself	[**proposed target**]	Equivalent to the unresolved special four-exponentials determinant.

46.1 Logical dependency graph

The new deductions can be summarized as

exact primitive monoid $+$ primitive depth

$\Downarrow$

synchronized Sturmian system and exact block valuation profiles

$\Downarrow$

ordinary all-prime fraction

mixed Gini conservation law

full-grid transport problem

$\Downarrow$

$0.871049\ldots$ ordinary ceiling, $4/5$ mixed ceiling, $\pi/4$ full-grid ceiling

$\Downarrow$

only cancellation beyond tropical minima or a sharper HCIZ bound can finish these routes.

Baker–Matveev separation and the finite scan are parallel inputs: they sharpen the possible counterexample but do not enter the determinant ceiling proofs.

47 Conclusions

The Alaoglu–Erdős conjecture remains unproved. The continuation nevertheless changes the shape of the search in several concrete ways.

First, the exact conditional monoid has a compact symbolic normal form. After removing the integer part of the primitive generator, one irrational Sturmian word controls both the binary and ternary denominator jumps of two rational recurrences. This exposes a possible route through finite-state rigidity, but it also prevents a false shortcut: rational multipliers do not by themselves make the word automatic.

Second, ordinary transcendence theory gives more Diophantine separation than the source report's elementary argument records. A fixed hypothetical generator is a ratio of two logarithms of integers, so Baker–W"ustholz/Matveev yields an effective polynomial irrationality measure. Continued-fraction denominators therefore grow at most polynomially from one stage to the next. This is useful quantitative structure, but it does not conflict with the $O(T)$ exact population of a dyadic block.

Third, the most direct p-adic determinant plan can now be audited with exact constants. For ordinary block Vandermondes, every valuation at every prime already supporting the inputs is accounted for, and primitive depth leaves at least $12.8950\ldots\%$ of the leading archimedean budget uncovered. For the balanced mixed determinant, all primes collapse into a finite family of slope vectors summing to zero. The triangular Gini norm then proves that entrywise divisibility misses at least 20% universally, at least 30% in the cross-coprime branch, and at least 70% if one uses only the structural places $2, 3$ there. The complete two-dimensional grid improves the geometry but still leaves a termwise ceiling of $\pi/4$.

These are no-go theorems for particular mechanisms, not evidence that the conjecture is false. They identify the exact remaining mechanisms:

1. a leading-order cascade of cancellations among many equal-valuation Leibniz terms;
2. an archimedean determinant estimate substantially smaller than its largest-term bound, most naturally through the HCIZ free energy;
3. a genuinely mixed real/p-adic construction that turns $\log M \log 3 = \log A \log 2$ into deep congruences; or
4. a rigidity theorem for the synchronized rational Sturmian system.

The first cancellation layer is provably too small, so any successful p-adic proof must propagate through many valuation layers rather than extracting one convenient Vandermonde factor.

Finally, the directed-rounding computation was extended by one full binary exponent. At the software-trust level, no non-power-of-two candidate occurs below 2^{27}, so a counterexample would satisfy $x > 27$. The formal Lean boundary remains $x \geq 12$ until a generated certificate is checked by the kernel. The old scanner-source hash printed in the source report is incorrect; the literal source, overlap replay, new log hash, and high-precision extremal replays are recorded explicitly in this document.

Bottom line. The conjecture is still open, but the continuation produces new exact mathematics rather than only a list of ideas: a synchronized Sturmian equivalence; effective polynomial separation; closed all-prime Vandermonde asymptotics; a triangular Gini norm governing mixed tropical divisibility; universal $4/5$, $7/10$, $3/10$, and $\pi/4$ ceilings; a first-layer cancellation theorem; exact sufficient thresholds for future work; and a reproducible software exclusion through 2^{27}. The next decisive calculation is the triangular HCIZ free energy or, on the arithmetic side, a proof of a genuinely leading tropical cancellation cascade.

48 Second continuation: scope and claim ledger

The problem is

This document is a sequel rather than a replacement for the preceding parts of this report. It restates every inherited fact used below, because several later arguments depend sensitively on the exact quantifiers and on whether a quantity is integral or merely rational. In particular, the slogan "one counterexample generates a grid" is proved here as an exact equality of monoids, not used as a heuristic.

No complete proof of Conjecture 2.1 is claimed. The strongest new unconditional outputs of this continuation are:

1. a streamlined proof of the exact free-monoid structure, including the denominator-elimination step that is easy to understate;
2. a universal N^2-scale prime-divisibility lower bound for exact-grid interpolation determinants;
3. the triangular determinant inequality of Theorem 53.1;
4. a rational-basis, product-formula extension that quantifies exactly how denominators defeat naive lattice reduction;
5. additional shared-prime, congruence, counting, and formalization consequences.

48.1 Claim ledger

Claim	Status	Main input
A counterexample is positive and transcendental	proved	elementary reductions; Gelfond–Schneider
All solutions lie in a rational affine line through 1 and one nonintegral solution	proved	Six Exponentials Theorem
The complete solution set is $\mathbb{N}_0 + \mathbb{N}_0\beta$	proved	valuations and minimality
A counterexample has finite effective irrationality measure	proved	linear forms in two logarithms
Universal square-grid divisor $6^{(1+\beta)L_m}$	proved	tropical determinant valuation
Triangular inequality $g(\sqrt{\beta \log 2 \log 3}) \geq 0$	proved	HCIZ bound, assignment asymptotics, Baker uniformity
Rational-basis denominator inequality	proved	product formula and signed assignment bounds
A complete contradiction for all β	not obtained	requires further N^2-scale arithmetic gain

The elementary reductions, the affine rank bound, the primitive least counterexample and the exact free-monoid theorem restated at this point in the source document are already established earlier in this report, and in kernel-verified form: see Section 5 and the modules `LeastSolution`, `PrimitiveGenerator` and `RationalPowerIndex`. They are not repeated here. What follows is the new material of the second continuation, which takes the exact grid as given and asks how much arithmetic a determinant built on it can extract.

49 Effective Diophantine and counting consequences

The two logarithmic descriptions of β imply more than transcendence. Because A is not a power of 2, the linear form $q\log A - p\log 2$ is nonzero whenever $(p,q)\neq(0,0)$. Effective lower bounds for linear forms in logarithms therefore give constants $c>0$ and $\tau>0$, effectively computable from A, such that

$$\|q\beta\| \geq cq^{-\tau} \qquad (q\geq 1). \tag{204}$$

The analogous representation $\beta = \log B/\log 3$ gives another such bound. No useful small absolute exponent is asserted here; standard explicit bounds are enormous. The qualitative conclusion is nevertheless important.

Corollary 49.1 (Finite irrationality measure)**.** *A counterexample β is not a Liouville number. Its irrationality measure is finite and effectively bounded in terms of either A or B.*

The exact monoid also gives an exact counting formula. Let

$$N_{\mathcal{S}}(T) := \#(\mathcal{S}\cap[0,T]).$$

Then, with $K=\lfloor T/\beta\rfloor$,

$$N_{\mathcal{S}}(T) = \sum_{k=0}^{K}\big(\lfloor T-k\beta\rfloor+1\big). \tag{205}$$

Proposition 49.2 (Quadratic counting with a power saving)**.** *There is an effective $\delta>0$ such that*

$$N_{\mathcal{S}}(T) = \frac{T^2}{2\beta} + \frac{\beta+1}{2\beta}T + O_\beta(T^{1-\delta}).$$

Proof sketch. Equation (204), the Erdős–Turán inequality, and the geometric-series estimate for $\sum_{k\leq K} e^{2\pi i hk\beta}$ give a power saving for the discrepancy of the rotation sequence $\{k\beta\}$. In particular,

$$\sum_{k=0}^{K}\{T-k\beta\} = \frac{K+1}{2} + O(K^{1-\delta})$$

for some effective $\delta>0$. Insert this in Equation (205). Writing $K=T/\beta-\{T/\beta\}$, the apparently oscillatory terms of order T cancel, leaving the displayed linear coefficient. □

The same argument gives quantitative equidistribution of the dense orbit

$$\left(2^{\{k\beta\}}, 3^{\{k\beta\}}\right) = \left(\frac{A^k}{2^{\lfloor k\beta\rfloor}}, \frac{B^k}{3^{\lfloor k\beta\rfloor}}\right).$$

Thus a counterexample would generate a dense family of rational points on the arc $t\mapsto(2^t,3^t)$, with pure base-power denominators. Their heights grow exponentially in k, so this observation alone is far below the density needed for Pila–Wilkie-type counting to force algebraicity; it is nevertheless a useful reformulation for S-unit methods.

50 Valuation-lattice saturation and an exact duality

The denominator argument has a useful invariant formulation. It also has a symmetric consequence that is easy to miss if one works only with the least counterexample β.

Write

$$a_p = v_p(A), \qquad b_p = v_p(B),$$

and form the two-column integer matrix whose rows are

$$(\mathbf{1}_{p=2}, a_p) \quad \text{and} \quad (\mathbf{1}_{p=3}, b_p) \qquad (p \text{ prime}).$$

Only finitely many entries in the second column are nonzero. Let $\mathcal{L}$ be the rank-two lattice generated by the columns in the direct sum of all row copies of $\mathbb{Z}$.

Theorem 50.1 (Saturated valuation lattice)**.** *The lattice $\mathcal{L}$ is primitive (saturated):*

$$(\mathcal{L} \otimes_{\mathbb{Z}} \mathbb{Q}) \cap \bigoplus_p (\mathbb{Z} \oplus \mathbb{Z}) = \mathcal{L}.$$

Equivalently, the greatest common divisor of all maximal minors of the matrix is 1*. In the present sparse matrix this is exactly*

$$\gcd(\{a_p : p \neq 2\} \cup \{b_p : p \neq 3\} \cup \{a_2 - b_3\}) = 1.$$

Proof. The nonzero maximal minors are, up to sign, exactly the displayed a_p, b_p, and $a_2 - b_3$: the first column is nonzero only in the row belonging to $p = 2$ in the first valuation block and the row belonging to $p = 3$ in the second. It remains to prove that their gcd is 1.

If a number $d > 1$ divided all these minors, choose $t \in \{0, \dots, d-1\}$ with $t \equiv -a_2 \pmod d$. The congruence $a_2 \equiv b_3 \pmod d$ then gives $t \equiv -b_3 \pmod d$, while every other valuation of A or B is already divisible by d. Consequently both $2^t A$ and $3^t B$ are perfect dth powers in $\mathbb{N}$. Hence

$$\frac{\beta + t}{d} \in S.$$

This number is nonintegral, and it is strictly between 0 and β because $\beta > 1$ and $0 \leq t \leq d - 1$, contradicting the choice of β. Thus the gcd of the maximal minors is 1. The rank-two Smith normal form criterion then says exactly that the column lattice is saturated. □

A particularly concrete form of the theorem records exactly how extraction of common roots behaves throughout the solution monoid.

Corollary 50.2 (Perfect-power content is preserved)**.** *For $u, v \in \mathbb{N}_0$, not both zero, define*

$$X_{u,v} = 2^u A^v, \qquad Y_{u,v} = 3^u B^v.$$

Then

$$\gcd(\{v_p(X_{u,v}) : p \text{ prime}\} \cup \{v_p(Y_{u,v}) : p \text{ prime}\}) = \gcd(u, v).$$

Equivalently, for every $d \geq 1$, the two integers $X_{u,v}$ and $Y_{u,v}$ are simultaneously perfect dth powers if and only if $d \mid u$ and $d \mid v$.

Proof. Every divisor of $\gcd(u,v)$ plainly divides all displayed valuations. In the other direction, if d divides all of them, then $X_{u,v} = C^d$ and $Y_{u,v} = D^d$ for positive integers C, D. Thus $(u + v\beta)/d \in S$. Uniqueness of the representation in the irrational basis $(1,\beta)$ gives $u/d, v/d \in \mathbb{N}_0$. □

For example, A and B cannot themselves be simultaneous proper powers; and for every $t \geq 0$, neither pair

$$(2^t A, 3^t B), \qquad (2A^t, 3B^t)$$

can consist of simultaneous proper powers of the same degree. These are not independent ad hoc restrictions: all of them are specializations of one saturated rank-two valuation lattice.

There is also an exact reciprocal description. Put

$$S_{A,B} = \{y \in \mathbb{R} : A^y \in \mathbb{Z},\ B^y \in \mathbb{Z}\}.$$

Theorem 50.3 (Reciprocal solution monoid)**.** *Under the counterexample hypothesis,*

$$S_{A,B} = \mathbb{N}_0 + \frac{1}{\beta}\mathbb{N}_0.$$

In particular $1/\beta$ is the least positive nonintegral element of $S_{A,B}$.

Proof. If $y \in S_{A,B}$, then $x = \beta y$ satisfies

$$2^x = A^y \in \mathbb{Z}, \qquad 3^x = B^y \in \mathbb{Z}.$$

Hence $x = n + k\beta$ with $n, k \in \mathbb{N}_0$, and $y = k + n/\beta$. Conversely,

$$A^{k+n/\beta} = A^k 2^n, \qquad B^{k+n/\beta} = B^k 3^n$$

are integers for all $n, k \geq 0$. □

This reciprocal monoid shows that the two columns of the valuation matrix may be interchanged without losing arithmetic information: the original and dual root-extraction problems are the same saturated lattice viewed in opposite bases.

51 Exact-grid interpolation determinants

The free monoid of the exact primitive-generator theorem (Section 5) supplies two exact rank-two grids: one in the logarithms of the bases and one in the solution exponents. Retaining both coordinates is the central step of this continuation.

For finite sets $\mathcal{R}, \mathcal{C} \subset \mathbb{N}_0^2$ with $|\mathcal{R}| = |\mathcal{C}| = N$, define

$$\begin{aligned} u_{a,b} &:= a\alpha + b\gamma, \qquad (a,b) \in \mathcal{R}, \\ v_{n,k} &:= n + k\beta, \qquad (n,k) \in \mathcal{C}, \end{aligned}$$

and the matrix

$$M_{(a,b),(n,k)} := \exp(u_{a,b} v_{n,k}) = 2^{an} 3^{bn} A^{ak} B^{bk}. \tag{206}$$

Every entry is a positive integer.

51.1 Nonvanishing and total positivity

Lemma 51.1 (Strict positivity)**.** *Order the u-values and the v-values increasingly. Then*

$$D(\mathcal{R},\mathcal{C}) := \det\bigl(\exp(u_i v_j)\bigr)_{i,j=1}^{N} > 0.$$

In particular it is a positive integer and hence at least 1.

Proof. The u-values are distinct because 2 and 3 are multiplicatively independent, and the v-values are distinct because β is irrational. The kernel $K(u,v)=e^{uv}$ is strictly totally positive on $\mathbb{R}\times\mathbb{R}$. Equivalently, the functions $e^{v_1 u},\dots,e^{v_N u}$ with $v_1<\cdots<v_N$ form an extended complete Chebyshev system: their Wronskian is

$$e^{u(v_1+\cdots+v_N)}\prod_{i<j}(v_j-v_i) > 0.$$

The generalized mean-value theorem for Chebyshev systems gives the desired sign. □

51.2 Archimedean upper bound

Write $\Delta(u)=\prod_{i<j}(u_j-u_i)$ and similarly for v, and let $G(N+1)=\prod_{j=1}^{N-1} j!$ be the relevant Barnes product.

Lemma 51.2 (HCIZ determinant inequality)**.** *For increasingly ordered real vectors u,v,*

$$0<\det(e^{u_i v_j}) \le \frac{\Delta(u)\,\Delta(v)}{G(N+1)}\exp\left(\sum_{i=1}^{N} u_i v_i\right). \tag{207}$$

Proof. The Harish–Chandra–Itzykson–Zuber formula with normalized Haar measure is

$$\int_{U(N)} e^{\operatorname{tr}(\operatorname{diag}(u)U\operatorname{diag}(v)U^*)}\,dU = G(N+1)\frac{\det(e^{u_i v_j})}{\Delta(u)\,\Delta(v)}.$$

Von Neumann's trace inequality bounds the exponent in the integral above by $\sum_i u_i v_i$. □

The factor $G(N+1)$ is crucial. When both node sets come from two-dimensional lattice regions, the leading $N^2\log N$ contributions of the two Vandermonde products cancel the $N^2\log N$ contribution of $G(N+1)$ exactly. The problem therefore lives at the next, N^2-constant, scale. This is the determinant manifestation of the critical nature of four exponentials.

51.3 Finite-prime lower bound as an assignment problem

For a prime p, the valuation of an entry in (206) is

$$w_p((a,b),(n,k)) = \mathbf{1}_{p=2}an + \mathbf{1}_{p=3}bn + a_p ak + b_p bk. \tag{208}$$

Lemma 51.3 (Tropical determinant bound)**.** *For every prime p,*

$$v_p(D(\mathcal{R},\mathcal{C})) \ge \min_{\sigma\in S_N}\sum_{i=1}^{N} w_p(i,\sigma(i)).$$

Proof. Each Leibniz term has the displayed summed valuation. The nonarchimedean triangle inequality says that the valuation of their signed sum is at least the minimum of their valuations. □

This converts divisibility into a finite optimal-transport problem. Even a coarse componentwise use of it already produces an N^2-scale divisor.

52 Square grids: a universal divisor and the critical cancellation

Take $\mathcal{R} = \mathcal{C} = I_m := \{0, \dots, m-1\}^2$ and $N = m^2$. Denote the corresponding determinant by D_m.

Let L_m be the minimum of $\sum_i a_i n_{\sigma(i)}$ over bijections between the two copies of I_m. The multiset of first coordinates consists of each integer $0, \dots, m-1$ repeated m times. The rearrangement inequality gives

$$L_m = m \sum_{j=0}^{m-1} j(m-1-j) = \frac{m^2(m-1)(m-2)}{6}. \tag{209}$$

The same value applies to each of the four pairings an, bn, ak, bk.

Theorem 52.1 (Universal square-grid divisor)**.** *For every $m \geq 1$,*

$$\log D_m \geq L_m(\log 2 + \log 3 + \log A + \log B) = L_m(1+\beta)\log 6. \tag{210}$$

Equivalently, the integer D_m is at least $6^{(1+\beta)L_m}$ in logarithmic size.

Proof. For a fixed prime, the minimum of a sum of nonnegative cost components is at least the sum of their separate minima. Apply Lemma 51.3 to (208); each present component has minimum L_m. Sum the resulting lower bounds with weights $\log p$ and use $\sum_p a_p \log p = \log A$ and $\sum_p b_p \log p = \log B$. □

The estimate is much stronger than the bare fact $D_m \geq 1$. Nevertheless, a balanced rectangular asymptotic does not yet contradict it. Let $z = \sqrt{\alpha\gamma\beta}$. After choosing rectangle aspect ratios so that the two contributions to each projected logarithm are equal, the normalized node distribution is that of $Z = (X+Y)/2$ for independent $X, Y \sim \mathrm{Unif}[0,1]$. One has

$$\mathbb{E}Z^2 = \frac{7}{24}, \qquad \iint \log|s-t| \, d\,\mathrm{Law}(Z)(s) d\,\mathrm{Law}(Z)(t) = \frac{1}{3}\log 2 - \frac{25}{12}.$$

Combining these with Lemma 51.2 and Theorem 52.1 yields the necessary rectangular inequality

$$h(z) := \frac{1}{2}\log(4z) + \frac{1}{3}\log 2 - \frac{4}{3} + \frac{z}{2} \geq 0. \tag{211}$$

For the range allowed even by $A \geq 3$, this inequality is already positive. The square-grid divisor is therefore real progress in the arithmetic estimate, but rectangular geometry spends too much archimedean size.

53 Triangular grids and a new exclusion inequality

The natural way to reduce the archimedean cost is to take the first N nodes rather than a rectangular block.

Let $\mathcal{R}_N$ consist of the N pairs $(a,b) \in \mathbb{N}_0^2$ with the smallest values of $u = a\alpha + b\gamma$, and let $\mathcal{C}_N$ consist of the N pairs $(n,k) \in \mathbb{N}_0^2$ with the smallest values of $v = n + k\beta$. Ties do not occur. Let D_N be the positive integer determinant from these sets.

Define

$$U := \sqrt{2\alpha\gamma}, \qquad V := \sqrt{2\beta}, \qquad z := \sqrt{\alpha\gamma\beta},$$

so $UV = 2z$.

53.1 Limiting distributions

Elementary lattice-point counting gives

$$\#\{(a,b) \in \mathbb{N}_0^2 : a\alpha + b\gamma \le T\} = \frac{T^2}{2\alpha\gamma} + O(T),$$

and similarly for $n + k\beta$. Thus the Nth thresholds are asymptotic to $U\sqrt{N}$ and $V\sqrt{N}$. After division by $\sqrt{N}$, the projected nodes converge to the distributions

$$d\mu_U(t) = \frac{2t}{U^2}\,dt \quad (0 \le t \le U), \qquad d\nu_V(t) = \frac{2t}{V^2}\,dt \quad (0 \le t \le V).$$

Their quantile functions are

$$Q_{\mu_U}(s) = U\sqrt{s}, \qquad Q_{\nu_V}(s) = V\sqrt{s}.$$

The logarithmic energies are

$$\mathcal{E}(\mu_U) = \log U - \frac{7}{4}, \qquad \mathcal{E}(\nu_V) = \log V - \frac{7}{4}. \tag{212}$$

The finite irrationality type supplied by linear forms in logarithms ensures that exceptionally close projected lattice points do not spoil convergence of the logarithmic energies; details are given in Appendix M.

53.2 The archimedean constant

The Barnes asymptotic is

$$\log G(N+1) = \frac{N^2}{2}\log N - \frac{3N^2}{4} + o(N^2).$$

Using (212), the two Vandermonde terms in Lemma 51.2, and comonotone coupling for the trace term, gives

$$\limsup_{N\to\infty} \frac{\log D_N}{N^2} \le \frac{1}{2}\log(UV) - 1 + \frac{UV}{2}. \tag{213}$$

Indeed,

$$\int_0^1 Q_{\mu_U}(s) Q_{\nu_V}(s)\,ds = UV\int_0^1 s\,ds = \frac{UV}{2}.$$

53.3 The finite-prime constant

Uniform measure on the row triangle can be written as

$$a = \frac{U}{\alpha}X, \qquad b = \frac{U}{\gamma}Y, \qquad X, Y \geq 0, \quad X + Y \leq 1,$$

where (X, Y) is uniform on the standard simplex. Each coordinate has density $2(1-x)$ and quantile

$$q(s) = 1 - \sqrt{1-s}.$$

The column coordinates have the corresponding scales $n = VX'$ and $k = (V/\beta)Y'$. The counter-monotone assignment constant is

$$J := \int_0^1 q(s)q(1-s)\,ds = \frac{\pi}{8} - \frac{1}{3}. \tag{214}$$

Consequently each of the four logarithmically weighted components $an\log 2$, $bn\log 3$, $ak\log A$, and $bk\log B$ contributes asymptotically $UVJN^2$ to the componentwise valuation lower bound. Therefore

$$\liminf_{N\to\infty} \frac{\log D_N}{N^2} \geq 4UVJ. \tag{215}$$

Theorem 53.1 (Triangular determinant inequality)**.** *Every nonintegral counterexample β satisfies*

$$\boxed{g\left(\sqrt{\beta\log 2\log 3}\right) \geq 0, \qquad g(z) = \frac{1}{2}\log(2z) - 1 + \left(\frac{11}{3} - \pi\right)z.} \tag{216}$$

Proof. Subtract the lower bound (215) from the upper bound (213). Since $UV = 2z$ and $4J = \pi/2 - 4/3$, the resulting necessary inequality is exactly (216). ☐

The derivative $g'(z) = 1/(2z) + 11/3 - \pi$ is positive for $z > 0$, so g has a unique positive zero

$$z_0 = 1.12895010749713, \qquad \beta_\triangle := \frac{z_0^2}{\log 2\log 3} = 1.67370758736671.$$

At the first arithmetically possible nonintegral value $\beta = \log_2 3$, one has $z = \log 3$ and

$$g(\log 3) = -0.029549732685145 < 0.$$

The numerical consequence of Theorem 53.1 is weaker than what is already known: it gives $\beta \geq \beta_\triangle = 1.6737\ldots$, whereas the kernel-verified finite check of Section 25 gives $\beta \geq 12$ and the directed-rounding scan reaches 2^{27}. The inequality is recorded for its method, not for its bound: it is the first exclusion obtained from a uniform asymptotic determinant estimate rather than from a per-value certificate, and it exposes the arithmetic constant that would have to improve in order to reach all A at once.

53.4 Numerical anatomy of the two grid geometries

The table compares the triangular gap g with the balanced-rectangle gap h from (211). Negative values would contradict a counterexample at that β.

β	$z = \sqrt{\alpha\gamma\beta}$	$g(z)$	$h(z)$
1	0.8726396796	-0.2633422644	-0.04093352603
$\log_2 3$	1.098612289	-0.02954973269	0.1871929656
$\beta_\triangle$	1.128950107	-1.054219794e-81	0.2159820074
2	1.23409887	0.09973735836	0.3130828643
$\log_2 5$	1.329717364	0.1872568287	0.3982047952
3	1.511456262	0.3467367942	0.5531278372
5	1.951281643	0.7053840783	0.900746934
10	2.759528964	1.303060538	1.47815739

The triangular region improves the constant substantially, but g is increasing and becomes comfortably positive in the range $\beta \geq \log_2 5$. To continue, one must either improve the finite-prime lower bound, find a better variational region, or enrich the determinant.

54 Rational changes of basis and the product formula

A natural reaction to the positive gap is to replace 3 by a ratio such as 3/2, whose logarithm is smaller. Ignoring denominators makes this look like an immediate proof. The product formula shows exactly why that argument is false and, more usefully, turns it into quantitative denominator constraints.

54.1 A signed arithmetic functional

For a rational $q > 1$, write it in lowest terms as $q = \mathrm{num}(q)/\mathrm{den}(q)$ and define

$$\rho(q) := \frac{J \log \mathrm{num}(q) - K \log \mathrm{den}(q)}{\log q}, \qquad J = \frac{\pi}{8} - \frac{1}{3}, \quad K = \frac{1}{6}. \tag{217}$$

The constant J is the countermonotone coordinate cost in a simplex, while K is the comonotone cost, since a simplex coordinate has second moment 1/6. If q is an integer, then $\rho(q) = J$. A denominator contributes with K, not J, because a negative valuation reverses the relevant assignment bound.

Theorem 54.1 (Adelic triangular inequality)**.** *Let $r_1, r_2 > 1$ be multiplicatively independent positive rationals and suppose $R_i = r_i^\beta \in \mathbb{Q}_{>1}$ for $i = 1, 2$. Put*

$$z_r := \sqrt{\beta \log r_1 \log r_2}.$$

Then the existence of these four rational exponentials implies

$$\frac{1}{2}\log(2z_r) - 1 + z_r - 2z_r(\rho(r_1) + \rho(r_2) + \rho(R_1) + \rho(R_2)) \geq 0. \tag{218}$$

Equivalently, if

$$\Delta_{\mathrm{den}} := \sum_{q \in \{r_1, r_2, R_1, R_2\}} \frac{\log \mathrm{den}(q)}{\log q},$$

then

$$g(z_r) + 2z_r \left(\frac{1}{2} - \frac{\pi}{8}\right) \Delta_{\mathrm{den}} \geq 0. \tag{219}$$

Proof. Use the same first-N triangular sets for the positive logarithms $\log r_1, \log r_2$ and for $1, \beta$. The determinant entries are now rational. For a fixed prime, split each signed valuation coefficient into its positive and negative parts. In the tropical determinant bound, a positive coefficient is multiplied by the countermonotone minimum J, while a negative coefficient is multiplied by the comonotone maximum K. Summing over primes and using the product formula for the nonzero rational determinant gives the lower constant

$$2z_r(\rho(r_1) + \rho(r_2) + \rho(R_1) + \rho(R_2)).$$

The HCIZ upper constant remains $\frac{1}{2}\log(2z_r) - 1 + z_r$. This proves (218). Finally,

$$\rho(q) = J - (K - J)\frac{\log \mathrm{den}(q)}{\log q},$$

and substitution gives (219). □

54.2 All unimodular bases of $\langle 2, 3 \rangle$

Let

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{Z})$$

and suppose

$$\lambda_1 = a\alpha + b\gamma > 0, \qquad \lambda_2 = c\alpha + d\gamma > 0.$$

Set

$$r_1 = 2^a 3^b, \quad r_2 = 2^c 3^d, \qquad R_1 = A^a B^b, \quad R_2 = A^c B^d.$$

Then Theorem 54.1 applies. This supplies an infinite family of necessary inequalities indexed by unimodular changes of basis. Continued-fraction vectors make one λ_i small, but the corresponding rational base has a large denominator; Equation (219) measures the precise price.

Proposition 54.2 (Optimality of the integral basis)**.** *Among unimodular bases for which both transformed bases are integers, the product $\lambda_1\lambda_2$ is at least $\alpha\gamma$. Equality occurs only for the original basis $(2, 3)$ up to permutation. Likewise, among unimodular bases of the solution lattice whose two transformed exponents remain in $\mathcal{S}$, the product of the positive exponent generators is at least β.*

Proof. If $2^a 3^b$ is an integer, then $a, b \geq 0$. Thus an integral unimodular basis has nonnegative entries. Expanding,

$$(a\alpha + b\gamma)(c\alpha + d\gamma) \geq (ad + bc)\alpha\gamma \geq \alpha\gamma,$$

because $|ad - bc| = 1$. Equality forces the standard basis up to permutation. For the exponent lattice, the exact primitive-generator theorem says that a transformed exponent is simultaneously integral at bases 2 and 3 exactly when its coefficients in $(1, \beta)$ are nonnegative. The same argument applies. □

This proposition explains why a denominator-free lattice reduction cannot improve the triangular inequality: the exact monoid has already selected the smallest integral basis.

54.3 The basis $(2, 3/2)$ and a gcd constraint

Take

$$r_1 = 2, \qquad r_2 = \frac{3}{2}, \qquad R_1 = A, \qquad R_2 = \frac{B}{A}.$$

Let $d = A/\gcd(A, B)$ be the denominator of B/A in lowest terms and put $\delta = \log(3/2)$. Then

$$\Delta_{\mathrm{den}} = \frac{\alpha}{\delta} + \frac{\log d}{\beta\delta}. \tag{220}$$

Theorem 54.1 therefore gives a rigorous lower bound on d, whenever the right side obtained by solving (219) is positive. In words: if $3/2$ substantially improves the archimedean configuration, then B/A must carry enough denominator to pay for that improvement. A counterexample cannot have excessive divisibility $A \mid B$ in the range where the reduced-basis gap is negative.

For orientation, the next table substitutes the hypothetical value $\beta = \log_2 A$. The column $d_{\min}$ is the least integer denominator not excluded by the inequality alone; it is not a claim that the corresponding B exists.

A	$\log_2 A$	z_r	gap before $\log d$	$d_{\min}$
3	1.5849625	0.667419621	-0.260297461	4
5	2.32192809	0.807818616	-0.0396109308	2
6	2.5849625	0.852347316	0.0269341919	1
7	2.80735492	0.88825597	0.0795954278	1
9	3.169925	0.94387388	0.159569469	1
10	3.32192809	0.966239056	0.191227186	1
11	3.45943162	0.986033907	0.219022705	1
12	3.5849625	1.00376439	0.243748126	1
13	3.70043972	1.01980266	0.265979188	1
14	3.80735492	1.03443012	0.286146736	1
15	3.9068906	1.04786443	0.304581094	1
17	4.08746284	1.07180649	0.337231592	1
18	4.169925	1.08256404	0.351820061	1
19	4.24792751	1.09264233	0.36544257	1
20	4.32192809	1.10211837	0.378212214	1
21	4.39231742	1.11105698	0.39022374	1
22	4.45943162	1.11951323	0.401557272	1
23	4.52356196	1.12753426	0.412281139	1

A second useful basis is

$$r_1 = \frac{3}{2}, \qquad r_2 = \frac{4}{3}, \qquad R_1 = \frac{B}{A}, \qquad R_2 = \frac{A^2}{B}.$$

Its two logarithms are smaller, but Equation (219) now charges the denominators of all four displayed ratios. It therefore constrains the possibility that B/A and A^2/B are simultaneously close to integers. More generally, continued-fraction bases produce a network of constraints on

$$\frac{B^q}{A^p} \quad \text{when} \quad q\log 3 - p\log 2 \text{ is small.}$$

This is a concrete future route: prove that the denominator requirements for a carefully chosen finite family of bases are incompatible with the prime valuation vectors of A and B.

55 Joint prime transport and shared-factor gains

The componentwise lower bounds used above are intentionally conservative. At a fixed prime, a single permutation must handle all valuation components at once. If a prime occurs in more than one generator, the separate countermonotone assignments can conflict, making the true minimum strictly larger.

For the original triangular determinant, define

$$\mathcal{T}_p(N) := \frac{1}{N^2}\min_{\sigma\in S_N}\sum_i w_p(i,\sigma(i)).$$

Then the strongest direct product-formula lower bound is

$$\liminf_{N\to\infty}\frac{\log D_N}{N^2} \geq \sum_p (\log p)\liminf_{N\to\infty}\mathcal{T}_p(N).$$

The baseline $4UVJ$ replaces each $\mathcal{T}_p$ by the sum of four independent one-coordinate minima.

A simple rigorous gain estimate is available for a prime $p\notin\{2,3\}$ that divides both A and B. Put $r = v_p(A)$, $s = v_p(B)$ and

$$R = \frac{r}{\alpha}, \qquad S = \frac{s}{\gamma}.$$

On the normalized row simplex, the relevant scalar is $RX + SY$, while the column coordinate k has the marginal law of a simplex coordinate. The countermonotone cost is at least the product of the means minus the product of the standard deviations. Since

$$\mathbb{E}X = \mathbb{E}Y = \frac{1}{3}, \quad \operatorname{Var}X = \operatorname{Var}Y = \frac{1}{18}, \quad \operatorname{Cov}(X,Y) = -\frac{1}{36},$$

one obtains

$$\liminf \mathcal{T}_p(N) \geq \frac{UV}{\beta}\left(\frac{R+S}{9} - \frac{\sqrt{R^2+S^2-RS}}{18}\right). \tag{221}$$

The componentwise baseline for the same two terms is

$$\frac{UV}{\beta}J(R+S).$$

Taking the larger of these two lower bounds gives a fully explicit shared-prime improvement. Summed over common prime divisors of A and B, it yields a sufficient contradiction criterion whenever the gain exceeds the positive gap $g(z)$.

The estimate is strongest when r/α and s/γ are comparable and vanishes continuously as one valuation goes to zero. It therefore does not supply a universal proof: the worst arithmetic configuration may have nearly disjoint prime support. But it turns “a large gcd should help” into a quantitative theorem and identifies exact valuation patterns worth targeting.

56 Universal congruence divisibility is lower order

There is also divisibility at primes not dividing $6AB$. Modulo such a prime p, a row of the matrix depends on (a, b) only modulo $p-1$, by Fermat's little theorem. Hence there are at most $(p-1)^2$ distinct rows modulo p. For an $N \times N$ determinant,

$$v_p(D_N) \geq N - \operatorname{rank}_{\mathbb{F}_p}(M) \geq N - (p-1)^2$$

whenever the right side is positive. Summing over primes $p \leq \sqrt{N}$ and using the prime number theorem gives the universal extra estimate

$$\sum_{\substack{p \leq \sqrt{N} \\ p \nmid 6AB}} (N - (p-1)^2) \log p = \left(\frac{2}{3} + o(1)\right) N^{3/2}. \tag{222}$$

Higher prime powers and column congruences can improve the lower-order term. They do not alter the N^2 constant in Theorem 53.1. This provides a useful barrier result: congruence collisions that depend only on finite-group periodicity are too small by a factor of $\sqrt{N}$. A complete determinant proof needs counterexample-specific valuation structure or a different determinant geometry.

57 Why several tempting routes still stop short

57.1 Continued fractions and integer gaps

For integers $p, q > 0$,

$$\frac{A^q}{2^p} = 2^{q\beta - p}, \qquad \frac{B^q}{3^p} = 3^{q\beta - p}.$$

When p/q is a convergent to β, both ratios are close to 1. Since the cross-multiplied differences are nonzero integers, one obtains exponential lower bounds such as

$$|q\beta - p| \gg 2^{-p}.$$

These are far weaker than the polynomial lower bounds already supplied by linear forms in logarithms. Conversely, Baker's polynomial lower bounds do not contradict the q^{-2} approximations supplied by continued fractions. The two representations of β give proportional linear forms, not two independent small quantities.

57.2 Catalan, *abc*, and small differences

The equations $A^q - 2^p = \pm 1$ and $B^q - 3^p = \pm 1$ would be highly constrained, but continued fractions only make the relative difference small, not the absolute difference. The absolute scale is exponential in q. Standard *abc* heuristics do not bridge that exponential gap without an additional source of unusually good approximation.

57.3 Counting rational points

The orbit of fractional parts gives $O(\log H)$ rational points of height at most H on a fixed transcendental arc. This is compatible with every known subpolynomial upper bound for rational points on definable transcendental curves. A counting theorem would need to exploit the pure 2-power and 3-power denominator restrictions, not merely rationality.

57.4 The four-exponentials boundary

If x is irrational, the pairs $(\log 2, \log 3)$ and $(1, x)$ are both $\mathbb{Q}$-linearly independent, while all four exponentials

$$2, \quad 3, \quad 2^x, \quad 3^x$$

are algebraic. Thus the Four Exponentials Conjecture would prove Conjecture 2.1 immediately. The exact cancellation of the $N^2 \log N$ terms in our determinant estimates is the quantitative shadow of being exactly at this boundary. Six Exponentials is sufficient to collapse all possible exceptions to one affine direction, but not to remove that last direction.

58 A variational reformulation of the remaining determinant problem

The preceding calculations extend to scaled lattice regions $R, C \subset \mathbb{R}^2_{\geq 0}$ of equal area. Push forward uniform measure on R by $(a, b) \mapsto a\alpha + b\gamma$, and uniform measure on C by $(n, k) \mapsto n + k\beta$. The HCIZ side is determined by their logarithmic energies and comonotone transport. The finite-prime side is a sum of countermonotone or, more accurately, joint optimal-transport costs for the valuation bilinear forms (208).

This suggests the following concrete variational program.

1. Choose regions R, C and compute the archimedean functional

$$\mathcal{A}(R, C) = \frac{1}{2}\mathcal{E}(\mu_R) + \frac{1}{2}\mathcal{E}(\nu_C) + \frac{3}{4} + \int_0^1 Q_{\mu_R}(t) Q_{\nu_C}(t)\, dt,$$

   with the scale normalization determined by $\operatorname{area}(R) = \operatorname{area}(C)$.

2. Compute or bound the exact prime transport functional

$$\mathcal{P}(R, C; A, B) = \sum_p \log p \inf_\pi \int w_p(r, \pi(r))\, dr.$$

3. Any counterexample must satisfy $\mathcal{A}(R, C) \geq \mathcal{P}(R, C; A, B)$ for every admissible pair of regions. A single strict reverse inequality proves a contradiction.

Rectangles and origin triangles are analytically soluble test cases. The next step is not merely to scan aspect ratios: it is to optimize jointly over region shape and valuation configurations allowed by the two-three unit corollary and the primitive-gcd lemma of Section 5. The latter constraints are discrete and may prevent the valuation directions that make the transport cost artificially small.

59 Confluent and integer-valued determinant extensions

Ordinary evaluations use one row per lattice point. A potentially stronger construction uses integer jets. View (206) as evaluating the monomials X^nY^k at the integer points

$$(X,Y) = (2^a3^b, A^aB^b).$$

The Euler operators

$$D_X = X\frac{\partial}{\partial X}, \qquad D_Y = Y\frac{\partial}{\partial Y}$$

act by

$$D_X^r D_Y^s(X^nY^k) = n^r k^s X^nY^k,$$

which remains integral at the lattice points. Hasse derivatives give the alternative integer values

$$\binom{n}{r}\binom{k}{s}X^{n-r}Y^{k-s}.$$

Thus one can build confluent evaluation determinants without introducing the irrational factors $n + k\beta$ that ordinary derivatives along the real curve would produce.

The unresolved issues are nonvanishing and optimal scaling. A useful target is an integer-jet determinant whose archimedean limit behaves like a confluent HCIZ determinant while its p-adic assignment problem acquires an additional N^2 contribution. This is more promising than generic factorial normalization: the elementary divisibility obtained from binomial bases alone is only of order $N^{3/2}\log N$ for balanced two-dimensional grids.

60 Conditional resolutions and exact reformulations

60.1 Four Exponentials and Schanuel

The Four Exponentials Conjecture immediately implies Conjecture 2.1. Schanuel's conjecture implies Four Exponentials and therefore also suffices. The present problem is thus a particularly concrete integral special case of a central algebraic-independence conjecture.

60.2 Logarithm determinant formulation

With $A = 2^x$ and $B = 3^x$, the condition is

$$\det\begin{pmatrix}\log 2 & \log A\\ \log 3 & \log B\end{pmatrix} = 0.$$

The conjecture says that if A, B are positive integers and this determinant vanishes, then the two columns are related by a nonnegative integer factor. Baker's theorem controls nonzero *linear* forms in logarithms, whereas this is a bilinear relation among four logarithms; that distinction is exactly where standard transcendence machinery stops.

60.3 Order-preserving monoid formulation

A counterexample would give an order-preserving monoid isomorphism

$$\langle 2, A\rangle_{\mathbb{N}_0^2} \longrightarrow \langle 3, B\rangle_{\mathbb{N}_0^2}, \qquad 2^n A^k \mapsto 3^n B^k,$$

because both sides have logarithm $n + k\beta$ up to a fixed positive scale. The exact primitive-generator theorem says this monoid is the entire domain on which $m \mapsto m^{\log_2 3}$ is integer-valued. A purely arithmetic classification of such order-preserving power isomorphisms would prove the conjecture without explicit transcendence estimates.

61 Formalization roadmap

A substantial portion of the new work is suitable for formalization in Lean, with the deep analytic inputs isolated behind explicit interfaces.

The first three items of the source roadmap are already discharged in the repository and are therefore not repeated as targets: the range, rationality, additive-monoid and local-finiteness lemmas are `IntegerExponent` and `LeastSolution`; the affine rank-two proposition is `MultiplicativeRank` and `AlgebraicExponentLocus`; and the primitive-gcd and denominator-elimination lemmas, together with the exact monoid identity they yield, are `PrimitiveOddCore`, `ThreeDenominatorNormalization`, `RationalPowerIndex` and `PrimitiveGenerator`. What remains is the following.

1. Define finite matrices (206). The tropical valuation lemma follows directly from the Leibniz formula and valuation of a sum. The square-grid assignment value (209) is a finite rearrangement theorem.
2. Treat strict total positivity, HCIZ, logarithmic-energy convergence, and asymptotic transport as separate analytic interfaces. Finite versions of all determinant inequalities can still be machine checked once supplied with rational interval certificates.
3. For computational searches over region shapes or rational bases, emit proof certificates consisting of finite node lists, exact valuation assignment dual variables, and rational upper bounds for the real determinant.

This separation would let the ProveIt development verify every new arithmetic step even before the transcendence and asymptotic libraries are available.

62 Prioritized research directions

1. **Joint valuation optimization.** Replace the componentwise assignment lower bound by the exact transport cost for each prime, and optimize over valuation vectors satisfying the primitive-gcd and base-prime-unit conditions. This is the most direct way to seek the missing N^2 constant.
2. **Rational-basis denominator network.** Apply Theorem 54.1 to a finite set of continued-fraction bases. Translate the resulting inequalities into lower bounds on denominators of B^q/A^p. Try to show that unique factorization makes the bounds mutually incompatible.

3. **Variational region search.** Numerically optimize convex and nonconvex lattice regions using the exact archimedean energy and assignment functionals. Any promising negative gap should then be converted into a rigorous interval proof.
4. **Integer jets and confluent total positivity.** Establish a nonvanishing theorem for Euler/Hasse-jet determinants and determine whether their prime transport grows faster than their archimedean capacity.
5. **Exploit minimal-generator valuations.** The conditions $\min(v_2(A), v_3(B)) = 0$ and $g_0 = 1$ have not yet been fully inserted into the variational optimization. They should be imposed from the beginning rather than used only as post-processing constraints.
6. **Effective finite certificates.** For candidate ranges of A, combine rigorous interval arithmetic with determinant and gcd constraints. Although finite verification cannot settle the conjecture alone, it can reveal which valuation patterns come closest to saturating the theoretical bounds.

63 A proof-producing finite verification

The structural and determinant arguments above are uniform, but they leave an infinite range of possible values of $A = 2^\beta$. A complementary finite calculation can remove an initial segment without assuming any unproved transcendence statement. The following computation is deliberately based on outward-rounded intervals and an explicit positive-series remainder, rather than on a test such as "the floating-point value is not visually close to an integer."

Theorem 63.1 (Certified finite range)**.** *For every integer A with*

$$2 \le A \le 100{,}000,$$

the number $A^{\log_2 3}$ is an integer if and only if A is a power of 2. Consequently, a nonintegral Alaoglu–Erdős counterexample must satisfy

$$2^\beta > 100{,}000, \qquad \beta > \log_2(100{,}000) \approx 16.609640474437.$$

Certificate method. For fixed A, put

$$F_A(b) = (\log b)(\log 2) - (\log A)(\log 3), \qquad b > 0.$$

The function F_A is strictly increasing, and $A^{\log_2 3} = b$ is equivalent to $F_A(b) = 0$. Every logarithm is enclosed without calling a transcendental floating-point routine in the decisive comparison. For $0 \le z \le 1/3$ we use

$$\log \frac{1+z}{1-z} = 2\sum_{j=0}^{M-1} \frac{z^{2j+1}}{2j+1} + R_M(z), \qquad 0 < R_M(z) < \frac{2z^{2M+1}}{(2M+1)(1-z^2)}.$$

For an integer $n \ge 1$, choose $k = \lfloor \log_2 n \rfloor$ and write

$$\log n = k\log 2 + \log \frac{1+z_n}{1-z_n}, \qquad z_n = \frac{n-2^k}{n+2^k} \in [0, 1/3).$$

Thus all terms are positive rational operations, and directed rounding gives certified lower and upper endpoints. The run used 70 decimal digits and $M = 28$ terms. A binary64 value was used only as an initial guess for the integer immediately below the root; the program then moved monotonically until it had certified, for consecutive integers $b, b+1$,

$$F_A(b) < 0 < F_A(b+1).$$

Hence no integer can be the root. Powers of 2 are separated at the start, and for $A = 2^m$ the corresponding value is exactly 3^m.

The completed run checked 99,983 nonpowers of 2. The largest number of integer steps needed to repair an initial binary64 guess was 0. The smallest absolute interval endpoint encountered was `6.263278510736485942331545732E-13`, far outside the interval uncertainty. Runtime on the working container was approximately 15.238 seconds. The exact verifier used for the run is reproduced in Appendix Q; its SHA–256 digest is

`acc099fb776f830b243a2421bc7ce27d5bedc63a38c3a7b190accb3c621cbadd`.

□

Remark 63.2. The numerical bound is not presented as evidence that near misses cannot occur later. Its role is narrower and rigorous: it turns all A in the displayed finite interval into closed cases, while the determinant inequalities address whole infinite parameter ranges and arithmetic configurations at once.

64 Conclusion

The continuation does not close the final four-exponentials gap, but it substantially narrows and clarifies the problem.

First, the hypothetical exceptional set is completely rigid: one least counterexample β would generate every solution, and only solutions, as $n + k\beta$ with $n, k \geq 0$. Second, that exact grid supports determinants that are simultaneously strictly totally positive over the reals and highly divisible at finite primes. The square-grid divisor and the triangular inequality quantify both sides at the correct N^2 scale. Third, rational basis reduction is no longer a vague temptation: the adelic inequality shows exactly how denominators pay for every smaller logarithm. Finally, the remaining obstacle has a precise form—one needs an additional N^2-scale arithmetic gain, not merely more lattice points or a sharper polynomial error term.

The most promising immediate attack is therefore a joint one: optimize the triangular or variational determinant using the full prime valuation vectors, while applying several rational bases to force incompatible denominator and gcd requirements. That program is concrete, computationally testable, and amenable to partial formal verification.

65 Varying the antecedent, the domain, and the codomain

The conjecture fixes three choices that look incidental: the values are asked to be *integers*, the exponent is asked to be *real*, and the codomain is $\mathbb{Z}$ rather than a larger ring. This section records what each choice is worth. The short answer is that the conjecture predicts the same thing in every reading; what changes is only which wall stands guard.

65.1 The antecedent ladder

Write the three antecedents

$$(\mathbb{A}_2)\ 2^x, 3^x \in \overline{\mathbb{Q}} \Rightarrow x \in \mathbb{Q}, \qquad (\mathbb{Q}_2)\ 2^x, 3^x \in \mathbb{Q} \Rightarrow x \in \mathbb{Z}, \qquad (\mathbb{Z}_2)\ 2^x, 3^x \in \mathbb{Z} \Rightarrow x \in \mathbb{Z}.$$

The three-base theorem decomposes into two fully decoupled mechanisms, and only one of them ever sees the difference between $\mathbb{Z}$, $\mathbb{Q}$ and $\overline{\mathbb{Q}}$.

Mechanism one: transcendence. If $2^x, 3^x, 5^x \in \overline{\mathbb{Q}}$ and $x \notin \mathbb{Q}$ then $\{1, x\}$ is $\mathbb{Q}$-linearly independent, $\{\log 2, \log 3, \log 5\}$ is $\mathbb{Q}$-linearly independent by unique factorization, and the six products exponentiate to $2, 3, 5, 2^x, 3^x, 5^x$, all algebraic, contradicting six exponentials. This step is blind to integrality: it uses only algebraicity. It is the entire transcendence content of the three-base theorem.

Mechanism two: descent from $\mathbb{Q}$ to $\mathbb{Z}$. If the values are rational, write $x = p/q$ in lowest terms. Then $2^{p/q} \in \mathbb{Q}$ is a rational root of the monic $X^q - 2^p$, hence an integer m, and comparing 2-adic valuations in $2^p = m^q$ gives $p = q\, v_2(m)$, so $q \mid p$ and $q = 1$. One rational value at one base kills the denominator, and the same works for values in any subring of $\mathbb{Q}$. The dividing line is simply whether the value ring sits inside $\mathbb{Q}$ or contains irrational algebraic numbers.

Status: [classical/open background] for mechanism one; [kernel-verified Lean] for mechanism two, which is `integer_of_rational_of_two_rpow_integer` in `IntegerExponent`. The implications $(\mathbb{A}_2) \Rightarrow (\mathbb{Q}_2) \Rightarrow (\mathbb{Z}_2)$ follow, the first by mechanism two and the second trivially.

The $\overline{\mathbb{Q}}$-conclusion is sharp. Every rational $x = p/q$ genuinely occurs, since $2^{p/q}, 3^{p/q}, 5^{p/q}$ are algebraic of degree exactly q: the binomial $X^q - 2^p$ is irreducible by Capelli's criterion for $\gcd(p, q) = 1$, because 2^p is positive and is not an ℓ-th power for any $\ell \mid q$. So the $\overline{\mathbb{Q}}$-antecedent solution set is exactly $\mathbb{Q}$, and no strengthening of six exponentials can improve the conclusion. The same degree computation gives a graded refinement: if $2^x, 3^x, 5^x$ are algebraic of degree at most d, then $x \in \mathbb{Q}$ with denominator at most d. The kernel-verified exact radical degree of `RadicalDegree` is the two-base conditional shadow of exactly this statement.

Remark 65.1 (Where primality is used). Mechanism one needs only that the bases be multiplicatively independent, but mechanism two needs a base that is not a perfect power. The bases $4, 9, 25$ are multiplicatively independent, yet $x = 1/2$ gives $4^x = 2$, $9^x = 3$, $25^x = 5$, all integers. For general multiplicatively independent integer bases the correct statement is: rational values force $x \in \mathbb{Q}$ with a denominator q such that every base is a perfect q-th power. This is exactly the phenomenon that the power-primitive normalization of Section 5 exists to quotient away.

The rational form $(\mathbb{Q}_2)$ deserves separate billing: it is the conjecture as Alaoglu and Erdős posed it, and it is the form the colossally abundant application consumes. Its solution set is exactly $\mathbb{Z}$, negative exponents included, which restores the $x \leftrightarrow -x$ symmetry that the $\mathbb{Z}$ form breaks by truncating to $\mathbb{N}_0$.

Proposition 65.2 (The exact cost of the rational form)**.** $(\mathbb{Q}_2)$ *is equivalent to* $(\mathbb{Z}_2)$ *together with a denominator-support lemma: if* $2^x = a/b$ *and* $3^x = c/d$ *in lowest terms, then* b *is a power of* 2 *and* d *is a power of* 3.

Proof. Given the support lemma, $y = x + s$ for large s has $2^y = 2^s a/b \in \mathbb{Z}$ and $3^y = 3^s c/d \in \mathbb{Z}$, with y nonintegral exactly when x is, so $(\mathbb{Z}_2)$ applies. Conversely $(\mathbb{Q}_2)$ makes the support lemma vacuous, every solution being integral. $\square$

Status: [paper proof in this report] — absent the support lemma no reduction is known in either direction. It is the same kind of statement as the kernel-verified rational gateway of `RationalThirdBase`, and the modules `ThreeDenominatorNormalization` and `RationalPowerIndex` operate inside exactly this gap without stating the equivalence.

65.2 Complexifying the exponent is free

On the principal branch the restriction to real exponents is cosmetic.

Theorem 65.3 (Every complex solution is real)**.** *Let $x \in \mathbb{C}$ and let $2^x = \exp(x \log 2)$, $3^x = \exp(x \log 3)$ be the principal determinations. If $2^x, 3^x \in \mathbb{Z}$ then $x \in \mathbb{R}$. The same holds with $\mathbb{Z}$ replaced by $\mathbb{Q}$.*

Proof. Write $x = s + it$. The modulus of 2^x is 2^s, which reproduces the real hypothesis, and its argument is $t \log 2$; an integer value is real, so $t \log 2 \in \pi\mathbb{Z}$, and likewise $t \log 3 \in \pi\mathbb{Z}$. Write $t \log 2 = \pi a$ and $t \log 3 = \pi b$ with $a, b \in \mathbb{Z}$. If $t \neq 0$ then $a \neq 0$, and dividing gives $\vartheta = b/a \in \mathbb{Q}$, contradicting irrationality of ϑ. $\square$

Status: [kernel-verified Lean] — `im_eq_zero_of_complexTwoBaseIntegralSolution` and `twoBaseIntegralSolution_re_of_complex` in `ComplexSolutions`. Only irrationality of ϑ enters; no transcendence input is used.

Geometrically the sets $\{x : 2^x \in \mathbb{Z}\}$ and $\{x : 3^x \in \mathbb{Z}\}$ lie on horizontal gratings with spacings $\pi/\log 2$ and $\pi/\log 3$, and the two spacings are incommensurable *precisely because* ϑ is irrational, so the gratings meet only on the real axis. The conjecture may therefore be stated over $\mathbb{C}$ at no mathematical cost.

65.3 The algebraic antecedent over the complex numbers

Here the reduction to $\mathbb{R}$ genuinely fails — algebraic values need not have argument in $\{0, \pi\}$ — but something cleaner is true. Since $\overline{\mathbb{Q}}$ is closed under conjugation and division, $z \in \overline{\mathbb{Q}} \setminus \{0\}$ has $|z| = \sqrt{z\bar{z}} \in \overline{\mathbb{Q}}$. So the complex solution set splits as a direct sum

$$\{x \in \mathbb{C} : 2^x, 3^x \in \overline{\mathbb{Q}}\} = \{s \in \mathbb{R} : 2^s, 3^s \in \overline{\mathbb{Q}}\} + i\,\{t \in \mathbb{R} : 2^{it}, 3^{it} \in \overline{\mathbb{Q}}\}.$$

The real slice is $(\mathbb{A}_2)$; the imaginary slice is the unit-circle problem. Both are four-exponentials instances — genuinely different instances, with rows $\{1, s\}$ and $\{1, it\}$ against the same columns $\{\log 2, \log 3\}$ — so the complex two-base algebraic problem is exactly two decoupled instances of the same conjecture, and under it the conclusion is still $x \in \mathbb{Q}$, still sharp.

For three bases everything closes unconditionally: six exponentials is a statement over $\mathbb{C}$, the real slice gives $s \in \mathbb{Q}$, and the imaginary slice is the standard corollary that $2^{it}, 3^{it}, 5^{it}$ cannot all be algebraic for real $t \neq 0$. The three-base algebraic solution set over $\mathbb{C}$ is therefore exactly $\mathbb{Q}$; complexification adds nothing. One unconditional crumb survives for two bases: nonreal *algebraic* x are excluded, because Gelfond–Schneider holds for complex algebraic irrational exponents, so the transcendence theorem of Section 4 extends to $\mathbb{C}$ verbatim.

65.4 Multivalued branches widen the wall by one prime

Demand only that *some* determination of each power be an integer: $e^{xL_2} = m$ and $e^{xL_3} = m'$ with $L_2 = \log 2 + 2\pi i J$ and $L_3 = \log 3 + 2\pi i K$. Writing $xL_2 = \log|m| + i\pi a$ and $xL_3 = \log|m'| + i\pi b$ and eliminating x, the imaginary part gives a $\mathbb{Z}$-linear relation among logarithms, hence by unique factorization the multiplicative constraint $|m|^{2K}3^a = |m'|^{2J}2^b$, while the real part gives

$$\log|m|\log 3 - \log|m'|\log 2 = 2\pi^2\,(aK - bJ). \tag{223}$$

If $aK-bJ = 0$ the left side vanishes, which is the log-product wall of Section 12: either $|m| = |m'| = 1$, forcing $x = 0$, or $(|m|, |m'|)$ is a real counterexample pair. If $aK - bJ \neq 0$ one needs the log-product form to equal a nonzero integer multiple of $2\pi^2$. Since $i\pi = \log(-1)$, this is exactly the same wall over the prime set extended by the “prime” -1: the bilinear lattice gains the directions $\pi^2 = -(\log(-1))^2$ and $i\pi \log p$.

Status: [paper proof in this report] — the independence criterion of Section 12, extended to this larger lattice, covers the multivalued reading as well, and it remains a consequence of Schanuel’s conjecture. The four exponentials conjecture masters the multivalued reading wholesale, because the branch logarithms stay $\mathbb{Q}$-independent: it gives $x \in \mathbb{Q}$, and then the modulus argument forces $x \in \mathbb{Z}$ with all phases trivial. The predicted answer is unchanged; only the wall widens.

65.5 Complexifying the codomain: the Gaussian version is equivalent

Since extending the domain is free, the honest place to look for new content is the codomain.

Theorem 65.4 (Gaussian equivalence)**.** *The statement “$2^x, 3^x \in \mathbb{Z}[i] \Rightarrow x \in \mathbb{Z}$” is equivalent to Conjecture 2.1.*

Proof. One direction is trivial since $\mathbb{Z} \subset \mathbb{Z}[i]$. Conversely assume the conjecture and let $2^x, 3^x \in \mathbb{Z}[i]$ with $x = s + it$. Taking norms, $N(2^x) = 2^{2s}$ and $N(3^x) = 3^{2s}$ are positive integers, which is the real hypothesis at exponent $2s$, so $2s \in \mathbb{Z}$. Now 3 is inert in $\mathbb{Z}[i]$, so a Gaussian integer of norm 3^{2s} is a unit times 3^k with $9^k = 3^{2s}$, forcing $s = k \in \mathbb{Z}$: the inert prime upgrades the half-integer ambiguity to integrality. Then $2^{2s} = 4^s$ shows that 2^x is divisible only by the ramified prime $1 + i$, so $2^x = u(1 + i)^{2s} = 2^s i^s u$. In both bases the values are 2^s and 3^s times fourth roots of unity, the only Gaussian integers of modulus one, so the argument conditions read $t \log 2,\ t \log 3 \in (\pi/2)\mathbb{Z}$, and irrationality of ϑ kills t as in Theorem 65.3. ☐

Status: [paper proof in this report] — a self-contained equivalence whose only inputs are Gaussian factorization, irrationality of ϑ, and the real conjecture as a hypothesis. It is a low-cost Lean target on top of `ComplexSolutions`.

Two boundary markers make the picture sharp. Replacing $\mathbb{Z}[i]$ by $\mathbb{Q}(i)$ breaks the argument at exactly one point: unit-modulus elements of $\mathbb{Q}(i)$ need not be roots of unity, as $(3+4i)/5$ shows, so the imaginary slice reopens as a genuine unit-circle four-exponentials instance. It is integrality up to units, not rationality, that closes the argument direction. Replacing $\mathbb{Z}[i]$ by the integers of an imaginary quadratic field in which 2 or 3 *splits* — for instance $\mathbb{Q}(\sqrt{-2})$, where 3 splits and $1+\sqrt{-2}$ has modulus $\sqrt{3}$ — destroys the root-of-unity conclusion, and eliminating t produces relations mixing $\vartheta\pi$ with logarithms of complex algebraic numbers: the product-of-logarithms wall again, with more rooms. These split-prime variants are recorded as open rather than pursued; they meet the same wall with worse constants.

65.6 Summary

Reading	Status
Exponent in $\mathbb{C}$, principal branch, values in $\mathbb{Z}$ or $\mathbb{Q}$	equivalent to the original, **[kernel-verified Lean]**
Values in $\overline{\mathbb{Q}}$, exponent real	two-base: four exponentials; conclusion $\mathbb{Q}$, sharp
Values in $\overline{\mathbb{Q}}$, exponent in $\mathbb{C}$	splits into real and unit-circle slices
Three bases, values in $\overline{\mathbb{Q}}$, exponent in $\mathbb{C}$	solution set exactly $\mathbb{Q}$, **[classical/open background]**
Multivalued branches	wall widened by $\log(-1)$, **[paper proof in this report]**
Values in $\mathbb{Z}[i]$	unconditionally equivalent, **[paper proof in this report]**
Values in $\mathbb{Q}(i)$ or split-prime rings	open

In every reading the answer the conjecture predicts is the same.

A Status matrix for major statements

The status tags have the following meanings. **[kernel-verified Lean]** marks a statement that the Lean kernel accepts at the current repository state, with no `sorry`, no extra axioms, and no `native_decide`. **[paper proof in this report]** marks a paper proof given in this report that has not yet been formalized. **[Lean draft, not yet accepted]** marks a statement for which Lean source exists in the repository but has not yet been accepted by the kernel. **[interval computation]** marks a finite interval computation with a stated software trust boundary, which is not a kernel certificate. **[classical/open background]** marks classical or open background. No entry below asserts the full Alaoglu–Erdős conjecture; it remains open.

No row of the table carries **[Lean draft, not yet accepted]** any longer. Every module that earlier revisions of this report listed as a pending draft has since been accepted by the kernel and placed on the aggregate import surface, so the tag survives only as a definition for future use.

Statement	Status	Location
Classical frame		
$2^x, 3^x, 5^x \in \mathbb{Z} \Rightarrow x \in \mathbb{Z}$	**[kernel-verified Lean]**	`IntegerExponent`
Gelfond–Schneider theorem	**[kernel-verified Lean]**	`GelfondSchneider`
Four exponentials conjecture (hypothesis only)	**[classical/open background]**	[13, 14]

Statement	Status	Location
Full Alaoglu–Erdős conjecture	Open	Section 6, Section 27
Transcendence and reduction		
Nonintegral solution irrational and transcendental	[kernel-verified Lean]	`TwoBaseIntegerExponent`, `Transcendence`
$\vartheta = \log_2 3$ transcendental	[kernel-verified Lean]	`Transcendence`
Four-exponentials conjecture ⇒ Conjecture 2.1	[kernel-verified Lean]	`FourExponentialsReduction`
Prime-log-product independence ⇒ Conjecture 2.1	[kernel-verified Lean]	Section 12; `LogProductIndependence`
Prime-log-product independence ⇒ the rational-output form (22)	[kernel-verified Lean]	Section 12; `LogProductIndependence`
Two-prime case of the independence criterion, and its sharpness	[kernel-verified Lean]	Section 12.5
Zero-free regions and finite searches		
$2^x < 100$ and $2^x < 256 \Rightarrow x \in \mathbb{Z}$	[kernel-verified Lean]	`FiniteCheck`, `FiniteCheck256`
$2^x < 4096 \Rightarrow x \in \mathbb{Z}$ (so $x \geq 12$)	[kernel-verified Lean]	`FiniteCheck4096`
No nonintegral candidate with $2^x < 2^{20}$	[interval computation]	Section 25, Appendix F
No nonintegral candidate with $2^x < 2^{26}$	[interval computation]	Section 25, Appendix I
Exact conditional structure		
Rank of candidate exponent vectors ≤ 2	[kernel-verified Lean]	`MultiplicativeRank`
Cross-valuation identity and conditional box growth	[kernel-verified Lean]	`CandidateStructure`
Common odd core $2^i w^j$ (existence)	[kernel-verified Lean]	`OddCore`
Counting $c \log_2 N \cdot c \log_3 N$; linear dyadic blocks	[kernel-verified Lean]	`OddCore`
Primitive (canonical) odd core; unique coordinates	[kernel-verified Lean]	`PrimitiveOddCore`
Unique least solution; affine primitive output pair	[kernel-verified Lean]	`LeastSolution`
Simultaneous primitive output normalization; gcd one; size dichotomy	[kernel-verified Lean]	`OutputNormalization`
Power-of-three denominator normalization	[kernel-verified Lean]	`ThreeDenominatorNormalization`
Rational-power index and exact solution monoid $\mathbb{N}_0 + \mathbb{N}_0\beta$	[kernel-verified Lean]	`RationalPowerIndex`, `PrimitiveGenerator`
Canonical primitive generator; exact candidate monoid $\{2^n M^k\}$	[kernel-verified Lean]	`PrimitiveGenerator`
Denominator-controlled rational third output and determinant margin	[kernel-verified Lean]	`RationalThirdBase`
Unrestricted rational-third-output determinant grid	[kernel-verified Lean]	`RationalSixExponentials`, `RationalBaseThird`
3-smooth classification of further bases	[kernel-verified Lean]	`ThreeSmooth`
Both outputs 3-smooth $\Rightarrow x \in \mathbb{Z}$ (unconditional exclusion)	[kernel-verified Lean]	`SmoothOutputs`
Sharp form: a 3-smooth output forces a partner with a prime factor ≥ 5	[kernel-verified Lean]	`SmoothOutputs`

Statement	Status	Location
Primitive output cores are never exactly 3 and 2; exact core rationality classification	[kernel-verified Lean]	`CoreIndependence`
Algebraic auxiliary bases and output loci		
Six exponentials for three positive rational bases and rational outputs	[kernel-verified Lean]	`RationalSixExponentials`
Field-norm determinant bound over rings of integers	[kernel-verified Lean]	`AlgebraicIntegerDeterminant`
One number field and one natural denominator for a finite algebraic family	[kernel-verified Lean]	`AlgebraicOutputBridge`, `FiniteAlgebraicOutputBridge`
Six exponentials for three positive algebraic real bases and algebraic outputs	[kernel-verified Lean]	`AlgebraicSixExponentials`
Algebraic output at a natural third base	[kernel-verified Lean]	`AlgebraicThirdBase`
Exact rational-base spectrum: $q^x \in \mathbb{Q}$ iff $x \in \mathbb{Z}$ or q is a $2,3$-unit	[kernel-verified Lean]	`RationalBaseThird`
Exact algebraic-output form of the same classification	[kernel-verified Lean]	`AlgebraicRationalBase`
Divisible-hull criterion: an algebraic base with algebraic power has a positive $2,3$-unit power	[kernel-verified Lean]	`AlgebraicBaseUnitPower`
Pointwise algebraic natural-base classification and iterated-output transcendence	[kernel-verified Lean]	`AlgebraicConsequences`
Exact algebraic output monomial locus is a coordinate face or the origin	[kernel-verified Lean]	`AlgebraicOutputLocus`
Algebraic output ratios lie on one external p-adic line (lattice rank ≤ 1)	[kernel-verified Lean]	`AlgebraicOutputLocus`
Common algebraic-output exponent plane: $y \in \mathbb{Q} + \mathbb{Q}x$	[kernel-verified Lean]	`AlgebraicExponentLocus`
General two-base saturated spectrum $\Gamma(a, b) = \{a^r b^s : r, s \in \mathbb{Q}\}$	[paper proof in this report]	algebraic spectrum (this report)
Strong spectrum: $x \log \alpha \notin \mathcal{L}$ for $\alpha \in \overline{\mathbb{Q}}_{>0} \setminus \Gamma$	[paper proof in this report]	algebraic spectrum (this report)
Every $\alpha \in \overline{\mathbb{Q}}_{>0} \setminus \Gamma$ is a one-output detector for Conjecture 2.1	[paper proof in this report]	algebraic spectrum (this report)
One-base equivalent formulations		
Conjecture 2.1 $\iff$ every solution has $5^x \in \overline{\mathbb{Q}}$	[kernel-verified Lean]	`AlgebraicConsequences`
Conjecture 2.1 $\iff$ every solution has $2^{x^2}, 3^{x^2} \in \overline{\mathbb{Q}}$	[kernel-verified Lean]	`AlgebraicConsequences`
Conjecture 2.1 $\iff$ every solution has $6^{x^2} \in \overline{\mathbb{Q}}$	[kernel-verified Lean]	`AlgebraicConsequences`
Conjecture 2.1 $\iff$ the algebraic locus among $(2^i/3^j)^{x^2}$ has rank two	[kernel-verified Lean]	`AlgebraicOutputLocus`
Conjecture 2.1 $\iff$ the algebraic real-power output locus has rank three	[kernel-verified Lean]	`AlgebraicRankThreeEquivalence`
Conjecture 2.1 $\iff$ no squared-monomial algebraic coordinate face	[kernel-verified Lean]	`AlgebraicOutputLocus`

Statement	Status	Location
Conjecture 2.1 $\iff$ $2^y, 3^y \in \overline{\mathbb{Q}}$ at every natural power $y = x^n$	[kernel-verified Lean]	`AlgebraicExponentLocus`
Canonical radical: valuation, degree, and hull		
Odd-prime valuation gcd; outer-power exponents of the canonical radical	[kernel-verified Lean]	`CanonicalRadicalValuation`
Exact rational outer-power criterion; the triple gcd is not replaceable by either pairwise gcd	[kernel-verified Lean]	`CanonicalRadicalSharpness`
Norm and trace of the pure-radical generator; the base-swapped relation blocks monomial injectivity	[kernel-verified Lean]	`CanonicalRadicalFieldNorm`
Algebraic integrality of the radical iff trivial reduced denominator	[kernel-verified Lean]	`CanonicalRadicalThirdOutput`
Algebraicity of the radical's β-th power iff $2, 3$-smooth numerator	[kernel-verified Lean]	`CanonicalRadicalAlgebraicOutput`
External numerator prime excludes the positive $2, 3$-unit divisible hull	[kernel-verified Lean]	`CanonicalRadicalDivisibleHull`
Mixed radical monomials retain an external prime; transcendental outputs	[kernel-verified Lean]	`CanonicalRadicalMixedOutput`
Exact rational-power indices and radical degrees of the simultaneous radicals; product degree de	[kernel-verified Lean]	`SimultaneousRadicalDegrees`
Localized-radical conjecture target itself	Open	Section 6
Valuation lattice, kernel, and towers		
Rational-output lattice Λ_x of the prime-valuation matrix V_x	[paper proof in this report]	valuation lattice (this report)
Smith classification: $\Lambda_x/\mathbb{Z}^2 \cong \mathbb{Z}/d_1\mathbb{Z} \oplus \mathbb{Z}/d_2\mathbb{Z}$ and the finite rational-output defect	[paper proof in this report]	valuation lattice (this report)
Second-iterate kernel $K_x = \{(r, s) \in \mathbb{Q}^2 : M^r A^s \in \Gamma\}$; $\dim_{\mathbb{Q}} K_x \le 1$	[kernel-verified Lean]	`SecondIterateKernel`
Kernel invariance $K_x = K_y$ across nonintegral solutions	[kernel-verified Lean]	`SecondIterateKernel`
Möbius criterion: $K \ne \{0\}$ iff $x \in \mathrm{PGL}_2(\mathbb{Q}) \cdot \vartheta$, iff $1, \vartheta, x, x\vartheta$ are $\mathbb{Q}$-dependent	[kernel-verified Lean]	`KernelDichotomy`
Multiplicative-dependence form: $K \ne \{0\}$ iff $M, A, 2, 3$ are multiplicatively dependent	[kernel-verified Lean]	`KernelDichotomy`
Four-type classification of counterexamples by external valuation rank	[paper proof in this report]	second-iterate kernel (this report)
No nontrivial two-step return inside Γ	[paper proof in this report]	depth-three tower (this report)
Depth-three tower rigidity: some α^{x^j}, $j \le 3$, is transcendental, with exact first depth	[paper proof in this report]	depth-three tower (this report)
Universal three-level detector: Conjecture 2.1 $\iff$ $\alpha^x, \alpha^{x^2}, \alpha^{x^3} \in \overline{\mathbb{Q}}$ for one fixed $\alpha \ne 1$	[paper proof in this report]	depth-three tower (this report)

Statement	Status	Location
Base-2 cubic-tower equivalence: Conjecture 2.1 $\iff$ $2^{x^2}, 2^{x^3} \in \overline{\mathbb{Q}}$	**[kernel-verified Lean]**	`TowerRigidity`
Base-2 depth-two dichotomy: $2^{x^2} \in \overline{\mathbb{Q}}$ iff 2^x is 3-smooth, and then $2^{x^2} \in \mathbb{N}$ while 2^{x^3} is transcendental	**[kernel-verified Lean]**	`TowerRigidity`
Symmetric tower constants: at least one of $\log E_3$, $\log E_2$ lies outside $\mathcal{L}$	**[paper proof in this report]**	symmetric tower constants (this report)
Exact conditional classification of $E_3 = 3^{\vartheta}$ and $E_2 = 2^{1/\vartheta}$	**[kernel-verified Lean]**	`SymmetricTowerConstants`
E_3 and E_2 are never both algebraic under a nonintegral solution	**[kernel-verified Lean]**	`SymmetricTowerConstants`
Counting, growth, and equidistribution		
Exact block counts and cumulative finite sum	**[kernel-verified Lean]**	`ExactCounting`
Cumulative $T^2/(2\beta) + O(T)$ asymptotic; subquadratic-growth equivalence	**[kernel-verified Lean]**	`QuadraticAsymptotic`
Vanishing nonzero block Fourier modes	**[kernel-verified Lean]**	`BlockWeyl`
Exact block averages for finite trigonometric polynomials	**[kernel-verified Lean]**	`BlockEquidistribution`
Full continuous-test/interval blockwise equidistribution	**[paper proof in this report]**	Section 5
Density dichotomy, gap equivalences, infinitude	**[kernel-verified Lean]**	`FractionalPartDensity`
Nonintegral candidate refinement; density zero	**[kernel-verified Lean]**	`NonintegerDensity`
Shrinking-target Diophantine bounds	**[kernel-verified Lean]**	`ShrinkingTarget`
Convergent-denominator growth $q_{n+1} < 2m^{q_n}$	**[kernel-verified Lean]**	`DenominatorGrowth`
Difference and progression obstructions		
Affine descent, $r + 8k \le x$	**[kernel-verified Lean]**	`ArithmeticRigidity`
Three-term AP obstruction under $h^5 \le n$	**[kernel-verified Lean]**	`ZeroDensity`
Sharp three-term and four-term AP obstructions ($r = 2, 3$)	**[kernel-verified Lean]**	`SharperProgressions`, `ThirdDifference`
Uniform arbitrary-order AP obstruction ($r \ge 2$)	**[kernel-verified Lean]**	`HigherDifference`
Effective rational three-term criterion $h^{400} \le n^{83}$	**[kernel-verified Lean]**	`SharpProgression`
Radical and localized reformulations		
Localized-radical and symmetric reformulations	**[kernel-verified Lean]**	`RadicalReformulation`, Appendix C
Primitive localized-radical reduction	**[kernel-verified Lean]**	`PrimitiveRadical`
Exact radical degree and binomial irreducibility	**[kernel-verified Lean]**	`RadicalDegree`
Candidate-form localization equivalence	**[kernel-verified Lean]**	`Localization`
New results of this report		
Exponential-polynomial zero count; generalized Vandermonde nonvanishing at ordered nodes	**[kernel-verified Lean]**	`ExponentialPolynomialZeros`

Statement	Status	Location
Generalized Vandermonde positivity (ordered chamber)	**[paper proof in this report]**	Section 19, Appendix D
Arbitrary-node determinant integrality and divided-difference repulsion bound	**[kernel-verified Lean]**	`ArbitraryNodeDeterminant`
Arbitrary-node repulsion theorem and its packing corollary	**[kernel-verified Lean]**	`ArbitraryNodeRepulsion`, conditional on its named divided-difference mean-value hypothesis
Newton divided-difference factorization of the interpolation determinant	**[kernel-verified Lean]**	`ArbitraryNodeRepulsion`, unconditional
Semigroup generalized-Vandermonde hierarchy (integrality half)	**[kernel-verified Lean]**	`MixedExponentSemigroup`, `SemigroupDeterminant`
Falling-factorial coefficient bound; dyadic-strip divided-difference bound	**[kernel-verified Lean]**	`FallingFactorialBound`
Semigroup gap theorem in quantitative form	**[kernel-verified Lean]**	`SemigroupGapBound`
Leibniz determinant bound from separated row/column bounds	**[kernel-verified Lean]**	`SemigroupGapBound`, unconditional
Window-span counting bridge for mixed exponents	**[kernel-verified Lean]**	`SemigroupWeightBound`
Explicit dyadic $O((\log X)^2)$ bound from that hierarchy	**[kernel-verified Lean]**	`SemigroupGapBound`, conditional on the named hypothesis `NewtonFactorization`
One-logarithm deficit analysis	**[paper proof in this report]**	Section 22
Min-plus ceiling: term-wise p-adic content never exceeds a correct archimedean bound	**[kernel-verified Lean]**	`MinPlusCeiling`
Exact termwise-divisor ceilings (4/5, 7/10, 3/10, $\pi/4$, 0.8710)	**[paper proof in this report]**	Section 21
A unique tropical minimizer forbids cancellation	**[kernel-verified Lean]**	`MinPlusCeiling`
Compact synchronized-orbit reformulation	**[kernel-verified Lean]**	`CompactOrbit`
Block-growth reformulations (bounded, $o(T)$, $\liminf$, $\equiv 1$)	**[kernel-verified Lean]**	`BlockGrowthReformulations`
Rational-function rigidity, multiplication and quotient criteria, closure reformulations	**[paper proof in this report]**	Section 8
Perfect-power content, matched affine decompositions, primitive-output statistics	**[paper proof in this report]**	Section 10
Candidate gcd/lcm lattice; prime-support rigidity of the whole branch	**[kernel-verified Lean]**	`CandidateLattice`
Ambient root saturation index d	**[paper proof in this report]**	Section 11
Operator formulation on the prime-logarithm space	**[kernel-verified Lean]**	`PrimeLogSpaceOperator`
Every complex solution is real (principal branch)	**[kernel-verified Lean]**	`ComplexSolutions`

Statement	Status	Location
Gaussian-integer version is equivalent to the conjecture	[paper proof in this report]	Section 65
Multivalued branches: the log-product wall widened by $\log(-1)$	[paper proof in this report]	Section 65
Exact cost of the rational antecedent (denominator-support lemma)	[paper proof in this report]	Section 65

B Formalization inventory

All modules live under `Analysis/ExponentialIdentities/Lean/ExponentialIdentities/`, in namespace `LeanProofs`. Every module listed below is reachable from the aggregate import surface, and the merged tree builds with no errors. None of them contains `sorry`, an extra axiom, or `native_decide`; certificate tables are replayed by the kernel.

B.1 Kernel-accepted modules

Module	Contents
`IntegerExponent` (+3 modules)	three-base theorem, interpolation determinants
`TwoBaseIntegerExponent`	conjecture statement, candidates, gaps, $2^x < 10$
`.../FiniteCheck`, `FiniteCheck256`	zero-free regions $2^x < 100$, $2^x < 256$
`.../FiniteCheck4096`	zero-free region $2^x < 4096$ (kernel `decide` table)
`.../ZeroDensity`	AP obstruction, Roth-number counting, $o(N)$
`.../NonintegerDensity`	nonintegral candidate refinement, density zero
`.../MultiplicativeRank`	rank ≤ 2, ϑ irrational, $(c\log_2 N)^2$
`.../CandidateStructure`	cross-valuation identity, conditional box growth
`.../OddCore`	common odd core, $c\log_2 N\ c\log_3 N$, dyadic blocks
`.../PrimitiveOddCore`	gcd-normalized canonical core, unique coordinates
`.../LeastSolution`	unique positive least solution, exact affine primitivity
`.../ThreeDenominatorNormalization`	reduced denominator supported at 3
`.../RationalPowerIndex`	least rational-power index, exact solution monoid
`.../PrimitiveGenerator`	canonical primitive generator and unique coordinates
`.../ExactCounting`	exact unit/dyadic block counts and cumulative finite sum
`.../QuadraticAsymptotic`	sharp quadratic limit; subquadratic-growth equivalence
`.../BlockWeyl`	exact-block geometric sums and vanishing nonzero Fourier modes
`.../BlockEquidistribution`	finite trigonometric-polynomial block averages
`.../OutputNormalization`	simultaneous primitive outputs, gcd one, size dichotomy
`.../RationalThirdBase`	denominator divisibility and determinant margin
`.../RadicalReformulation`	localized-radical equivalences, base-swapped form
`.../PrimitiveRadical`	canonical primitive-power reduction of radical target
`.../RadicalDegree`	irreducible binomial, exact minimal polynomial and degree
`.../SharperProgressions`	sharp three-candidate curvature obstruction
`.../ThirdDifference`	exact third difference, four-candidate obstruction
`.../HigherDifference`	uniform arbitrary-order forward-difference obstruction
`.../SharpProgression`	effective rational three-term criterion $h^{400} \leq n^{83}$
`.../ArithmeticRigidity`	descent, $r + 8k \leq x$
`.../ThreeSmooth`	3-smooth classification of further bases
`.../ShrinkingTarget`	fractional-part shrinking targets, Diophantine lower bounds
`.../DenominatorGrowth`	convergent-denominator growth cap $q_{n+1} < 2m^{q_n}$
`.../Transcendence`	Gelfond–Schneider applications: solutions and $\log_2 3$

Module	Contents
`.../FractionalPartDensity`	density dichotomy, gap equivalences, infinitude
`.../Localization`	candidate-form localization equivalence
`.../FourExponentialsReduction`	four exponentials $\Rightarrow$ the conjecture
`GelfondSchneider/...`	Gelfond–Schneider theorem (port of Karatarakis)
Rational and algebraic auxiliary bases	
`.../RationalSixExponentials`	six exponentials for three positive rational bases and rational outputs
`.../RationalBaseThird`	specialization to bases $2, 3$; exact $2, 3$-unit classification of positive rational third bases
`.../AlgebraicOutputBridge`	a number field, an embedding, and one natural denominator clearing an algebraic value to an algebraic integer
`.../FiniteAlgebraicOutputBridge`	one common number field and one denominator for a finite family of algebraic values
`.../AlgebraicIntegerDeterminant`	field-norm lower bound for a nonzero algebraic-integer determinant; house and factorial bounds on the n^{11} scale
`.../AlgebraicSixExponentials`	field-norm six-exponentials determinant for three positive algebraic real bases and three algebraic outputs
`.../AlgebraicThirdBase`	three-base determinant for algebraic output at a natural third base, first over a number field and then by clearing denominators
`.../AlgebraicRationalBase`	algebraic outputs at positive rational bases; the algebraic locus is exactly the $2, 3$-units unless the exponent is integral
`.../AlgebraicBaseUnitPower`	positive $2, 3$-unit powers and the multiplicative rank of an algebraic auxiliary base
`.../AlgebraicConsequences`	pointwise algebraic natural-base classification; iterated-output transcendence; the 5^x, 6^{x^2} and both-iterate equivalences
`.../AlgebraicOutputLocus`	exact algebraic output monomial locus (coordinate face or origin); external p-adic line for output ratios; coordinate-face and rank-two equivalences
`.../AlgebraicRankThreeEquivalence`	rank three in the algebraic real-power output locus forces integrality; the resulting equivalence
`.../AlgebraicExponentLocus`	the common algebraic-output exponent plane $\mathbb{Q} + \mathbb{Q}x$, and its natural-power equivalence
Canonical radical arithmetic	
`.../CanonicalRadicalValuation`	odd-prime valuation gcd; divisors of the gcd are honest outer-power exponents
`.../CanonicalRadicalSharpness`	exact criterion for a reduced $c/3^a$ to be a rational outer power; the triple gcd resists both pairwise relaxations
`.../CanonicalRadicalFieldNorm`	norm and trace of the pure-radical generator; the base-swapped relation blocks monomial injectivity
`.../CanonicalRadicalThirdOutput`	algebraic integrality of the normalized radical is equivalent to a trivial reduced denominator
`.../CanonicalRadicalAlgebraicOutput`	algebraicity of the radical's β-th power is equivalent to $2, 3$-smoothness of the normalized numerator
`.../CanonicalRadicalDivisibleHull`	an external numerator prime excludes the whole positive $2, 3$-unit divisible hull
`.../CanonicalRadicalMixedOutput`	genuinely mixed radical monomials keep an external prime, so all their outputs are transcendental
`.../SimultaneousRadicalDegrees`	the displayed exponents are the least rational-power indices; exact radical degrees and the product degree de
`.../CoreIndependence`	the two primitive output cores are never exactly 3 and 2; exact rationality classification of the simultaneous cores
Smoothness and log-products	
`.../SmoothOutputs`	both outputs 3-smooth forces an integral exponent; sharp form supplying a partner with a prime factor ≥ 5

Module	Contents
`.../LogProductIndependence`	prime-log-product independence as a sufficient condition distinct from four exponentials, in both the integer-output and rational-output forms
Determinant hierarchy and reformulations	
`.../ArbitraryNodeDeterminant`	interpolation determinant at arbitrary ordered nodes; integrality and the divided-difference repulsion bound
`.../ExponentialPolynomialZeros`	zero-count theorem for real exponential polynomials and nonvanishing of the generalized Vandermonde determinant
`.../MixedExponentSemigroup`	pair-indexed exponent $a + b\vartheta$, injectivity, and integrality of $m^{a+b\vartheta}$
`.../SemigroupDeterminant`	integrality half of the semigroup generalized-Vandermonde hierarchy, with $\lvert\det\rvert \geq 1$
`.../FallingFactorialBound`	falling-factorial coefficient bound and dyadic-strip divided-difference bound
`.../SemigroupWeightBound`	mixed-exponent family construction and the window-summation counting bridge
`.../BlockGrowthReformulations`	boundedness, $o(T)$, $\liminf$, and $\equiv 1$ reformulations of the conjecture
`.../CompactOrbit`	compact synchronized-orbit reformulation with shared floor exponent, in both directions
`.../ArbitraryNodeRepulsion`	divided differences, the Newton factorization of the interpolation determinant, and the repulsion and packing bounds conditional on the named mean-value hypothesis
`.../SemigroupGapBound`	Leibniz determinant bound, semigroup span inequality, and the explicit dyadic $10^4(\log X)^2$ block bound conditional on the named Newton factorization
Second iterates, towers, and the candidate lattice	
`.../CandidateLattice`	coordinatewise gcd and lcm for $2^s w^t$; unconditional closure of the candidate set under gcd and lcm; prime-support rigidity of the branch
`.../KernelDichotomy`	multiplicative dependence of $M, A, 2, 3$ is equivalent to a rational linear relation among $1, \vartheta, x, x\vartheta$ and to the Möbius orbit
`.../SecondIterateKernel`	the second-iterate kernel as a subgroup of $\mathbb{Z}^2$; no same-sign vectors, rank at most one, smooth coordinate cases, and solution-independence
`.../TowerRigidity`	base-two second iterate as a natural number; the depth-two algebraic dichotomy; the depth-three transcendence and the cubic-tower equivalence
`.../SymmetricTowerConstants`	$E_3 = 3^\vartheta$ and $E_2 = 2^{1/\vartheta}$, their conditional rational powers, and their mutual exclusion
`.../PrimeLogSpaceOperator`	the ϑ-pullback of the prime-logarithm span is exactly $\mathbb{Q}\log 2$ if and only if the rational-output conjecture holds
`.../ComplexSolutions`	on the principal branch every complex solution is real, by incommensurability of the two argument gratings
`.../MinPlusCeiling`	the $\{2,3\}$-part of an integer divides it, hence term-wise p-adic content can never exceed a correct archimedean bound
`.../AuditNewModules`	`#print axioms` audit of the headline theorems of the recent formalization rounds

B.2 Drafted modules not yet accepted by the kernel

There are none. Earlier revisions of this report listed nine modules whose Lean sources existed in the repository but had not been accepted:

`ArbitraryNodeDeterminant`, `ExponentialPolynomialZeros`, `MixedExponentSemigroup`, `SemigroupDeterminant`, `FallingFactorialBound`, `SemigroupWeightBound`, `BlockGrowthReformulations`, `CompactOrbit`, `LogProductIndependence`.

All nine have since been accepted by the kernel, placed on the aggregate import surface, and moved into the table above. Statements that those modules actually contain are therefore **[kernel-verified Lean]**.

A second round has since added eight further modules: `CandidateLattice`, `KernelDichotomy`, `SecondIterateKernel`, `TowerRigidity`, `SymmetricTowerConstants`, `PrimeLogSpaceOperator`, `ArbitraryNodeRepulsion` and `SemigroupGapBound`, all on the aggregate import surface and all audited with `#print axioms`.

Two cautions survive both promotions, and the status matrix reflects them. First, a module being accepted certifies exactly the theorems it states. The analytic halves of the repulsion and gap theorems are now formal, but each carries its missing analytic input as an explicit, named *hypothesis* — `RpowDividedDifferenceMeanValue` and `NewtonFactorization` — rather than as an axiom, so those two conclusions are conditional in a way the reader can see in the theorem statement; everything upstream and downstream of them is unconditional. `ExponentialPolynomialZeros` still delivers nonvanishing rather than the ordered-chamber positivity proved on paper in Appendix D. Second, the finite scans through 2^{20} and 2^{26} are **[interval computation]**, not kernel certificates; the kernel-verified exponent bound is still $x \geq 12$ from `FiniteCheck4096`.

Each further headline theorem should be added to the aggregate import surface and checked with `#print axioms` before its status is promoted. Numerical constants in the tables of Appendix E are explanatory only; the formal statements should use exact expressions or safe rational constants such as 1/24 and 10000.

C A symmetric formulation

Everything above has a base-swapped version. Put $\eta = \log_3 2 = 1/\vartheta$. Failure of the conjecture is also equivalent to the existence of an integer $v > 1$ with $3 \nmid v$ and

$$v^\eta \in \mathbb{Z}[1/2].$$

Status: **[kernel-verified Lean]** — `not_alaogluErdosConjecture_iff_exists_threeFree_localized_radical` and `alaogluErdosConjecture_iff_threeFreeLocalizedRadicalExclusion` in `RadicalReformulation`.

For the primitive pair (M, A), factoring A into a power of 3 times a 3-coprime part produces a symmetric primitive core and rational-power index; the two normalizations give the same least exponent β, and their compatibility condition is exactly $\log M \log 3 = \log A \log 2$. The symmetry is useful in formalization — some divisibility arguments are easier on one side — and confirms that the localized-radical reduction of Section 6 is not an artifact of privileging the base 2.

D Detailed proof of the generalized Vandermonde sign

This appendix expands the generalized Vandermonde positivity lemma of Section 19 to make its analytic input fully transparent.

Let $0 \le \lambda_0 < \cdots < \lambda_r$ and define $f_j(x) = x^{\lambda_j}$ on $(0, \infty)$. For $k \le r$,

$$f_j^{(k)}(x) = (\lambda_j)_{\underline{k}}\, x^{\lambda_j - k}.$$

The kth initial Wronskian is

$$\begin{aligned} W_k(x) &= \det\bigl(f_j^{(i)}(x)\bigr)_{0\le i,j\le k} \\ &= x^{\sum_{j=0}^{k} \lambda_j - k(k+1)/2} \det\bigl((\lambda_j)_{\underline{i}}\bigr)_{0\le i,j\le k}. \end{aligned}$$

The polynomial $(z)_{\underline{i}}$ is monic of degree i. Replacing the monomial basis $1, z, \dots, z^k$ by the falling-factorial basis is unit lower-triangular. Therefore

$$\det\bigl((\lambda_j)_{\underline{i}}\bigr)_{i,j=0}^{k} = \prod_{0\le i<j\le k} (\lambda_j - \lambda_i) > 0.$$

Hence every initial Wronskian is positive.

One standard characterization of extended complete Chebyshev systems [6] now implies that for $x_0 < \cdots < x_r$,

$$\det(f_j(x_i))_{i,j=0}^{r} > 0.$$

To avoid using that characterization as a black box, proceed inductively. Suppose a nontrivial combination

$$F(x) = \sum_{j=0}^{r} c_j x^{\lambda_j}$$

has $r+1$ distinct positive zeros. Put $x = e^t$ and factor the lowest exponential:

$$e^{-\lambda_0 t} F(e^t) = c_0 + \sum_{j=1}^{r} c_j e^{(\lambda_j - \lambda_0)t}.$$

Rolle's theorem gives at least r zeros of its derivative, an exponential polynomial with at most r terms. Induction gives a contradiction unless all coefficients vanish. Thus the evaluation determinant is nonzero. Its sign cannot change on the connected chamber $x_0 < \cdots < x_r$. Let the nodes coalesce as $x_i = x + i\varepsilon$ and divide by the positive ordinary Vandermonde. As $\varepsilon \downarrow 0$, the quotient tends to $W_r(x)/\prod_{k=0}^{r} k! > 0$. Hence the original determinant is positive everywhere in the ordered chamber.

Status: [paper proof in this report] — paper proof in this report. The zero-count engine is drafted in `ExponentialPolynomialZeros` but has not been accepted by the kernel ([Lean draft, not yet accepted]).

E Numerical constants

For reference,

$$\vartheta = 1.58496250072115618145\ldots, \qquad \vartheta(\vartheta-1) = 0.92714362797110482756\ldots.$$

The three-point span constant is

$$\left(\frac{8}{\vartheta(\vartheta-1)}\right)^{1/3} = 2.051072395661874\ldots.$$

For the ordinary-power order r, the raw span constant and exponent are

$$C_r = \left(\frac{r!}{|\,(\vartheta)_{\underline{r}}\,|}\right)^{2/(r(r+1))}, \qquad \alpha_r = \frac{2(r-\vartheta)}{r(r+1)}.$$

The continuous critical point for α_r solves

$$-r^2 + 2\vartheta r + \vartheta = 0,$$

so

$$r_* = \vartheta + \sqrt{\vartheta^2+\vartheta} = 3.6090841940\ldots,$$

confirming that $r = 4$ is the optimal integer order.

For the semigroup hierarchy of Section 20, the first exponents of $\Lambda_\vartheta = \{a + b\vartheta : a, b \in \mathbb{N}_0\}$ and their pair labels are:

Index	Pair (a,b)	Exponent $a+b\vartheta$
0	$(0,0)$	0
1	$(1,0)$	1
2	$(0,1)$	ϑ
3	$(2,0)$	2
4	$(1,1)$	$1+\vartheta$
5	$(3,0)$	3
6	$(0,2)$	2ϑ
7	$(2,1)$	$2+\vartheta$
8	$(4,0)$	4
9	$(1,2)$	$1+2\vartheta$
10	$(3,1)$	$3+\vartheta$
11	$(0,3)$	3ϑ
12	$(5,0)$	5

These decimal values are explanatory. They are not part of any machine-checked statement, and formal versions of the corresponding theorems should carry exact expressions instead.

F Interval computation: script and outputs

The script below verifies one range. Set `START` to 1 or 2^{19} and `LIMIT` to 2^{19} or 2^{20} to obtain the two halves reported afterwards.

```python
import math, time, hashlib
import mpmath as mp

START = 1
LIMIT = 1 << 19
DPS = 40

mp.iv.dps = DPS
ln2 = mp.iv.log(2)
ln3 = mp.iv.log(3)
theta_float = math.log(3.0) / math.log(2.0)

checked = 0
failures = []
min_lower = (float("inf"), None, None)
min_upper = (float("inf"), None, None)
manifest = hashlib.sha256()
start_time = time.time()

for m in range(START, LIMIT):
    if (m & (m - 1)) == 0:      # integral exponent: m is a power of two
        continue

    # Only proposes a neighboring integer.  The interval tests below certify it.
    n = math.floor(math.exp(theta_float * math.log(m)))

    lhs = mp.iv.log(m) * ln3
    lower_diff = lhs - mp.iv.log(n) * ln2
    upper_diff = mp.iv.log(n + 1) * ln2 - lhs

    lower_lb = float(lower_diff.a)
    upper_lb = float(upper_diff.a)
    if not (lower_lb > 0.0 and upper_lb > 0.0):
        failures.append((m, n, str(lower_diff), str(upper_diff)))
        break

    min_lower = min(min_lower, (lower_lb, m, n))
    min_upper = min(min_upper, (upper_lb, m, n))
    manifest.update(f"{m}:{n};".encode())
    checked += 1

print("checked_nonpowers=", checked)
print("failures=", len(failures))
print("min_lower_log_margin=", min_lower)
print("min_upper_log_margin=", min_upper)
print("sha256_m_n_manifest=", manifest.hexdigest())
print("elapsed_seconds=", time.time() - start_time)
assert not failures
```

Listing 12: Interval verification

The two 40-digit runs produced:

```
LIMIT=524288
DPS=40
checked_nonpowers=524268
failures=0
min_lower_log_margin=(1.4147089634746718e-15, 189427, 231498127)
```

```
min_upper_log_margin=(3.0841010256409534e-15, 306982, 497586055)
sha256_m_n_manifest=02b589461fa8c1721bc251575b98b1e3d58001f54cb7faa2cd6c59edb8c819a5
elapsed_seconds=18.505
```

Listing 13: Lower half output.

```
LIMIT=1048576
DPS=40
checked_nonpowers=524287
failures=0
min_lower_log_margin=(9.490232037370244e-16, 958460, 3023921230)
min_upper_log_margin=(2.3019781003173366e-15, 556061, 1275861767)
sha256_m_n_manifest=fccccf5de8673379d37ff0fc840ab091d9ff7a2352ff64a93091f01c21bf4cb4
elapsed_seconds=18.904
```

Listing 14: Upper half output.

The two SHA-256 manifests are

lower: `02b589461fa8c1721bc251575b98b1e3d58001f54cb7faa2cd6c59edb8c819a5`,

upper: `fccccf5de8673379d37ff0fc840ab091d9ff7a2352ff64a93091f01c21bf4cb4`.

Elapsed times are machine-dependent sanity checks; the counts, zero-failure results, extremal tuples, and hashes are the reproducibility data. The hashes cover the ASCII stream `m:n;` of proposed-and-certified pairs and make accidental changes to the candidate sequence detectable across reruns.

Status: [interval computation] — interval arithmetic in `mpmath` [15] at 40 digits. This is a software trust boundary, not a kernel certificate. The kernel-verified zero-free region remains $2^x < 4096$ in `FiniteCheck4096`.

G Suggested experimental determinant families

Before developing a full p-adic theory, several finite experiments can test whether the required amplification exists.

G.1 Rectangular row and column boxes

Choose rows (n, k) with $0 \le n < N_0$, $0 \le k < K_0$ and columns (a, b) with $0 \le a < A_0$, $0 \le b < B_0$. The entry is

$$(2^a 3^b)^n (M^a A^b)^k.$$

For complete rectangles, the matrix is a rearranged Kronecker product only when the row and column cardinalities match in a compatible way. Its determinant or maximal minors can be factored symbolically for small boxes. Record valuations at 2 and 3 as polynomials in $a_0 = v_2(M)$ and $b_0 = v_3(A)$.

G.2 Triangular boxes adapted to height

Use

$$n + k\beta \le T, \qquad a + b\vartheta \le L.$$

These are the natural row and column shapes from the exact counting theorem and the mixed-exponent semigroup. Search for square minors with minimal archimedean weight and maximal forced valuation. The linear boundary slopes are irrational, so floor-sum fluctuations may prevent cancellation patterns present in rectangles.

G.3 Finite-difference row operations

Order rows by n for fixed k, or by k for fixed n, and apply repeated difference operators. Differences introduce factors such as

$$2^a 3^b - 1, \qquad M^a A^b - 1.$$

These may have systematic 2- or 3-adic valuations for carefully selected column pairs. Lifting-the-exponent lemmas can then provide exact lower bounds. The risk is that the archimedean size of the same factors grows too quickly; experiments should track both sides.

G.4 Confluent columns

Instead of only distinct mixed monomials, consider derivatives with respect to an auxiliary continuous exponent parameter before specializing. Confluent generalized Vandermonde matrices introduce logarithmic factors, which are not integral, but common-denominator or linear-form constructions may restore arithmetic control. This is closer to Laurent's interpolation determinant method [10] and may reduce total exponent weight.

H Dependency map and Lean-oriented proof sketches

The conditional arithmetic criteria of this report were written with formalization in mind: each one is a short argument over an already kernel-verified base. This appendix records the shortest route from the existing corpus to each of them, and then gives skeleton statements for the four criteria whose formalization is least obvious.

Status: Design note. The snippets below are schematic ASCII transliterations and are not claimed to compile unchanged. The mathematics they target is **[paper proof in this report]**; none of it has been accepted by the kernel, and nothing here may be read as a claim of verification.

H.1 Dependency map

Each row names one new result, the principal existing input it is built on, and the extra mathematics a formalization would have to supply. All of the new results are conditional: they describe the structure forced on $\mathcal{S}$ and $\mathcal{C}$ by a hypothetical counterexample to Conjecture 2.1, and they are vacuous if the conjecture holds.

New result	Principal existing input	Additional mathematics
rational-function rigidity	canonical primitive generator; transcendence of β	injectivity of polynomial evaluation at a transcendental element
product and power criteria	unique (n, k) coordinates	quadratic coefficient comparison
quotient criterion	unique (n, k) coordinates	denominator clearing and coefficient comparison
three-smooth output test	transcendence of ϑ	unique factorization in $\mathbb{N}$
product transcendence	three-smooth-base theorem	classical six exponentials
common power degree	unique (n, k) coordinates	positive root uniqueness; divisibility
primitive statistics	total lattice count	Möbius inversion
gcd and lcm lattice	exact candidate monoid	prime-valuation min and max
ambient root saturation	primitive valuation gcd of w	Bézout and linear cone arithmetic
optimized differences	arbitrary-order difference theorem	gamma identity and Stirling estimate

Every entry in the middle column is **[kernel-verified Lean]** in the corpus catalogued in Section 4. In the right-hand column, only the six exponentials theorem is external background (**[classical/open background]**); the remaining items are ordinary mathlib-level algebra and analysis, which is what makes these criteria cheap to formalize once the base is in place.

Transliteration conventions. The listings write Unicode in plain ASCII: `R`, `Q`, `Z` and `N` are the real, rational, integer and natural types; arrows are `->` and `<->`; membership is `in` and divisibility is `dvd`; anonymous constructors use angle brackets; the two logical connectives use their two-character ASCII spellings; and the focusing dot is written as a period. Coercions are written as explicit casts such as `Int.cast` rather than as the coercion arrow.

H.2 Multiplication skeleton

A direct statement can use the repository predicate `TwoBaseIntegralSolution`, which is the defining conjunction $2^x \in \mathbb{Z}$ and $3^x \in \mathbb{Z}$.

```
theorem product_solution_iff_integer_factor
    {x y : R}
    (hx : TwoBaseIntegralSolution x)
    (hy : TwoBaseIntegralSolution y) :
    TwoBaseIntegralSolution (x * y) <->
      x in Set.range (Int.cast : Z -> R) \/
      y in Set.range (Int.cast : Z -> R) := by
  classical
  constructor
  . intro hxy
    by_cases hAE : AlaogluErdosConjecture
    . left
      exact hAE x hx
    . obtain <w, d, a, c, beta, ..., hcoord, ...> :=
        exists_canonical_primitiveGenerator_of_not_alaogluErdosConjecture hAE
```

```
      obtain <nx, kx, hxcoord> := (hcoord x).mp hx
      obtain <ny, ky, hycoord> := (hcoord y).mp hy
      obtain <np, kp, hpcoord> := (hcoord (x * y)).mp hxy
      -- If kx and ky are both positive, expanding the product gives a
      -- nonzero quadratic polynomial over Q vanishing at beta, which
      -- contradicts transcendence of beta.  Hence kx = 0 or ky = 0.
      ...
  . rintro (hxint | hyint)
    . exact solution_nsmul_left hxint hy
    . exact solution_nsmul_right hyint hx
```

Listing 15: Product criterion: skeleton.

In the finished theorem the integer range can be narrowed to $\mathbb{N}_0$, because every solution is nonnegative by `nonneg_of_two_rpow_integer`. The algebraic step should be isolated, since three of the four criteria reduce to it. Note that the coefficients must be rational — with arbitrary real a, b, c the statement is false — and that is exactly how the applications supply them, as differences of the integer coordinates:

```
lemma transcendental_quadratic_coeff_eq_zero
    {beta : R} {a b c : Q}
    (hbeta : Transcendental Q beta)
    (h : (a : R) * beta ^ 2 + (b : R) * beta + (c : R) = 0) :
    a = 0 /\ b = 0 /\ c = 0 := ...
```

Listing 16: The shared coefficient lemma.

Mathlib's polynomial evaluation API is probably the cleanest route: build the polynomial $aX^2 + bX + c$, rewrite `h` as an evaluation, and apply the definition of transcendence to conclude that the polynomial is zero.

H.3 Output smoothness skeleton

Define the predicate

$$\mathrm{ThreeSmooth}(m) :\Longleftrightarrow \exists u, v \in \mathbb{N}_0,\ m = 2^u 3^v.$$

A practical implementation should avoid a partial "natural output" function and instead carry explicit witnesses $M, A \in \mathbb{N}$ for the two integer powers:

```
theorem integer_of_outputs_threeSmooth
    {x : R} {M A : N}
    (hM : (2 : R) ^ x = (M : R))
    (hA : (3 : R) ^ x = (A : R))
    (hMs : ThreeSmooth M) (hAs : ThreeSmooth A) :
    x in Set.range (Int.cast : Z -> R) := by
  obtain <a, b, rfl> := hMs
  obtain <c, d, rfl> := hAs
  -- Taking logarithms base two: x = a + b * theta and x * theta = c + d * theta.
  have hpoly :
      (b : R) * logThreeDivLogTwo ^ 2
        + ((a : R) - (d : R)) * logThreeDivLogTwo - (c : R) = 0 := by
    ...
  have htheta := transcendental_logThreeDivLogTwo
  -- The coefficients are integers, so all three vanish:
  -- b = 0, a = d, c = 0, whence 2 ^ x = 2 ^ a and x = a.
  ...
```

Listing 17: Output smoothness: skeleton.

The converse direction is immediate: for integral $x = a$ the outputs are 2^a and 3^a. The interesting content is that three-smoothness of *both* outputs is already enough to force integrality, with no appeal to the conditional structure theory.

H.4 Common-power skeleton

The formal theorem is most reusable when it is stated for explicit output witnesses and for a fixed canonical generator, so that the coordinate comparison is available as a hypothesis rather than reconstructed inside the proof.

```
theorem simultaneous_perfectPower_iff_divides_coordinates
    (hfail : not AlaogluErdosConjecture)
    {x beta : R} {n k r : N}
    (hr : 0 < r)
    (hbeta : CanonicalPrimitiveGenerator beta)
    (hxcoord : x = (n : R) + (k : R) * beta) :
    (Exists fun U : N => (U : R) ^ r = (2 : R) ^ x) /\
    (Exists fun V : N => (V : R) ^ r = (3 : R) ^ x) <->
      r dvd n /\ r dvd k := by
  -- Convert the two perfect powers into a solution at x / r,
  -- apply uniqueness of coordinates, and compare.
  ...
```

Listing 18: Common powers: skeleton.

The positive-root step can be discharged either by strict monotonicity of $t \mapsto t^r$ on the nonnegative reals or by injectivity of the logarithm; the first is usually shorter in mathlib.

H.5 Ambient saturation skeleton

The only genuinely arithmetic lemma in the saturation argument is the reconstruction of a root of a power of a primitive number:

```
lemma primitive_core_root
    {w u q r : N}
    (hw : oddPrimeValuationGCD w = 1)
    (hr : 0 < r)
    (hpow : u ^ r = w ^ q) :
    Exists fun t : N => q = r * t /\ u = w ^ t := by
  -- Compare every coordinate of the prime factorizations.
  -- A Bezout combination of the positive valuations of w gives r | q;
  -- factorization extensionality then gives u = w ^ t.
  ...
```

Listing 19: Primitive-core root lemma.

The proof in this report only needs the case $q = dk$, but the general lemma is worth stating once: the same reconstruction is used implicitly in `PrimitiveOddCore` and in `OutputNormalization`, and a shared version would remove that duplication.

H.6 Suggested theorem order for a pull request

A low-conflict sequence, each step reviewable on its own, is:

1. add a small `ThreeSmooth` API together with the output-smoothness theorem;
2. add coordinate comparison lemmas for affine, quadratic and quotient expressions;
3. prove the product, power and quotient criteria;
4. prove the common-power and affine-descent criteria;
5. add the candidate divisibility and support results;
6. add primitive-core root reconstruction and ambient saturation;
7. only then add the asymptotic and six-exponentials extensions.

This ordering keeps the first several commits elementary, and it defers everything that depends on external transcendence input to the end, where it can be reviewed against the status discipline of Appendix A rather than mixed into routine algebra.

I Directed-rounding verifier

Appendix F gives the interval-arithmetic script behind the 2^{20} verification. This appendix gives the second, independent verifier: a C program that uses MPFR directed rounding rather than an interval library, and that extends the scanned range sixty-fourfold, to 2^{26}.

Status: [interval computation] — MPFR 4.2.2 at 192 bits with directed rounding modes. Like the 2^{20} scan this is a software trust boundary, not a kernel certificate. The kernel-verified zero-free region remains $2^x < 4096$ in `FiniteCheck4096`.

I.1 What the verifier certifies

For a non-power of two m, proving $m^\vartheta \notin \mathbb{N}$ amounts to exhibiting $n \in \mathbb{N}$ with $n < m^\vartheta < n+1$, and, as in Section 25, that pair of strict inequalities is certified in logarithmic form so that no exponentiation is ever performed. At precision $p = 192$ the program computes directed enclosures

$$L_m^- \le \log m \le L_m^+, \qquad L_2^- \le \log 2 \le L_2^+, \qquad L_3^- \le \log 3 \le L_3^+,$$

together with $L_n^+ \ge \log n$ and $L_{n+1}^- \le \log(n+1)$, and then rounds *downward* in

$$\delta_- = L_m^- L_3^- - L_n^+ L_2^+, \qquad \delta_+ = L_{n+1}^- L_2^- - L_m^+ L_3^+.$$

Positivity of δ_- proves the lower inequality and positivity of δ_+ proves the upper one. Every input on the n side is below 2^{42} and is therefore exactly representable at 192 bits.

An ordinary `long double` evaluation proposes $n \approx \lfloor m^\vartheta \rfloor$ and the offsets $0, -1, 1, -2, 2, -3, 3$ are tried in that order. The proposal has no logical role: a proposal is accepted only when both directed lower bounds are strictly positive, and a failure to locate any admissible n is reported as a failure rather than silently reclassified. Locator error can therefore produce a false alarm but not a false proof. Powers of two are skipped rather than tested, since $(2^q)^\vartheta = 3^q$ is the known integral candidate; an exact integral value at a non-power of two would satisfy neither strict inequality for any n and would be reported.

I.2 Source identity and environment

The recorded run used MPFR 4.2.2 linked as `libmpfr.so.6`, GCC 14.2.0 with `-O3 -fopenmp`, GNU OpenMP with five worker threads, Linux on x86-64, and 192-bit precision with the directed modes `MPFR_RNDD` and `MPFR_RNDU`. The SHA-256 digest of the exact executed source reproduced below is

`f888c0d89be9366f01e2d26f339dcbe446556d39d4c5dbcf8f579ec0d0052e1d`

The execution image contained the MPFR shared library but not its development header, so the source carries a guarded minimal declaration of the MPFR 4.x ABI; on a system with `mpfr.h` present the standard header branch is taken instead. That ABI detail is part of the software trust boundary and is disclosed here rather than hidden.

I.3 Source

The listing below is the executed source verbatim. Its language is C99; it is typeset in this report's generic code style, which has no C-specific highlighting.

```
#define _GNU_SOURCE
#include <stdio.h>
#include <stdlib.h>
#include <stdint.h>
#include <inttypes.h>
#include <math.h>
#include <omp.h>

#if defined(__has_include)
#  if __has_include(<mpfr.h>)
#    include <mpfr.h>
#    define HAVE_SYSTEM_MPFR_HEADER 1
#  endif
#endif

#ifndef HAVE_SYSTEM_MPFR_HEADER
/* Minimal declarations for the MPFR 4.x ABI.  The execution environment had
   libmpfr.so.6 but not the development header package.  A normal build with
   libmpfr-dev installed takes the branch above instead. */
typedef unsigned long mp_limb_t;
typedef long mpfr_prec_t;
typedef long mpfr_exp_t;
typedef int mpfr_sign_t;
typedef struct {
  mpfr_prec_t _mpfr_prec;
  mpfr_sign_t _mpfr_sign;
  mpfr_exp_t _mpfr_exp;
  mp_limb_t *_mpfr_d;
} __mpfr_struct;
```

```
typedef __mpfr_struct mpfr_t[1];
typedef __mpfr_struct *mpfr_ptr;
typedef const __mpfr_struct *mpfr_srcptr;
typedef enum {
  MPFR_RNDN = 0, MPFR_RNDZ = 1, MPFR_RNDU = 2,
  MPFR_RNDD = 3, MPFR_RNDA = 4, MPFR_RNDF = 5,
  MPFR_RNDNA = -1
} mpfr_rnd_t;
extern void mpfr_init2(mpfr_ptr, mpfr_prec_t);
extern void mpfr_clear(mpfr_ptr);
extern int mpfr_set_ui(mpfr_ptr, unsigned long, mpfr_rnd_t);
extern int mpfr_log(mpfr_ptr, mpfr_srcptr, mpfr_rnd_t);
extern int mpfr_mul(mpfr_ptr, mpfr_srcptr, mpfr_srcptr, mpfr_rnd_t);
extern int mpfr_sub(mpfr_ptr, mpfr_srcptr, mpfr_srcptr, mpfr_rnd_t);
extern int mpfr_cmp_ui(mpfr_srcptr, unsigned long);
extern double mpfr_get_d(mpfr_srcptr, mpfr_rnd_t);
extern const char *mpfr_get_version(void);
#endif

static inline int is_power_of_two_u64(uint64_t x) {
  return x && ((x & (x - 1)) == 0);
}

typedef struct {
  uint64_t checked;
  uint64_t failures;
  double min_lower;
  double min_upper;
  uint64_t min_lower_m, min_lower_n;
  uint64_t min_upper_m, min_upper_n;
  uint64_t first_fail_m, first_fail_guess;
} Stats;

int main(int argc, char **argv) {
  uint64_t start = 1;
  uint64_t limit = (1ULL << 20);
  int bits = 192;
  if (argc > 1) start = strtoull(argv[1], NULL, 0);
  if (argc > 2) limit = strtoull(argv[2], NULL, 0);
  if (argc > 3) bits = atoi(argv[3]);
  if (start < 1 || limit <= start) {
    fprintf(stderr, "bad range\n");
    return 2;
  }

  int nt = omp_get_max_threads();
  Stats *stats = calloc((size_t)nt, sizeof(Stats));
  if (!stats) return 3;
  for (int i = 0; i < nt; ++i) {
    stats[i].min_lower = INFINITY;
    stats[i].min_upper = INFINITY;
  }

  const long double theta = logl(3.0L) / logl(2.0L);
  double t0 = omp_get_wtime();

  mpfr_t two, three, ln2_lo, ln2_hi, ln3_lo, ln3_hi;
  mpfr_init2(two, bits); mpfr_init2(three, bits);
  mpfr_init2(ln2_lo, bits); mpfr_init2(ln2_hi, bits);
```

```c
  mpfr_init2(ln3_lo, bits); mpfr_init2(ln3_hi, bits);
  mpfr_set_ui(two, 2, MPFR_RNDN);
  mpfr_set_ui(three, 3, MPFR_RNDN);
  mpfr_log(ln2_lo, two, MPFR_RNDD);
  mpfr_log(ln2_hi, two, MPFR_RNDU);
  mpfr_log(ln3_lo, three, MPFR_RNDD);
  mpfr_log(ln3_hi, three, MPFR_RNDU);

#pragma omp parallel
  {
    int tid = omp_get_thread_num();
    Stats *st = &stats[tid];
    mpfr_t xm, xn, xnp1;
    mpfr_t logm_lo, logm_hi, logn_hi, lognp1_lo;
    mpfr_t lhs_lo, lhs_hi, rhs_lo, rhs_hi, dl, du;
    mpfr_init2(xm, bits); mpfr_init2(xn, bits); mpfr_init2(xnp1, bits);
    mpfr_init2(logm_lo, bits); mpfr_init2(logm_hi, bits);
    mpfr_init2(logn_hi, bits); mpfr_init2(lognp1_lo, bits);
    mpfr_init2(lhs_lo, bits); mpfr_init2(lhs_hi, bits);
    mpfr_init2(rhs_lo, bits); mpfr_init2(rhs_hi, bits);
    mpfr_init2(dl, bits); mpfr_init2(du, bits);

#pragma omp for schedule(static)
    for (uint64_t m = start; m < limit; ++m) {
      if (is_power_of_two_u64(m)) continue;

      /* This is only a locator.  The MPFR inequalities below are the proof. */
      uint64_t guess = (uint64_t)floorl(expl(theta * logl((long double)m)));

      mpfr_set_ui(xm, (unsigned long)m, MPFR_RNDN);
      mpfr_log(logm_lo, xm, MPFR_RNDD);
      mpfr_log(logm_hi, xm, MPFR_RNDU);
      mpfr_mul(lhs_lo, logm_lo, ln3_lo, MPFR_RNDD);
      mpfr_mul(lhs_hi, logm_hi, ln3_hi, MPFR_RNDU);

      int ok = 0;
      uint64_t certn = 0;
      double dl_d = 0.0, du_d = 0.0;
      static const int offsets[7] = {0, -1, 1, -2, 2, -3, 3};
      for (int oi = 0; oi < 7; ++oi) {
        int off = offsets[oi];
        if (off < 0 && guess < (uint64_t)(-off) + 1) continue;
        uint64_t n = (uint64_t)((int64_t)guess + off);
        if (n < 1) continue;

        mpfr_set_ui(xn, (unsigned long)n, MPFR_RNDN);
        mpfr_set_ui(xnp1, (unsigned long)(n + 1), MPFR_RNDN);

        /* Directed lower bound for log(m)log(3)-log(n)log(2). */
        mpfr_log(logn_hi, xn, MPFR_RNDU);
        mpfr_mul(rhs_hi, logn_hi, ln2_hi, MPFR_RNDU);
        mpfr_sub(dl, lhs_lo, rhs_hi, MPFR_RNDD);
        if (mpfr_cmp_ui(dl, 0) <= 0) continue;

        /* Directed lower bound for log(n+1)log(2)-log(m)log(3). */
        mpfr_log(lognp1_lo, xnp1, MPFR_RNDD);
        mpfr_mul(rhs_lo, lognp1_lo, ln2_lo, MPFR_RNDD);
        mpfr_sub(du, rhs_lo, lhs_hi, MPFR_RNDD);
        if (mpfr_cmp_ui(du, 0) <= 0) continue;
```

```

      ok = 1;
      certn = n;
      dl_d = mpfr_get_d(dl, MPFR_RNDD);
      du_d = mpfr_get_d(du, MPFR_RNDD);
      break;
    }

    if (!ok) {
      ++st->failures;
      if (!st->first_fail_m || m < st->first_fail_m) {
        st->first_fail_m = m;
        st->first_fail_guess = guess;
      }
      continue;
    }

    ++st->checked;
    if (dl_d < st->min_lower) {
      st->min_lower = dl_d;
      st->min_lower_m = m;
      st->min_lower_n = certn;
    }
    if (du_d < st->min_upper) {
      st->min_upper = du_d;
      st->min_upper_m = m;
      st->min_upper_n = certn;
    }
  }

  mpfr_clear(xm); mpfr_clear(xn); mpfr_clear(xnp1);
  mpfr_clear(logm_lo); mpfr_clear(logm_hi);
  mpfr_clear(logn_hi); mpfr_clear(lognp1_lo);
  mpfr_clear(lhs_lo); mpfr_clear(lhs_hi);
  mpfr_clear(rhs_lo); mpfr_clear(rhs_hi);
  mpfr_clear(dl); mpfr_clear(du);
}

Stats total = {0};
total.min_lower = INFINITY;
total.min_upper = INFINITY;
for (int i = 0; i < nt; ++i) {
  Stats *s = &stats[i];
  total.checked += s->checked;
  total.failures += s->failures;
  if (s->min_lower < total.min_lower) {
    total.min_lower = s->min_lower;
    total.min_lower_m = s->min_lower_m;
    total.min_lower_n = s->min_lower_n;
  }
  if (s->min_upper < total.min_upper) {
    total.min_upper = s->min_upper;
    total.min_upper_m = s->min_upper_m;
    total.min_upper_n = s->min_upper_n;
  }
  if (s->first_fail_m &&
      (!total.first_fail_m || s->first_fail_m < total.first_fail_m)) {
    total.first_fail_m = s->first_fail_m;
    total.first_fail_guess = s->first_fail_guess;
```

```
    }
  }

  double elapsed = omp_get_wtime() - t0;
  printf("mpfr_version=%s\n", mpfr_get_version());
  printf("range=[%" PRIu64 ",%" PRIu64 ")\n", start, limit);
  printf("precision_bits=%d\nthreads=%d\n", bits, nt);
  printf("checked_nonpowers=%" PRIu64 "\n", total.checked);
  printf("failures=%" PRIu64 "\n", total.failures);
  printf("min_lower_log_margin=%.17g at (m,n)=(%" PRIu64 ",%" PRIu64 ")\n",
         total.min_lower, total.min_lower_m, total.min_lower_n);
  printf("min_upper_log_margin=%.17g at (m,n)=(%" PRIu64 ",%" PRIu64 ")\n",
         total.min_upper, total.min_upper_m, total.min_upper_n);
  if (total.failures) {
    printf("first_failure=(m,guess)=(%" PRIu64 ",%" PRIu64 ")\n",
           total.first_fail_m, total.first_fail_guess);
  }
  printf("elapsed_seconds=%.6f\n", elapsed);

  mpfr_clear(two); mpfr_clear(three);
  mpfr_clear(ln2_lo); mpfr_clear(ln2_hi);
  mpfr_clear(ln3_lo); mpfr_clear(ln3_hi);
  free(stats);
  return total.failures ? 1 : 0;
}
```

Listing 20: Directed-rounding verifier (C99, MPFR, OpenMP).

I.4 Build and invocation

On a system with the MPFR development headers installed, the whole build and a single full-range run are:

```
gcc -O3 -fopenmp mpfr_scan.c -lmpfr -lm -o mpfr_scan
OMP_NUM_THREADS=5 ./mpfr_scan 1 67108864 192
```

Listing 21: Build and invocation (POSIX shell).

The recorded audit instead used sixteen shorter invocations over half-open subranges, so that each chunk produces an independently inspectable log. The exit status is nonzero exactly when some input failed to be certified, which makes the chunked run scriptable without parsing the output.

J Chunk audit and reproducibility

Status: [interval computation] — the numbers in this appendix are the output of the verifier of Appendix I. They are reproducible data at a software trust level, not kernel certificates; the kernel-verified exponent bound remains $x \geq 12$.

There are $2^{26} - 1 = 67{,}108{,}863$ positive integers below 2^{26}, of which exactly 26 are powers of two, namely 2^0 through 2^{25}. The verifier therefore tests $67{,}108{,}863 - 26 = 67{,}108{,}837$ inputs. The range was split at multiples of 2^{22} into one initial chunk $[1, 2^{22})$ and fifteen chunks of width 2^{22}. The canonical chunk log holds the full standard output of every invocation and has SHA-256 digest

`dcaa7f3b7e5be44acaee9015a2045da14ee42d8fa2d96b9619cdd44c072a6fd7`

Table 16 reports, for each chunk, the exact scanned integer interval, the number of non-powers certified, and the smallest directed lower and upper logarithmic margins recorded in that chunk. Every chunk returned a failure count of zero.

Table 16: Directed-rounding audit through 2^{26}.

Chunk	Integer range	Checked	Smallest lower gap	Smallest upper gap
0	1–4,194,303	4,194,281	1.1539×10^{-17}	3.0076×10^{-17}
1	4,194,304–8,388,607	4,194,303	4.6319×10^{-20}	1.3515×10^{-19}
2	8,388,608–12,582,911	4,194,303	1.2811×10^{-18}	1.3515×10^{-19}
3	12,582,912–16,777,215	4,194,304	4.6319×10^{-20}	2.9082×10^{-19}
4	16,777,216–20,971,519	4,194,303	1.1898×10^{-18}	1.3515×10^{-19}
5	20,971,520–25,165,823	4,194,304	5.4212×10^{-19}	1.5111×10^{-19}
6	25,165,824–29,360,127	4,194,304	4.6319×10^{-20}	1.7669×10^{-19}
7	29,360,128–33,554,431	4,194,304	1.4439×10^{-19}	5.4132×10^{-21}
8	33,554,432–37,748,735	4,194,303	4.2839×10^{-19}	1.3515×10^{-19}
9	37,748,736–41,943,039	4,194,304	3.9292×10^{-19}	1.0460×10^{-19}
10	41,943,040–46,137,343	4,194,304	1.9177×10^{-19}	1.5111×10^{-19}
11	46,137,344–50,331,647	4,194,304	1.0117×10^{-19}	3.4145×10^{-19}
12	50,331,648–54,525,951	4,194,304	5.6927×10^{-20}	1.2322×10^{-19}
13	54,525,952–58,720,255	4,194,304	3.6447×10^{-20}	1.1345×10^{-19}
14	58,720,256–62,914,559	4,194,304	6.7779×10^{-20}	5.4132×10^{-21}
15	62,914,560–67,108,863	4,194,304	5.0923×10^{-20}	3.9197×10^{-20}
	total	**67,108,837**		

J.1 Extremal records

The global lower-gap record is the chunk 13 value 3.6447×10^{-20}, attained at

$$(m, n) = (58{,}570{,}438,\ 2{,}048{,}703{,}434{,}093),$$

where a separate 100-digit evaluation gives $m^{\vartheta} - n = 1.07725158750919017548\ldots \times 10^{-7}$. The global upper-gap record is the chunk 7 value 5.4132×10^{-21}, attained at

$$(m, n) = (29{,}411{,}704,\ 687{,}581{,}893{,}443),$$

where $n + 1 - m^{\vartheta} = 5.36974926710988874953\ldots \times 10^{-9}$. The same upper margin reappears in chunk 14 at the scaled pair (58,823,408, 2,062,745,680,331); this is not a coincidence but the identity $(2m)^{\vartheta} = 3\, m^{\vartheta}$, which multiplies the input by two and its ϑ-power by three while leaving the logarithmic margin unchanged.

Both extremal inequalities were reevaluated independently at 100 decimal digits. That secondary check is not an independent proof system — it shares the same floating-point philosophy — but it does guard against transcription and aggregation mistakes, and the certified margins sit many orders of magnitude above the nominal 192-bit rounding scale of the logarithmic operations.

The 2^{20} run of Appendix F and the present run are independent implementations, one using interval arithmetic in `mpmath` and the other MPFR directed rounding, and they agree on the overlapping range $1 \le m < 2^{20}$: no failures. Their recorded extremal metadata are not directly comparable, because the chunk boundaries differ.

J.2 Reproducibility checklist

To reproduce the finite claim independently:

1. obtain MPFR 4.2.x and GMP on a 64-bit platform;
2. save the source of Appendix I and verify its SHA-256 digest;
3. compile with optimization and OpenMP, or delete the OpenMP pragmas for a serial run;
4. run the sixteen half-open intervals of Table 16 at 192 bits;
5. verify that every invocation reports `failures=0`;
6. sum the reported `checked_nonpowers` and obtain 67,108,837;
7. compare the two global extremal records above;
8. optionally rerun at 256 or 320 bits, and on a second MPFR build;
9. for a theorem-grade result, translate the intervals into exact rational certificates and replay them in Lean rather than trusting the executable.

The finite statement is deterministic. Thread count changes execution time and the reduction order that produces the recorded minimum metadata, but it cannot change whether an individual directed inequality is certified. Step nine is the one that would change the status of the result rather than its range, and it is the reason the scan is presented here as data for certificate generation — in the sense of Section 27 — and not as an extension of the kernel-verified zero-free region.

K Classical transcendence inputs and their exact roles

Status: [classical/open background] — this appendix states external theorems, not new ones. Exactly one of them, Gelfond–Schneider, is formalized in this repository (Theorem 4.11); the remaining ones enter only through paper proofs, which are tagged **[paper proof in this report]** where they appear. The corpus does not introduce any of them as an axiom: where a conjectural or unformalized statement is needed in Lean it is defined as a proposition and the implication is proved conditionally on it. Deleting six exponentials, five exponentials, Baker's theorems, or Roy's theorem from the report would change which paper results stand, and would change none of the kernel-verified results catalogued in Appendix A.

The purpose of this appendix is auditability. A reader who wants to know how much of the report is elementary consequence of the repository hypotheses, and how much rests on major transcendence theory, should be able to answer that question by looking in one place. For each classical input below we give the statement in the form actually used, the exact places where the report uses it, and the exact reason it stops short of Conjecture 2.1. The last reason is the same in every case, seen from a different angle: every input listed here constrains a *linear* configuration of logarithms or an *algebraic* exponent, while the hypothetical counterexample is a *bilinear* relation between logarithms with a transcendental exponent.

Gelfond–Schneider (Hilbert's seventh problem)

Statement. Let $\alpha, \beta \in \overline{\mathbb{Q}}$ with $\alpha \neq 0, 1$ and $\beta \notin \mathbb{Q}$. Then every value of α^β is transcendental [4, 9, 3]. **[classical/open background]**

Role here. This is the one classical transcendence theorem the corpus proves rather than cites; the Lean proof follows the classical exposition [4] and is recorded as Theorem 4.11. It is used twice. First, a nonintegral solution x is shown irrational by the elementary rational-exponent argument, and then must be transcendental because 2^x is an algebraic number; this is the algebraic–integral dichotomy of Corollary 4.12, and it is what makes the conditional primitive generator β and the odd-core exponent $\log_2 w$ transcendental throughout Section 5 and Section 6. Second, $\vartheta = \log_2 3$ is irrational, hence transcendental, because $2^\vartheta = 3$.

Why it does not settle Conjecture 2.1. The theorem speaks only about algebraic irrational exponents. Once the dichotomy has removed algebraic exponents, the remaining hypothetical exponent is transcendental, and Gelfond–Schneider has nothing further to say about 2^x for transcendental x. The theorem closes the easy half of the problem and defines the hard half.

Six exponentials

Statement. Let $u_1, u_2, u_3 \in \mathbb{C}$ be linearly independent over $\mathbb{Q}$ and let $v_1, v_2 \in \mathbb{C}$ be linearly independent over $\mathbb{Q}$. Then at least one of the six numbers

$$e^{u_i v_j} \qquad (1 \le i \le 3,\ 1 \le j \le 2)$$

is transcendental [10, 11, 12]. **[classical/open background]**

Role here. It is the analytic engine behind every exact spectrum statement in the report. The saturated algebraic-base spectrum — for multiplicatively independent $a, b \in \overline{\mathbb{Q}}_{>0}$ with $a^x, b^x \in \overline{\mathbb{Q}}$ and x irrational, $\alpha^x \in \overline{\mathbb{Q}}$ if and only if $\alpha \in \Gamma(a, b)$ — is the case with rows $\log a, \log b, \log \alpha$ and columns $1, x$. The conditional classification of the symmetric constant $E_3 = 3^\vartheta$ is the case with rows $1, x, \vartheta$ and columns $\log 2, \log 3$, and the classification of $E_2 = 2^\eta$ is its mirror image under $\vartheta \mapsto \eta = 1/\vartheta$. The ordinary form of the unconditional E_3/E_2 dichotomy uses the same configuration. The kernel-verified 3-smooth classification of auxiliary bases and the three-base theorem $2^x, 3^x, 5^x \in \mathbb{Z} \Rightarrow x \in \mathbb{Z}$ are the formalized integer-level shadows of this theorem; the Lean proofs proceed by interpolation determinants and do not cite it.

Why it does not settle Conjecture 2.1. Six exponentials needs three independent directions on one side of the configuration. The two-base problem supplies exactly two logarithmic directions, $\log 2$ and $\log 3$, and every auxiliary base built from 2 and 3 has logarithm in their rational span, so it creates no third column. The remaining 2×2 configuration of Section 7 is precisely the four exponentials conjecture, which is open.

Five exponentials

Statement. Let u_1, u_2 be linearly independent over $\mathbb{Q}$, let v_1, v_2 be linearly independent over $\mathbb{Q}$, and let $\gamma \in \overline{\mathbb{Q}}$ with $\gamma \neq 0$. Then at least one of the five numbers

$$e^{u_1 v_1}, \quad e^{u_1 v_2}, \quad e^{u_2 v_1}, \quad e^{u_2 v_2}, \quad e^{\gamma u_2 / u_1}$$

is transcendental [11, 8]. **[classical/open background]**

Role here. It supplies the two forbidden exponential rays of the hypothetical world: for a nonintegral solution x and any nonzero algebraic γ, both $e^{\gamma x}$ and $e^{\gamma\vartheta}$ are transcendental. For the first, apply the theorem to the independent pairs $(1, x)$ and $(\log 2, \log 3)$, whose four rectangular exponentials are the algebraic numbers $2, 3, M, A$, leaving the fifth transcendental; the second is the same argument with the roles of the two logarithms exchanged. Formalizing it would also upgrade the denominator-controlled auxiliary-base theorem in `RationalThirdBase` from a controlled denominator to an unrestricted rational output, which is why it appears as a research priority in Section 27.

Why it does not settle Conjecture 2.1. The theorem does not remove the fifth exponential, so it does not prove four exponentials. More concretely, the natural auxiliary quantities in this problem are $p^x = e^{x\log p}$ for a prime p, whose coefficient $\log p$ is transcendental rather than an admissible algebraic γ. The forbidden rays therefore never convert a prime divisor of the integer outputs into a contradiction, as Section 26 records.

Baker's theorem on linear forms in logarithms

Statement. Let $\gamma_1, \ldots, \gamma_n$ be nonzero algebraic numbers of degree at most d and height at most H, and let $\alpha_1, \ldots, \alpha_n$ be algebraic numbers of height at most B, not all zero. If

$$\Lambda = \alpha_1 \log\gamma_1 + \cdots + \alpha_n \log\gamma_n \neq 0,$$

then $|\Lambda| > B^{-C}$ for an effectively computable $C > 0$ depending only on n, d, H and the degree of the α_i [3, 10]. **[classical/open background]**

Role here. The effective form is used as a measuring stick rather than as a step in any proof. In the rational-approximation analysis of Section 26 it fixes the scale of what is available: Baker-type lower bounds are polynomial in the relevant heights, whereas the integer gaps produced by the hypotheses only assert $|\varepsilon| \gtrsim 2^{-p}$, exponentially weaker, so the two scales are consistent and no contradiction arises. It is also the theorem that effective irrationality measures for ϑ would refine.

Why it does not settle Conjecture 2.1. Baker's theory bounds forms whose coefficients are *algebraic*. The decisive relation under failure is $\log M \,\log 3 = \log A \,\log 2$, and its radical form is $d \log w \,\log 3 = (\log A - a\log 3)\log 2$: each side is a product of two logarithms, so the coefficient of every logarithm is itself a logarithm. Expanded over prime factorizations $m = \prod_p p^{m_p}$ and $A = \prod_p p^{a_p}$, failure is the nontrivial vanishing of

$$D = \sum_p \bigl(m_p \log 3 - a_p \log 2\bigr)\log p,$$

an integer-coefficient linear form in the *products* $\log p \cdot \log 2$ and $\log p \cdot \log 3$. No lower-bound technology exists for such forms, and even the $\mathbb{Q}$-linear independence of $\{\log p \cdot \log q\}$ is open; Section 12 turns exactly that independence into a sufficient condition.

Baker's homogeneous theorem

Statement. Let $\gamma_1, \ldots, \gamma_n$ be nonzero algebraic numbers. If $\log\gamma_1, \ldots, \log\gamma_n$ are linearly independent over $\mathbb{Q}$, then $1, \log\gamma_1, \ldots, \log\gamma_n$ are linearly independent over $\overline{\mathbb{Q}}$. The homogeneous consequence used here is the shorter one: logarithms of algebraic numbers that are linearly independent over $\mathbb{Q}$ are linearly independent over $\overline{\mathbb{Q}}$ [3, 10]. **[classical/open background]**

Role here. It is the bridge from a rational-independence hypothesis to the $\overline{\mathbb{Q}}$-independence hypothesis that Roy's theorem requires. In the strong spectrum, the three logarithms $\log a, \log b, \log\alpha$

are first shown $\mathbb{Q}$-independent by an elementary saturation argument, and Baker's homogeneous theorem upgrades them to $\overline{\mathbb{Q}}$-independence; the same upgrade produces the logarithmic independence used in the strong forms of the E_3 and E_2 classifications. The ordinary spectrum does not need it, since rational independence already suffices for six exponentials.

Why it does not settle Conjecture 2.1. It is a statement about linear relations among logarithms. No independence statement about the family $\{\log p\}$, over $\mathbb{Q}$ or over $\overline{\mathbb{Q}}$, implies anything about the family of products $\{\log p \cdot \log q\}$, and it is the products that carry the obstruction.

Roy's strong and sharp six exponentials theorems

Statement. Let $\mathcal{L}$ be the $\overline{\mathbb{Q}}$-vector space spanned by 1 together with all logarithms of nonzero algebraic numbers. If u_1, u_2, u_3 are linearly independent over $\overline{\mathbb{Q}}$ and v_1, v_2 are linearly independent over $\overline{\mathbb{Q}}$, then at least one of the six products $u_i v_j$ lies outside $\mathcal{L}$. The sharp form replaces the independence hypotheses by a rank condition on a matrix whose entries are linear forms in logarithms of algebraic numbers [8, 11]. **[classical/open background]**

Role here. It produces the strengthened conclusions only. Concretely: the strong auxiliary-base spectrum, which asserts $x \log \alpha \notin \mathcal{L}$ for $\alpha \notin \Gamma(a, b)$ rather than merely $\alpha^x \notin \overline{\mathbb{Q}}$; the strong form of the symmetric-constant dichotomy, which places $\vartheta \log 3$ and $\eta \log 2$ outside $\mathcal{L}$; and the log-strong transcendence clauses in the second-iterate kernel and depth-three tower theorems.

Why it does not settle Conjecture 2.1. Every ordinary transcendence conclusion in the report survives the removal of this theorem; only the "lies outside $\mathcal{L}$" clauses disappear. Structurally it inherits the six exponentials dimension count and still demands three independent directions on one side, so it does not reach the 2×2 configuration where the conjecture actually lives.

Lindemann–Weierstrass

Statement. If $\alpha_1, \ldots, \alpha_n$ are distinct algebraic numbers, then $e^{\alpha_1}, \ldots, e^{\alpha_n}$ are linearly independent over $\overline{\mathbb{Q}}$. The special case used here is Hermite–Lindemann: for algebraic $\alpha \neq 0$ the value e^{α} is transcendental, equivalently every nonzero value of $\log \gamma$ is transcendental for algebraic $\gamma \neq 0, 1$ [3]. **[classical/open background]**

Role here. It is the standing background fact that each $\log p$, and in particular $\log 2$ and $\log 3$, is transcendental. That fact is what makes the coefficients in the log-product form D above transcendental rather than algebraic, and therefore what places the wall outside Baker's reach; it is also the precise content of the audit warning against treating $\log 2$ and $\log 3$ as algebraic coefficients (Remark L.3).

Why it does not settle Conjecture 2.1. Lindemann–Weierstrass constrains exponentials of *algebraic* arguments. Under failure the exponent x is transcendental and the arguments $x \log 2$ and $x \log 3$ are transcendental, so none of the four corner exponentials $2, 3, 2^x, 3^x$ is within its scope, and it gives no information whatever about products of two logarithms.

K.1 Logical dependency graph

The table records, for each family of results in the report, the classical input it actually consumes. The rows whose input column reports no classical theorem are elementary or kernel-verified, and depend on no transcendence theory beyond the Gelfond–Schneider dichotomy that the corpus already formalizes.

Table 17: Classical transcendence input consumed by each family of results.

Result	Required classical input
Nonintegral solution is transcendental	Rational-exponent lemma and Gelfond–Schneider; both kernel-verified
Reduction to real four exponentials	None beyond $\mathbb{Q}$-independence of $1, x$ and of $\log 2, \log 3$; kernel-verified
3-smooth auxiliary-base classification and the three-base theorem	None; interpolation determinants, kernel-verified
Exact algebraic-base spectrum for an independent pair	Six exponentials
Strong $\mathcal{L}$ spectrum	Exact row independence, Baker's homogeneous theorem, and Roy
Valuation lattice and Smith normal form of rational outputs	Exact spectrum and unique factorization
Second-iterate kernel: dimension and the four branch types	Exact spectrum and elementary prime valuations
Kernel invariance across all nonintegral solutions	Exact spectrum applied twice
Möbius orbit criterion	Kernel definition and real logarithms
Depth-three tower theorem	Exact spectrum and kernel dimension; Roy only for the strong form
Base-2 tower equivalence	Natural-base algebraicity classification and smooth branch algebra
Symmetric constants E_3 and E_2	Six exponentials; Baker's homogeneous theorem and Roy for the strong form
Two forbidden exponential rays $e^{\gamma x}$, $e^{\gamma \vartheta}$	Five exponentials
Transcendence of the coefficients $\log p$ in the log-product form	Lindemann–Weierstrass, in its Hermite–Lindemann case

L Common false shortcuts and audit warnings

Status: [classical/open background] — this appendix records negative facts and standard cautions. Nothing in it is a new result, and nothing in it changes the status of any statement elsewhere in the report.

The problem attracts several plausible but invalid one-line arguments. Recording them is part of the audit trail: each is a specific way in which a proposed proof can be circular, can silently assume the conclusion, or can mistake numerical evidence for exclusion. Future paper and formal work should be checked against this list before a short argument is believed.

Remark L.1 (Fractional-part error). Writing $x = n + \alpha$ with $n \in \mathbb{Z}$ and $0 < \alpha < 1$, and assuming $2^x \in \mathbb{Z}$, does *not* imply $2^\alpha \in \mathbb{Z}$. It implies only $2^\alpha \in \mathbb{Z}[1/2]$, and an irrational α is not an ordinary radical exponent $1/q$. The integrality of the output does not descend to the fractional part.

Remark L.2 (Rational-root circularity). From $M^\vartheta \in \mathbb{Z}$ one cannot conclude that ϑ is rational, nor that M is a perfect power. Gelfond–Schneider constrains algebraic irrational exponents, whereas ϑ is transcendental. An argument that quietly treats ϑ as a rational exponent has assumed the conclusion.

Remark L.3 (Baker is linear, not bilinear). The relation $\log M \, \log 3 = \log A \, \log 2$ is not a Baker linear form. Treating $\log 2$ and $\log 3$ as algebraic coefficients is invalid: by Hermite–Lindemann both are transcendental. The same objection applies to the radical form $d \log w \log 3 = (\log A - a \log 3) \log 2$ and to the prime-expanded log-product form of Section 12.

Remark L.4 (No modular real exponent). The expression $2^x \bmod p$ has no canonical meaning for a general real x. A p-adic exponential requires a p-adic exponent and a unit in its convergence domain; the real equality $2^x \in \mathbb{Z}$ supplies neither. Congruence arguments must therefore be stated for the integer outputs M and A, never for the real exponent.

Remark L.5 (Candidate quotients need not be candidates). If $m_1, m_2 \in \mathcal{C}$ and $m_1 \mid m_2$, it does not follow that $m_2/m_1 \in \mathcal{C}$, because the corresponding quotient of the outputs need not be integral. Multiplication is safe; division requires a separate valuation check. This is the reason $\mathcal{C}$ is treated as a multiplicative monoid rather than as a divisor-closed set.

Remark L.6 (Density alone is compatible). The orbit $\{kx\}$ is dense modulo one for irrational x, but the associated rational points have exponential height. A sequence of $O(\log H)$ points up to height H is compatible with generic transcendental-curve counting, so density by itself yields no contradiction without a denominator-sensitive refinement.

Remark L.7 (Numerical tolerance is not exclusion). Computing M^ϑ in floating point and comparing its distance to the nearest integer against a fixed epsilon cannot certify a zero-free region, however wide. Outward directed interval bounds or an exact certificate are required, which is why the finite scans in this report are tagged **[interval computation]** and the kernel-verified exponent bound remains $x \geq 12$.

Remark L.8 (Algebraicity of a power is branch-sensitive). From $z^d \in \overline{\mathbb{Q}}$ with an integer $d > 0$ one may conclude $z \in \overline{\mathbb{Q}}$, but from $z^x \in \overline{\mathbb{Q}}$ with transcendental x no such algebraic closure argument is available. The finite-degree argument uses the polynomial $X^d - z^d$, which has no analogue when the exponent is not an integer.

Every warning above was re-derived independently during the preparation of the present unified report: each was proposed at some point as a short route to the conjecture, and each failed on exactly the ground stated here.

M Logarithmic-energy convergence for projected lattice triangles

We record the technical point needed in the triangular determinant limit. Let $\lambda_1, \lambda_2 > 0$ and assume their ratio has finite irrationality type: there are $c, \tau > 0$ such that

$$|r\lambda_1 + s\lambda_2| \geq c(1 + |r| + |s|)^{-\tau}$$

for all nonzero $(r, s) \in \mathbb{Z}^2$. Let $t_{1,N} < \cdots < t_{N,N}$ be the first N values of $a\lambda_1 + b\lambda_2$ with $a, b \in \mathbb{N}_0$, and set $\widetilde{t}_{i,N} = t_{i,N}/\sqrt{N}$. Then the empirical measures converge to

$$d\mu(t) = \frac{2t}{U^2}\, dt, \qquad U = \sqrt{2\lambda_1\lambda_2},$$

and

$$\frac{2}{N^2}\sum_{i<j}\log|\widetilde{t}_{j,N}-\widetilde{t}_{i,N}| \longrightarrow \iint \log|x-y|\,d\mu(x)d\mu(y). \tag{224}$$

The weak convergence follows from a Riemann-sum lattice count in the triangle $\lambda_1 a+\lambda_2 b\le U\sqrt{N}$. For the singular kernel, truncate $\log|x-y|$ below at $-M$. Weak convergence handles the truncated kernel. Uniform integrability of the negative tail follows by counting difference vectors (r,s) in thin strips around $r\lambda_1+s\lambda_2=0$; the finite-type lower bound prevents an individual difference from being smaller than a fixed power of N, while the strip count shows that the proportion below ε is $O(\varepsilon)$ up to a power-saving discrepancy term. Integrating the distribution function of $-\log|x-y|$ and then sending $M\to\infty$ proves (224).

For $(\lambda_1,\lambda_2)=(\log 2,\log 3)$, the required finite type follows from linear forms in logarithms. For $(1,\beta)$, it follows from $\beta=\log A/\log 2$ and the multiplicative independence of A and 2.

N Integral evaluations used in the determinant constants

N.1 The beta-two logarithmic energy

For the probability density $f(x)=2x$ on $[0,1]$,

$$\begin{aligned}\iint_0^1 4xy\log|x-y|\,dx\,dy &= 8\int_0^1\int_0^x xy\log(x-y)\,dy\,dx\\ &= 8\int_0^1 x^3\left(\frac{1}{2}\log x-\frac{3}{4}\right)dx\\ &= -\frac{7}{4}.\end{aligned}$$

Scaling by U adds $\log U$, proving Equation (212).

N.2 The simplex assignment constant

A coordinate of a point uniformly distributed on the standard two-simplex has quantile $q(s)=1-\sqrt{1-s}$. Therefore

$$\begin{aligned}J &= \int_0^1 q(s)q(1-s)\,ds\\ &= \int_0^1\left(1-\sqrt{s}-\sqrt{1-s}+\sqrt{s(1-s)}\right)ds\\ &= 1-\frac{4}{3}+\frac{\pi}{8}=\frac{\pi}{8}-\frac{1}{3}.\end{aligned}$$

Its comonotone second-moment constant is $K=\int_0^1 q(s)^2ds=1/6$.

N.3 The balanced-rectangle energy

Let $Z = (X+Y)/2$ with independent uniform $X, Y \in [0,1]$. The density of $S = X+Y$ is triangular, and the density of the difference of two independent copies of S is

$$h(t) = \begin{cases} \frac{2}{3} - t^2 + \frac{1}{2}t^3, & 0 \le t \le 1, \\ \frac{1}{6}(2-t)^3, & 1 \le t \le 2. \end{cases}$$

A direct integration gives

$$\mathbb{E} \log |S - S'| = \frac{4}{3} \log 2 - \frac{25}{12}.$$

Since $Z - Z' = (S - S')/2$,

$$\mathbb{E} \log |Z - Z'| = \frac{1}{3} \log 2 - \frac{25}{12}.$$

Also $\mathbb{E} Z^2 = 7/24$.

O General two-prime version

The structural and determinant arguments extend verbatim to distinct primes p, q. If a nonintegral β satisfies $p^\beta, q^\beta \in \mathbb{Z}$, choose the least such β. Six Exponentials gives the same affine-rank collapse, and valuation minimality yields the exact monoid $\mathbb{N}_0 + \mathbb{N}_0\beta$. The triangular determinant then gives

$$g\left(\sqrt{\beta \log p \log q}\right) \ge 0.$$

This form may be useful for comparing the strength of the method across base pairs and for testing variational improvements.

P Finite determinant certificates

The asymptotic argument can be converted into finite certificates. For chosen finite node sets, a certificate consists of:

1. the exact integer or rational matrix entries;
2. sorted rational intervals for u_i, v_i and their differences;
3. an upper bound from Equation (207);
4. for each relevant prime, dual potentials for the assignment problem in Lemma 51.3, proving the claimed minimum;
5. a product-formula comparison showing the upper bound is smaller than the certified divisor.

Such certificates are compact enough for independent checking and eventual Lean verification. They also avoid trusting floating-point determinant computations, which are extremely ill-conditioned in the near-singular regime.

Q Source of the finite verifier

The following is the complete source used to establish Theorem 63.1. Its only non-rigorous arithmetic—the binary64 call used to suggest a starting integer—cannot affect the result: the subsequent directed-interval sign search is monotone and independently certifies the final pair of inequalities.

```
#!/usr/bin/env python3
"""Rigorous finite verification for the Alaoglu--Erdos problem.

For every integer A in [2, LIMIT], this proves that A^(log(3)/log(2))
is an integer only when A is a power of two.  All decisive comparisons use
Decimal directed rounding plus the positive atanh series for logarithms.
The ordinary binary64 approximation is used only to choose an initial nearby
integer; a monotone certified search repairs any inaccurate guess.
"""
from __future__ import annotations

from dataclasses import dataclass
from decimal import Decimal, Context, ROUND_FLOOR, ROUND_CEILING, localcontext
import json
import math
import sys
import time

LIMIT = int(sys.argv[1]) if len(sys.argv) > 1 else 100_000
PREC = 70
TERMS = 28

DOWN = Context(prec=PREC, rounding=ROUND_FLOOR)
UP = Context(prec=PREC, rounding=ROUND_CEILING)

@dataclass(frozen=True)
class Interval:
    lo: Decimal
    hi: Decimal


def _atanh_log_ratio(p: int, q: int, terms: int = TERMS) -> Interval:
    """Enclose 2*atanh(p/q), for 0 <= p/q <= 1/3."""
    assert 0 <= p <= q // 3 + (1 if 3 * p == q else 0)
    if p == 0:
        z = Decimal(0)
        return Interval(z, z)

    with localcontext(DOWN):
        z_lo = Decimal(p) / Decimal(q)
        z2_lo = z_lo * z_lo
        power = z_lo
        total = Decimal(0)
        for j in range(terms):
            total = total + power / Decimal(2 * j + 1)
            power = power * z2_lo
        lo = Decimal(2) * total

    with localcontext(UP):
        z_hi = Decimal(p) / Decimal(q)
        z2_hi = z_hi * z_hi
        power = z_hi
        total = Decimal(0)
        for j in range(terms):
            total = total + power / Decimal(2 * j + 1)
            power = power * z2_hi
        partial = Decimal(2) * total
        # power is now z^(2*terms+1).  Since all omitted terms are positive,
        # replace every odd denominator by the first one and sum geometrically.
```

```
        rem = Decimal(2) * power / (
            Decimal(2 * terms + 1) * (Decimal(1) - z2_hi)
        )
        hi = partial + rem
    return Interval(lo, hi)


LOG2 = _atanh_log_ratio(1, 3)


def log_integer(n: int, terms: int = TERMS) -> Interval:
    """Rigorous interval for log(n), n >= 1, by binary range reduction."""
    assert n >= 1
    if n == 1:
        z = Decimal(0)
        return Interval(z, z)
    k = n.bit_length() - 1
    two_k = 1 << k
    # n/2^k in [1,2), and z=(y-1)/(y+1) lies in [0,1/3).
    local = _atanh_log_ratio(n - two_k, n + two_k, terms)
    with localcontext(DOWN):
        lo = Decimal(k) * LOG2.lo + local.lo
    with localcontext(UP):
        hi = Decimal(k) * LOG2.hi + local.hi
    return Interval(lo, hi)


LOG3 = log_integer(3)


def determinant_log_interval(log_a: Interval, b: int) -> Interval:
    """Enclose F(A,b)=log(b)log(2)-log(A)log(3)."""
    log_b = log_integer(b)
    with localcontext(DOWN):
        lo = log_b.lo * LOG2.lo - log_a.hi * LOG3.hi
    with localcontext(UP):
        hi = log_b.hi * LOG2.hi - log_a.lo * LOG3.lo
    return Interval(lo, hi)


def certified_sign(log_a: Interval, b: int) -> tuple[int, Interval]:
    f = determinant_log_interval(log_a, b)
    if f.hi < 0:
        return -1, f
    if f.lo > 0:
        return +1, f
    # At this finite precision an interval meeting zero is not silently accepted.
    raise ArithmeticError(f"inconclusive interval at A-log={log_a}, b={b}: {f}")


def is_power_of_two(n: int) -> bool:
    return n > 0 and (n & (n - 1)) == 0


def main() -> None:
    start = time.monotonic()
    rho_approx = math.log(3.0) / math.log(2.0)
    min_abs_endpoint = Decimal('Infinity')
    max_repairs = 0
    checked = 0

    for a in range(2, LIMIT + 1):
        if is_power_of_two(a):
            continue
        checked += 1
        log_a = log_integer(a)
        guess = max(1, int(math.floor(math.exp(rho_approx * math.log(float(a))))))
        repairs = 0
```

```
        # Find consecutive integers b,b+1 on opposite sides of the unique real root.
        while True:
            s0, f0 = certified_sign(log_a, guess)
            s1, f1 = certified_sign(log_a, guess + 1)
            min_abs_endpoint = min(
                min_abs_endpoint,
                abs(f0.lo), abs(f0.hi), abs(f1.lo), abs(f1.hi)
            )
            if s0 < 0 < s1:
                break
            if s1 < 0:
                guess += 1
            elif s0 > 0:
                guess -= 1
                assert guess >= 1
            else:
                raise AssertionError((a, guess, s0, s1))
            repairs += 1
            if repairs > 100:
                raise AssertionError(f"unexpectedly bad initial approximation at A={a}")
        max_repairs = max(max_repairs, repairs)

    elapsed = time.monotonic() - start
    result = {
        "limit": LIMIT,
        "checked_nonpowers": checked,
        "precision_decimal_digits": PREC,
        "atanh_terms": TERMS,
        "max_initial_guess_repairs": max_repairs,
        "minimum_absolute_interval_endpoint": str(min_abs_endpoint),
        "elapsed_seconds": elapsed,
        "conclusion": (
            f"For every integer 2 <= A <= {LIMIT}, "
            "A^(log(3)/log(2)) is integral only if A is a power of two."
        ),
    }
    print(json.dumps(result, indent=2, sort_keys=True))


if __name__ == '__main__':
    main()
```

R External literature reconnaissance

This appendix merges two literature-and-strategy surveys of the same corpus. It records what the surrounding literature does and does not supply, which imports are immediately usable, and which are techniques that would need adaptation. Two claims made in the source surveys are retracted here; both are flagged where they occur.

Citations are given in full in the text rather than through the bibliography, because the purpose of the appendix is provenance rather than argument: nothing below is used as a step in any proof in this report.

R.1 Effective measures for the two logarithms

The sharpest linear-independence measure for the relevant triple is Qiang Wu, *On the linear independence measure of logarithms of rational numbers*, Math. Comp. **72** (2003), 901–911, which gives $\mu(1, \log 2, \log 3) \le 7.6155$, improving Rhin's 7.616. For the single logarithm, Wu and Wang, J. Number Theory **142** (2014), 264–273, give $\mu(\log 3) \le 5.1163051$, improving Salikhov's 5.125. For two-logarithm forms, Laurent–Mignotte–Nesterenko, J. Number Theory **55** (1995), 285–321, and Laurent, Acta Arith. **133** (2008), 325–348, give explicit $\log|b_1 \log 2 + b_2 \log 3| \ge -C(\log B)^2$ with C of the order of a few tens. On the non-archimedean side, Bugeaud–Laurent (1996) and Bugeaud (1999) are the two-logarithm workhorses, and K. C. Chim, J. Number Theory (2025), improves the $(\log B)^2$ dependence to $\log B$ for two p-adic logarithms.

Caution R.1 (Retracted claim)**.** The first survey proposed that Wu's measure would yield a *uniform* zero-free criterion replacing the per-value certificates of Section 25. That is wrong, and the error is instructive. Near-integrality of U^ϑ is not a rational approximation to ϑ; it is the vanishing of the bilinear form $\log 3 \log U - \log 2 \log V$, which is exactly the wall of Section 7. No linear-in-logarithms measure touches it. The finite verification stays per-value.

What the measure does give unconditionally is an effective irrationality exponent $\mu_{\text{eff}}(\vartheta) \le 8.62$, hence a polynomial recurrence $q_{n+1} \ll q_n^{7.62}$ for the convergent denominators of ϑ. That is the strongest known bound of its kind and supersedes the exponential estimate; the kernel-verified module `DenominatorGrowth` proves only the weaker exponential form, which remains the strongest **[kernel-verified Lean]** statement here.

R.2 Interpolation determinants with non-archimedean amplification

The closest existing realization of the amplification idea audited in Section 21 is Bombieri, *Effective Diophantine approximation on* $\mathbb{G}_m$, Ann. Scuola Norm. Sup. Pisa **20** (1993), 61–89, and Bombieri–Cohen, *ibid.* **24** (1997), 205–225, with the elementary account in LMS Lecture Notes **303** (2003), 41–62. These combine archimedean and non-archimedean interpolation determinants through the equivariant Thue–Siegel principle and are the only non-Baker route to effective results on the multiplicative group. Laurent's interpolation-determinant papers (Acta Arith. **66** (1994), 181–199, and **133** (2008)) supply the determinant method in the form this report already uses.

This is a technique needing adaptation, not a plug-in. Its relevance is sharpened by Corollary 21.2: any amplification must obtain its archimedean bound from genuine cancellation, and Bombieri–Cohen is where that is done systematically.

R.3 Small-value estimates and the four-exponentials frontier

Roy, *An arithmetic criterion for the values of the exponential function*, Acta Arith. **97** (2001), 183–194, constructs a small-value criterion on $\mathbb{G}_a \times \mathbb{G}_m$ and proves it *equivalent* to Schanuel's conjecture. His subsequent estimates — Acta Arith. **135** (2008), 357–393; Int. J. Number Theory **6** (2010), 919–956; Mathematika **59** (2013), 333–363 — and Nguyen–Roy, Int. J. Number Theory **12** (2016), 1273–1293, are the deepest partial progress toward the four-exponentials and Schanuel circle. Roy's Strong Six Exponentials Theorem, J. Number Theory **41** (1992), 22–47, is the object whose formalization would most directly strengthen the algebraic-base spectrum of Section 13.

Butler, *Some cases of Wilkie's conjecture*, Bull. London Math. Soc. **44** (2012), 642–660, shows that certain cases of Wilkie's conjecture are equivalent to real forms of the three- and four-exponentials conjectures. That is a direct bridge between the counting route below and the transcendence core, and it was not previously cited in this corpus.

R.4 Counting rational points on the transcendental arc

A counterexample produces $\asymp T$ *rational* points

$$P_k = \left(\frac{M^k}{2^{r_k}},\ \frac{A^k}{3^{r_k}}\right) = (2^{\{k\beta\}}, 3^{\{k\beta\}})$$

of logarithmic height at most T, lying simultaneously in a *fixed* rank-two multiplicative subgroup $\Gamma \subset \mathbb{G}_m^2(\overline{\mathbb{Q}})$ and on the *fixed* transcendental Pfaffian arc $\mathcal{C} = \{(u, u^{\vartheta}) : 1 \le u < 2\}$. This unlikely-intersection formulation is the most promising framing the surveys contribute, and it is not used elsewhere in this report.

The counting technology is Binyamini–Novikov, Ann. of Math. **186** (2017), 237–275, and Binyamini–Novikov–Zak, Ann. of Math. **199** (2024), 795–821, which prove Wilkie's conjecture for restricted elementary and Pfaffian structures with $(\log H)^{\kappa}$ counting; and, on the non-archimedean side, Binyamini–Cluckers–Novikov (Duke Math. J.), Cluckers–Comte–Loeser, Forum Math. Pi **3** (2015), and *Counting rational points on transcendental curves in valued fields*, arXiv:2506.19411 (2025). Boxall–Jones–Schmidt, J. Eur. Math. Soc. **24** (2022), 1567–1592, combine archimedean counting with p-adic methods.

The honest verdict is negative in a familiar way. Polylogarithmic counting gives $(\log H)^{\kappa}$ where a counterexample supplies $\asymp \log H$ points, so the framing reproduces the one-logarithm deficit of Section 22 in counting form: a third independent route arriving at the same shortfall. Moreover no cited theorem is sensitive to *which* primes divide the denominators, which is precisely what the compact-arc reformulation of Section 23 needs. The statement required — that a compact transcendental arc of nonzero curvature carries $o(H)$ points of height at most H whose first coordinate has 2-power denominator and second 3-power denominator — is not implied by any of them and would be a new theorem.

R.5 Multiplicative dependence, S-units, and rigidity

Bugeaud–Corvaja–Zannier, Math. Z. **243** (2003), 79–84, prove that for multiplicatively independent $a, b \ge 2$ and every $\varepsilon > 0$ one has $\log\gcd(a^n - 1, b^n - 1) \le \varepsilon n$ for large n, and that this is sharp. Together with the Corvaja–Zannier subspace-theorem machinery and Evertse–Győry's effective S-unit theory, this confirms that S-unit methods deliver finiteness, not nonexistence, and cannot supply the exponential-in-q divisibility the rational-approximation route would need.

For the $\times 2 \times 3$ side, Bourgain–Lindenstrauss–Michel–Venkatesh, Ergodic Theory Dynam. Systems **29** (2009), 1705–1722, give the only *effective* Furstenberg-type density result, with a density gap of order $(\log\log L)^{-\kappa}$. Shmerkin and Wu, Ann. of Math. **189** (2019), resolved Furstenberg's intersection conjecture. None of these can force visits to targets shrinking like M^{-k}; the target-width mismatch is fatal for a direct application, exactly as Section 23 anticipates. The effective Bourgain–Lindenstrauss–Michel–Venkatesh rate is the one usable quantitative handle and confirms, rather than defeats, the shrinking-target obstruction.

R.6 Kummer theory and the radical reformulation

Perucca–Sgobba–Tronto, Int. J. Number Theory **17** (2021), 1091–1110, and *Kummer theory for number fields via entanglement groups*, Manuscripta Math. **169** (2022), 251–270, compute degrees of

Kummer extensions exactly through entanglement groups, and Chan–Pajaziti–Perissinotto–Perucca, arXiv:2508.19211 (2025), complete Kneser's theorem on additive relations among radicals. These make the algebraic half of the radical reformulation of Section 6 rigorous and formalizable. They do not see the special value ϑ: the mixed archimedean–radical invariant that the reformulation needs is supplied by no Kummer-theoretic result found.

R.7 Why the generic-power model theory excludes this exponent

Bays, Kirby and Wilkie, J. London Math. Soc. (2010), prove a Schanuel property for raising to an *exponentially transcendental* power: for such λ and multiplicatively independent $\bar{y}$ one has $\mathrm{td}(\bar{y}, \bar{y}^{\lambda}) \geq n$. The result does not apply here, and the reason is worth recording precisely, because it explains a pattern rather than merely reporting a failure.

The exponent ϑ is exponentially *algebraic*. The pair $(z, w) = (\log 2, \vartheta)$ is a nonsingular solution of the system

$$e^z - 2 = 0, \qquad e^{zw} - 3 = 0,$$

whose Jacobian determinant is $6 \log 2 \neq 0$. So ϑ lies in exactly the exceptional set that the generic-power theorems are stated to avoid. The same holds for every version of the problem in this report, since all of them involve the same exponent. Generic o-minimal and model-theoretic machinery should therefore be treated as a structural guide here, not as a black box expected to apply.

R.8 Prioritized directions, and what the ceiling changes

The second survey graded the located literature by how directly it attaches to the obstruction. Its ranking, compressed, was: highest priority to the rank-two multiplicative group combined with specialized o-minimal or algebraic covering and S-unit rigidity; to a Laurent-style interpolation-determinant redesign of the semigroup determinant of Section 20; and to tropical or p-adic determinant amplification with modern p-adic logarithmic-form control. High priority to the Corvaja–Zannier recurrence and power-sum machinery, once an integral quotient has been manufactured, and to the Mahler–Ridout–Corvaja–Zannier exponential-nearness criterion, which supplies a sharp conditional target: construct one non-Pisot algebraic power sequence exponentially close to the integers. Medium priority to the explicit logarithmic-form technology of Matveev, Laurent, Gouillon and Mignotte–Voutier, excellent for finite verification and auxiliary lemmas but unlikely to cross the bilinear barrier alone, and to integer-valued o-minimal function theory, methodologically close but not fitted to a sparse multiplicative grid. Low priority, as a direct attack, to Skolem–Mahler–Lech, generic-power model theory and general Zilber–Pink.

Every item in that survey was recorded as entirely absent from the corpus at the time, and the present report's bibliography still does not carry most of them; they are listed here so that the gap is visible rather than implicit.

One entry of the ranking has since been settled negatively. The third highest-priority direction, tropical or p-adic determinant amplification, is retired at the term-wise level by Corollary 21.2: automatic divisibility can never contradict a correct archimedean bound, whatever the primes, nodes or columns. What survives of that direction is the cancellation problem localized by Proposition 21.3, which is a different and harder target than the one the ranking had in view.

R.9 Provenance of the conjecture

Alaoglu and Erdős proved that the ratio of consecutive superabundant numbers is prime, and, using the three-prime case that Siegel had claimed, that the ratio of consecutive colossally abundant numbers is a prime or a semiprime. Erdős and Nicolas later showed that exactly one, two, or four integers can attain the maximum in the colossally abundant construction, so a positive answer to the rational-power form of the conjecture bounds this to at most two and makes the ratio of consecutive colossally abundant numbers prime. Lagarias, Amer. Math. Monthly **109** (2002), 534–543, records the Robin/Riemann reformulation of the surrounding circle.

Caution R.2 (Unverified figure)**.** The first survey quoted a frequently repeated count of colossally abundant numbers below 10^{18}, attributed to Briggs, Experiment. Math. **15** (2006), 251–256. That figure could not be verified against the accessible text and is not repeated here. Any use of it should re-derive it.

R.10 The formalization landscape

Karatarakis and Wiedijk formalized the Gelfond–Schneider theorem in Lean 4, including a formalized Siegel lemma for algebraic integers and the auxiliary-function machinery; that development is the basis of the port used by this corpus and cited in Appendix K. Mathlib's Lindemann–Weierstrass development supplies analytic scaffolding only. No formalization of Baker's theorem, of the six exponentials theorem, or of linear forms in logarithms was found in any proof assistant, so the transcendence inputs isolated behind interfaces in this report would be greenfield work.

R.11 Summary of what the literature does not supply

Three gaps are unanimous across the surveyed material. No lower-bound technology handles linear or bilinear forms whose *coefficients* are themselves logarithms, which is the wall of Section 7. No counting theorem is sensitive to the arithmetic of denominators, which is what the compact-arc reformulation needs. And no unconditional algebraic-independence result reaches products of two logarithms, which is what the criterion of Section 12 would need. The conjecture remains open, and every route surveyed inherits the same rank-two-to-rank-one collapse.

Status: [classical/open background] throughout, with the two retractions marked as such. Nothing in this appendix is used as a step in any argument of this report; it records provenance and the state of the surrounding literature.

R1 Third continuation: outcome and claim ledger

This continuation starts from the complete inherited report rather than replacing it. In particular, it treats the exact free-monoid theorem, the determinant ceilings, the second-iterate kernel dichotomy, and the software verification through 2^{27} as established inputs with the trust labels already assigned there. The primary goal was again an unconditional proof of the Alaoglu–Erdős conjecture. No such proof was obtained. The new work does, however, isolate the conjecture inside the recent Matrix Coefficient framework for logarithm matrices, determine an exact Newton-polytope interpolation threshold for the hypothetical counterexample grid, and extend the directed-rounding exclusion range.

The most useful conceptual change is that the four-exponentials wall can now be stated as an *exact factor-four auxiliary-polynomial gap*. A hypothetical counterexample gives a singular 2×2 matrix of logarithms satisfying the "weak" condition of Dasgupta–Kakde, while its irrational rank-one factorization forbids every rational matrix coefficient from vanishing. On the associated $(2N) \times (2N)$ multiplicative grid, the least possible Bernstein–Kushnirenko degree of a nonzero vanishing polynomial is exactly $4N^2 - 1$. The auxiliary-polynomial condition that would imply the Matrix Coefficient Conjecture asks for degree below N^2. Thus the missing implication must gain a full factor of four over exact interpolation; no rearrangement of a generic interpolation determinant can supply it.

Result	Content	Status
Fixed-base matrix-coefficient equivalence	The Alaoglu–Erdős conjecture is exactly the Matrix Coefficient Conjecture restricted to logarithm matrices whose first column is $(\log 2, \log 3)^T$ and whose second column consists of logarithms of positive integers.	**[paper proof in this report]**
No rational coefficient	For a nonintegral solution β, the singular logarithm matrix factors as $(\log 2, \log 3)^T(1, \beta)$, and every rational matrix coefficient factors into two nonzero terms.	**[paper proof in this report]**
Exact sparse-interpolation threshold	On any $R \geq 2$ rational points of the irrational power curve $y = x^\beta$, the minimum two-dimensional BK-degree of a nonzero rational polynomial vanishing at all points is exactly $R - 1$.	**[paper proof in this report]**
Boundary-defect compression	The degree-$K^2 - 1$ graph interpolant has dyadic and triadic dilation defects divisible by the entire interior grid polynomial, leaving two residual polynomials of degree only $K - 1$ and an exact boundary commutator identity.	**[paper proof in this report]**
Factor-four obstruction	On the Dasgupta–Kakde grid $X(2N)$ attached to a counterexample, the exact minimum is $4N^2 - 1$, whereas their condition (m) requires a polynomial of BK-degree $< N^2$.	**[paper proof in this report]**
Directed finite extension	The same MPFR directed-rounding verifier used in the inherited report was replayed on the next dyadic ranges; the precise endpoint, chunk audit, and extremal records are stated in Section R5.	**[interval computation]**

The first four statements are conventional proofs in this report. They use no unproved transcendence conjecture. Their role is diagnostic: they do not prove that the singular matrix cannot exist, but they identify exactly what a successful Matrix Coefficient or auxiliary polynomial argument would have to manufacture.

R2 The fixed-base Matrix Coefficient formulation

R2.1 The logarithm matrix of a hypothetical counterexample

Let $M, A \in \mathbb{N}$ and put

$$L(M, A) := \begin{pmatrix} \log 2 & \log M \\ \log 3 & \log A \end{pmatrix}. \tag{225}$$

If $M = 2^\beta$ and $A = 3^\beta$, then

$$L(M,A) = \begin{pmatrix} \log 2 \\ \log 3 \end{pmatrix} \begin{pmatrix} 1 & \beta \end{pmatrix}, \qquad \det L(M,A) = 0. \tag{226}$$

The matrix has entries in the vector space of logarithms of algebraic numbers. It is therefore a particularly rigid 2×2 instance of the Matrix Coefficient Conjecture of Dasgupta and Kakde [37]. That conjecture says that a singular square matrix L of logarithms should admit nonzero rational vectors w, v with $w^T L v = 0$; in dimension two it is equivalent to the Four Exponentials Conjecture.

The rank-one factorization in (226) makes the coefficient condition completely explicit.

Proposition R2.1 (Exact coefficient factorization)**.** *Let $\beta \in \mathbb{R}$, $M = 2^\beta$, and $A = 3^\beta$. For $w = (r,s)^T$ and $v = (u,z)^T$ one has*

$$w^T L(M,A) v = (r \log 2 + s \log 3)(u + z\beta). \tag{227}$$

If $\beta \notin \mathbb{Q}$, then this number is nonzero for every pair of nonzero vectors $w, v \in \mathbb{Q}^2$.

Proof. The displayed identity follows immediately from (226). If $r \log 2 + s \log 3 = 0$ with $r, s \in \mathbb{Q}$, clearing denominators and exponentiating gives a multiplicative relation $2^a 3^b = 1$, hence $a = b = 0$ by unique factorization; thus $r = s = 0$. Likewise $u + z\beta = 0$ with $u, z \in \mathbb{Q}$ and $(u,z) \neq (0,0)$ would make β rational. □

A counterexample is irrational by the elementary rational-exponent argument (and transcendental by Gelfond–Schneider, as formalized in the inherited corpus). It would therefore give a singular logarithm matrix with *no* vanishing rational coefficient: not merely an application of the Matrix Coefficient Conjecture, but a literal counterexample to its 2×2 case.

R2.2 An exact equivalence, not only a conditional implication

Define the fixed-base matrix-coefficient statement $\mathrm{MC}_{2,3}$ as follows:

> For all positive integers M, A with $\det L(M,A) = 0$, there exist nonzero $w, v \in \mathbb{Q}^2$ such that $w^T L(M,A) v = 0$.

Theorem R2.2 (Fixed-base Matrix Coefficient equivalence)**.** *The Alaoglu–Erdős conjecture is equivalent to* $\mathrm{MC}_{2,3}$.

Proof. Assume first $\mathrm{MC}_{2,3}$. Let $2^x = M \in \mathbb{N}$ and $3^x = A \in \mathbb{N}$. Then $\det L(M,A) = 0$. By (227), a vanishing rational coefficient forces either a rational relation between $\log 2$ and $\log 3$, which is impossible, or a relation $u + zx = 0$ with $(u,z) \neq (0,0)$, which makes $x \in \mathbb{Q}$. Writing $x = p/q$ in lowest terms, integrality of both $2^{p/q}$ and $3^{p/q}$ forces $q = 1$, so $x \in \mathbb{Z}$.

Conversely assume the Alaoglu–Erdős conjecture and suppose $M, A \in \mathbb{N}$ make $L(M,A)$ singular. Put $x = \log M / \log 2$. Singularity gives $x = \log A / \log 3$, hence $2^x = M$ and $3^x = A$. The conjecture gives $x \in \mathbb{Z}$, and then $v = (-x, 1)^T$ is a nonzero rational vector satisfying $L(M,A)v = 0$; take any nonzero w. (The degenerate case $M = A = 1$ has $x = 0$ and is included.) □

This equivalence narrows the open transcendence input more sharply than the usual statement "Four Exponentials implies the conjecture." Only one fixed first column is needed, and the second column is restricted to logarithms of positive integers. Conversely, even this very thin positive-integral rank-one family already contains the full Alaoglu–Erdős problem.

R2.3 The multiplicative pairing and the absence of orthogonality

The Matrix Coefficient strategy is formulated multiplicatively. Let

$$x_1 = (2, M), \qquad x_2 = (3, A) \in (\mathbb{Q}^{\times}_{>0})^2 \tag{228}$$

and, for $a = (a_1, a_2)$ and $b = (b_1, b_2)$ in $\mathbb{Z}^2$, set

$$\langle a, b\rangle_X := \prod_{i,j=1}^{2} x_{i,j}^{a_i b_j}. \tag{229}$$

Under $M = 2^\beta$, $A = 3^\beta$ this pairing factors as

$$\langle a, b\rangle_X = \exp((a_1 \log 2 + a_2 \log 3)(b_1 + \beta b_2)). \tag{230}$$

Thus, when $\beta \notin \mathbb{Q}$, no nonzero $a, b \in \mathbb{Z}^2$ are orthogonal. In the notation of Dasgupta–Kakde, a hypothetical counterexample satisfies condition (w) (the logarithm matrix is singular) and fails condition (o) (there is no nontrivial multiplicative orthogonality). Their general theorem proves (m) $\Rightarrow$ (o); the missing implication (w) $\Rightarrow$ (m) is exactly what would rule out the counterexample.

The next section specializes their support-based condition (m) to the present rank-one grid and computes its exact interpolation threshold.

R3 Exact Newton-polytope complexity of the counterexample grid

R3.1 The grid as an irrational power curve

Assume for this section that $\beta \notin \mathbb{Q}$ and that $M = 2^\beta$, $A = 3^\beta$ are positive rational numbers; in the conjectural application they are integers. For $K \geq 1$ define

$$X_\beta(K) := \Big\{(2^i 3^j, M^i A^j) : 0 \leq i, j < K\Big\}. \tag{231}$$

Writing $t_{ij} = i \log 2 + j \log 3$ gives

$$(2^i 3^j, M^i A^j) = (e^{t_{ij}}, e^{\beta t_{ij}}). \tag{232}$$

The K^2 parameters t_{ij} are distinct, because equality would give $2^{i-i'} 3^{j-j'} = 1$.

For a nonzero polynomial

$$P(X, Y) = \sum_{(u,v) \in S(P)} c_{u,v} X^u Y^v$$

its restriction to the curve is the exponential polynomial

$$F_P(t) := P(e^t, e^{\beta t}) = \sum_{(u,v) \in S(P)} c_{u,v} e^{(u+\beta v)t}. \tag{233}$$

Irrationality of β makes all frequencies $u + \beta v$ distinct.

Lemma R3.1 (Support controls zeros on an irrational power curve)**.** *Let $\beta \notin \mathbb{Q}$ and let $P \in \mathbb{C}[X^{\pm 1}, Y^{\pm 1}]$ be nonzero with r nonzero monomials. Then F_P has at most $r-1$ distinct real zeros. Consequently, if P vanishes on R distinct positive points of $y = x^{\beta}$, then*

$$|S(P)| \geq R + 1. \tag{234}$$

Proof. Order the distinct real frequencies as $\lambda_1 < \cdots < \lambda_r$. The functions $e^{\lambda_1 t}, \ldots, e^{\lambda_r t}$ form an extended complete Chebyshev system on $\mathbb{R}$: for any $t_1 < \cdots < t_r$, the generalized exponential Vandermonde determinant $\det(e^{\lambda_j t_i})$ is strictly positive. If a nontrivial linear combination vanished at r distinct real points, its coefficient vector would lie in the kernel of this invertible evaluation matrix. This is impossible. The same argument works for complex coefficients. □

This is the one-dimensional zero estimate hidden inside the generalized-Vandermonde machinery of the inherited report. Here it is used not to estimate gaps between nodes but to constrain the Newton support of an auxiliary polynomial.

R3.2 BK-degree and a sharp lattice-point bound

Following Dasgupta–Kakde, divide P by its greatest common monomial, which does not change its zero set in $(\mathbb{C}^{\times})^2$. Let $\Delta(P)$ be the convex hull of the remaining support and write

$$\mathrm{bkd}(P) := \max\{\deg P,\ 2\,\mathrm{Area}\,\Delta(P)\}. \tag{235}$$

For two variables this is exactly their Bernstein–Kushnirenko degree: the first term is the mixed volume with one generic linear polynomial, and the second is the normalized area $\mathrm{BK}(P) = 2!\,\mathrm{Area}\,\Delta(P)$.

Lemma R3.2 (Support cardinality versus BK-degree)**.** *Every nonzero two-variable polynomial satisfies*

$$|S(P)| \leq \mathrm{bkd}(P) + 2. \tag{236}$$

If $\Delta(P)$ is one-dimensional, the sharper bound $|S(P)| \leq \mathrm{bkd}(P) + 1$ holds.

Proof. Suppose first that $\Delta(P)$ is a two-dimensional lattice polygon. Let I and B be its numbers of interior and boundary lattice points. Pick's theorem gives

$$2\,\mathrm{Area}\,\Delta(P) = 2I + B - 2.$$

The support is contained in the lattice points of its convex hull, so

$$|S(P)| \leq I + B = 2\,\mathrm{Area}\,\Delta(P) + 2 - I \leq 2\,\mathrm{Area}\,\Delta(P) + 2 \leq \mathrm{bkd}(P) + 2.$$

If the hull is a segment, the number of lattice points on it is its lattice length plus one. After monomial normalization both endpoint coordinates are nonnegative and bounded by the total degree; hence the lattice length is at most $\deg P$. The point case is immediate. □

The inequality is nearly best possible for the precise reason relevant here: a thin triangle with vertices $(0,0)$, $(d,0)$, $(0,1)$ has normalized area and degree both equal to d and contains $d+2$ lattice points.

R3.3 The exact interpolation threshold

Theorem R3.3 (Exact BK interpolation threshold)**.** *Let $R \geq 2$, let $\beta \notin \mathbb{Q}$, and let $\xi_1, \dots, \xi_R$ be distinct positive rational numbers such that $\xi_i^\beta \in \mathbb{Q}$ for every i. Among all nonzero polynomials $P \in \mathbb{Q}[X, Y]$ vanishing at all points (ξ_i, ξ_i^β), the minimum possible BK-degree is exactly*

$$\min_P \operatorname{bkd}(P) = R - 1. \tag{237}$$

After clearing denominators, the same minimum is attained by a polynomial in $\mathbb{Z}[X, Y]$.

Proof. For the lower bound, Lemma R3.1 gives $|S(P)| \geq R + 1$. Lemma R3.2 therefore gives $R + 1 \leq \operatorname{bkd}(P) + 2$, or $\operatorname{bkd}(P) \geq R - 1$.

For the upper bound, let $Q \in \mathbb{Q}[X]$ be the Lagrange interpolation polynomial satisfying $Q(\xi_i) = \xi_i^\beta$ for all i. It has degree at most $R - 1$, and $P(X, Y) = Y - Q(X)$ vanishes at all the required points. Its Newton polygon lies in the thin triangle with vertices $(0, 0)$, $(R - 1, 0)$, and $(0, 1)$, so both its degree and normalized area are at most $R - 1$. Hence $\operatorname{bkd}(P) \leq R - 1$, proving equality. Rational coefficients may be cleared without changing the support. □

The theorem is an exact result, not an asymptotic estimate. It also explains why the constant cannot be improved by a more clever choice of support: the elementary interpolant $Y - Q(X)$ already attains the optimum. Total positivity says that every polynomial with smaller Newton complexity has too few independent frequencies to vanish at all the points.

Corollary R3.4 (Full support and alternating coefficient signs)**.** *Under the hypotheses of Theorem R3.3, the interpolation polynomial*

$$Q(X) = q_0 + q_1 X + \cdots + q_{R-1} X^{R-1}$$

has degree exactly $R - 1$ and every coefficient q_k is nonzero. Moreover, if the frequencies $0, 1, \dots, R - 1, \beta$ are put in increasing order, then the coefficients of $Y - Q(X)$, put in the corresponding order, alternate in sign.

Proof. The polynomial $Y - Q(X)$ has at most $R + 1$ monomials, while Lemma R3.1 says that every vanishing polynomial has at least $R + 1$. Thus all R possible coefficients of Q occur and the leading one is nonzero. For the sign statement, form the $R \times (R + 1)$ evaluation matrix whose columns are the functions x^λ for the ordered frequencies. Every maximal minor is strictly positive by total positivity. The signed maximal minors give a generator of its one-dimensional kernel, and therefore alternate in sign. □

This sign statement is the support-theoretic shadow of the arbitrary-node divided-difference identities in the inherited report. Here it follows without estimating any gap: the entire coefficient vector is forced by strict total positivity.

Corollary R3.5 (Exact threshold on the counterexample grid)**.** *For $K \geq 2$, if $M = 2^\beta$ and $A = 3^\beta$ are rational and $\beta \notin \mathbb{Q}$, then*

$$\min\left\{\operatorname{bkd}(P) : 0 \neq P \in \mathbb{Q}[X, Y],\ P|_{X_\beta(K)} = 0\right\} = K^2 - 1. \tag{238}$$

Proof. The first coordinates $2^i 3^j$ are K^2 distinct positive rationals and their β-th powers are the rational second coordinates $M^i A^j$. Apply Theorem R3.3 with $R = K^2$. □

R3.4 Boundary compression of the multiplicative interpolation problem

The optimal interpolant contains more structure than its Newton polygon records. Let Q_K be the unique polynomial of degree at most K^2-1 satisfying

$$Q_K(2^i3^j) = M^iA^j, \qquad 0 \le i,j < K. \tag{239}$$

Under the irrational power-curve hypothesis, Corollary R3.4 shows that $\deg Q_K = K^2-1$. Define the rectangular node polynomials

$$\Pi_{a,b}(X) := \prod_{0\le i<a}\prod_{0\le j<b}(X-2^i3^j), \qquad \Pi_{0,b} = \Pi_{a,0} = 1. \tag{240}$$

Theorem R3.6 (Dilation-defect factorization)**.** *For every $K \ge 2$ there are unique polynomials $U_K, V_K \in \mathbb{Q}[X]$ of degree exactly $K-1$ such that*

$$Q_K(2X) - MQ_K(X) = \Pi_{K-1,K}(X)U_K(X), \tag{241}$$
$$Q_K(3X) - AQ_K(X) = \Pi_{K,K-1}(X)V_K(X). \tag{242}$$

If q_K is the leading coefficient of Q_K, then the leading coefficients of U_K, V_K are

$$q_K(2^{K^2-1}-M), \qquad q_K(3^{K^2-1}-A), \tag{243}$$

respectively.

Proof. For $0 \le i < K-1$ and $0 \le j < K$, equation (239) gives

$$Q_K(2\cdot 2^i3^j) - MQ_K(2^i3^j) = M^{i+1}A^j - MM^iA^j = 0.$$

The $K(K-1)$ roots are distinct, proving divisibility by $\Pi_{K-1,K}$. The left side of (241) has degree K^2-1, because its leading coefficient is $q_K(2^{K^2-1}-M)$ and $M = 2^\beta$ with $\beta \notin \mathbb{Z}$. Subtracting the divisor degree $K(K-1)$ leaves degree $K-1$. The triadic identity is symmetric. Uniqueness follows because the divisors are nonzero. □

The two dilation operators commute. Cancelling the common $(K-1)\times(K-1)$ interior node polynomial turns this tautological commutation into a low-degree boundary identity. Put

$$L_3^-(X) := \prod_{i=0}^{K-2}\left(X - \frac{2^i}{3}\right), \qquad L_3^+(X) := \prod_{i=0}^{K-2}\left(X - 2^i3^{K-1}\right),$$
$$L_2^-(X) := \prod_{j=0}^{K-2}\left(X - \frac{3^j}{2}\right), \qquad L_2^+(X) := \prod_{j=0}^{K-2}\left(X - 2^{K-1}3^j\right).$$

Theorem R3.7 (Boundary commutator identity)**.** *The residual polynomials of Theorem R3.6 satisfy*

$$\begin{aligned}3^{K(K-1)}L_3^-(X)U_K(3X) &- AL_3^+(X)U_K(X)\\ &= 2^{K(K-1)}L_2^-(X)V_K(2X) - ML_2^+(X)V_K(X).\end{aligned} \tag{244}$$

Both sides have degree at most $2K-2$.

Proof. Let $D_2 f(X) = f(2X) - M f(X)$ and $D_3 f(X) = f(3X) - A f(X)$. Direct expansion gives $D_2 D_3 = D_3 D_2$. Apply this to Q_K and substitute (241)–(242). The four node polynomials factor as

$$\begin{aligned}
\Pi_{K-1,K}(3X) &= 3^{K(K-1)}\Pi_{K-1,K-1}(X)L_3^-(X),\\
\Pi_{K-1,K}(X) &= \Pi_{K-1,K-1}(X)L_3^+(X),\\
\Pi_{K,K-1}(2X) &= 2^{K(K-1)}\Pi_{K-1,K-1}(X)L_2^-(X),\\
\Pi_{K,K-1}(X) &= \Pi_{K-1,K-1}(X)L_2^+(X).
\end{aligned}$$

Cancel the common nonzero factor $\Pi_{K-1,K-1}(X)$. □

This compression is exact: K^2 graph values leave only two degree-$(K-1)$ boundary defects. It does not by itself give a contradiction—the leading coefficients of the two sides of (244) agree identically—but it exposes a smaller arithmetic object than the full interpolation determinant. A promising next step is to study the primitive integer contents and the 2- and 3-adic Newton polygons of U_K and V_K. Any leading-order divisibility forced on these residuals would be genuinely boundary-sensitive and would not be covered by the inherited termwise min-plus ceiling.

R3.5 The factor-four obstruction to the support strategy

For a 2×2 logarithm matrix, condition (m) in Dasgupta–Kakde [37] asks, for all sufficiently large N, for a polynomial vanishing on $X(2N)$ and satisfying

$$\operatorname{bkd}(P) < N^2. \tag{245}$$

Under a hypothetical Alaoglu–Erdős counterexample, however, Corollary R3.5 gives the exact minimum

$$\operatorname{bkd}_{\min}(X_\beta(2N)) = 4N^2 - 1. \tag{246}$$

Thus condition (m) lies asymptotically a factor of four below the exact interpolation floor. This is the quantitative form, for this problem, of the missing implication (w) $\Rightarrow$ (m).

There is no paradox. Dasgupta–Kakde prove (m) $\Rightarrow$ (o), while Proposition R2.1 proves that condition (o) fails for an irrational counterexample. Therefore condition (m) must fail, and (246) says exactly how badly it fails. A proof of (w) $\Rightarrow$ (m) would derive a polynomial below this sharp floor from the singularity of the logarithm matrix, thereby contradicting the existence of the counterexample.

Interpretation. Exact interpolation, Siegel-lemma dimension counting, changing the shape of the Newton polygon, and adding generic monomials cannot bridge the factor four: the optimum $4N^2 - 1$ is already attained by the thin interpolant $Y - Q(X)$. The missing gain must be genuinely analytic or adelic. It must turn the vanishing determinant into simultaneous archimedean and non-archimedean smallness strong enough that a product-formula argument forces an auxiliary polynomial to vanish, rather than merely choosing a polynomial that vanishes by linear algebra. This is precisely the distinction emphasized by the “analytic reasons” question in Dasgupta–Kakde.

R3.6 Relation to the inherited determinant ceilings

The new threshold and the inherited min-plus ceiling are complementary.

- The min-plus ceiling says that entrywise p-adic divisibility of an integer determinant cannot, by itself, beat a correct archimedean determinant bound. Any successful determinant must create additional cancellation.
- The exact BK threshold says that support counting cannot, by itself, create the sparse auxiliary polynomial required by the Matrix Coefficient strategy. Any successful support argument must create analytic smallness that the product formula upgrades to exact vanishing.
- Both failures are sharp. The termwise divisor is an assignment minimum that is attained by a Leibniz term, while the BK threshold is attained by the Lagrange interpolant $Y - Q(X)$.

The two diagnostics therefore point to the same surviving object: a support-controlled interpolation determinant with *cancellation beyond tropical minima* at selected finite places and a simultaneous archimedean small-value estimate. Merely enlarging the node grid or repacking the exponent support cannot help; both operations scale the two sides together.

R4 Research targets exposed by the exact threshold

R4.1 A support-preserving analytic determinant

The most direct target is a two-variable analogue of the classical auxiliary-function construction in which the Newton support is fixed first and coefficient heights are controlled second.

Research target R4.1 (Analytic factor-four theorem)**.** Let $L(M, A)$ be singular. Construct, for all sufficiently large N, a nonzero $P_N \in \mathbb{Z}[X, Y]$ with $\operatorname{bkd}(P_N) < N^2$ such that a product-formula estimate forces P_N to vanish on $X(2N)$.

The word "forces" is essential. Theorem R3.3 proves that no such polynomial exists under an irrational counterexample, so the construction must use singularity to derive an impossible object. A linear-algebra interpolant begins at BK-degree $4N^2 - 1$ and therefore cannot be massaged into the target without a new source of equations or smallness.

A plausible architecture is:

1. choose a Newton polygon of normalized area $< N^2$ and solve a Siegel-lemma problem for a polynomial whose values, rather than being zero, are extremely small on $X(2N)$;
2. use the rank-one real logarithm relation to amplify archimedean smallness along the ordered curve;
3. use the exact prime-valuation grid to obtain p-adic smallness from cancellation, not from the termwise assignment divisor excluded by the min-plus ceiling;
4. apply the product formula to each algebraic value $P(x)$.

The difficult step is the third. The inherited report's first tied-layer calculation shows that the most obvious cancellation is lower order, so a successful construction must align many tied layers coherently.

R4.2 A coefficient-height version of the exact threshold

The exact interpolant $P = Y - Q(X)$ has optimal BK-degree but potentially enormous coefficient height. The next quantitative invariant to compute is therefore

$$H_K := \min\Big\{h(P) : P \in \mathbb{Z}[X,Y] \setminus \{0\},\ P|_{X_\beta(K)} = 0,\ \mathrm{bkd}(P) = K^2 - 1\Big\}, \tag{247}$$

where $h(P)$ is logarithmic projective coefficient height. The leading coefficient of the Lagrange interpolant is the arbitrary-node divided difference already studied in the inherited report, so its denominator is controlled by a generalized Vandermonde product. What is new in (247) is the optimization over *all* thin Newton polygons and all integer kernel vectors, not only the standard monomial basis.

A lower bound of order K^3 or larger, uniform over supports, would make precise why exact interpolation is arithmetically expensive. More importantly, a support for which the height drops anomalously could identify the cancellation geometry needed in Target R4.1.

R4.3 Formalization targets

The matrix-coefficient reduction is elementary enough for immediate formalization. A natural module order is:

1. `MatrixCoefficientReduction`: define $L(M, A)$, prove (227), and prove Theorem R2.2;
2. `IrrationalPowerCurveZeros`: package the existing generalized-Vandermonde positivity as the support zero bound of Lemma R3.1;
3. `NewtonPolygonSupport`: formalize the lattice-point estimate $|S(P)| \le \mathrm{bkd}(P)+2$, using Pick's theorem or an equivalent lattice-polygon count;
4. `ExactBKInterpolation`: formalize Lagrange interpolation and Theorem R3.3;
5. `MatrixCoefficientGrid`: instantiate the theorem on $X_\beta(K)$ and record the factor-four corollary.

The only potentially substantial library gap is Pick's theorem. For the exact threshold it can be avoided by proving the weaker but sufficient inequality directly for convex lattice polygons through a triangulation into primitive lattice triangles; the rest is elementary linear algebra and the Chebyshev-system theorem already represented in the corpus.

R5 Directed-rounding extension through 2^{29}

The inherited report certified all non-power inputs $m < 2^{27}$ at the MPFR software-trust level. I extracted the literal C verifier reproduced in Appendix I, verified that its SHA-256 digest is `f888c0d89be9366f01e2d26f339dcbe446556d39d4c5dbcf8f579ec0d0052e1d` ,

compiled it against MPFR 4.2.2, and scanned the two next dyadic ranges. The first range $[2^{27}, 2^{28})$ was run at 128 bits; the second, $[2^{28}, 2^{29})$, at 96 bits. Both precisions are logically exact for the claim being made: MPFR rounds every endpoint in the specified outward direction, and an input is counted only after both strict logarithmic inequalities are positive. Reducing the precision can widen an enclosure and create a reported failure, but cannot turn an uncertified input into a proof. The ordinary `long double` value remains only a locator for the neighboring integer and has no deductive role.

The recorded environment was Linux on x86-64, GCC 14.2.0, MPFR 4.2.2, and five OpenMP worker threads. All integers passed to MPFR were exactly representable at the selected precisions: through the endpoint 2^{29}, the neighboring output integer is below $3^{29} < 2^{46}$.

Status: [interval computation] — the extension is a directed-rounding software proof, not a Lean kernel theorem. The trusted base is the displayed C source, GCC, OpenMP, GMP/MPFR, the minimal MPFR ABI declarations used when the development header is absent, and the host hardware. The source digest agrees with the corrected literal-source digest in the inherited continuation.

R5.1 Audit of $[2^{27}, 2^{28})$

The range contains 2^{27} integers and exactly one power of two, its left endpoint. Thus the required checked count is $2^{27} - 1 = 134{,}217{,}727$. Sixteen chunks of width 2^{23} were run at 128 bits. The concatenated canonical log has SHA-256 digest

`187d860b93ae9843a466a1613c115f2a4f899db8a00be22b963725953b30eca6` .

Every chunk returned zero failures, and the counts sum exactly to the required total.

Table 18: Directed-rounding audit on $[2^{27}, 2^{28})$.

Chunk	Integer range	Checked	Smallest lower gap	Smallest upper gap
0	134,217,728–142,606,335	8,388,607	2.8148×10^{-20}	2.8468×10^{-21}
1	142,606,336–150,994,943	8,388,608	3.5925×10^{-21}	6.2726×10^{-22}
2	150,994,944–159,383,551	8,388,608	3.7823×10^{-21}	1.6063×10^{-20}
3	159,383,552–167,772,159	8,388,608	3.8468×10^{-21}	7.5693×10^{-21}
4	167,772,160–176,160,767	8,388,608	8.6652×10^{-21}	1.4331×10^{-20}
5	176,160,768–184,549,375	8,388,608	1.8762×10^{-20}	5.4449×10^{-22}
6	184,549,376–192,937,983	8,388,608	3.0851×10^{-22}	7.8715×10^{-21}
7	192,937,984–201,326,591	8,388,608	1.6385×10^{-21}	1.1766×10^{-20}
8	201,326,592–209,715,199	8,388,608	9.3056×10^{-22}	1.3699×10^{-21}
9	209,715,200–218,103,807	8,388,608	4.1009×10^{-22}	1.6788×10^{-21}
10	218,103,808–226,492,415	8,388,608	3.6380×10^{-21}	2.8733×10^{-21}
11	226,492,416–234,881,023	8,388,608	1.9530×10^{-22}	6.3473×10^{-21}
12	234,881,024–243,269,631	8,388,608	3.8124×10^{-21}	3.7502×10^{-21}
13	243,269,632–251,658,239	8,388,608	3.9710×10^{-21}	5.0056×10^{-21}
14	251,658,240–260,046,847	8,388,608	6.9901×10^{-21}	8.3073×10^{-22}

Chunk	Integer range	Checked	Smallest lower gap	Smallest upper gap
15	260,046,848–268,435,455	8,388,608	1.3740×10^{-21}	9.4093×10^{-22}
	total	**134,217,727**		

R5.2 Audit of $[2^{28}, 2^{29})$

This range contains 2^{28} integers and again exactly one power of two. The required checked count is $2^{28} - 1 = 268{,}435{,}455$. Sixteen chunks of width 2^{24} were run at 96 bits. The concatenated log has SHA-256 digest

`5edd9a63f1229ad688be78faeff3b120bfcf557444194c5882f451ef3f68477c` .

Every chunk returned zero failures, and the checked counts again agree with the combinatorial total.

Table 19: Directed-rounding audit on $[2^{28}, 2^{29})$.

Chunk	Integer range	Checked	Smallest lower gap	Smallest upper gap
0	268,435,456–285,212,671	16,777,215	1.3357×10^{-21}	2.8468×10^{-21}
1	285,212,672–301,989,887	16,777,216	1.7754×10^{-21}	6.2726×10^{-22}
2	301,989,888–318,767,103	16,777,216	5.1906×10^{-22}	3.6531×10^{-22}
3	318,767,104–335,544,319	16,777,216	3.8468×10^{-21}	1.9512×10^{-21}
4	335,544,320–352,321,535	16,777,216	1.4372×10^{-22}	1.9892×10^{-21}
5	352,321,536–369,098,751	16,777,216	5.1428×10^{-22}	1.3160×10^{-22}
6	369,098,752–385,875,967	16,777,216	3.0851×10^{-22}	8.1046×10^{-22}
7	385,875,968–402,653,183	16,777,216	4.9130×10^{-24}	8.1143×10^{-22}
8	402,653,184–419,430,399	16,777,216	9.3056×10^{-22}	4.7107×10^{-22}
9	419,430,400–436,207,615	16,777,216	4.1009×10^{-22}	2.5129×10^{-22}
10	436,207,616–452,984,831	16,777,216	1.2750×10^{-21}	2.8218×10^{-22}
11	452,984,832–469,762,047	16,777,216	1.9530×10^{-22}	7.4586×10^{-22}
12	469,762,048–486,539,263	16,777,216	4.3653×10^{-22}	2.1994×10^{-22}
13	486,539,264–503,316,479	16,777,216	5.7804×10^{-22}	5.2103×10^{-23}
14	503,316,480–520,093,695	16,777,216	1.1154×10^{-21}	4.2736×10^{-22}
15	520,093,696–536,870,911	16,777,216	6.0230×10^{-22}	9.3067×10^{-22}
	total	**268,435,455**		

R5.3 Extremal records and high-precision replay

Across both new dyadic ranges, the smallest lower logarithmic margin recorded by the range scan is

$$4.9130004741674945 \times 10^{-24} \quad \text{at} \quad (m, n) = (388{,}909{,}543,\ 41{,}170{,}668{,}399{,}174), \tag{248}$$

and the smallest upper margin is

$$5.2102685068037388 \times 10^{-23} \quad \text{at} \quad (m, n) = (501{,}373{,}708,\ 61{,}578{,}600{,}753{,}798). \tag{249}$$

Both one-input cases were replayed at 512 bits. The lower-record replay gives the sharper directed margin

$$4.9141462461940518 \times 10^{-24},$$

and a 140-decimal evaluation gives

$$(388{,}909{,}543)^{\vartheta} - 41{,}170{,}668{,}399{,}174 = 2.91884164346830087671035555\ldots \times 10^{-10}.$$

The upper-record replay gives

$$5.2103670350428438 \times 10^{-23},$$

with actual gap

$$61{,}578{,}600{,}753{,}799 - (501{,}373{,}708)^{\vartheta} = 4.62884536546002680350749500\ldots \times 10^{-9}.$$

Thus the higher-precision runs preserve the neighboring integer and the strict sign while refining the last digits of the coarser 96-bit enclosures. Both replays report zero failures.

Theorem R5.1 (Software-level finite exclusion through 2^{29})**.** *At the stated MPFR software-trust level, for every integer* $1 \le m < 2^{29}$,

$$m^{\vartheta} \in \mathbb{N} \quad \Longrightarrow \quad m \textit{ is a power of two.}$$

Consequently every nonintegral real x *satisfying* $2^x, 3^x \in \mathbb{Z}$ *obeys*

$$x > 29. \tag{250}$$

Proof. The inherited scans cover $m < 2^{27}$, Table 18 covers the next dyadic range, and Table 19 covers the following one. At each non-power input the directed interval for $\vartheta \log m$ lies strictly between the directed logarithms of two consecutive integers. If $m = 2^x$ came from a nonintegral solution with $x < 29$, then $m < 2^{29}$ and $m^{\vartheta} = 3^x$ would be an integer at a non-power input, contradicting the scan. Equality $m = 2^{29}$ forces $x = 29$ by injectivity of the exponential, so a nonintegral solution must satisfy $x > 29$. □

The improvement from 27 to 29 remains computational rather than conceptual. Its value is that every hypothetical primitive generator in the exact conditional theory now begins beyond 29, which can be inserted into all finite-order spacing and determinant optimizations. It does not change the kernel-verified floor $x \ge 12$ until the directed inequalities are converted to exact rational certificates and replayed in Lean.

R6 Conclusions of the third continuation

The Alaoglu–Erdős conjecture remains open. The work in this continuation does not hide that fact behind a conditional theorem or a large finite computation. Its strongest unconditional mathematical conclusions are instead structural.

First, the conjecture is exactly the positive-integral, fixed-first-column instance $\mathrm{MC}_{2,3}$ of the Matrix Coefficient Conjecture. A hypothetical counterexample produces the rank-one matrix

$$\begin{pmatrix} \log 2 \\ \log 3 \end{pmatrix} (1, \beta),$$

and every rational matrix coefficient splits into two factors that are separately nonzero. This pinpoints the obstruction more narrowly than the generic phrase "a case of Four Exponentials."

Second, the support-based auxiliary-polynomial strategy now has an exact calibration. On R rational points of the irrational power curve, the minimum BK-degree of a rational vanishing polynomial is $R-1$, attained by the thin Lagrange interpolant $Y-Q(X)$. On the $(2N)^2$ Matrix Coefficient grid this is $4N^2-1$, while the desired support condition is $< N^2$. The factor four is therefore not a defect of a particular determinant estimate; it is the exact interpolation barrier. Any proof along this route must use analytic or adelic smallness to manufacture an impossible polynomial, not merely optimize the monomial set.

Third, the optimal graph interpolant has a new boundary normal form. Its dyadic and triadic dilation defects contain the complete interior grid polynomial, leaving residuals of degree only $K-1$ tied by the boundary commutator identity (244). This is the most concrete new arithmetic object produced by the continuation. It separates the exact multiplicative recurrence in the interior from the $O(K)$ boundary data where finite interpolation must fail. The primitive contents and local Newton polygons of these residuals should be computed before returning to larger determinant families.

Finally, the directed-rounding scan moves the software-level lower bound from $x > 27$ to $x > 29$. This is useful for calibrating finite-order inequalities and future certificates, but it remains qualitatively separate from the transcendence problem and does not alter the kernel-verified bound $x \geq 12$.

The resulting priority order is:

1. determine exact or asymptotic p-adic valuations of the primitive integer forms of U_K and V_K, especially whether the boundary commutator forces cancellation beyond the termwise min-plus ceiling;
2. attack the singular-to-sparse implication of the Matrix Coefficient program with a support-preserving product-formula construction, using the exact factor-four threshold as the required quantitative benchmark;
3. formalize the fixed-base matrix-coefficient equivalence, the zero/support theorem, and the exact BK interpolation threshold in Lean;
4. convert the finite MPFR intervals into compact exact rational certificates and replay them in the kernel rather than extending the raw scan indefinitely.

Bottom line. No complete proof was found. The continuation nevertheless adds three exact pieces of mathematics—the fixed-base Matrix Coefficient equivalence, the sharp Newton-polytope interpolation threshold, and the boundary-defect compression—together with a reproducible finite exclusion through 2^{29}. The remaining wall is now visible both algebraically and quantitatively: one must turn logarithmic singularity into an auxiliary object lying a factor of four below the ordinary interpolation floor, or extract leading p-adic cancellation from the $O(K)$ boundary residuals.

P1 Fourth continuation: polynomial approximation, outcome, and claim ledger

This continuation investigates whether approximation theory can convert the integrality of m^ϑ at candidate integers into an auxiliary-polynomial contradiction. The starting point is deliberately broader than ordinary Lagrange interpolation. Uniform minimax approximation, direct-scale rational polynomials, integer-valued polynomials, Müntz systems, Padé and Hermite–Padé contact, and constrained Remez methods are all examined with the arithmetic denominator included in the estimate rather than appended at the end.

No complete proof of the Alaoglu–Erdős conjecture was obtained. The approximation program nevertheless produces three exact results which materially sharpen the inherited failure analysis.

1. A rational polynomial approximant on the normalized interval $[1,2]$ must beat the lattice spacing
$$\frac{1}{H3^T2^{Td}}$$
when its degree is d and its coefficient denominator is H. A finite-difference lower bound proves that this is impossible for every degree as soon as $T \geq 4$. Thus the most natural "normalize, approximate, clear denominators" strategy is not merely quantitatively weak; its required inequality has the wrong sign in every degree.

2. If $P \in \mathbb{Q}[X,Y]$ and $(a,b) \in \mathbb{Q}^2_{>0}$, then restriction to the transcendental one-parameter curve
$$z \longmapsto (ae^z, be^{\vartheta z})$$
has order of vanishing exactly equal to the ordinary algebraic multiplicity of P at (a,b). Consequently rational polynomials have no anomalous tangency to $Y = X^\vartheta$ at rational points. In a bidegree (d,e) rectangle, the largest possible rational Hermite–Padé contact is exactly $d+e$, not $(d+1)(e+1)-1$. This is an exact area-to-diameter collapse caused by the transcendence of ϑ.

3. There is an exact arithmetic Müntz certificate: an integer linear combination of the functions $x^{i+j\vartheta}$ which has norm below one on a dyadic interval vanishes at every candidate there, and hence bounds their number by the size of its support minus one. This is the approximation-theoretic dual of the generalized-Vandermonde determinant method. It identifies the viable remaining target: not ordinary real minimax approximation, but a small vector in an integral coefficient lattice with simultaneous archimedean and finite-prime control.

The following table separates exact results from diagnostics and proposed targets.

Item	Content	Status
Dyadic arithmetic-resolution lemma	Exact integer obtained by clearing a rational approximant at a candidate point; both polynomial and rational-residual versions are given.	[paper proof in this report]
Finite-difference approximation floor	An explicit all-degree lower bound for $\|t^\vartheta - p(t)\|_{[1,2]}$.	[paper proof in this report]
Normalization-tax theorem	For every $T \geq 4$, every degree d, and every rational coefficient denominator H, the approximation error is at least the arithmetic forcing threshold.	[paper proof in this report]
Direct-scale degree floor	Any rational polynomial which is close enough on $[2^T, 2^{T+1}]$ to force equality at candidates must have degree linear in T; the sharp asymptotic slope is governed by the Bernstein ellipse through the branch point at zero.	[paper proof in this report] plus classical approximation theory
Contact equals multiplicity	Rational algebraic curves have no excess contact with $Y = X^\vartheta$ at positive rational points.	[paper proof in this report]
Area-to-diameter collapse	Rational bidegree (d, e) Hermite–Padé contact is exactly $d+e$, while coefficients in $\mathbb{Q}(\vartheta)$ permit the generic order $(d+1)(e+1) - 1$.	[paper proof in this report]
Integer Müntz certificate	A norm-below-one integral exponential polynomial gives an exact candidate-count bound.	[paper proof in this report]
Recent literature audit	Approximation, Hermite–Padé, effective Diophantine approximation, and modern transcendence-measure papers from 2024–2026 are compared with the fixed-base obstruction.	[classical/open background]
Arithmetic minimax construction	Construct a nonzero integer Müntz polynomial or a rational numerator residual which is small enough adelically to cross the product-formula threshold.	[proposed target]

The proofs below use only the already established transcendence of $\vartheta = \log_2 3$, elementary finite differences, standard Chebyshev-system zero counting, and classical polynomial approximation. No unproved transcendence conjecture enters any statement tagged [paper proof in this report].

P2 Arithmetic resolution on a dyadic block

P2.1 The normalized candidate equation

Fix an integer $T \geq 0$, and let m be a candidate in the half-open dyadic block $[2^T, 2^{T+1})$. Put

$$u := \frac{m}{2^T} \in [1, 2), \qquad n := m^\vartheta \in \mathbb{N}. \tag{251}$$

Since $2^{T\vartheta} = 3^T$, the candidate identity becomes

$$n = 3^T u^\vartheta. \tag{252}$$

Analytically this normalization is ideal: the approximation interval no longer grows with T. Arithmetically it is expensive: a degree-d monomial in $u = m/2^T$ introduces a denominator as large as 2^{Td}. The next lemma records the exact cost.

Lemma P2.1 (Normalized polynomial arithmetic resolution). *Let $p \in \mathbb{Q}[t]$ have degree at most d, and let $H \in \mathbb{N}$ satisfy $Hp \in \mathbb{Z}[t]$. For every candidate pair (m, n) satisfying (251),*

$$H2^{Td}(n - 3^T p(u)) \in \mathbb{Z}. \tag{253}$$

Consequently, if

$$\|t^\vartheta - p(t)\|_{[1,2]} < \frac{1}{H3^T 2^{Td}}, \tag{254}$$

then $p(u) = u^\vartheta$ at every candidate in the block.

Proof. Write

$$Hp(t) = \sum_{k=0}^{d} a_k t^k, \qquad a_k \in \mathbb{Z}.$$

Then

$$H2^{Td}p(m/2^T) = \sum_{k=0}^{d} a_k m^k 2^{T(d-k)} \in \mathbb{Z}.$$

The first term $H2^{Td}n$ is also integral, proving (253). By (252), the absolute value of that integer is

$$H3^T 2^{Td} |u^\vartheta - p(u)|.$$

Under (254) it is strictly below one, hence is zero. □

Because the functions

$$1, t, t^2, \ldots, t^d, t^\vartheta$$

form a Chebyshev system on $(0, \infty)$, a nonzero difference $t^\vartheta - p(t)$ has at most $d + 1$ positive zeros. Thus a polynomial meeting (254) would imply

$$\#(\mathcal{C} \cap [2^T, 2^{T+1}]) \le d + 1. \tag{255}$$

The point of Section P3 is that the required polynomial does not exist for $T \ge 4$, regardless of d.

P2.2 Direct-scale rational polynomials

Normalizing first is not the only possible arithmetic arrangement. One may instead choose a rational polynomial in the physical variable x, where evaluation at an integer does not create the factor 2^{Td}.

Lemma P2.2 (Direct-scale arithmetic forcing). *Let $q \in \mathbb{Q}[X]$, $\deg q \le d$, and suppose $Hq \in \mathbb{Z}[X]$. If*

$$\|X^\vartheta - q(X)\|_{[2^T,2^{T+1}]} < \frac{1}{H}, \tag{256}$$

then $q(m) = m^\vartheta$ for every candidate integer m in that block. Necessarily,

$$H3^T E_d(\vartheta) < 1, \qquad E_d(\vartheta) := \inf_{\deg p \le d} \|t^\vartheta - p(t)\|_{[1,2]}. \tag{257}$$

Proof. At a candidate m, the number

$$H(m^{\vartheta} - q(m))$$

is an integer. Inequality (256) makes its absolute value less than one, so it vanishes. For the necessary condition, put

$$p(t) := 3^{-T} q(2^T t).$$

Then p is a real polynomial of degree at most d, and

$$\|t^{\vartheta} - p(t)\|_{[1,2]} = 3^{-T} \|X^{\vartheta} - q(X)\|_{[2^T, 2^{T+1}]} < \frac{1}{H 3^T}.$$

Taking the infimum over all real polynomials proves (257). □

The direct-scale formulation avoids the generic 2^{Td} denominator tax, so it is not excluded by the normalization theorem below. It pays in a different currency: degree. To make an error of order one while the function has size 3^T, a polynomial must have degree linear in T. Section P3.3 shows that this degree is far larger than the conditional candidate population it is supposed to contradict.

P2.3 The correct residual for rational approximation

A quotient error $|t^{\vartheta} - A(t)/B(t)|$ is not itself an algebraic integer. The quantity compatible with the candidate lattice is the numerator residual.

Lemma P2.3 (Rational numerator-residual resolution). *Let $A, B \in \mathbb{Z}[t]$ have degree at most d. For a candidate $m = 2^T u$, $n = m^{\vartheta}$,*

$$2^{Td}(B(u)n - 3^T A(u)) = 3^T 2^{Td}(B(u)u^{\vartheta} - A(u)) \in \mathbb{Z}. \tag{258}$$

Hence

$$\|B(t)t^{\vartheta} - A(t)\|_{[1,2]} < \frac{1}{3^T 2^{Td}} \tag{259}$$

forces $B(u)u^{\vartheta} = A(u)$ at every candidate in the block.

Proof. Both $2^{Td}A(m/2^T)$ and $2^{Td}B(m/2^T)$ are integers. Multiply the first by 3^T and the second by the integer n, then use $n = 3^T u^{\vartheta}$. The strict inequality (259) makes the resulting integer have absolute value less than one. □

If A and B have rational coefficients with a common denominator H, the right side of (259) is divided by H. The analytic quotient error satisfies

$$\left| t^{\vartheta} - \frac{A(t)}{B(t)} \right| = \frac{|B(t)t^{\vartheta} - A(t)|}{|B(t)|};$$

therefore poles or a large denominator polynomial may make the quotient look excellent while the arithmetic numerator remains much too large. Any rational-approximation proof must control the numerator residual and coefficient height simultaneously.

P3 Uniform polynomial approximation and the normalization tax

P3.1 An explicit finite-difference lower bound

The next estimate is elementary but has exactly the dependence needed for the arithmetic comparison. Recall

$$(\vartheta)_{\underline{r}} = \vartheta(\vartheta-1)\cdots(\vartheta-r+1).$$

Theorem P3.1 (All-degree finite-difference lower bound). *For $d \geq 1$, every real polynomial p of degree at most d satisfies*

$$\left\|t^{\vartheta} - p(t)\right\|_{[1,2]} \geq c_d, \qquad c_d := \frac{|\,(\vartheta)_{\underline{d+1}}\,|\, 2^{\vartheta-2d-2}}{(d+1)^{d+1}}. \tag{260}$$

For $d = 0$, the lower bound $E_0(\vartheta) \geq 1$ holds.

Proof. Let $h = 1/(d+1)$, let $f(t) = t^{\vartheta}$, and put $E = \|f - p\|_{[1,2]}$. Since $\Delta_h^{d+1} p(1) = 0$,

$$|\Delta_h^{d+1} f(1)| = |\Delta_h^{d+1}(f-p)(1)| \leq \sum_{j=0}^{d+1} \binom{d+1}{j} E = 2^{d+1} E.$$

The finite-difference mean-value theorem gives a $\xi \in (1, 2)$ such that

$$\Delta_h^{d+1} f(1) = h^{d+1} f^{(d+1)}(\xi) = h^{d+1}\,(\vartheta)_{\underline{d+1}}\,\xi^{\vartheta-d-1}.$$

For $d \geq 1$, the exponent $\vartheta - d - 1$ is negative, so $\xi^{\vartheta-d-1} \geq 2^{\vartheta-d-1}$. Combining the two displays and dividing by 2^{d+1} gives (260). If p is constant, the endpoint values of f are 1 and 3, so one of the two endpoint errors is at least one. $\square$

The bound is weaker than the sharp Bernstein-ellipse asymptotics, but it is explicit, valid for every degree, and strong enough to settle the normalized arithmetic comparison.

P3.2 No normalized rational polynomial crosses the lattice threshold

The simple rational bounds

$$\frac{3}{2} < \vartheta < \frac{8}{5} \tag{261}$$

follow from $2^3 < 3^2$ and $3^5 < 2^8$. They are sufficient for a completely rational proof of the next theorem.

Theorem P3.2 (The normalization-tax obstruction). *Let $T \geq 4$, $d \geq 0$, $H \geq 1$, and let $p \in \mathbb{Q}[t]$ have degree at most d with $Hp \in \mathbb{Z}[t]$. Then*

$$H 3^T 2^{Td} \left\|t^{\vartheta} - p(t)\right\|_{[1,2]} \geq 1. \tag{262}$$

In particular, the arithmetic forcing inequality (254) is impossible in every degree once $T \geq 4$.

Proof. The case $d = 0$ follows from Theorem P3.1, since the left side is at least $3^T \geq 81$. For $d \geq 1$, it is enough to treat $T = 4$ and $H = 1$. Define

$$L_d := 3^4 2^{4d} c_d = 81\, 2^{\vartheta-2} \frac{4^d |\,(\vartheta)_{\underline{d+1}}\,|}{(d+1)^{d+1}}. \tag{263}$$

For $d \geq 1$,

$$\frac{L_{d+1}}{L_d} = 4(d+1-\vartheta) \frac{(d+1)^{d+1}}{(d+2)^{d+2}}. \tag{264}$$

Using $\vartheta > 3/2$, this ratio is less than one for $d = 1, 2, 3, 4$: respectively it is bounded above by

$$\frac{8}{27}, \qquad \frac{162}{256}, \qquad \frac{2560}{3125}, \qquad \frac{43750}{46656}.$$

Thus $L_1 > \cdots > L_5$.

Next, $2^{\vartheta-2} > 2^{-1/2} > 7/10$. From (261),

$$\begin{aligned} L_5 &> 81 \cdot \frac{7}{10} \cdot \frac{4^5}{6^6} \left(\frac{3}{2}\right) \left(\frac{1}{2}\right) \left(\frac{2}{5}\right) \left(\frac{7}{5}\right) \left(\frac{12}{5}\right) \left(\frac{17}{5}\right) \\ &= \frac{13328}{3125} > 4. \end{aligned}$$

For the next ratio, $\vartheta < 8/5$ gives

$$\frac{L_6}{L_5} > 4\left(6 - \frac{8}{5}\right) \frac{6^6}{7^7} = \frac{4105728}{4117715} > \frac{99}{100}.$$

Hence $L_6 > 1$. Also

$$\frac{L_7}{L_6} > 4\left(7 - \frac{8}{5}\right) \frac{7^7}{8^8} = \frac{22235661}{20971520} > 1.$$

For $d \geq 7$, put $n = d + 2$. The inequality

$$\log(1-y) > -\frac{y}{1-y} \qquad (0 < y < 1)$$

gives $(1 - 1/n)^{n-1} > e^{-1}$. Moreover

$$e = \sum_{k=0}^{\infty} \frac{1}{k!} < 2 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} \sum_{j=0}^{\infty} 4^{-j} = \frac{49}{18} < \frac{11}{4},$$

so $e^{-1} > 4/11$. Therefore

$$\begin{aligned} \frac{L_{d+1}}{L_d} &> 4 \frac{d+1-8/5}{d+2} \cdot \frac{4}{11} \\ &= \frac{16(5d-3)}{55(d+2)} \geq \frac{512}{495} > 1. \end{aligned}$$

The first five terms satisfy $L_1 > \cdots > L_5 > 4$; independently, the displayed lower bound gives $L_6 > 1$, and every ratio from L_6 onward is greater than one. Hence $L_d > 1$ for every $d \geq 1$. Increasing T or H can only increase the left side of (262). $\square$

Interpretation. Uniform approximation of t^ϑ on a fixed interval is exponentially good in the degree. But after the substitution $t = m/2^T$, clearing a generic rational degree-d polynomial costs 2^{Td}. Theorem P3.2 proves that the arithmetic resolution decays faster than the approximation error for every degree, already at $T = 4$. The hypothetical counterexample range begins beyond $T = 29$ by Theorem R5.1, so this route fails with an enormous margin where it matters.

For calibration, set $c_0 := 1$. Table 20 gives the base-ten logarithm of the lower bound $3^{29}2^{29d}c_d$. The table is not used in the proof.

d	$\log_{10} c_d$	$\log_{10}(3^{29}2^{29d}c_d)$
0	0.000	13.837
1	−1.362	21.204
2	−3.175	28.121
4	−5.909	42.847
5	−7.152	50.334
8	−10.692	72.984
16	−19.613	133.901
32	−36.823	256.369
64	−70.615	501.933

Table 20: The normalized arithmetic gap at the first currently admissible block scale.

P3.3 The direct-scale degree floor

The direct-scale strategy of Lemma P2.2 avoids the factor 2^{Td}, but Theorem P3.1 still gives an explicit necessary degree condition:

$$H3^T c_d < 1. \tag{265}$$

The gamma identity

$$|\,(\vartheta)_{\underline{d+1}}\,| = \frac{\Gamma(d+1-\vartheta)}{|\Gamma(-\vartheta)|} \tag{266}$$

and Stirling's formula yield

$$c_d = C_\vartheta\, d^{-\vartheta-1/2}(4e)^{-d}\big(1 + O(d^{-1})\big) \tag{267}$$

for an explicit positive constant C_ϑ. Consequently any direct-scale polynomial meeting (256) must satisfy

$$d \geq \frac{\log 3}{\log(4e)}T - O(\log T) = 0.460384\ldots\, T - O(\log T). \tag{268}$$

A larger denominator H only raises the required degree.

This elementary slope can be sharpened by classical complex approximation. Map $[1, 2]$ to $[-1, 1]$ by $s = 2t - 3$. The branch point $t = 0$ becomes $s = -3$. The largest Bernstein ellipse avoiding that singularity has parameter

$$\rho_* = 3 + 2\sqrt{2}. \tag{269}$$

Indeed, the leftmost point of the ellipse of parameter ρ is $-\frac{1}{2}(\rho+\rho^{-1})$, and equality with -3 gives $\rho = 3+2\sqrt{2}$. Bernstein's theorem therefore gives

$$\limsup_{d\to\infty} E_d(\vartheta)^{1/d} = \rho_*^{-1} \tag{270}$$

(and in this algebraic-branch-point case the root rate is a limit). See [38]. The sharp asymptotic degree floor for $3^T E_d(\vartheta) < 1/H$ is consequently

$$d \geq \frac{\log 3}{\log(3+2\sqrt{2})} T - o(T) = 0.623238\ldots\, T - o(T). \tag{271}$$

For the half-open block $B(T) := \#(\mathcal{C} \cap [2^T, 2^{T+1}))$, failure of the conjecture gives the inherited exact formula

$$B(T) = 1 + \left\lfloor \frac{T+1}{\beta} \right\rfloor, \tag{272}$$

where β is the least nonintegral solution. The directed finite exclusion gives $\beta > 29$, so

$$B(T) \leq 1 + \frac{T+1}{29}.$$

A direct-scale polynomial of degree d which forces equality at candidates only gives $B(T) \leq d+1$. But the necessary degree has slope at least $0.460384\ldots$, and the true approximation slope is $0.623238\ldots$, whereas the conditional population has slope below $1/29$. Thus the finite-difference calibration misses by a factor exceeding 13.35, and the sharp Bernstein calibration misses by a factor exceeding 18.07. Better minimax constants cannot reverse a linear-slope mismatch of this size.

At $T = 29$, the explicit lower bound already requires $d \geq 11$. Indeed, $c_0 > c_1 > c_2 > \cdots$ (use the quotient of consecutive terms in (260)), and

$$3^{29} c_{10} > 7.39, \qquad 3^{29} c_{11} < 0.557.$$

The exact numerical values are merely illustrative. The asymptotic comparison is the structural point.

P3.4 Using all trivial powers as interpolation nodes

There is a more aggressive polynomial idea. Instead of working block by block, interpolate through every known trivial solution

$$(1,1), (2,3), (2^2,3^2), \ldots, (2^N, 3^N).$$

Let $Q_N \in \mathbb{Q}[X]$ be the unique polynomial of degree at most N with

$$Q_N(2^j) = 3^j, \qquad 0 \leq j \leq N. \tag{273}$$

Its Newton expansion is

$$Q_N(X) = \sum_{r=0}^{N} a_r \prod_{j=0}^{r-1} (X - 2^j), \qquad a_r = \frac{\prod_{j=0}^{r-1}(3-2^j)}{\prod_{j=0}^{r-1}(2^r - 2^j)}, \tag{274}$$

with the empty products equal to one. Formula (274) is the Newton divided-difference formula for geometric nodes. One quick proof observes that, as a polynomial in a formal value q, the r-th divided difference of the data $(2^k, q^k)$ vanishes at $q = 1, 2, 2^2, \dots, 2^{r-1}$, because in those cases the data come from a polynomial in X of degree below r; its leading coefficient fixes the displayed denominator.

The system

$$1, X, \dots, X^N, X^\vartheta$$

is Chebyshev, so $X^\vartheta - Q_N(X)$ has at most $N+1$ positive zeros. It already has exactly the $N+1$ trivial zeros in (273). Hence

$$m^\vartheta \neq Q_N(m) \quad \text{for every positive nonpower of two } m. \tag{275}$$

This saturates the zero budget perfectly: an arithmetic estimate forcing equality at one additional candidate would prove the conjecture.

The obstruction is capacity. On the expanding interval $[1, X]$, the singular point zero maps to

$$-\frac{X+1}{X-1}$$

in the normalized variable. The associated maximal Bernstein parameter is

$$\rho_X = \frac{\sqrt{X}+1}{\sqrt{X}-1} = 1 + \frac{2}{\sqrt{X}-1}. \tag{276}$$

For $X = 2^N$, this tends to one exponentially fast in N. More precisely, $\rho_X^{-N} = \exp(-O(N/\sqrt{X})) = 1 - o(1)$, so degree $N = \log_2 X$ obtains no geometric decay from analyticity. Equivalently, after dividing by X^ϑ and rescaling to $[1/X, 1]$, the problem approaches approximation of the endpoint-singular function y^ϑ on $[0, 1]$, where classical best-polynomial convergence is algebraic rather than geometric. The exact Newton denominators in (274) grow rapidly at the same time. Thus the scheme spends its entire degree budget creating the known zeros while the approximation interval becomes exponentially ill-conditioned. This is a capacity diagnostic, not a new sharp asymptotic theorem; it shows that the algebraically optimal interpolant comes with no available arithmetic smallness estimate.

P4 Local rigidity: contact equals multiplicity

The approximation routes above are global. Padé and Hermite–Padé methods instead seek high-order contact at one point. Here the transcendence of ϑ produces an exact and unexpectedly strong rigidity theorem.

P4.1 The local contact theorem

For a polynomial P and a point (a, b), write $\operatorname{mult}_{(a,b)} P$ for the least total degree of a nonzero term in the Taylor expansion of $P(a+U, b+V)$.

Theorem P4.1 (Rational contact equals algebraic multiplicity)**.** *Let τ be transcendental, let $0 \neq P \in \mathbb{Q}[X, Y]$, and let $(a, b) \in \mathbb{Q}^2_{>0}$. If*

$$s = \operatorname{mult}_{(a,b)} P,$$

then

$$\operatorname{ord}_{z=0} P(ae^{z}, be^{\tau z}) = s. \tag{277}$$

In particular the statement applies to $\tau = \vartheta$.

Proof. Expand

$$P(a+U, b+V) = H_s(U,V) + H_{s+1}(U,V) + \cdots,$$

where $H_s \neq 0$ is homogeneous of degree s and has rational coefficients. Since

$$a(e^{z}-1) = az + O(z^2), \qquad b(e^{\tau z}-1) = b\tau z + O(z^2),$$

the coefficient of z^s in $P(ae^{z}, be^{\tau z})$ is

$$H_s(a, b\tau). \tag{278}$$

Writing

$$H_s(U,V) = \sum_{j=0}^{s} h_j U^{s-j} V^j$$

shows that $H_s(a, b\tau)$ is a nonzero polynomial in the transcendental number τ with rational coefficients: not all $h_j a^{s-j} b^j$ vanish. Hence (278) is nonzero, and the order is exactly s. ☐

The theorem says that a rational algebraic curve cannot be tangent to the irrational-power curve to an order not already visible in its algebraic singularity. Transcendence eliminates all cancellation among the different powers of ϑ.

Corollary P4.2 (Simple intersections at candidate points)**.** *Let $a > 0$ and $b = a^{\vartheta}$ be rational. If $Q \in \mathbb{Q}[X]$ and $Q(a) = b$, then*

$$\operatorname{ord}_{X=a}\big(X^{\vartheta} - Q(X)\big) = 1. \tag{279}$$

Equivalently, the graph polynomial $Y - Q(X)$ meets $Y = X^{\vartheta}$ transversely at every rational point of intersection.

Proof. The polynomial $P(X,Y) = Y - Q(X)$ has multiplicity one at (a,b), because $\partial P/\partial Y = 1$. Apply Theorem P4.1 after writing $X = ae^{z}$. Directly, the derivative of the difference at a is

$$\vartheta \frac{b}{a} - Q'(a),$$

a nonzero number because its first term is a nonzero rational multiple of the transcendental number ϑ, while the second is rational. ☐

Thus rational Hermite interpolation cannot double a candidate node "for free." Every additional prescribed derivative consumes genuine algebraic multiplicity and therefore ordinary degree.

P4.2 Exact rectangular contact

Theorem P4.3 (Exact rational bidegree contact)**.** *Let $a, b \in \mathbb{Q}_{>0}$, let τ be transcendental, and let $0 \neq P \in \mathbb{Q}[X, Y]$ satisfy*

$$\deg_X P \le d, \qquad \deg_Y P \le e.$$

Then

$$\operatorname{ord}_{z=0} P(ae^z, be^{\tau z}) \le d + e. \tag{280}$$

The bound is attained by $(X-a)^d(Y-b)^e$, and equality determines that polynomial up to a nonzero rational scalar.

Proof. The total degree of P is at most $d+e$, so its multiplicity at (a, b) is at most $d+e$. Theorem P4.1 gives (280). The displayed product has multiplicity $d+e$, so the bound is sharp. If equality holds, the shifted polynomial $P(a+U, b+V)$ contains no term of total degree below $d+e$. Since its U-degree is at most d and its V-degree at most e, the only possible term of total degree $d+e$ is a scalar multiple of U^dV^e. □

Corollary P4.4 (No nondegenerate rational Padé contact)**.** *Let $A, B \in \mathbb{Q}[t]$ have degree at most d. Then*

$$\operatorname{ord}_{t=1}\big(B(t)t^{\vartheta} - A(t)\big) \le d + 1. \tag{281}$$

Equality holds only when

$$A(t) = B(t) = c(t-1)^d$$

for some $c \in \mathbb{Q}^{\times}$. In particular, if A and B are coprime, $B(1) \neq 0$, and $A(1) = B(1)$, then the residual has order exactly one at $t = 1$.

Proof. Apply Theorem P4.3 to

$$P(X, Y) = B(X)Y - A(X),$$

which has bidegree at most $(d, 1)$. For the equality case, expand at $(1, 1)$:

$$P(1+U, 1+V) = B(1+U)V + \big(B(1+U) - A(1+U)\big).$$

Multiplicity $d+1$ forces $B(1+U) = cU^d$ and forces $B - A$, a polynomial of degree at most d, to vanish to order $d+1$; hence $A = B = c(t-1)^d$. If A, B are coprime and $B(1) \neq 0$, the derivative of the residual is

$$B(1)\vartheta + B'(1) - A'(1),$$

which is nonzero by transcendence of ϑ. □

Standard Padé approximants to $(1+z)^{\alpha}$ achieve high order because their coefficients are allowed to depend algebraically or rationally on the parameter α. When the coefficient field is fixed to $\mathbb{Q}$ and $\alpha = \vartheta$ is transcendental, even matching the first derivative at a rational point is impossible without inserting a common vanishing factor. This is not a defect of a particular Padé table; it is an exact field-of-coefficients obstruction.

P5 Rational Hermite–Padé approximation: area collapses to diameter

P5.1 Moment equations over $\mathbb{Q}$

Take

$$P(X,Y) = \sum_{0\le i\le d}\sum_{0\le j\le e} c_{ij}X^iY^j, \qquad c_{ij}\in\mathbb{Q}, \tag{282}$$

and restrict it to $(e^z, e^{\vartheta z})$:

$$F(z) = P(e^z, e^{\vartheta z}) = \sum_{i,j} c_{ij}e^{(i+j\vartheta)z}. \tag{283}$$

The k-th derivative at zero is

$$F^{(k)}(0) = \sum_{i,j} c_{ij}(i+j\vartheta)^k \tag{284}$$

$$= \sum_{r=0}^{k}\binom{k}{r}\vartheta^r\sum_{i,j} c_{ij}i^{k-r}j^r. \tag{285}$$

Because ϑ is transcendental, the scalar equation $F^{(k)}(0) = 0$ is equivalent to all $k+1$ rational moment equations

$$\sum_{i,j} c_{ij}i^{k-r}j^r = 0, \qquad 0\le r\le k. \tag{286}$$

Thus a single formal jet condition unfolds into an entire homogeneous layer of bivariate moment conditions over $\mathbb{Q}$.

Theorem P4.3 now gives the exact answer:

$$\max_{\substack{0\ne P\in\mathbb{Q}[X,Y]\\ \deg_X P\le d,\ \deg_Y P\le e}} \operatorname{ord}_{z=0} P(e^z, e^{\vartheta z}) = d+e. \tag{287}$$

The rectangle contains $(d+1)(e+1)$ coefficient slots, but rational coefficients extract only the sum of the side lengths as contact order.

P5.2 What changes over $\mathbb{Q}(\vartheta)$

Over the coefficient field $\mathbb{Q}(\vartheta)$, the frequencies $i+j\vartheta$ are distinct and the ordinary Vandermonde dimension count applies. With $M = (d+1)(e+1)$ unknown coefficients one can annihilate the first $M-1$ derivatives of (283); the M-th derivative cannot vanish as well because the full frequency Vandermonde is nonsingular. Therefore

$$\max_{\substack{0\ne P\in\mathbb{Q}(\vartheta)[X,Y]\\ \deg_X P\le d,\ \deg_Y P\le e}} \operatorname{ord}_{z=0} P(e^z, e^{\vartheta z}) = (d+1)(e+1)-1. \tag{288}$$

For a square $d = e = K-1$, the contrast is

$$\boxed{\begin{array}{rl} \text{coefficients in } \mathbb{Q}: & \operatorname{ord}\le 2K-2,\\ \text{coefficients in } \mathbb{Q}(\vartheta): & \operatorname{ord}\le K^2-1. \end{array}} \tag{289}$$

This is the *area-to-diameter collapse*. The coefficient count is quadratic, but the rational contact order is only linear.

The high-contact object over $\mathbb{Q}(\vartheta)$ is analytically legitimate but arithmetically useless for the present number-field argument: $\mathbb{Q}(\vartheta)$ is a transcendental function field, not a number field. It has no finite set of algebraic conjugates and no finite-degree algebraic norm to $\mathbb{Q}$; in particular there is no finite-rank number-field integer lattice on which the usual norm-of-a-nonzero-algebraic-integer contradiction can be based. Replacing ϑ by an algebraic parameter restores those tools, but then one must control the perturbation back to the transcendental value with enough precision to survive the rapidly growing coefficient height.

P5.3 Triangles, arbitrary support, and the one-logarithm deficit

For triangular support

$$\{(i,j) \in \mathbb{N}_0^2 : i + j \le L\},$$

there are $(L+1)(L+2)/2$ monomials, while rational local contact is at most L. The extremizer is simply $(X-1)^L$, or any homogeneous shifted polynomial of multiplicity L. More generally, a rational polynomial of total degree D has contact at most D at every positive rational point.

This local fact mirrors the global determinant ceiling of the inherited report. A $K \times K$ support contains K^2 monomials, but rational jet conditions behave as if the useful complexity were only the boundary length. In the semigroup determinant, quadratic support complexity buys only a linear scale in the logarithm of the nodes; here quadratic coefficient count buys only linear contact. These are two manifestations of the same field collapse: the irrational parameter supplies many distinct real frequencies but no algebraic coefficient field in which they can be canceled cheaply.

P6 Integer-valued and Müntz approximation

P6.1 Why the physical and normalized intervals pull in opposite directions

On a physical interval, integer coefficients have excellent arithmetic but terrible capacity.

Lemma P6.1 (No small low-degree integer polynomial on a long integral interval)**.** *Let $N \in \mathbb{N}$ and let $0 \neq P \in \mathbb{Z}[X]$ have degree $d < N$. Then*

$$\|P\|_{[N,2N]} \ge 1. \tag{290}$$

Proof. If $\|P\|_{[N,2N]} < 1$, then $P(k)$ is an integer of absolute value below one for each of the $N+1$ integers $k = N, N+1, \dots, 2N$. Hence all those values vanish, giving more roots than the degree. □

On the normalized interval $[1,2]$, analytic capacity is favorable, but substituting $X = m/2^T$ creates the denominator tax quantified by Theorem P3.2. This is a genuine capacity–integrality incompatibility: the scale on which approximation is efficient is not the scale on which the coefficient lattice is integral.

Integer-valued polynomials do not remove the conflict. If $G \in \mathrm{Int}(\mathbb{Z})$ has degree at most d, then the binomial-basis theorem gives

$$G(X) = \sum_{k=0}^{d} b_k \binom{X}{k}, \qquad b_k \in \mathbb{Z}, \tag{291}$$

so $d!G \in \mathbb{Z}[X]$. In the normalized arithmetic-resolution lemma one may therefore take $H = d!$, which makes the required threshold smaller, not larger.

There is also a global mismatch of targets. By the classical uniform-distribution theorem for nonintegral powers, the sequence $\{n^\vartheta\}$ is uniformly distributed modulo one [7]. Hence for all sufficiently large dyadic intervals there is an integer n with

$$\mathrm{dist}(n^\vartheta, \mathbb{Z}) \geq \frac{1}{4}.$$

For every integer-valued polynomial G,

$$|n^\vartheta - G(n)| \geq \frac{1}{4}$$

at such an input. Thus uniform integer-valued approximation spends most of its effort on the noncandidate integers, where no exact arithmetic identity is available. The candidate set is too sparse for global integer-valued approximation to be the correct object.

P6.2 An exact arithmetic Müntz certificate

The mixed exponents already present in the determinant method are better adapted to candidate integrality. Let $S \subset \mathbb{N}_0^2$ be finite, and for $c = (c_{ij})_{(i,j)\in S} \in \mathbb{Z}^S$ define

$$F_c(x) := \sum_{(i,j)\in S} c_{ij} x^{i+j\vartheta}. \tag{292}$$

At a candidate m, with $n = m^\vartheta \in \mathbb{Z}$,

$$F_c(m) = \sum_{(i,j)\in S} c_{ij} m^i n^j \in \mathbb{Z}. \tag{293}$$

Theorem P6.2 (Arithmetic Müntz certificate)**.** *Let $S \subset \mathbb{N}_0^2$ have r elements, and let $0 \neq c \in \mathbb{Z}^S$. If for some interval $I \subset (0, \infty)$*

$$\|F_c\|_I < 1, \tag{294}$$

then

$$\#(\mathcal{C} \cap I) \leq r - 1. \tag{295}$$

Proof. At every candidate in I, equations (293) and (294) force $F_c(m) = 0$. The exponents $i + j\vartheta$ are distinct because ϑ is irrational. The functions x^λ with distinct real exponents form an extended complete Chebyshev system on $(0, \infty)$, so a nonzero combination with r terms has at most $r - 1$ positive zeros. □

The theorem is exactly the approximation-theoretic dual of the generalized-Vandermonde method. A determinant lower bound says that evaluations of the mixed monomials at too many candidate nodes cannot be linearly dependent over the integers. Theorem P6.2 says that a short integer vector in the uniform evaluation norm would create precisely such a dependence.

The unresolved problem is therefore a shortest-vector problem in an anisotropic function lattice:

$$\text{find } 0 \neq c \in \mathbb{Z}^S \quad \text{with} \quad \sup_{x\in[X,2X]} \left| \sum_{(i,j)\in S} c_{ij} x^{i+j\vartheta} \right| < 1. \tag{296}$$

Ordinary Remez theory minimizes over real coefficient vectors. It does not control the integral coefficient lattice or the finite-prime size of the resulting vector. The recent work of Protasov and Kamalov develops generalized alternance and Remez algorithms for arbitrary finite systems, constraints, complex exponentials, and lacunary polynomials [39]; it is well suited to numerical reconnaissance for (296), but an arithmetic transference theorem is still needed before its output can enter a product-formula proof.

P6.3 A convex-body formulation

Choose $M \geq |S|$ distinct sample points $x_1, \ldots, x_M \in [X, 2X]$, and form the evaluation matrix

$$V_{\ell,(i,j)} = x_\ell^{i+j\vartheta}.$$

Distinct positive nodes make V full column rank, so the set

$$\mathcal{K}(\varepsilon) = \{c \in \mathbb{R}^S : |Vc|_\infty \leq \varepsilon\}$$

is a bounded symmetric convex body. Minkowski's theorem can produce a nonzero integer point only when its volume is large relative to the coefficient lattice. But the singular values of V encode the same generalized-Vandermonde determinants already estimated in the report. Using more sample points improves uniform control while shrinking the convex body; using fewer points enlarges the body but loses control between samples. Any successful transference must therefore exploit special prime divisibility or coefficient sparsity not visible in the real condition number alone.

A concrete computational program is nevertheless available:

1. use generalized Remez iterations to identify supports S and real minimax sign patterns with unusually small norm;
2. solve the corresponding closest-vector problem in $\mathbb{Z}^S$ using exact interval evaluations;
3. compute the complete content and local Newton polygons of the coefficient vector;
4. accept a candidate only if its archimedean gain survives the product of all coefficient heights and denominator factors.

This is substantially narrower than a blind search over auxiliary polynomials.

P7 Rational approximation and the residual-height barrier

P7.1 Why a better quotient is not yet a better arithmetic object

Rational approximation can converge much faster than polynomial approximation when poles are placed near a branch singularity. For the present problem, however, Lemma P2.3 shows that the decisive object is

$$R_{A,B}(t) := B(t)t^{\vartheta} - A(t),$$

not $t^{\vartheta} - A(t)/B(t)$. To force equality at a candidate in the T-th block one needs

$$\|R_{A,B}\|_{[1,2]} < 3^{-T}2^{-Td}$$

for integral A, B of degree at most d, with an additional common-denominator factor for rational coefficients. A small quotient error is useful only if B remains small in norm and height at the same time.

The zero count also remains linear. The residual is a generalized polynomial with exponents

$$0, 1, \ldots, d, \qquad \vartheta, \vartheta + 1, \ldots, \vartheta + d.$$

Unless it is identically zero, it has at most $2d + 1$ positive zeros. Thus a residual meeting the arithmetic threshold would give a block count bound of $2d + 1$. This can improve the constant relative to ordinary polynomials only if the numerator residual converges at a rate which survives both $\|B\|$ and coefficient height.

P7.2 Local Padé versus nonlocal rational approximation

Corollary P4.4 rules out high-order rational Padé contact over $\mathbb{Q}$ at a rational point. It does not rule out a nonlocal rational minimax approximant with many alternating extrema. The distinction is important:

- *local route:* exact Taylor cancellation requires coefficients in $\mathbb{Q}(\vartheta)$, and rational coefficients collapse to first-order contact after common factors are removed;
- *global route:* rational coefficients can approximate real minimax coefficients arbitrarily well, but the common denominator and numerator residual must be included in the arithmetic inequality;
- *adelic route:* one seeks coefficients chosen simultaneously for small real norm and favorable p-adic divisibility. This is the only version not already blocked by the local theorem or the normalization tax.

The appropriate target is therefore not a classical Padé table. It is a rational or integral residual with a product-formula budget:

$$\log \|B(t)t^{\vartheta} - A(t)\|_{[1,2]} + \sum_{p} \mathcal{H}_p(A, B) < -T \log 3 - Td \log 2, \tag{297}$$

where $\mathcal{H}_p$ records exactly the loss incurred when evaluations are cleared at candidate points. The inherited min-plus ceiling proves that termwise valuation minima cannot supply the missing amount. Any gain in (297) must come from cancellation among terms or from the boundary residuals of the multiplicative interpolant.

P8 Consequences for the inherited grid and boundary residuals

The third continuation constructed the degree-K^2-1 graph interpolant Q_K on the conditional $K \times K$ multiplicative grid and factored its dyadic and triadic dilation defects as

$$Q_K(2X) - MQ_K(X) = \Pi_{K-1,K}(X)U_K(X),$$

$$Q_K(3X) - AQ_K(X) = \Pi_{K,K-1}(X)V_K(X),$$

with $\deg U_K = \deg V_K = K-1$. A hypothetical least generator β is transcendental by the inherited results, so Theorem P4.1 applies here with $\tau = \beta$. It adds three exact pieces of information.

1. Every interpolation node is a simple zero of $X^\beta - Q_K(X)$, because Q_K has rational coefficients and each node has rational coordinates. No squared grid divisor can be recovered from hidden tangency.
2. More generally, if a rational auxiliary polynomial $P(X,Y)$ vanishes to algebraic multiplicity s at a grid point, then its restriction to the irrational-power curve has exactly order s. Confluent conditions do not acquire any bonus from the curve parameter; each derivative condition must be paid for by ordinary local multiplicity.
3. Therefore any extra divisibility of U_K or V_K is necessarily global. It must come from cancellation among boundary interpolation data, from a common arithmetic content, or from a local Newton-polygon phenomenon. It cannot be explained by an analytic high-contact zero inherited from the interior grid.

This focuses the proposed boundary computation. For moderate K, one should clear Q_K, U_K, V_K primitively and record:

$$\text{gcd of coefficients}, \qquad v_p\text{-Newton polygons}, \qquad \operatorname{Res}(U_K, V_K), \qquad \operatorname{Res}(U_K, L_2^\pm L_3^\pm), \tag{298}$$

for the primes dividing $2 \cdot 3 \cdot MA$ and for the large primes appearing in the node differences. A systematic excess over the termwise min-plus prediction would be genuine new information. Absence of such excess would retire the boundary route just as decisively as Theorem P3.2 retires generic normalized polynomial approximation.

P9 Recent literature, 2024–2026

The literature search was updated through August 21, 2026, with emphasis on primary sources and arXiv revisions. No paper located claims a proof of the fixed-base Alaoglu–Erdős conjecture. The relevant developments sharpen auxiliary-function methods around it but retain algebraicity hypotheses absent here.

P9.1 Matrix coefficients and logarithm matrices

Dasgupta and Kakde's Matrix Coefficient Conjecture asks that a singular matrix of logarithms of algebraic numbers acquire a zero coefficient after rational changes of basis [37]. Their paper appeared in *Advances in Mathematics* in 2026. As proved in the third continuation, the present conjecture is exactly a positive-integral, fixed-first-column instance in dimension two. Polynomial approximation does not bypass that identification: a successful arithmetic minimax construction would amount to an auxiliary object strong enough to prove this fixed-base matrix-coefficient case.

P9.2 Constrained Chebyshev and Remez algorithms

Protasov and Kamalov's revised 2025 paper develops best uniform approximation for arbitrary finite systems, including non-Chebyshev systems and linear constraints, by a generalized alternance criterion and modified Remez iteration [39]. Their examples include complex exponential systems and lacunary polynomial families. This is directly useful for exploring real minimax versions of (296). The missing ingredient is arithmetic: the algorithm optimizes over a real affine space, whereas the conjecture requires an integral coefficient vector whose complete height and p-adic content survive the final inequality.

P9.3 Hermite–Padé constructions with algebraic parameters

Fürnsinn and Pannier give an effective version of the order-one p-curvature theorem by making a Chudnovsky Hermite–Padé argument explicit [40]. Their construction uses approximants to consecutive powers of a binomial function with an algebraic parameter, coefficients in the corresponding algebraic field, and a norm from a splitting field to $\mathbb{Q}$. The algebraic norm is the final discreteness mechanism. For the present parameter ϑ, Theorem P4.3 shows what happens if one insists on rational coefficients, while (288) shows that the high-contact coefficients naturally lie in the transcendental field $\mathbb{Q}(\vartheta)$, where the norm argument no longer exists.

Fischler and Rivoal obtain new transcendence measures for e^{α} at algebraic arguments by optimizing an accessory parameter in Siegel's determinant method with Hermite–Padé approximants to powers of the exponential function [41]. Their work is a strong demonstration that auxiliary-parameter optimization can materially improve exponents. It does not apply directly here: a hypothetical counterexample forces the relevant exponent and logarithmic products into the transcendental, bilinear regime rather than the algebraic argument regime.

Their 2024 work on logarithms of E-function values proves transcendence and, in rational strict cases, quantitative irrationality for many values [42]. Faverjon and Adamczewski establish Liouville-type polynomial inequalities for values of arbitrary E-functions at nonzero algebraic points and analogous M-function results [43]. These papers enlarge the class of special values admitting quantitative lower bounds, but they do not supply lower bounds for the bilinear logarithm relation

$$\log 2 \log A - \log 3 \log M = 0$$

or for rational-coefficient Hermite–Padé cancellation involving the transcendental ratio ϑ.

P9.4 Arithmetic holonomy and effective approximation on $\mathbb{G}_m$

Calegari, Dimitrov, and Tang propose arithmetic holonomy bounds yielding effective irrationality and linear-independence measures on the projective line and the multiplicative group, with applications to high roots of algebraic numbers and classical Diophantine equations [44]. This is relevant methodologically because it packages analytic continuation and arithmetic height into a single effective framework. Its stated applications still begin with algebraic data in a number field or a finitely generated algebraic multiplicative group. A useful future test is whether the boundary residuals U_K, V_K can be embedded into such an algebraic holonomy problem before the transcendental parameter is introduced. No such embedding is presently known.

P9.5 Literature comparison table

Source	What it supplies	Why it does not presently close the conjecture
Dasgupta–Kakde	Structural conjecture and support-based route for singular logarithm matrices.	The fixed-base 2×2 instance is essentially the present problem.
Protasov–Kamalov	Generalized alternance and Remez algorithms for arbitrary finite systems and constraints.	No integral coefficient lattice, denominator height, or product-formula control.
Fürnsinn–Pannier	Effective Hermite–Padé plus algebraic field norm.	High-contact coefficients and norm require an algebraic parameter; ϑ is transcendental.
Fischler–Rivoal	Improved transcendence measures using optimized Hermite–Padé determinants; logarithms of many E-function values.	Arguments concern algebraic inputs and do not control bilinear products of logarithms.
Faverjon–Adamczewski	Polynomial Liouville inequalities for E- and M-function values at algebraic points.	The candidate relation is not presently representable as such a polynomial value with algebraic input.
Calegari–Dimitrov–Tang	Effective arithmetic holonomy and Diophantine approximation on $\mathbb{P}^1$ and $\mathbb{G}_m$.	Current framework remains algebraic; no fixed-base bilinear logarithm specialization is known.

P10 A focused approximation-theoretic research program

The negative theorems above do not eliminate approximation theory. They identify which version could still matter.

P10.1 Priority one: arithmetic minimax in a Müntz lattice

For supports S adapted to the conditional grid, solve the exact optimization problem

$$\lambda_1(S;X) := \min_{0 \neq c \in \mathbb{Z}^S} \left\| \sum_{(i,j)\in S} c_{ij} x^{i+j\vartheta} \right\|_{[X,2X]}.$$

The target is $\lambda_1(S;X) < 1$ with $|S|$ smaller than the conditional block count. A proof needs both an upper construction and a nonvanishing certificate for the coefficient vector. Real Remez output should be treated only as a guide to the active alternation set; integer-relation or lattice reduction must be followed by rigorous interval verification and full coefficient-height accounting.

Supports should not be chosen only by cardinality. Three families deserve separate tests:

1. thin rectangles, which maximize distinct mixed frequencies for a fixed side length;
2. boundary strips, suggested by the residuals U_K, V_K;
3. valuation-adapted staircases whose monomials share large powers of two or three at the conditional nodes.

The inherited termwise min-plus theorem predicts failure for generic shapes. The search is specifically for exceptional cancellation beyond that prediction.

P10.2 Priority two: rational residuals with controlled denominator polynomial

Search directly for primitive $A, B \in \mathbb{Z}[t]$ minimizing

$$\|B(t)t^{\vartheta} - A(t)\|_{[1,2]}$$

subject to simultaneous bounds on $\|B\|_{[1,2]}$, coefficient height, and local contents. The comparison object is the exact threshold $3^{-T}2^{-Td}$, not the quotient error. Candidate supports found by rational Remez or Zolotarev-type experiments must be rejected unless their numerator residual survives this threshold after clearing every coefficient.

A particularly sharp test is to require B to be nonvanishing on $[1, 2]$ and primitive, then compare

$$-\log \|Bt^{\vartheta} - A\|, \qquad dT \log 2 + T \log 3 + \log H(A, B).$$

Only a positive asymptotic gap is relevant. Constant-factor improvements in quotient error are immaterial if coefficient height grows at the same exponential rate.

P10.3 Priority three: algebraic-parameter deformation

Choose algebraic α close to ϑ, construct Hermite–Padé approximants over $K = \mathbb{Q}(\alpha)$, take norms, and then estimate the perturbation from α to ϑ. The local theorem predicts the central tradeoff. Contact of quadratic size is available over K, but the height and degree of α must grow as the approximation improves. A viable deformation theorem would need an inequality of the form

$$\text{analytic gain from contact} > \text{number-field norm cost} + \text{parameter perturbation cost.} \tag{299}$$

No current algebraic approximation theorem for ϑ gives such a balance, but (299) is precise enough for experimental optimization.

P10.4 Priority four: boundary cancellation before approximation

Compute the primitive boundary residuals first, then approximate them, rather than approximating x^{ϑ} directly. They already contain the complete interior grid factor, so any remaining smallness is concentrated in $O(K)$ coefficients. If their local Newton polygons show systematic excess, use those valuations as constraints in a generalized Remez or Hermite–Padé problem. If not, the calculation supplies a rigorous reason to stop investing in that branch.

P10.5 Formalization targets

The following new statements are suitable for Lean formalization without deep new libraries:

1. Lemma P2.1 and Lemma P2.3, which are exact denominator-clearing identities;
2. Theorem P3.1, using the existing higher finite-difference module;
3. the rational inequalities in Theorem P3.2;
4. Theorem P4.1, once multivariate polynomial multiplicity and formal power-series order are connected;
5. Theorem P6.2, reusing the generalized Chebyshev-system and exponential-polynomial zero-count interfaces already needed by the corpus.

The Bernstein-ellipse rate can remain a named analytic interface. The all-degree normalization-tax theorem does not depend on it.

P11 Conclusions of the fourth continuation

The Alaoglu–Erdős conjecture remains open. Polynomial approximation does not presently supply the missing proof, but the exploration resolves several ambiguities that remained in the inherited report.

First, the most obvious uniform strategy is now closed exactly. After normalizing a dyadic block to $[1, 2]$, a degree-d rational polynomial with denominator H must approximate t^{ϑ} to better than $(H3^{T}2^{Td})^{-1}$ before integrality can force equality. The finite-difference lower bound proves that no degree can meet this threshold for $T \geq 4$. This is stronger than saying that known estimates are insufficient: the desired inequality itself is false.

Second, avoiding normalization does not repair the zero-count method. Direct-scale polynomials need degree at least $0.460384\dots T$ by the explicit lower bound and asymptotically $0.623238\dots T$ by the sharp Bernstein ellipse. Under a hypothetical counterexample, the exact candidate count has slope below $1/29$. The approximation degree therefore exceeds the population to be contradicted by more than an order of magnitude. Interpolating through all trivial powers saturates the zero budget, but the expanding interval has Bernstein parameter tending exponentially to one, destroying analytic control.

Third, local Padé theory over rational coefficients collapses completely. At every positive rational point, contact with the irrational-power curve equals ordinary algebraic multiplicity. A $K \times K$ rational coefficient rectangle has maximal contact $2K - 2$, while allowing coefficients in $\mathbb{Q}(\vartheta)$ restores $K^2 - 1$. The high-contact field is transcendental and has no algebraic norm. This exact area-to-diameter collapse explains why importing Hermite–Padé machinery from algebraic-parameter problems cannot be done formally by substitution.

Finally, the viable approximation-theoretic object is no longer vague. It is an integral Müntz polynomial or a rational numerator residual which is small in the real norm and simultaneously favorable at every finite prime. Theorem P6.2 shows exactly what such an object would prove. The inherited determinant ceilings and the new local rigidity theorem show exactly where ordinary real optimization loses the arithmetic battle. The remaining opportunity is global adelic cancellation, most plausibly in the low-degree boundary residuals of the multiplicative grid.

Bottom line. No complete proof was found. The continuation contributes three exact advances: an all-degree normalization-tax obstruction for rational polynomial approximation; a local theorem that rational contact with $Y = X^{\vartheta}$ equals algebraic multiplicity; and an arithmetic Müntz certificate dual to the semigroup determinant. Together they retire generic polynomial and rational Padé approaches and reduce the approximation program to one precise open task: construct an auxiliary residual whose archimedean smallness comes from genuine adelic cancellation rather than from real minimax approximation alone.

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